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RAVI MATHS TUITION CENTER, NEAR VILLIVAKKAM RLY STATION, CHENNAI – 82.
WHATSAPP - 8056206308

MCQ BB

Date : 01-Jan-70

12th Standard

Maths

Exam Time : 04:10:00 Hrs

250 x 1 = 250

1) If |adj(adj A)| = $|A|^9$, then the order of the square matrix A is
(a) 3 (b) 4 (c) 2 (d) 5

2) If A is a 3 × 3 non-singular matrix such that $AA^T = A^TA$ and $B = A^{-1}A^T$, then BB^T =
(a) A (b) B (c) I (d) B^T

3) If A = $\begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$, B = adj A and C = 3A, then $\frac{|adjB|}{|C|}$ =
(a) $\frac{1}{3}$ (b) $\frac{1}{9}$ (c) $\frac{1}{4}$ (d) 1

4) If A $\begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$, then A =
(a) $\begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix}$ (c) $\begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 4 & -1 \\ 2 & 1 \end{bmatrix}$

5) If A = $\begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix}$, then 9I - A =
(a) A^{-1} (b) $\frac{A^{-1}}{2}$ (c) $3A^{-1}$ (d) $2A^{-1}$

6) If A = $\begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}$ and B = $\begin{bmatrix} 1 & 4 \\ 2 & 0 \end{bmatrix}$ then |adj (AB)| =
(a) -40 (b) -80 (c) -60 (d) -20

7) If P = $\begin{bmatrix} 1 & x & 0 \\ 1 & 3 & 0 \\ 2 & 4 & -2 \end{bmatrix}$ is the adjoint of 3 × 3 matrix A and |A| = 4, then x is
(a) 15 (b) 12 (c) 14 (d) 11

8) If A = $\begin{bmatrix} 3 & 1 & -1 \\ 2 & -2 & 0 \\ 1 & 2 & -1 \end{bmatrix}$ and A^{-1} = $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ then the value of a_{23} is
(a) 0 (b) -2 (c) -3 (d) -1

9) If A, B and C are invertible matrices of some order, then which one of the following is not true?
(a) adj A = $|A|A^-$ (b) adj(AB) = (adj A)(adj (c) det A^{-1} = (det (d) $(ABC)^{-1} = C^{-1}B^{-1}A^-$

1 B) A)$^{-1}$ 1

10) If $(AB)^{-1} = \begin{bmatrix} 12 & -17 \\ -19 & 27 \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$, then B^{-1} =

(a) $\begin{bmatrix} 2 & -5 \\ -3 & 8 \end{bmatrix}$ (b) $\begin{bmatrix} 8 & 5 \\ 3 & 2 \end{bmatrix}$ (c) $\begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 8 & -5 \\ -3 & 2 \end{bmatrix}$

11) If A^TA^{-1} is symmetric, then A^2 =

(a) A^{-1} (b) $(A^T)^2$ (c) A^T (d) $(A^{-1})^2$

12) If A is a non-singular matrix such that $A^{-1} = \begin{bmatrix} 5 & 3 \\ -2 & -1 \end{bmatrix}$, then $(A^T)^{-1}$ =

(a) $\begin{bmatrix} -5 & 3 \\ 2 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 5 & 3 \\ -2 & -1 \end{bmatrix}$ (c) $\begin{bmatrix} -1 & -3 \\ 2 & 5 \end{bmatrix}$ (d) $\begin{bmatrix} 5 & -2 \\ 3 & -1 \end{bmatrix}$

13) If A = $\begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ x & \frac{3}{5} \end{bmatrix}$ and $A^T = A^{-1}$, then the value of x is

(a) $\frac{-4}{5}$ (b) $\frac{-3}{5}$ (c) $\frac{3}{5}$ (d) $\frac{4}{5}$

14) If A = $\begin{bmatrix} 1 & \tan\frac{\theta}{2} \\ -\tan\frac{\theta}{2} & 1 \end{bmatrix}$ and AB = I , then B =

(a) $\left(\cos^2\frac{\theta}{2}\right)A$ (b) $\left(\cos^2\frac{\theta}{2}\right)A^T$ (c) $\left(\cos^2\theta\right)I$ (d) $(\text{Sin}_2\frac{\theta}{2})A$

15) If A = $\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ and A(adj A) = $\begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$ then adj (AB) is

(a) 0 (b) sin θ (c) cos θ (d) 1

16) If A = $\begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$ be such that λA^{-1} =A, then λ is

(a) 17 (b) 14 (c) 19 (d) 21

17) If adj A = $\begin{bmatrix} 2 & 3 \\ 4 & -1 \end{bmatrix}$ and adj B = $\begin{bmatrix} 1 & -2 \\ -3 & 1 \end{bmatrix}$ then adj (AB) is

(a) $\begin{bmatrix} -7 & -1 \\ 7 & -9 \end{bmatrix}$ (b) $\begin{bmatrix} -6 & 5 \\ -2 & -10 \end{bmatrix}$ (c) $\begin{bmatrix} -7 & 7 \\ -1 & -9 \end{bmatrix}$ (d) $\begin{bmatrix} -6 & -2 \\ 5 & -10 \end{bmatrix}$

18)

The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ -1 & -2 & -3 & -4 \end{bmatrix}$ is

(a) 1 (b) 2 (c) 4 (d) 3

19)

If $x^a y^b = e^m$, $x^c y^d = e^n$, $\Delta_1 = \begin{vmatrix} m & b \\ n & d \end{vmatrix}$, $\Delta_2 = \begin{vmatrix} a & m \\ c & n \end{vmatrix}$, $\Delta_3 = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$, then the values of x and y are respectively,

(a) $e^{(\Delta_2/\Delta_1)}$, $e^{(\Delta_3/\Delta_1)}$ (b) $\log(\Delta_1/\Delta_3)$, $\log(\Delta_2/\Delta_3)$ (c) $\log(\Delta_2/\Delta_1)$, $\log(\Delta_3/\Delta_1)$ (d) $e^{(\Delta_1/\Delta_3)}$, $e^{(\Delta_2/\Delta_3)}$

20) Which of the following is/are correct?

(i) Adjoint of a symmetric matrix is also a symmetric matrix.

(ii) Adjoint of a diagonal matrix is also a diagonal matrix.

(iii) If A is a square matrix of order n and λ is a scalar, then $\text{adj}(\lambda A) = \lambda^n \text{adj}(A)$.

(iv) A(adjA) = (adjA)A = |A| I

(a) Only (i) (b) (ii) and (iii) (c) (iii) and (iv) (d) (i), (ii) and (iv)

21) If ρ(A) = ρ([A | B]), then the system AX = B of linear equations is

(a) consistent and has a unique solution (b) consistent (c) consistent and has infinitely many solution (d) inconsistent

22) If $0 \le \theta \le \pi$ and the system of equations $x + (\sin\theta)y - (\cos\theta)z = 0$, $(\cos\theta)x - y + z = 0$, $(\sin\theta)x + y - z = 0$ has a non-trivial solution then θ is

(a) $\frac{2\pi}{3}$ (b) $\frac{3\pi}{4}$ (c) $\frac{5\pi}{6}$ (d) $\frac{\pi}{4}$

23)

The augmented matrix of a system of linear equations is $\begin{bmatrix} 1 & 2 & 7 & 3 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & \lambda-7 & \mu+5 \end{bmatrix}$. The system has infinitely many solutions if

(a) λ = 7, μ ≠ -5 (b) λ = 7, μ = 5 (c) λ ≠ 7, μ ≠ -5 (d) λ = 7, μ = -5

24)

Let A = $\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ and 4B = $\begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & x \\ -1 & 1 & 3 \end{bmatrix}$. If B is the inverse of A, then the value of x is

(a) 2 (b) 4 (c) 3 (d) 1

25)

If A = $\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$, then adj(adj A) is

(a) $\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 6 & -6 & 8 \\ 4 & -6 & 8 \\ 0 & -2 & 2 \end{bmatrix}$ (c) $\begin{bmatrix} -3 & 3 & -4 \\ -2 & 3 & -4 \\ 0 & 1 & -1 \end{bmatrix}$ (d) $\begin{bmatrix} 3 & -3 & 4 \\ 0 & -1 & 1 \\ 2 & -3 & 4 \end{bmatrix}$

26) $i^n+i^{n+1}+i^{n+2}+i^{n+3}$ is

(a) 0 (b) 1 (c) -1 (d) i

27) The value of $\sum_{i=1}^{13}\left(i^n + i^{n-1}\right)$ is

(a) 1+ i (b) i (c) 1 (d) 0

28) The area of the triangle formed by the complex numbers z,iz, and z+iz in the Argand's diagram is

(a) $\frac{1}{2}|z|^2$ (b) $|z|^2$ (c) $\frac{3}{2}|z|^2$ (d) $2|z|^2$

29) The conjugate of a complex number is $\frac{1}{i-2}$/Then the complex number is

(a) $\frac{1}{i+2}$ (b) $\frac{-1}{i+2}$ (c) $\frac{-1}{i-2}$ (d) $\frac{1}{i-2}$

30) If $z = \frac{\left(\sqrt{3}+i\right)^3(3i+4)^2}{(8+6i)^2}$, then |z| is equal to

(a) 0 (b) 1 (c) 2 (d) 3

31) If z is a non zero complex number, such that $2iz^2=\bar{z}$ then |z| is

(a) $\frac{1}{2}$ (b) 1 (c) 2 (d) 3

32) If $|z-2+i|\leq 2$, then the greatest value of |z| is

(a) $\sqrt{3}-2$ (b) $\sqrt{3}+2$ (c) $\sqrt{5}-2$ (d) $\sqrt{5}+2$

33) If $\left|z-\frac{3}{z}\right| = 2$ then the least value |z| is

(a) 1 (b) 2 (c) 3 (d) 5

34) If |z|=1, then the value of $\frac{1+z}{1+\bar{z}}$ is

(a) z (b) $\bar{z}$ (c) $\frac{1}{z}$ (d) 1

35) The solution of the equation |z|-z=1+2i is

(a) $\frac{3}{2} - 2i$ (b) $-\frac{3}{2} + 2i$ (c) $2 - \frac{3}{2}i$ (d) $2 + \frac{3}{2}i$

36) If $|z_1|=1, |z_2|=2|z_3|=3$ and $|9z_1z_2+4z_1z_3+z_2z_3|=12$, then the value of $|z_1+z_2+z_3|$ is
(a) 1 (b) 2 (c) 3 (d) 4

37)
If z is a complex number such that $z \varepsilon C/R$ and $z + \frac{1}{z} \epsilon R$ then|z| is

(a) 0 (b) 1 (c) 2 (d) 3

38) z_1, z_2 and z_3 are complex number such that $z_1+z_2+z_3=0$ and $|z_1|=|z_2|=|z_3|=1$ then $z_1^2+z_2^2+z_3^3$ is
(a) 3 (b) 2 (c) 1 (d) 0

39)
If $\frac{z-1}{z+1}$ is purely imaginary, then |z| is

(a) $\frac{1}{2}$ (b) 1 (c) 2 (d) 3

40) If z=x+iy is a complex number such that |z+2|=|z−2|, then the locus of z is
(a) real axis (b) imaginary axis (c) ellipse (d) circle

41)
The principal argument of $\frac{3}{-1+i}$

(a) $\frac{-5\pi}{6}$ (b) $\frac{-2\pi}{3}$ (c) $\frac{-3\pi}{4}$ (d) $\frac{-\pi}{2}$

42) The principal argument of $(\sin 40° + i \cos 40°)^5$ is
(a) −110° (b) −70° (c) 70° (d) 110°

43)
If (1+i)(1+2i)(1+3i)...(1+ni)=x+iy, then $2 \cdot 5 \cdot 10 \ldots \left(1+n^2\right)$ is

(a) 1 (b) i (c) x^2+y^2 (d) $1+n^2$

44) If $\omega \neq 1$ is a cubic root of unity and $(1+\omega)^7 = A + B\omega$,then (A,B) equals
(a) (1,0) (b) (−1,1) (c) (0,1) (d) (1,1)

45)
The principal argument of the complex number $\frac{\left(1+i\sqrt{3}\right)^2}{4i\left(1-i\sqrt{3}\right)}$ is

(a) $\frac{2\pi}{3}$ (b) $\frac{\pi}{6}$ (c) $\frac{5\pi}{6}$ (d) $\frac{\pi}{2}$

46) If α and β are the roots of $x^2+x+1=0$, then $\alpha^{2020} + \beta^{2020}$ is
(a) -2 (b) -1 (c) 1 (d) 2

47) The product of all four values of $\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)^{\frac{3}{4}}$ is

(a) -2 (b) -1 (c) 1 (d) 2

48) If $\omega \neq 1$ is a cubic root of unity and $\begin{vmatrix} 1 & 1 & 1 \\ 1 & -\omega^2 & \omega^2 \\ 1 & \omega^2 & \omega^2 \end{vmatrix}$ =3k, then k is equal to

(a) 1 (b) -1 (c) $\sqrt{3i}$ (d) $-\sqrt{3i}$

49) The value of $\left(\frac{1+3\sqrt{i}}{1-\sqrt{3i}}\right)^{10}$

(a) $cis\frac{2\pi}{3}$ (b) $cis\frac{4\pi}{3}$ (c) $-cis\frac{2\pi}{3}$ (d) $-cis\frac{4\pi}{3}$

50) If $\omega = cis\frac{2\pi}{3}$, then the number of distinct roots of $\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix}$

(a) 1 (b) 2 (c) 3 (d) 4

51) A zero of $x^3 + 64$ is

(a) 0 (b) 4 (c) 4i (d) -4

52) If f and g are polynomials of degrees m and n respectively, and if h(x) =(f 0 g)(x), then the degree of h is

(a) mn (b) m+n (c) m^n (d) n^m

53) A polynomial equation in x of degree n always has

(a) n distinct roots (b) n real roots (c) n imaginary roots (d) at most one root

54) If α,β and γ are the roots of x^3+px^2+qx+r, then $\Sigma\frac{1}{\alpha}$ is

(a) $-\frac{q}{r}$ (b) $\frac{p}{r}$ (c) $\frac{q}{r}$ (d) $-\frac{q}{p}$

55) According to the rational root theorem, which number is not possible rational root of $4x^7+2x^4-10x^3-5$?

(a) -1 (b) $\frac{5}{4}$ (c) $\frac{4}{5}$ (d) 5

56) The polynomial x^3-kx^2+9x has three real zeros if and only if, k satisfies

(a) $|k|\leq 6$ (b) k=0 (c) $|k| > 6$ (d) $|k|\geq 6$

57) The number of real numbers in [0,2π] satisfying $\sin^4x-2\sin^2x+1$ is

(a) 2 (b) 4 (c) 1 (d) ∞

58) If $x^3+12x^2+10ax+1999$ definitely has a positive zero, if and only if

(a) $a\geq 0$ (b) $a > 0$ (c) $a< 0$ (d) $a\leq 0$

59) The polynomial x^3+2x+3 has

(a) one negative and two (b) one positive and two (c) three real (d) no

real roots imaginary roots roots solution

60) The number of positive zeros of the polynomial $\Sigma_{j=0}^{n} n_{C_r}(-1)^r x^r$ is

(a) 0 (b) n (c) < n (d) r

61) The value of $\sin^{-1}(\cos x), 0 \le x \le \pi$ is

(a) $\pi - x$ (b) $x - \frac{\pi}{2}$ (c) $\frac{\pi}{2} - x$ (d) $\pi - x$

62) If $\sin^{-1} x + \sin^{-1} y = \frac{2\pi}{3}$; then $\cos^{-1}x + \cos^{-1} y$ is equal to

(a) $\frac{2\pi}{3}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{6}$ (d) π

63) $sin^{-1}\frac{3}{5} - cos^{-1}\frac{12}{13} + sec^{-1}\frac{5}{3} - cosec^{1-}\frac{13}{2}$ is equal to

(a) 2π (b) π (c) 0 (d) $\tan_{-1}\frac{12}{65}$

64) If $\sin^{-}1x = 2\sin^{-1}\alpha$ has a solution, then

(a) $|\alpha| \le \frac{1}{\sqrt{2}}$ (b) $|\alpha| \ge \frac{1}{\sqrt{2}}$ (c) $|\alpha| < \frac{1}{\sqrt{2}}$ (d) $|\alpha| > \frac{1}{\sqrt{2}}$

65) $\sin^{-1}(\cos x) = \frac{\pi}{2} - x$ is valid for

(a) $-\pi \le x \le 0$ (b) $0\pi \le x \le 0$ (c) $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$ (d) $-\frac{\pi}{4} \le x \le \frac{3\pi}{4}$

66) If $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$, the value of $x^{2017} + y^{2018} + z^{2019} - \frac{9}{x^{101} + y^{101} + z^{101}}$ is

(a) 0 (b) 1 (c) 2 (d) 3

67) If $\cot{-1}x = \frac{2\pi}{5}$ for some $x \in R$, the value of $\tan^{-1} x$ is

(a) $\frac{-\pi}{10}$ (b) $\frac{\pi}{5}$ (c) $\frac{\pi}{10}$ (d) $-\frac{\pi}{5}$

68) The domain of the function defined by $f(x) = \sin^{-1}\sqrt{x-1}$ is

(a) [1,2] (b) [-1,1] (c) [0,1] (d) [-1,0]

69) If $x = \frac{1}{5}$, the valur of $\cos(\cos^{-1}x + 2\sin^{-1}x)$ is

(a) $-\sqrt{\frac{24}{25}}$ (b) $\sqrt{\frac{24}{25}}$ (c) $\frac{1}{5}$ (d) $-\frac{1}{5}$

70) $tan^{-1}\left(\frac{1}{4}\right) + tan^{-1}\left(\frac{2}{3}\right)$ is equal to

(a) $\frac{1}{2}cos^{-1}\left(\frac{3}{5}\right)$ (b) $\frac{1}{2}sin^{-1}\left(\frac{3}{5}\right)$ (c) $\frac{1}{2}tan^{-1}\left(\frac{3}{5}\right)$ (d) $tan^{-1}\left(\frac{1}{2}\right)$

71) If the function $f(x)\sin^{-1}(x^2-3)$, then x belongs to

(a) [-1,1] (b) $[\sqrt{2},2]$ (c) $\left[-2, -\sqrt{2}\right] \cup \left[\sqrt{2}, 2\right]$ (d) $\left[-2, -\sqrt{2}\right] \cap \left[\sqrt{2}, 2\right]$

72) If $\cot^{-1} 2$ and $\cot^{-1} 3$ are two angles of a triangle, then the third angle is

(a) $\frac{\pi}{4}$ (b) $\frac{3\pi}{4}$ (c) $\frac{\pi}{6}$ (d) $\frac{\pi}{3}$

73) $sin^{-1}\left(tan\frac{\pi}{4}\right) - sin^{-1}\left(\sqrt{\frac{3}{x}}\right) = \frac{\pi}{6}$.Then x is a root of the equation

(a) $x^2-x-6=0$ (b) $x^2-x-12=0$ (c) $x^2+x-12=0$ (d) $x^2+x-6=0$

74) $\sin^{-1}(2\cos^2x-1)+\cos^{-1}(1-2\sin^2x)=$

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$

75) If $cot^{-1}\left(\sqrt{sin\alpha}\right) + tan^{-1}\left(\sqrt{sin\alpha}\right) = u$, then cos2u is equal to

(a) $\tan^2\alpha$ (b) 0 (c) -1 (d) $\tan 2\alpha$

76) If $|x| \leq 1$, then $2\tan^{-1} x-\sin^{-1} \frac{2x}{1+x^2}$ is equal to

(a) $\tan^{-1}x$ (b) $\sin^{-1}x$ (c) 0 (d) π

77) The equation $\tan^{-1} x-\cot^{-1} x=\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$has

(a) no solution (b) unique solution (c) two solutions (d) infinite number of solutions

78) If $\sin^{-1} x+\cot^{-1}(\frac{1}{2}) = \frac{\pi}{2}$, then x is equal to

(a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{5}}$ (c) $\frac{2}{\sqrt{5}}$ (d) $\frac{\sqrt{3}}{2}$

79) If $\sin^{-1} \frac{x}{5} + cosec^{-1}\frac{5}{4} = \frac{\pi}{2}$, then the value of x is

(a) 4 (b) 5 (c) 2 (d) 3

80) $\sin(\tan^{-1}x)$, $|x|< 1$ ia equal to

(a) $\frac{x}{\sqrt{1-x^2}}$ (b) $\frac{1}{\sqrt{1-x^2}}$ (c) $\frac{1}{\sqrt{1+x^2}}$ (d) $\frac{x}{\sqrt{1+x^2}}$

81) The equation of the circle passing through(1,5) and (4,1) and touching y -axis is $x^2+y^2-5x-6y+9+(4x+3y-19)=0$ wherelλ is equal to

(a) $0, -\frac{40}{9}$ (b) 0 (c) $\frac{40}{9}$ (d) $\frac{-40}{9}$

82) The eccentricity of the hyperbola whose latus rectum is 8 and conjugate axis is equal to half the distance between the foci is

(a) $\frac{4}{3}$ (b) $\frac{4}{\sqrt{3}}$ (c) $\frac{2}{\sqrt{3}}$ (d) $\frac{3}{2}$

83) The circle $x^2+y^2=4x+8y+5$intersects the line$3x-4y=m$ at two distinct points if

(a) $15< m < 65$ (b) $35< m < 85$ (c) $-85< m < -35$ (d) $-35< m < 15$

84) The length of the diameter of the circle which touches the x -axis at the point (1,0) and passes through the point (2,3) .

(a) $\frac{6}{5}$ (b) $\frac{5}{3}$ (c) $\frac{10}{5}$ (d) $\frac{3}{5}$

85) The radius of the circle$3x^2+by^2+4bx-6by+b^2 =0$ is

(a) 1 (b) 3 (c) $\sqrt{10}$ (d) $\sqrt{11}$

86) The centre of the circle inscribed in a square formed by the lines $x^2-8x-12=0$ and $y^2-14y+45 = 0$ is

(a) (4,7) (b) (7,4) (c) (9,4) (d) (4,9)

87) The equation of the normal to the circle $x^2+y^2-2x-2y+1=0$ which is parallel to the line $2x+4y=3$ is

(a) $x+2y=3$ (b) $x+2y+3=0$ (c) $2x+4y+3=0$ (d) $x-2y+3=0$

88) If P(x, y) be any point on $16x^2+25y^2=400$ with foci F1 (3,0) and F2 (-3,0) then PF_1 PF_2 +
is

(a) 8 (b) 6 (c) 10 (d) 12

89) The radius of the circle passing through the point(6,2) two of whose diameter arex+y=6
and x+2y=4 is

(a) 10 (b) $2\sqrt{5}$ (c) 6 (d) 4

90) The area of quadrilateral formed with foci of the hyperbolas $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ and $\frac{x^2}{a^2}-\frac{y^2}{b^2}=-1$

(a) $4(a^2+b^2)$ (b) $2(a^2+b^2)$ (c) a^2+b^2 (d) $\frac{1}{2}(a_2+b_2)$

91) If the normals of the parabola $y^2 = 4x$ drawn at the end points of its latus rectum are tangents to the circle $(x-3)^2+(y+2)^2=r^2$, then the value of r^2 is

(a) 2 (b) 3 (c) 1 (d) 4

92) If x+y=k is a normal to the parabola $y^2 =12x$, then the value of k is

(a) 3 (b) -1 (c) 1 (d) 9

93) The ellipse E1 $\frac{x^2}{9}+\frac{y^2}{4}=1$ is inscribed in a rectangle R whose sides are parallel to the coordinate axes. Another ellipse E2 passing through the point(0,4) circumscribes the rectangle R . The eccentricity of the ellipse is

(a) $\frac{\sqrt{2}}{2}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{1}{2}$ (d) $\frac{3}{4}$

94) Tangents are drawn to the hyperbola $\frac{x^2}{9}+\frac{y^2}{4}=1$ 1parallel to the straight line2x−y=1. One of the points of contact of tangents on the hyperbola is

(a) $\frac{9}{2\sqrt{2}}, \frac{-1}{\sqrt{2}}$ (b) $\frac{-9}{2\sqrt{2}}, \frac{1}{\sqrt{2}}$ (c) $\frac{9}{2\sqrt{2}}, \frac{1}{\sqrt{2}}$ (d) $\left(3\sqrt{3}, -2\sqrt{2}\right)$

95) The equation of the circle passing through the foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ 1having centre at
(0,3) is

(a) $x^2+y^2-6y-7=0$ (b) $x^2+y^2-6y+7=0$ (c) $x^2+y^2-6y-5=0$ (d) $x^2+y^2-6y+5=0$

96) Let C be the circle with centre at(1,1) and radius =1. If T is the circle centered at(0, y) passing through the origin and touching the circleC externally, then the radius of T is equal to

(a) $\frac{\sqrt{3}}{2}$ (b) $\frac{\sqrt{3}}{\sqrt{2}}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

97) Consider an ellipse whose centre is of the origin and its major axis is along x-axis. If its eccentrcity is $\frac{3}{5}$ and the distance between its foci is 6, then the area of the

quadrilateral inscribed in the ellipse with diagonals as major and minor axis of the ellipse is

(a) 8 (b) 32 (c) 80 (d) 40

98) Area of the greatest rectangle inscribed in the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1.$ is

(a) 2ab (b) ab (c) $\sqrt{ab}$ (d) $\frac{a}{b}$

99) An ellipse hasOB as semi minor axes, F and F′ its foci and the angle FBF′ is a right angle. Then the eccentricity of the ellipse is

(a) $\frac{1}{\sqrt{2}}$ (b) $\frac{1}{2}$ (c) $\frac{1}{4}$ (d) $\frac{1}{\sqrt{3}}$

100) The eccentricity of the ellipse $(x-3)^2+(y-4)^2=\frac{y^2}{9}$ is

(a) $\frac{\sqrt{3}}{2}$ (b) $\frac{1}{3}$ (c) $\frac{1}{3\sqrt{2}}$ (d) $\frac{1}{\sqrt{3}}$

101) If the two tangents drawn from a point P to the parabolay2 = 4x are at right angles then the locus of P is

(a) 2x+1=0 (b) x = −1 (c) 2x−1=0 (d) x =1

102) The circle passing through(1,-2) and touching the axis of x at (3,0) passing through the point

(a) (-5,2) (b) (2,-5) (c) (5,-2) (d) (-2,5)

103) The locus of a point whose distance from (-2,0) is $\frac{2}{3}$ times its distance from the line x = $\frac{-9}{2}$ is

(a) a parabola (b) a hyperbola (c) an ellipse (d) a circle

104) The values of m for which the line y=mx+ $2\sqrt{5}$ touches the hyperbola $16x^2-9y^2=144$ are the roots of $x^2-(a+b)x-4=0$, then the value of (a+b) is

(a) 2 (b) 4 (c) 0 (d) -2

105) If the coordinates at one end of a diameter of the circle $x^2+y^2-8x-4y+c=0$ are (11,2) ,
the coordinates of the other end are

(a) (-5,2) (b) (2,-5) (c) (5,-2) (d) (-2,5)

106) If $\vec{a}$ and $\vec{b}$ are parallel vectors, then $[\vec{a},\vec{c},\vec{b}]$ is equal to

(a) 2 (b) -1 (c) 1 (d) 0

107) If a vector $\vec{\alpha}$ lies in the plane of $\vec{\beta}$ and $\vec{\gamma}$, then

(a) $[\vec{\alpha},\vec{\beta},\vec{\gamma}]$ = 1 (b) $[\vec{\alpha},\vec{\beta},\vec{\gamma}]$ = -1 (c) $[\vec{\alpha},\vec{\beta},\vec{\gamma}]$ = 0 (d) $[\vec{\alpha},\vec{\beta},\vec{\gamma}]$ = 2

108) $\vec{a}.\vec{b}=\vec{b}.\vec{c}=\vec{c}.\vec{a}=0$, then the value of $[\vec{a},\vec{b},\vec{c}]$ is

(a) $|\vec{a}||\vec{b}||\vec{c}|$ (b) $\frac{1}{3}|\vec{a}||\vec{b}||\vec{c}|$ (c) 1 (d) -1

109) If $\vec{a},\vec{b},\vec{c}$ are three unit vectors such that $\vec{a}$ is perpendicular to $\vec{b}$ and is parallel to $\vec{c}$ then $\vec{a}\times(\vec{b}\times\vec{c})$ is equal to

(a) $\vec{a}$ (c) $\vec{c}$

(b) $\vec{b}$ (d) $\vec{0}$

110) If $[\vec{a}, \vec{b}, \vec{c}] = 1$, $\frac{\vec{a}.(\vec{b}\times\vec{c})}{(\vec{c}\times\vec{a}).\vec{b})} + \frac{\vec{b}.(\vec{c}\times\vec{a})}{(\vec{a}\times\vec{b}).\vec{c}} + \frac{\vec{c}.(\vec{a}\times\vec{b})}{(\vec{c}\times\vec{b}).\vec{a}}$ is

(a) 1 (b) -1 (c) 2 (d) 3

111) The volume of the parallelepiped with its edges represented by the vectors $\hat{i}+\hat{j}, \hat{i}+2\hat{j}, \hat{i}+\hat{j}+\pi\hat{k}$ is

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$ (c) π (d) $\frac{\pi}{4}$

112) If $\vec{a}$ and $\vec{b}$ are unit vectors such that $[\vec{a}, \vec{b}, \vec{a}\times\vec{b}] = \frac{\pi}{4}$, then the angle between $\vec{a}$ and $\vec{b}$ is

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

113) If $\vec{a} = \hat{i}+\hat{j}+\hat{k}, \vec{b} = \hat{i}+\hat{j}, \vec{c} = \hat{i}$ and $(\vec{a}\times\vec{b})\times\vec{c} = \lambda\vec{a}+\mu\vec{b}$ then the value of $\lambda+\mu$ is

(a) 0 (b) 1 (c) 6 (d) 3

114) If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar, non-zero vectors such that $[\vec{a}, \vec{b}, \vec{c}] = 3$, then $\{[\vec{a}\times\vec{b}, \vec{b}\times\vec{c}, \vec{c}\times\vec{a}]\}^2$ is equal to

(a) 81 (b) 9 (c) 27 (d) 18

115) If $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar vectors such that $\vec{a}\times(\vec{b}\times\vec{c}) = \frac{\vec{b}+\vec{c}}{\sqrt{2}}$, then the angle between

(a) $\frac{\pi}{2}$ (b) $\frac{3\pi}{6}$ (c) $\frac{\pi}{4}$ (d) π

116) If the volume of the parallelepiped with $\vec{a}\times\vec{b}, \vec{b}\times\vec{c}, \vec{c}\times\vec{a}$ as coterminous edges is 8 cubic units, then the volume of the parallelepiped with $(\vec{a}\times\vec{b})\times(\vec{b}\times\vec{c}), (\vec{b}\times\vec{c})\times(\vec{c}\times\vec{a})$ and $(\vec{c}\times\vec{a})\times(\vec{a}\times\vec{b})$as coterminous edges is,

(a) 8 cubic units (b) 512 cubic units (c) 64 cubic units (d) 24 cubic units

117) Consider the vectors $\vec{a}, \vec{b}, \vec{c}, \vec{c}$ such that $(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d}) = \vec{0}$ Let P_1 and P_2 be the planes determined by the pairs of vectors $\vec{a}, \vec{b}$ and $\vec{c}, \vec{d}$ respectively. Then the angle between P_1 and P_2 is

(a) 0° (b) 45° (c) 60° (d) 90°

118) If $\vec{a}\times(\vec{b}\times\vec{c}) = (\vec{a}\times\vec{b})\times\vec{c}$ where $\vec{a}, \vec{b}, \vec{c}$ are any three vectors such that $\vec{a}, \vec{b} \neq 0$ and $\vec{a}.\vec{b} \neq 0$ then $\vec{a}$ and $\vec{c}$ are

(a) perpendicular (b) parallel (c) inclined at an angle $\frac{\pi}{3}$ (d) inclined at an angle $\frac{\pi}{6}$

119) If $\vec{a} = 2\hat{i}+3\hat{j}-\hat{k}, \vec{b} = \hat{i}+2\hat{j}+5\hat{k}, \vec{c} = 3\hat{i}+5\hat{j}-\hat{k}$ then a vector perpendicular to $\vec{a}$ and lies in the plane containing $\vec{b}$ and $\vec{c}$ is

(a) $-17\hat{i}+21\hat{j}-97\hat{k}$ (b) $17\hat{i}+21\hat{j}-123\hat{k}$ (c) $-17\hat{i}-21\hat{j}+197\hat{k}$ (d) $-17\hat{i}-21\hat{j}-197\hat{k}$

120) The angle between the lines $\frac{x-2}{3}=\frac{y+1}{-2}$, z=2 and $\frac{x-1}{1}=\frac{2y+3}{3}=\frac{z+5}{2}$

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

121) If the line $\frac{x-2}{3}=\frac{y-1}{-5}=\frac{x+2}{2}$ lies in the plane x + 3y + - αz + β = 0, then (α, β) is

(a) (-5, 5) (b) (-6, 7) (c) (5, 5) (d) (6, -7)

122) The angle between the line $\vec{r}=(\hat{i}+2\hat{j}-3\hat{k})+t(2\hat{i}+\hat{j}-2\hat{k})$ and the plane $\vec{r}.(\hat{i}+\hat{j})+4=0$ is

(a) 0° (b) 30° (c) 45° (d) 90°

123) The coordinates of the point where the line $\vec{r}=(6\hat{i}-\hat{j}-3\hat{k})+t(\hat{i}+4\hat{j})$ meets the plane $\vec{r}=(\hat{i}+\hat{j}-\hat{k})$ = 3 are

(a) (2,1,0) (b) (7,1,7) (c) (1,2,6) (d) (5,1,1)

124) Distance from the origin to the plane 3x - 6y + 2z 7 = 0 is

(a) 0 (b) 1 (c) 2 (d) 3

125) The distance between the planes x + 2y + 3z + 7 = 0 and 2x + 4y + 6z + 7 = 0

(a) $\frac{\sqrt{7}}{2\sqrt{2}}$ (b) $\frac{7}{2}$ (c) $\frac{\sqrt{7}}{2}$ (d) $\frac{7}{2\sqrt{2}}$

126) If the direction cosines of a line are $\frac{1}{c},\frac{1}{c},\frac{1}{c}$, then

(a) $c=\pm 3$ (b) $c=\pm\sqrt{3}$ (c) c > 0 (d) 0 < c < 1

127) The vector equation $\vec{r}=(\hat{i}-2\hat{j}-\hat{k})+t(6\hat{i}-\hat{k})$ represents a straight line passing through the points

(a) (0,6,1)− and (1,2,1) (b) (0,6,-1) and (1,4,2) (c) (1,-2,-1) and (1,4,-2) (d) (1,-2,-1) and (0,-6,1)

128) If the distance of the point (1,1,1) from the origin is half of its distance from the plane x + y + z + k =0, then the values of k are

(a) ± 3 (b) ± 6 (c) -3, 9 (d) 3, 9

129) If the planes $\vec{r}=(2\hat{i}-\lambda\hat{j}+\hat{k})=3$ and $\vec{r}=(4+\hat{j}-\mu\hat{k})=5$ are parallel, then the value of λ and μ are

(a) $\frac{1}{2},-2$ (b) $-\frac{1}{2},2$ (c) $-\frac{1}{2},-2$ (d) $\frac{1}{2},2$

130) If the length of the perpendicular from the origin to the plane 2x + 3y + λz =1, λ > 0 is $\frac{1}{5}$ then the value of λ is

(a) $2\sqrt{3}$ (b) $3\sqrt{2}$ (c) 0 (d) 1

131) The volume of a sphere is increasing in volume at the rate of 3 πcm^3 sec. The rate of change of its radius when radius is $\frac{1}{2}$ cm

(a) 3 cm/s (b) 2 cm/s (c) 1 cm/s

(d) $\frac{1}{2} cm/s$

132) A balloon rises straight up at 10 m/s. An observer is 40 m away from the spot where the balloon left the ground. Find the rate of change of the balloon's angle of elevation in radian per second when the balloon is 30 metres above the ground.

(a) $\frac{3}{25} radians/sec$ (b) $\frac{4}{25} radians/sec$ (c) $\frac{1}{5} radians/sec$ (d) $\frac{1}{3} radians/sec$

133) The position of a particle moving along a horizontal line of any time t is given by set) = $3t^2$ -2t- 8. The time at which the particle is at rest is

(a) t= 0 (b) $t = \frac{1}{3}$ (c) t =1 (d) t = 3

134) A stone is thrown up vertically. The height it reaches at time t seconds is given by x = 80t -$16t^2$. The stone reaches the maximum height in time t seconds is given by

(a) 2 (b) 2.5 (c) 3 (d) 3.5

135) Find the point on the curve 6y = .x^3 + 2 at which y-coordinate changes 8 times as fast as x-coordinate is

(a) (4,11) (b) (4,-11) (c) (-4,11) (d) (-4,-11)

136) The abscissa of the point on the curve $f(x) = \sqrt{8-2x}$ at which the slope of the tangent is -0.25 ?

(a) -8 (b) -4 (c) -2 (d) 0

137)

The slope of the line normal to the curve f(x) = 2cos 4x at $x = \frac{\pi}{12}$

(a) $-4\sqrt{3}$ (b) -4 (c) $\frac{\sqrt{3}}{12}$ (d) $4\sqrt{3}$

138) The tangent to the curve y^2 - xy + 9 = 0 is vertical when

(a) y = 0 (b) $y = \pm\sqrt{3}$ (c) $y = \frac{1}{2}$ (d) $y = \pm 3$

139) Angle between y^2 = x and.x^2= y at the origin is

(a) $tan^{-1}\frac{3}{4}$ (b) $tan^{-1}\left(\frac{4}{3}\right)$ (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{4}$

140)

The value of the limit $lim_{x\to 0}\left(cotx - \frac{1}{x}\right)$

(a) 0 (b) 1 (c) 2 (d) ∞

141) The function $\sin^4$ x + $\cos^4$X is increasing in the interval

(a) $\left[\frac{5\pi}{8}, \frac{3\pi}{4}\right]$ (b) $\left[\frac{\pi}{2}, \frac{5\pi}{8}\right]$ (c) $\left[\frac{\pi}{4}, \frac{\pi}{2}\right]$ (d) $\left[0, \frac{\pi}{4}\right]$

142) The number given by the Rolle's theorem for the functlon x^3-3x^2, $x\in[0,3]$ is

(a) 1 (b) $\sqrt{2}$ (c) $\frac{3}{2}$ (d) 2

143) The number given by the Mean value theorem for the function $\frac{1}{x}$,$x\in[1,9]$ is

(a) 2 (b) 2.5 (c) 3 (d) 3.5

144) The minimum value ofthe function |3-x|+9 is

(a) 0 (b) 3 (c) 6 (d) 9

145) The maximum slope of the tangent to the curve y = t:r sin x, x ∈ [0, 2π] is at

(a) $x = \frac{\pi}{4}$ (b) $x = \frac{\pi}{2}$ (c) $x = \pi$ (d) $x = \frac{3\pi}{2}$

146) The maximum value of the function $x^2 e^{-2x}$,

(a) $\frac{1}{e}$ (b) $\frac{1}{2e}$ (c) $\frac{1}{e^2}$ (d) $\frac{4}{e^4}$

147) One of the closest points on the curve $x^2 - y^2$.= 4 to the point (6, 0) is

(a) (2,0) (b) $\left(\sqrt{5}, 1\right)$ (c) $\left(3, \sqrt{5}\right)$ (d) $\left(\sqrt{13}, -\sqrt{3}\right)$

148) The maximum value of the product of two positive numbers, when their sum' of the squares is 200, is

(a) 100 (b) $25\sqrt{7}$ (c) 28 (d) $24\sqrt{14}$

149) The curve y= $ax^4 + bx^2$ with ab > 0

(a) has, no horizontal tangent (b) is concave up (c) isconeavedown (d) has no points of inflection

150) The point of inflection of the curve y = $(x - 1)^3$ is

(a) (0,0) (b) (0,1) (c) (1,0) (d) (1,1)

151) A circular template has a radius of 10 cm. The measurement of radius has an approximate error of 0.02 cm. Then the percentage error in calculating area of this template is

(a) 0.2% (b) 0.4% (c) 0.04% (d) 0.08%

152) The percentage error of fifth root of 31 is approximately how many times the percentage error in 31?

(a) $\frac{1}{31}$ (b) $\frac{1}{5}$ (c) 5 (d) 31

153) If u (x, y) = e^{x2+y2}, then $\frac{\partial u}{\partial x}$ is equal to

(a) e^{x2+y2} (b) 2xu (c) x^2u (d) y^2u

154) If v (x, y) = log (ex + ev), then $\frac{\partial v}{\partial x} + \frac{\partial v}{\partial y}$ is equal to

(a) $e^x + e^y$ (c) 2 (d) 1

(b) $\frac{1}{e^x+e^y}$

155) If w (x, y) = xy, x > 0, then $\frac{\partial w}{\partial x}$ is equal to

(a) $x^y \log x$ (b) y log x (c) yx^{y-1} (d) x log y

156) If f (x, y) = e^{xy} then $\frac{\partial^2 f}{\partial x \partial y}$ is equal to

(a) xye^{xy} (b) $(1 + xy)e^{xy}$ (c) $(1 + y)e^{xy}$ (d) $(1 + x)e^{xy}$

157) If we measure the side of a cube to be 4 cm with an error of 0.1 cm, then the error in our calculation of the volume is

(a) 0.4 cu.cm (b) 0.45 cu.cm (c) 2 cu.cm (d) 4.8 cu.cm

158) The change in the surface area S = $6x^2$ of a cube when the edge length varies from x_o to x_o+ dx is

(a) 12 x_o+dx (b) 12x_o dx (c) 6x_o dx (d) 6x_o+ dx

159) The approximate change in the volume V of a cube of side x metres caused by increasing the side by 1% is

(a) 0.3xdx m^3 (b) 0.03 xm^3 (c) 0.03.x^2 m^3 (d) 0.03x^3m^3

160) If g(x, y) = $3x^2$ - 5y + 2y, x(t) = e^t and y(t) = cos t, then $\frac{dg}{dt}$ is equal to

(a) $6e^{2t}$+5 sin t - 4 cos t sin t (b) $6e^{2t}$- 5 sin t + 4 cos t sin t (c) $3e^{2t}$+ 5 sin t + 4 cos t sin t (d) $3e^{2t}$ - 5 sin t + 4 cos t sin t

161) If f(x) = $\frac{x}{x+1}$ then its differential is given by

(a) $\frac{-1}{(x+1)^2}dx$ (b) $\frac{1}{(x+1)^2}dx$ (c) $\frac{1}{1+x}dx$ (d) $\frac{-1}{1+x}dx$

162) If u(x, y) = x^2+ 3xy + y - 2019, then $\frac{\partial u}{\partial x}(4, -5)$ is equal to

(a) -4 (b) -3 (c) -7 (d) 13

163) Linear approximation for g(x) = cos x at x=$\frac{-\pi}{2}$ is

(a) x + $\frac{-\pi}{2}$ (b) - x + $\frac{\pi}{2}$ (c) x - $\frac{\pi}{2}$ (d) - x + $\frac{\pi}{2}$

164) If w (x, y, z) = x^2 (v - z) + y^2 (z - x) + z^2(x - y), then $\frac{\partial w}{\partial x}+\frac{\partial w}{\partial y}+\frac{\partial w}{\partial z}$ is

(a) xy + yz + zx (b) x(y + z) (c) y(z + x) (d) 0

165) If (x,y, z) = xy +yz +zx, then f_x - f_z is equal to

(a) z - x (b) y - z (c) x - z (d) y - x

166)

The value of $\int_{-4}^{4}\left[tan^{-1}\left(\frac{x^2}{x^4+1}\right)+tan^{-1}\left(\frac{x^4+1}{x^2}\right)\right]dx$ is

(a) π (b) 2π (c) 3π (d) 4π

167)

The value of $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\left(\frac{2x^7-3x^5+7x^3-x+1}{cos^2x}\right)dx$ is

(a) 4 (b) 3 (c) 2 (d) 0

168) If $f(x) = \int_0^x t\ cos\ t\ dt,\ then\ \frac{dx}{dx}$

(a) cos x-x sin x (b) sin x+x cos x (c) x cos x (d) x sin x

169) The area between $y^2 x = 4$ and its latus rectum is

(a) $\frac{2}{3}$ (b) $\frac{4}{3}$ (c) $\frac{8}{3}$ (d) $\frac{5}{3}$

170) The value of $\int_0^1 x(1-x)^{99}dx$ is

(a) $\frac{1}{11000}$ (b) $\frac{1}{10100}$ (c) $\frac{1}{10010}$ (d) $\frac{1}{10001}$

171) The value of $\int_0^{\pi} \frac{dx}{1+5^{\cos x}}$ is

(a) $\frac{\pi}{2}$ (b) π (c) $\frac{3\pi}{2}$ (d) 2π

172) The value of $\frac{(n+2)}{(n)} = 90$ then n is

(a) 10 (b) 5 (c) 8 (d) 9

173) The value of $\int_0^{\frac{\pi}{6}} \cos^3 3x dx$

(a) $\frac{2}{3}$ (b) $\frac{2}{9}$ (c) $\frac{1}{9}$ (d) $\frac{1}{3}$

174) The value of $\int_0^{\pi} \sin^4 x dx$ is

(a) $\frac{3\pi}{10}$ (b) $\frac{3\pi}{8}$ (c) $\frac{3\pi}{4}$ (d) $\frac{3\pi}{2}$

175) The value of $\int_0^{\infty} e^{-3x}x^2 dx$ is

(a) $\frac{7}{27}$ (b) $\frac{5}{27}$ (c) $\frac{4}{27}$ (d) $\frac{2}{27}$

176) If $\int_a^a \frac{1}{4+x^2}dx = \frac{\pi}{8}$ then a is

(a) 4 (b) 1 (c) 3 (d) 2

177) The volume of solid of revolution of the region bounded by $y^2 = x(a - x)$ about x-axis is

(a) πa^2 (b) $\frac{\pi a^2}{4}$ (c) $\frac{\pi a^2}{5}$ (d) $\frac{\pi a^2}{6}$

178) If f(x)$f(x) = \int_1^x \frac{e^{\sin^u}}{u} du, x > 1$ *and* $\int_1^3 \frac{e^{\sin x^2}}{x} dx = \frac{1}{2}[f(a) - f(1)]$, then one of the possible value of a is

(a) 3 (b) 6 (c) 9 (d) 5

179) The value of $\int_0^1 (\sin^{-1}x)^2 dx$

(a) $\frac{\pi^2}{4} - 1$ (b) $\frac{\pi^2}{4} + 2$ (c) $\frac{\pi^2}{4} + 1$ (d) $\frac{\pi^2}{4} - 2$

180) The value of $\int_0^a (\sqrt{a^2 - x^2})^2 dx$

(a) $\frac{\pi a^2}{16}$ (b) $\frac{3\pi a^4}{16}$ (c) $\frac{3\pi a^2}{8}$ (d) $\frac{3\pi a^4}{8}$

181) If $\int_0^x f(t)dt = x + \int_x^1 tf(t)dt$ then the value of f (1) is

(a) $\frac{1}{2}$ (b) 2 (c) 1 (d) $\frac{3}{4}$

182) The value of $\int_0^{\frac{2}{3}} \frac{dx}{\sqrt{4-9x^2}}$ is

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{4}$ (d) π

183) The value of $\int_{-1}^{2} |x|\, dx$

(a) $\frac{1}{2}$ (b) $\frac{3}{2}$ (c) $\frac{5}{2}$ (d) $\frac{7}{2}$

184) For any value of n∈Z, $\int_0^{\pi} e^{\cos^2 x} \cos^3[(2n+1)x]$ is

(a) $\frac{\pi}{2}$ (b) π (c) 0 (d) 2

185) The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \pi \sin^2 x \cos x dx$ is

(a) $\frac{3}{2}$ (b) $\frac{1}{2}$ (c) 0 (d) $\frac{2}{3}$

186) The order and degree of the differential equation $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^{1/3} + x^{1/4} = 0$are respectively

(a) 2, 3 (b) 3, 3 (c) 2, 6 (d) 2, 4

187) The differential equation representing the family of curves y = Acos(x + B), where A and B are parameters, is

(a) $\frac{d^2y}{dx^2} - y = 0$ (b) $\frac{d^2y}{dx^2} + y = 0$ (c) $\frac{d^2y}{dx^2} = 0$ (d) $\frac{d^2x}{dy^2} = 0$

188) The order and degree of the differential equation $\sqrt{\sin \quad x}(dx + dy) = \sqrt{\cos \quad x}(dx - dy)$

(a) 1,2 (b) 2,2 (c) 1,1 (d) 2,1

189) The order of the differential equation of all circles with centre at (h, k) and radius 'a' is

(a) 2 (b) 3 (c) 4 (d) 1

190) The differential equation of the family of curves y = $Ae^x + Be^{-x}$, where A and B are arbitrary constants is

(a) $\frac{d^2y}{dx^2} + y = 0$ (b) $\frac{d^2y}{dx^2} - y = 0$ (c) $\frac{dy}{dx} + y = 0$ (d) $\frac{dy}{dx} - y = 0$

191) The general solution of the differential equation $\frac{dy}{dx} = \frac{y}{x}$ is

(a) xy = k (b) y = k log x (c) y = kx (d) log y = kx

192) The solution of the differential equation 2x$\frac{dy}{dx} - y = 3$represents

(a) straight lines (b) circles (c) parabola (d) ellipse

193) The solution of $\frac{dy}{dx}$ + p(x)y=0 is

(a) $y = ce^{\int pdx}$ (b) $y = ce^{-\int pdx}$ (c) $x = ce^{-\int pdx}$ (d) $xce^{\int pdx}$

194) The integrating factor of the differential equation $\frac{dy}{dx} + y = \frac{1+y}{\lambda}$ is

(a) $\frac{x}{e^{\lambda}}$ (b) $\frac{e^{\lambda}}{x}$ (c) λe^x (d) e^x

195) The integrating factor of the differential equation $\frac{dy}{dx}$+P(x)y=Q(x)is x, then P(x)

(a) x (b) $\frac{x^2}{2}$ (c) $\frac{1}{x}$ (d) $\frac{1}{x^2}$

196) The degree of the differential equation y $y(x) = 1 + \frac{dy}{dx} + \frac{1}{1.2}\left(\frac{dy}{dx}\right)^2 + \frac{1}{1.2.3}\left(\frac{dy}{dx}\right)^3 + \ldots.$ is

(a) 2 (b) 3 (c) 1 (d) 4

197) If p and q are the order and degree of the differential equation

$y = \frac{dy}{dx} + x^3\left(\frac{d^2y}{dx^2}\right) + xy = cosx$, When

(a) p < q (b) p = q (c) p > q (d) p exists and q does not exist

198) The solution of the differential equation $\frac{dy}{dx} + \frac{1}{\sqrt{1-x^2}} = 0$

(a) y + $\sin^{-1}$ x = c (b) x + $\sin^{-1}$ y = 0 (c) y^2+ 2 $\sin^{-1}$ x = c (d) x^2+ 2 $\sin^{-1}$y= c

199) The solution of the differential equation $\frac{dy}{dx} = 2xy$is

(a) y = Ce^{x^2} (b) y= $2x^2$ +C (c) y = Ce^{-x^2} +C (d) y = x^2 +C

200) The general solution of the differential equation $log\left(\frac{dy}{dx}\right) = x + y$ is

(a) e^x+e^y=C (b) e^x+e^{-y}=C (c) $e^{-x}+e^y$=C (d) $e^{-x}+e^{-y}$=C

201) The solution of $\frac{dy}{dx} = 2^{y-x}$is

(a) 2^x+2^y=C (b) 2^x-2^y=C (c) $\frac{1}{2^x} - \frac{1}{2^y} = C$ (d) x+y=C

202) The solution of the differential equation $\frac{dy}{dx} = \frac{y}{x} + \frac{\phi\left(\frac{y}{x}\right)}{\phi'\left(\frac{y}{x}\right)}$is

(a) $x\phi\left(\frac{y}{x}\right) = k$ (b) $\phi\left(\frac{y}{x}\right) = kx$ (c) $y\phi\left(\frac{y}{x}\right) = k$ (d) $\phi\left(\frac{y}{x}\right) = ky$

203) If sin x is the integrating factor of the linear differential equation $\frac{dy}{dx} + Pt = Q$, Then P is

(a) log sin x (b) cos x (c) tan x (d) cot x

204) The number of arbitrary constants in the general solutions of order n and n +1are respectively

(a) n-1,n (b) n,n+1 (c) n+1,n+2 (d) n+1,n

205) The number of arbitrary constants in the particular solution of a differential equation of third order is

(a) 3 (b) 2 (c) 1 (d) 0

206) Integrating factor of the differential equation $\frac{dy}{dx} = \frac{x+y+1}{x+1}$ is

(a) $\frac{1}{x+1}$ (b) x+1 (c) $\frac{1}{\sqrt{x+1}}$ (d) $\sqrt{x+1}$

207) The population P in any year t is such that the rate of increase in the population is proportional to the population. Then

(a) $P=Ce^{kt}$ (b) $P=Ce^{-kt}$ (c) P=Ckt (d) P=C

208) P is the amount of certain substance left in after time t. If the rate of evaporation of the substance is proportional to the amount remaining, then

(a) $P=Ce^{kt}$ (b) $P=ce^{-kt}$ (c) P=Ckt (d) Pt=C

209) If the solution of the differential equation $\frac{dy}{dx} = \frac{ax+3}{2y+f}$ represents a circle, then the value of a is

(a) 2 (b) -2 (c) 1 (d) -1

210) The slope at any point of a curve y = f (x) is given by $\frac{dy}{dx} = 3x^2$ and it passes through (-1,1). Then the equation of the curve is

(a) $y=x^3+2$ (b) $y=3x^2+4$ (c) $y=3x^4+4$ (d) $y=3x^2+5$

211) Let X be random variable with probability density function

$$f(x) = \begin{cases} \frac{2}{x^3} & 0 < x \geq l \\ 0 & 1 \leq x < 2l \end{cases}$$

Which of the following statement is correct

(a) both mean and variance exist (b) mean exists but variance does not exist (c) both mean and variance do not exist (d) variance exists but Mean does uot exist

212) A rod of length 2l is broken into two pieces at random. The probability density function of the shorter of the two pieces is

$$f(x) = \begin{cases} \frac{2}{x^3} & 0 < x > l \\ 0 & 1 \leq x < 2l \end{cases}$$

(a) $\frac{l}{2}, \frac{l^2}{3}$ (b) $\frac{l}{2}, \frac{l^2}{6}$ (c) $1, \frac{l^2}{12}$ (d) $\frac{1}{2}, \frac{l^2}{12}$

213) Consider a game where the player tosses a sixsided fair die. H the face that comes up is 6, the player wins Rs.36, otherwise he loses Rs. k^2 , where k is the face that comes up k = {l, 2,3,4, 5}.

The expected amount to win at this game in Rs is

(a) $\frac{19}{6}$ (b) $-\frac{19}{6}$ (c) $\frac{3}{2}$ (d) $-\frac{3}{2}$

214) A pair of dice numbered 1, 2, 3, 4, 5, 6 of a six-sided die and 1, 2, 3, 4 of a four-sided die is roUed and the sum is determined. Let the random variable X denote this

sum. Then the number of elements in the inverse image of 7 is

(a) 1 (b) 2 (c) 3 (d) 4

215) A random variable X has binomial distribution with n = 25 and p = 0.8 then standard deviation of X is

(a) 6 (b) 4 (c) 3 (d) 2

216) Let X represent the difference between the number of heads and the number of tails obtained when a coin is tossed n times. Then the possible values of X are

(a) i + 2n, i = 0,1,2... n (b) 2i- n, i = 0,1,2... n (c) n - i, i = 0,1,2... n (d) 2i + 2n, i = 0, 1, 2...n

217)

If the function $f(x) = \frac{1}{12}$ for. a < x < b, represents a probability density function of a continuous random variable X, then which of the followingcannot be the value of a and b?

(a) 0 and 12 (b) 5 and 17 (c) 7 and 19 (d) 16 and 24

218) Four buses carrying 160 students from the same school arrive at a football stadium. The buses carry, respectively, 42, 36, 34, and 48 students. One of the students is randomly selected. Let X denote the number of students that were on the bus carrying the randomly selected student One of the 4 bus drivers is also randomly selected. Let Y denote the number of students on that bus. Then E[X] and E[Y] respectively are

(a) 50,40 (b) 40,50 (c) 40.75,40 (d) 41,41

219) Two coins are to be flipped. The first coin will land on heads with probability 0.6, the second with probability 0.5. Assume that the results of the flips are independent, and let X equal the total number of heads that result The value of E[X] is

(a) 0.11 (b) 1.1 (c) 1.1 (d) 1

220) On a multiple-choice exam with 3 possible destructives for each of the 5 questions, the probability that a student will get 4 or more correct answers just by guessing is

(a) $\frac{11}{243}$ (b) $\frac{3}{8}$ (c) $\frac{1}{243}$ (d) $\frac{5}{243}$

221) If P{X = 0} = 1- P{X = I}. IfE[X) = 3Var(X), then P{X = 0}.

(a) $\frac{2}{3}$ (b) $\frac{2}{5}$ (c) $\frac{1}{5}$ (d) $\frac{1}{3}$

222) If X is a binomial randam variable with expected value 6 and variance 2.4, then P(X=5) is

(a) $\left(\frac{10}{5}\right)^2\left(\frac{3}{5}\right)^6\left(\frac{2}{5}\right)^4$ (b) $\left(\frac{10}{5}\right)\left(\frac{3}{5}\right)^5$ (c) $\left(\frac{10}{5}\right)\left(\frac{3}{5}\right)^4\left(\frac{2}{5}\right)^6$ (d) $\left(\frac{10}{5}\right)\left(\frac{3}{5}\right)^5\left(\frac{2}{5}\right)^5$

223)

The random variable X has the probability density function $f(x) = \begin{cases} ax+b & 0<x<1 \\ 0 & otherwise \end{cases}$

and $E(X) = \frac{7}{12}$ then a and b are respectively.

(c) 2 and 1 (d) 1 and 2

(a) 1 and $\frac{1}{2}$ (b) $\frac{1}{2}$ and 1

224) Suppose that X takes on one of the values 0, 1, and 2. If for some constant k, P(X = I) = k P(X = i-I) i = 1, 2 and P(X = 0) = $\frac{1}{7}$ then the value of k is

(a) 1 (b) 2 (c) 3 (d) 4

225) Which of the following is a discrete random variable?
I. The number of cars crossing a particular signal in a day
II.The number of customers in a queue-to buy train tickets at a moment.
III.The time taken to complete a telephone call.
(a) I and II (b) II only (c) III only (d) II and III

226) If f(x) = is a probability density function of a random variable, then the value of a is
(a) 1 (b) 2 (c) 3 (d) 4

227) The probability mass function of a random variable is defined as:

x	-2	-1	0	1	2
f(x)	k	2k	3k	4k	5k

(a) $\frac{1}{15}$ (b) $\frac{1}{10}$ (c) $\frac{1}{3}$ (d) $\frac{2}{3}$

228) Let X have a Bernoulli distribution with mean 0.4, then the variance of (2X - 3) is
(a) 0.24 (b) 0.48 (c) 0.6 (d) 0.96

229) If in 6 trials, X is a binomial variate which foUows the relation 9P(X = 4) = P(X = 2), then the probability of success is
(a) 0.125 (b) 0.25 (c) 0.375 (d) 0.75

230) A computer salesperson knows from his past experience that he seUs computers to one in every twenty customers who enter the showroom. What is the probability that he will seU a computer to exactly two of the next three customers?

(a) $\frac{57}{20^3}$ (b) $\frac{57}{20^2}$ (c) $\frac{19^3}{20^3}$ (d) $\frac{57}{20}$

231) A binary operation on a set S is a function from
(a) S ⟶ S (b) (SxS) ⟶ S (c) S⟶ (SxS) (d) (SxS) ⟶ (SxS)

232) Subtraction is not a binary operation in
(a) R (b) Z (c) N (d) Q

233) Which one of the following is a binary operation on N?
(a) Subtraction (b) Multiplication (c) Division (d) All the above

234) In the set R of real numbers '*' is defined as follows. Which one of the following is not a binary operation on R?
(a) a*b=min (a.b) (b) a*b= max (a,b) (c) a*b= a (d) a*b=a^b

235) The operation * defined by a*b = $\frac{ab}{7}$ is not a binary operation on
(a) Q^+ (b) Z (c) R (d) C

236) In the set Q define a⊙b= a+b+ab. For what value of y, 3⊙(y⊙5)=7?
(d) y=4

(a) $y=\frac{2}{3}$ (b) $y=\frac{-2}{3}$ (c) $y=\frac{-3}{2}$

237) If a*b=$\sqrt{a^2+b^2}$ on the real numbers then * is

(a) commutative but not associative (b) associative but not commutative (c) both commutative and associative (d) neither commutative nor associative

238) Which one of the following statements has the truth value T?

(a) sin x is an even function (b) Every square matrix is non-singular (c) The product of complex number and its conjugate is purely imaginary (d) $\sqrt{5}$ is an irrational number

239) Which one of the following statements has truth value F?

(a) Chennai is in India or $\sqrt{2}$ is an integer (b) Chennai is in India or $\sqrt{2}$ is an irrational number (c) Chennai is in China or $\sqrt{2}$ is an integer (d) Chennai is in China or $\sqrt{2}$ is an irrational number

240) If a compound statement involves 3 simple statements, then the number of rows in the truth table is

(a) 9 (b) 8 (c) 6 (d) 3

241) Which one is the inverse of the statement (PVq)→(p∧q)?

(a) (p∧q)→(pVq) (b) ¬(pvq)→(p∧q) (c) (¬pv¬q)→(¬p∧¬q) (d) (¬p∧¬q)→(¬pV¬q)

242) Which one is the contrapositive of the statement (pVq)⟶r?

(a) ¬r→(¬p∧¬q) (b) ¬r⟶(p∨q) (c) r⟶(p∧q) (d) p⟶(q∨r)

243) The truth table for (p ∧ q) ∨ ¬q is given below

p	q	(p ∧ q) ∨ (¬q)
T	T	(a)
T	F	(b)
F	T	(c)
F	F	(d)

Which one of the following is true?

(a)

(a)	(b)	(c)	(d)
T	T	T	T

(b)

(a)	(b)	(c)	(d)
T	F	T	T

(c)

(a)	(b)	(c)	(d)
T	T	F	F

(d)

(a)	(b)	(c)	(d)
T	F	F	F

244) In the last column of the truth table for ¬(p ∨ ¬q) the number of final outcomes of the truth value 'F' are

(a) 1 (b) 2 (c) 3 (d) 4

245) Which one of the following is incorrect? For any two propositions p and q, we have

(a) ¬ (p∨q) ≡ ¬ p ∧ ¬q (b) ¬ (p∧ q)≡¬p ∨ ¬q (c) ¬ (p ∨ q)≡¬p∨¬q (d) ¬(¬p)≡ p

246)

p	q	(p ∧ q) ⟶ ¬q
T	T	(a)
T	F	(b)
F	T	(c)
F	F	(d)

Which one of the following is correct for the truth value of (p ∧ q)⟶ ¬p p?

(a)

(a)	(b)	(c)	(d)
T	T	T	T

(b)

(a)	(b)	(c)	(d)
F	T	T	T

(c)

(a)	(b)	(c)	(d)
F	F	T	T

(d)

(a)	(b)	(c)	(d)
T	T	T	F

247) The dual of $\neg(p \vee q) \vee [p \vee (p \wedge \neg r)]$ is

(a) $\neg(p \wedge q) \wedge [p \vee (p \wedge \neg r)]$
(b) $(p \wedge q) \wedge [p \wedge (p \vee \neg r)]$
(c) $\neg(p \wedge q) \wedge [p \wedge (p \wedge r)]$
(d) $\neg(p \wedge q) \wedge [p \wedge (p \vee \neg r)]$

248) The proposition $p \wedge (\neg p \vee q)$ is

(a) a tautology
(b) a contradiction
(c) logically equivalent to $p \wedge q$
(d) logically equivalent to $p \vee q$

249) Determine the truth value of each of the following statements:

(a) 4+2=5 and 6+3=9
(b) 3+2=5 and 6+1=7
(c) 4+5=9 and1+2= 4
(d) 3+2=5 and 4+7=11

(a)

(a)	(b)	(c)	(d)
F	T	T	T

(b)

(a)	(b)	(c)	(d)
T	F	T	F

(c)

(a)	(b)	(c)	(d)
T	T	F	F

(d)

(a)	(b)	(c)	(d)
F	F	T	T

250) Which one of the following is not true?

(a) Negation of a negation of a statement is the statement itself
(b) If the last column of the truth table contains only T then it is a tautology.
(c) If the last column of its truth table contains only F then it is a contradiction
(d) If p and q are any two statements then $p \leftrightarrow q$ is a tautology.

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MCQ CREATIVE

12th Standard

Maths

270 x 1 = 270

1) The system of linear equations x + y + z = 6, x + 2y + 3z =14 and 2x + 5y + λz =μ (λ, μ ∈ R) is consistent with unique solution if
(a) λ = 8 (b) λ = 8, μ ≠ 36 (c) λ ≠ 8 (d) none

2) If the system of equations x = cy + bz, y = az + cx and z = bx + ay has a non - trivial solution then
(a) $a^2 + b^2 + c^2 = 1$ (b) abc ≠ 1 (c) a + b + c =0 (d) $a^2 + b^2 + c^2 + 2abc = 1$

3) Let A be a 3 x 3 matrix and B its adjoint matrix If |B|=64, then |A|=
(a) ±2 (b) ±4 (c) ±8 (d) ±12

4) If A^T is the transpose of a square matrix A, then
(a) $|A| \neq |A^T|$ (b) $|A| = |A^T|$ (c) $|A| + |A^T| = 0$ (d) $|A| = |A^T|$ only

5) The number of solutions of the system of equations 2x+y = 4, x - 2y = 2, 3x + 5y = 6 is
(a) 0 (b) 1 (c) 2 (d) infinitely many

6) If A is a square matrix that IAI = 2, than for any positive integer n, $|A^n|$ =
(a) 0 (b) 2n (c) 2^n (d) n^2

7) The system of linear equations x + y + z = 2, 2x + y - z = 3, 3x + 2y + kz = has a unique solution if
(a) k ≠ 0 (b) -1 < k < 1 (c) -2 < k < 2 (d) k=0

8) If A is a square matrix of order n, then |adj A| =
(a) $|A|^{n-1}$ (b) $|A|^{n-2}$ (c) $|A|^n$ (d) None

9) If the system of equations x + 2y - 3x = 2, (k + 3) z = 3, (2k + 1) y + z = 2. is inconsistent then k is
(a) -3, $-\frac{1}{2}$ (b) $-\frac{1}{2}$ (c) 1 (d) 2

10) If A = $\begin{pmatrix} cosx & sinx \\ -sinx & cosx \end{pmatrix}$ and A(adj A) =$\lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ then λ is
(a) sinx cosx (b) 1 (c) 2 (d) none

11) If A is a matrix of order m x n, then ρ(A) is
(a) m (b) n (c) ≤ min (m,n) (d) ≥ min (m,n)

12) The system of equations x + 2y + 3z = 1, x - y + 4z = 0, 2x + y + 7z = 1 has
(a) One solution (b) Two solution (c) No solution (d) Infinitely many solution

13) If ρ(A) = ρ([A/B]) = number of unknowns, then the system is
(a) consistent and has infinitely many solutions (b) consistent inconsistent (c) unique solution (d) consistent and has

14) Which of the following is not an elementary transformation?
(a) $R_i \leftrightarrow R_j$ (b) $R_i \longrightarrow 2R_i + R_j$ (c) $C_j \longrightarrow C_j + C_i$ (d) $R_i \longrightarrow R_i + C_j$

15) If ρ(A) = r then which of the following is correct?
(a) all the minors of order n which do not vanish (b) 'A' has at least one minor "of order r which does not vanish and all higher order minors vanish (c) 'A' has at least one (r + 1) order minor which vanish (d) all (r + 1) and higher order minors should not vanish

16) Every homogeneous system _____
(a) Is always (b) Has only trivial (c) Has infinitely many (d) Need not be

consistent solution solution consistent

17) If $\rho(A) \neq \rho([A|B])$, then the system is
(a) consistent and has infinitely many solutions (b) consistent and has a unique solution (c) consistent (d) inconsistent

18) In the non - homogeneous system of equations with 3 unknowns if $\rho(A) = \rho([A|B]) = 2$, then the system has _____
(a) unique solution (b) one parameter family of solution (c) two parameter family of solutions (d) in consistent

19) Cramer's rule is applicable only when _____
(a) $\Delta \neq 0$ (b) $\Delta = 0$ (c) $\Delta = 0, \Delta_x = 0$ (d) $\Delta_x = \Delta_y = \Delta_z = 0$

20) In a homogeneous system if $\rho(A) = \rho([A|0]) <$ the number of unknouns then the system has _____
(a) trivial solution (b) only non - trivial solution (c) no solution (d) trivial solution and infinitely many non - trivial solutions

21) In the system of equations with 3 unknowns, if $\Delta = 0$, and one of Δ_x, Δ_y of Δ_z is non zero then the system is _____
(a) Consistent (b) inconsistent (c) consistent with one parameter family of solutions (d) consistent with two parameter family of solutions

22) In the system of liner equations with 3 unknowns If $\rho(A) = \rho([A|B]) = 1$, the system has _____
(a) unique solution (b) inconsistent (c) consistent with 2 parameter -family of solution (d) consistent with one parameter family of solution.

23) If A = [2 0 1] then the rank of AA^T is _____
(a) 1 (b) 2 (c) 3 (d) 0

24) If A is a non-singular matrix then $|A^{-1}| =$ _____
(a) $\left|\frac{1}{A^2}\right|$ (b) $\frac{1}{|A^2|}$ (c) $\left|\frac{1}{A}\right|$ (d) $\frac{1}{|A|}$

25) In a square matrix the minor M_{ij} and the' co-factor A_{ij} of and element a_{ij} are related by _____
(a) $A_{ij} = -M_{ij}$ (b) $A_{ij} = M_{ij}$ (c) $A_{ij} = (-1)^{i+j} M_{ij}$ (d) $A_{ij} = (-1)^{i-j} M_{ij}$

26) The value of $(1+i)(1+i^2)(1+i^3)(1+i^4)$ is
(a) 2 (b) 0 (c) 1 (d) i

27) If $\sqrt{a+ib} = x+iy$, then possible value of $\sqrt{a-ib}$ is
(a) x^2+y^2 (b) $\sqrt{x^2+y^2}$ (c) x+iy (d) x-iy

28) If, $i^2 = -1$, then $i^1 + i^2 + i^3 + ...+$ up to 1000 terms is equal to
(a) 1 (b) -1 (c) i (d) 0

29) If $z = \cos\frac{\pi}{4} + i \sin\frac{\pi}{6}$, then
(a) $|z| = 1$, $\arg(z) = \frac{\pi}{4}$ (b) $|z| = 1$, $\arg(z) = \frac{\pi}{6}$ (c) $|z| = \frac{\sqrt{3}}{2}$, $\arg(z) = \frac{5\pi}{24}$ (d) $|z| = \frac{\sqrt{3}}{2}$, $\arg(z) = \tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$

30) If $a = \cos\theta + i \sin\theta$, then $\frac{1+a}{1-a} =$
(b) cot θ

(a) $\cot\frac{\theta}{2}$ (c) $i\cot\frac{\theta}{2}$ (d) $i\tan\frac{\theta}{2}$

31) The principal value of the amplitude of (1+i) is

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{12}$ (c) $\frac{3\pi}{4}$ (d) π

32) The least positive integer n such that $\left(\frac{2i}{1+i}\right)^n$ is a positive integer is

(a) 16 (b) 8 (c) 4 (d) 2

33) If a = 1+i, then a^2 equals

(a) 1-i (b) 2i (c) (1+i)(1-i) (d) i-1

34) If z = $\frac{1}{(2+3i)^2}$ then |z| =

(a) $\frac{1}{13}$ (b) $\frac{1}{5}$ (c) $\frac{1}{12}$ (d) none of these

35) If z=1-cosθ + i sinθ, then |z| =

(a) $2\sin\frac{1}{3}$ (b) $2\cos\frac{\theta}{2}$ (c) $2|\sin\frac{\theta}{2}|$ (d) $2|\cos\frac{\theta}{2}|$

36) If z=$\frac{1}{1-\cos\theta-i\sin\theta}$, the Re(z) =

(a) 0 (b) $\frac{1}{2}$ (c) $\cot\frac{\theta}{2}$ (d) $\frac{1}{2}\cot\frac{\theta}{2}$

37) If x+iy =$\frac{3+5i}{7-6i}$, they y =

(a) $\frac{9}{85}$ (b) $-\frac{9}{85}$ (c) $\frac{53}{85}$ (d) none of these

38) The amplitude of $\frac{1}{i}$ is equal to

(a) 0 (b) $\frac{\pi}{2}$ (c) $-\frac{\pi}{2}$ (d) π

39) The value of $(1+i)^4 + (1-i)^4$ is

(a) 8 (b) 4 (c) -8 (d) -4

40) The complex number z which satisfies the condition $\left|\frac{1+z}{1-z}\right|$ =1 lies on

(a) circle x^2+y^2 =1 (b) x-axis (c) y-axis (d) the lines x+y=1

41) If z = a + ib lies in quadrant then $\frac{\bar{z}}{z}$ also lies in the III quadrant if

(a) a > b > 0 (b) a < b < 0 (c) b < a < 0 (d) b > a > 0

42) $\frac{1+e^{-i\theta}}{1+e^{i\theta}}$ =

(a) cosθ + i sinθ (b) cosθ - i sinθ (c) sinθ - i cosθ (d) sinθ + icosθ

43) If $z^n = \cos\frac{n\pi}{3} + i\sin\frac{n\pi}{3}$, then $z_1, z_2 \ldots z_6$ is

(a) 1 (b) -1 (c) i (d) -i

44) If x =cosθ + i sinθ, then the value of $x^n+\frac{1}{x^n}$ is

(a) 2 cosθ (b) 2i sin nθ (c) 2i sin nθ (d) 2i cos nθ

45) If ω is the cube root of unity, then the value of $(1-\omega)(1-\omega^2)(1-\omega^4)(1-\omega^8)$ is

(a) 9 (b) -9 (c) 16 (d) 32

46) The points represented by 3 - 3i, 4 - 2i, 3 - i and 2 - 2i form _____ in the argand plane.

(a) collinear points (b) Vertices of a parallelogram (c) Vertices of a rectangle (d) Vertices of a square

47) $(1+i)^3$ = _____

(a) 3 + 3i (b) 1 + 3i (c) 3 - 3i (d) 2i - 2

48) $\frac{(cos\theta+isin\theta)^6}{(cos\theta-isin\theta)^5}$ = _____

(a) cos 11θ - isin 11θ (b) cos 11θ + isin 11θ (c) cosθ + i sinθ (d) $cos\frac{6\theta}{5}+isin\frac{6\theta}{5}$

49) If a =cosα + i sinα, b= -cosβ + i sinβ then $\left(ab-\frac{1}{ab}\right)$ is _____

(a) -2i sin(α - β) (b) 2i sin(α - β) (c) 2 cos(α - β) (d) -2 cos(α - β)

50) The conjugate of $\frac{1+2i}{1-(1-i)^2}$ is _____

(a) $\frac{1+2i}{1-(1-i)^2}$ (b) $\frac{5}{1-(1-i)^2}$ (c) $\frac{1-2i}{1+(1+i)^2}$ (d) $\frac{1+2i}{1+(1-i)^2}$

51) The modular of $\frac{(-1+i)(1-i)}{1+i\sqrt{3}}$ is _____

(a) $\sqrt{2}$ (b) 2 (c) 1 (d) $\frac{1}{2}$

52) The value of $\frac{(cos45^0+isin45^0)^2(cos30^0-isin30^0)}{cos30^0+isin30^0}$ is

(a) $\frac{1}{2}+i\frac{\sqrt{3}}{2}$ (b) $\frac{1}{2}-i\frac{\sqrt{3}}{2}$ (c) $-\frac{\sqrt{3}}{2}+\frac{1}{2}$ (d) $\frac{\sqrt{3}}{2}+\frac{1}{2}$

53) If x=cosθ + i sinθ, then $x^n+\frac{1}{x^n}$ is _____

(a) 2 cos nθ (b) 2 i sin nθ (c) 2^n cosθ (d) 2^n i sinθ

54) If z_1, z_2, z_3 are the vertices of a parallelogram, then the fourth vertex z_4 opposite to z_2 is _____

(a) $z_1 + z_2 - z_2$ (b) $z_1 + z_2 - z_3$ (c) $z_1 + z_2 - z_3$ (d) $z_1 - z_2 - z_3$

55) If $x_r=cos\left(\frac{\pi}{2^r}\right)+isin\left(\frac{\pi}{2^r}\right)$ then x_1, x_2 ... x_∞ is

(a) -∞ (b) -2 (c) -1 (d) 0

56) If a, b, c ∈ Q and p +√q (p,q ∈ Q) is an irrational root of $ax^2+bx+c=0$ then the other root is

(a) -p+√q (b) p-iq (c) p-√q (d) -p-√q

57) The quadratic equation whose roots are ∝ and β is

(a) (x - ∝)(x -β) =0 (b) (x - ∝)(x + β) =0 (c) ∝+β=$\frac{b}{a}$ (d) ∝.β=$\frac{-c}{a}$

58) Ifj(x) = 0 has n roots, thenf'(x) = 0 has _____ roots

(a) n (b) n -1 (c) n+1 (d) (n-r)

59) If x is real and $\frac{x^2-x+1}{x^2+x+1}$ then

(a) $\frac{1}{3}$ ≤k≤ (b) k≥5 (c) k≤0 (d) none

60) Let a > 0, b > 0, c > 0. h n both th root of th quatlon $ax^2+b+C= 0$ are
(a) real and negative (b) real and positive (c) rational numb rs (d) none

61) The equation $\sqrt{x+1}-\sqrt{x-1}=\sqrt{4x-1}$ has
(a) no solution (b) one solution (c) two solution (d) more than one solution

62) If the root of the equation $x^3+bx^2+cx-1=0$ form an Increasing G.P, then
(a) one of the roots is 2 (b) one of the rots is 1 (c) one of the rots is -1 (d) one of the rots is -2

63) For real x, the equation $\left|\frac{x}{x-1}\right|+|x|=\frac{x^2}{|x-1|}$ has
(a) one solution (b) two solution (c) at least two solution (d) no solution

64) If the equation ax2+ bx+c=0(a > 0) has two roots ∝ and β such that ∝< - 2 and β > 2, then
(a) b2-4ac=0 (b) $b^2 - 4ac < 0$ (c) $b^2 - 4ac > 0$ (d) $b^2 - 4ac \geq 0$

65) If $(2+\sqrt{3})x^2-2x+1+(2-\sqrt{3})x^2-2x-1=\frac{2}{2-\sqrt{3}}$ then x=
(a) 0,2 (b) 0,1 (c) 0,3 (d) 0, √3

66) If ∝, β, ɤ are the roots of the equation $x^3-3x+11=0$, then ∝+β+ɤ is _____.
(a) 0 (b) 3 (c) -11 (d) -3

67) If ∝, β,ɤ are the roots of 9x3-7x+6=0, then ∝ β ɤ is _____
(a) $\frac{-7}{9}$ (b) $\frac{7}{9}$ (c) 0 (d) $\frac{-2}{3}$

68) If $x^2 - hx - 21 = 0$ and $x^2 - 3hx + 35 = 0$ (h > 0) have a common root, then h = _____
(a) 0 (b) 1 (c) 4 (d) 3

69) If $ax^2 + bx + c = 0$, a, b, c E R has no real zeros, and if a + b + c < 0, then _____
(a) c > 0 (b) c< 0 (c) c=0 (d) c≥0

70) If $p(x) = ax^2 + bx + c$ and $Q(x) = -ax^2 + dx + c$ where ac ≠ 0 then p(x). Q(x) = 0 has at least _____ real roots.
(a) no (b) 1 (c) 2 (d) infinite

71) If $tan^{-1}\left\{\frac{\sqrt{1+x^2}-\sqrt{1-x^2}}{\sqrt{1+x^2}+\sqrt{1-x^2}}\right\}=\alpha$ then x^2 =
(a) $sin2\alpha$ (b) $sin\alpha$ (c) $cos2\alpha$ (d) $cos\alpha$

72) If $sin^{-1}x-cos^{-1}x=\frac{\pi}{6}$ then
(a) $\frac{1}{2}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{-1}{2}$ (d) none of these

73) The number of solutions of the equation $tan^{-1}2x+tan^{-1}3x=\frac{\pi}{4}$
(a) 2 (b) 3 (c) 1 (d) none

74) If $\alpha = tan^{-1}\left(tan\frac{5\pi}{4}\right)$ and $\beta = tan^{-1}\left(-tan\frac{2\pi}{3}\right)$ then

(a) $4\alpha = 3\beta$ (b) $3\alpha = 4\beta$ (c) $\alpha - \beta = \frac{7\pi}{12}$ (d) none

75) The number of real solutions of the equation $\sqrt{1 + cos2x} = 2sin^{-1}(sinx),\ -\pi < x < \pi$ is

(a) 0 (b) 1 (c) 2 (d) infinte

76) ·If $\alpha = tan^{-1}\left(\frac{\sqrt{3}}{2y - x}\right), \beta = tan^{-1}\left(\frac{2x - y}{\sqrt{3y}}\right)$ then $\alpha - \beta$

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{2}$ (d) $\frac{-\pi}{3}$

77) $tan^{-1}\left(\frac{1}{4}\right) + tan^{-1}\left(\frac{2}{11}\right) =$

(a) 0 (b) $\frac{1}{2}$ (c) -1 (d) none

78) If $\tan^{-1}(3)+\tan^{-1}(x)=\tan^{-1}(8)$then x=

(a) 5 (b) $\frac{1}{5}$ (c) $\frac{5}{14}$ (d) $\frac{14}{5}$

79) The value of $cos^{-1}\left(\frac{cos5\pi}{3}\right) + sin^{-1}\left(\frac{sin5\pi}{3}\right)$ is

(a) $\frac{\pi}{2}$ (b) $\frac{5\pi}{3}$ (c) $\frac{10\pi}{3}$ (d) 0

80) $sin\left\{2cos^{-1}\left(\frac{-3}{5}\right)\right\} =$

(a) $\frac{6}{15}$ (b) $\frac{24}{25}$ (c) $\frac{4}{5}$ (d) $\frac{-24}{25}$

81) If $4cos^{-1}x + sin^{-1}x = \pi$ then x is

(a) $\frac{3}{2}$ (b) $\frac{1}{\sqrt{2}}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{2}{\sqrt{3}}$

82) If $tan^{-1}\left(\frac{x+1}{x-1}\right) + tan^{-1}\left(\frac{x-1}{x}\right) = tan^{-1}(-7)$ then x Is

(a) 0 (b) -2 (c) 1 (d) 2

83) If $cos^{-1}x > x > sin^{-1}x$ then

(a) $\frac{1}{\sqrt{2}} < x \le 1$ (b) $0 \le x < \frac{1}{\sqrt{2}}$ (c) $-1 \le x < \frac{1}{\sqrt{2}}$ (d) x > 0

84)

In a ΔABC if C is a right angle, then $tan^{-1}\left(\frac{a}{b+c}\right) + tan^{-1}\left(\frac{b}{c+a}\right) =$

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{4}$ (c) $\frac{5\pi}{2}$ (d) $\frac{\pi}{6}$

85)

$cot\left(\frac{\pi}{4} - cot^{-1}3\right)$

(a) 7 (b) 6 (c) 5 (d) none

86) If $\tan^{-1}(\cot\theta) = 2\theta$, then θ = _____

(a) ± 3 (b) $\pm\frac{\pi}{4}$ (c) $\pm\frac{\pi}{6}$ (d) none

87) The domain of $\cos^{-1}(x^2 - 4)$ is_____

(a) [3, 5] (b) [-1, 1] (c) $\left[-\sqrt{5}, -\sqrt{3}\right] \cup \left[\sqrt{3}, \sqrt{5}\right]$ (d) [0, 1]

88)

The value of tan $\left(cos^{-1}\frac{3}{5} + tan^{-1}\frac{1}{4}\right)$ is _____

(a) $\frac{19}{8}$ (b) $\frac{8}{19}$ (c) $\frac{19}{12}$ (d) $\frac{3}{4}$

89) The value of $\sin(2(\tan^{-1} 0.75)$ is_____

(a) 0.75 (b) 1.5 (c) 0.96 (d) $\sin^{-1}(1.5)$

90)

If x > 1,then $2tan^{-1}x + sin^{-1}\left(\frac{2x}{1+x^2}\right)$ _____

(a) $4\tan^{-1}x$ (b) 0 (c) $\frac{\pi}{2}$ (d) π

91) If $\theta = sin^{-1}\left(sin(-60^0)\right)$ then one of the possible values of θ is _____

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$ (c) $\frac{2\pi}{3}$ (d) $\frac{-2\pi}{3}$

92)

The value of $sin^{-1}\left(cos\frac{33\pi}{5}\right)$ is_____

(a) $\frac{3\pi}{5}$ (b) $\frac{-\pi}{10}$ (c) $\frac{\pi}{10}$ (d) $\frac{7\pi}{5}$

93) If x < 0, y < 0 such that xy = 1, then $\tan^{-1}(x) + \tan^{-1}(y) =$_____

(a) $\frac{\pi}{2}$ (b) $\frac{-\pi}{2}$ (c) $-\pi$ (d) none

94) The pricipal value of $sin^{-1}\left(\frac{-1}{2}\right)$ is _____

(a) $\frac{\pi}{6}$ (b) $\frac{-\pi}{6}$ (c) $\frac{\pi}{3}$ (d) $\frac{-\pi}{3}$

95) $tan^{-1}\left(tan\frac{9\pi}{8}\right)$

(a) $\frac{9\pi}{8}$ (b) $\frac{9\pi}{8}$ (c) $\frac{\pi}{8}$ (d) $\frac{-\pi}{8}$

96) If (0, 4) and (0, 2) are the vertex and focus of a parabola then its equation is
(a) $x^2 + 8y = 32$ (b) $y^2 + 8x = 32$ (c) $x^2 - 8y = 32$ (d) $y^2 - 8x = 32$

97) The equation of the directrix of the parabola $y^2 + 4y + 4x + 2 = 0$ is
(a) x = -1 (b) x = 1 (c) $x = \frac{-3}{2}$ (d) $x = \frac{3}{2}$

98) Equation of tangent at (-4, -4) on $x^2 = -4y$ is
(a) 2x - y + 4 = 0 (b) 2x + y - 4 = 0 (c) 2x - y - 12 = 0 (d) 2x + y + 4 = 0

99) $y^2 - 2x - 2y + 5 = 0$ is a
(a) circle (b) parabola (c) ellipse (d) hyperbola

100) If a parabolic reflector is 20 em in diameter and 5 em deep, then its focus is
(a) (0,5) (b) (5,0) (c) (10,0) (d) (0, 10)

101) The eccentricity of the ellipse $9x^2 + 5y^2 - 30y = 0$ is
(a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) $\frac{3}{4}$ (d) none of these

102) The length of the latus rectum of the ellipse $\frac{x^2}{36} + \frac{y^2}{49} = 1$ is
(a) $\frac{98}{6}$ (b) $\frac{72}{7}$ (c) $\frac{72}{14}$ (d) $\frac{98}{12}$

103) If the distance between the foci is 2 and the distance between the direction is 5, then the equation of the ellipse is
(a) $6x^2 + 10y^2 = 5$ (b) $6x^2 + 10y^2 = 15$ (c) $x^2 + 3y^2 = 10$ (d) none

104) In an ellipse, the distance between its foci is 6 and its minor axis is 8, then e is
(a) $\frac{4}{5}$ (b) $\frac{1}{\sqrt{52}}$ (c) $\frac{3}{5}$ (d) $\frac{1}{2}$

105) The equation $7x^2 - 6\sqrt{3}\,xy + 13y^2 - 4\sqrt{3}\,x - 4y - 12 = 0$ represents
(a) parabola (b) ellipse (c) hyperbola (d) rectangular hyperbola

106) The distance between the foci of a hyperbola is 16 and $e = \sqrt{2}$ Its equation is

(a) $x^2 - y^2 = 32$ (b) $y^2 - x^2 = 32$ (c) $x^2 - y^2 = 16$ (d) $y^2 - x^2 = 16$

107) If the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$ coincide then b^2 is

(a) 1 (b) 5 (c) 7 (d) 9

108) When the eccentricity of a ellipse becomes zero, then it becomes a

(a) straight line (b) circle (c) point (d) parabola

109) The director circle of the ellipse $\frac{x^2}{9} - \frac{y^2}{5} = 1$ is

(a) $x^2 + y^2 = 4$ (b) $x^2 + y^2 = 9$ (c) $x^2 + y^2 = 45$ (d) $x^2 + y^2 = 14$

110) The auxiliary circle of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ is

(a) $x^2 + y^2 = 25$ (b) $x^2 + y^2 = 16$ (c) $x^2 + y^2 = 41$ (d) $x^2 + y^2 = 5$

111) The length of the diameter of a circle with centre (1, 2) and passing through (5, 5) is

(a) 5 (b) $\sqrt{45}$ (c) 10 (d) $\sqrt{50}$

112) If (1, -3) is the centre of the circle $x^2 + y^2 + ax + by + 9 = 0$ its radius is

(a) $\sqrt{10}$ (b) 1 (c) 5 (d) $\sqrt{19}$

113) The area of the circle $(x - 2)^2 + (y - k)^2 = 25$ is

(a) 25π (b) 5π (c) 10π (d) 25

114) The equation of tangent at (1, 2) to the circle $x^2 + y^2 = 5$ is

(a) $x+y=3$ (b) $x + 2y = 3$ (c) $x- y= 5$ (d) $x - 2y = 5$

115) If $y = 2x + c$ is a tangent to the circle $x^2 + y^2 = 5$, then c is

(a) ± 5 (b) $\pm\sqrt{5}$ (c) $\pm 5\sqrt{2}$ (d) $\pm 2\sqrt{5}$

116) The line $y = mx +1$ is a tangent to the parabola $y^2 = 4x$ if m = _____

(a) 1 (b) 2 (c) 3 (d) 4

117) The angle between the tangents drawn from (1, 4) to the parabola $y^2 = 4x$ is _____

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{5}$ (d) $\frac{\pi}{5}$

118) In an ellipse $5x^2 + 7y^2 = 11$, the point (4, -3) lies _____ the ellipse

(a) on (b) outside (c) inside (d) none

119) If e_1, e_2 are eccentricities of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ then

(a) $e_1^2 - e_2^2 = 1$ (b) $e_1^2 + e_2^2 = 1$ (c) $e_1^2 - e_2^2 = 2$ (d) $e_1^2 - e_2^2 = 2$

120) The equation of the director circle of the hyperbola $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is _____

(a) $x^2 + y^2 = a^2 + b^2$ (b) $x^2 + y^2 = a^2$ (c) $x^2 + y^2 = b^2$ (d) $x^2 + y^2 = a^2 - b^2$

121) The number of normals to the hyperbola $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ from an external point is

(a) 2 (b) 4 (c) 6 (d) 5

122) The point of contact of $y^2 = 4ax$ and the tangent $y = mx + c$ is

(a) $\left(\frac{2a}{m^2}, \frac{a}{m}\right)$ (b) $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$ (c) $\left(\frac{a}{m}, \frac{2a}{m^2}\right)$ (d) $\left(\frac{-a}{m^2}, \frac{-2a}{m}\right)$

123) If B, B_1 are the ends of minor axis, F_1 ,F_2 are foci of the ellipse $\frac{x^2}{8}+\frac{y^2}{4}$ = 1 then area of $F_1BF_2B^1$ is

(a) 16 (b) 8 (c) $16\sqrt{2}$ (d) $32\sqrt{2}$

124) The length of major and minor axes of $4x^2 + 3y^2 = 12$ are _____

(a) $4, 2\sqrt{3}$ (b) $2, \sqrt{3}$ (c) $2\sqrt{3}, 4$ (d) $\sqrt{3}, 2$

125) The tangent at any point P on the ellipse $\frac{x^2}{6}+\frac{y^2}{3}$ = 1 whose centre C meets the major axis at T and PN is the perpendicular to the major axis; The CN CT = _____

(a) $\sqrt{6}$ (b) 3 (c) $\sqrt{3}$ (d) 6

126) The locus of the point of Inter eetton of perpendicular tangents to the hyperbela $\frac{x^2}{16}+\frac{y^2}{9}$ = 1 is _____

(a) $x^2+y^2= 25$ (b) $x^2 +y^2 = 4$. (c) $x^2 +y^2 = 3$ (d) $x^2+y^2=7$

127) If t_1 and t_2 are the extremities of any focal chord of $y^2 = 4ax$ then t_1t_2 is _____

(a) -1 (b) 0 (c) ±1 (d) $\frac{1}{2}$

128) The locus of the foot of perpendicular from the focus on any tangent to $y^2 = 4ax$ is

(a) $x^2 + y^2 = a^2 - b^2$ (b) $x^2 + y^2 = a^2$ (c) $x^2 + y^2 = a^2 - b^2$ (d) x = 0

129) Th point of curve $y = 2x^2 - 6x - 4$ at which the targent is parallel to x - axis is

(a) $\left(\frac{5}{2}, \frac{-7}{12}\right)$ (b) $\left(\frac{-5}{2}, \frac{-7}{2}\right)$ (c) $\left(\frac{-5}{2}, \frac{17}{12}\right)$ (d) $\left(\frac{3}{2}, \frac{-7}{2}\right)$

130) The locus of the point of intersection of perpendicular tangents of the parabola $y^2 = 4ax$ is

(a) latus rectum (b) directrix (c) tangent at the vertex (d) axis of the parabola

131) The vector, d$\hat{i}+\hat{j}+2\hat{k}, \hat{i}+\lambda\hat{j}-\hat{k}$ t and $2\hat{i}-\hat{j}+\lambda\hat{k}$ are co-planar if

(a) λ = -2 (b) λ = $1+\sqrt{3}$ (c) λ = $1 - \sqrt{3}$ (d) λ = $-2, 1\pm\sqrt{3}$

132) Let $\vec{a},\vec{b}$ and $\vec{c}$ be three non- coplanar vectors and let $\vec{p}, \vec{q}, \vec{r}$ be the vectors defined by the relations $\vec{P}=\frac{\vec{b}\times\vec{c}}{\left[\vec{a}\vec{b}\vec{c}\right]}, \vec{q}=\frac{\vec{c}\times\vec{a}}{\left[\vec{a}\vec{b}\vec{c}\right]}, \vec{r}=\frac{\vec{a}\times\vec{b}}{\left[\vec{a}\vec{b}\vec{c}\right]}$ Then the value of

$\left(\vec{a}+\vec{b}\right).\vec{p}+\left(\vec{b}+\vec{c}\right).\vec{q}+\left(\vec{c}+\vec{a}\right).\vec{r}=$

(a) 0 (b) 1 (c) 2 (d) 3

133) The number of vectors of unit length perpendicular to the vectors $\left(\hat{i}+\hat{j}\right)$ and $\left(\hat{j}+\hat{k}\right)$ is

(a) 1 (b) 2 (c) 3 (d) ∞

134) If $\vec{d} = \vec{a} \times \left(\vec{b} \times \vec{c}\right) + \vec{b} \times \left(\vec{c} \times \vec{a}\right) + \vec{c} \times \left(\vec{a} + \vec{b}\right)$, then

(a) $\left|\vec{d}\right|$ (b) $\vec{d} = \vec{a} + \vec{b} + \vec{c}$ (c) $\vec{d} = \vec{0}$ (d) a, b, c are coplanar

135) If $\vec{a}$ and $\vec{b}$ are two unit vectors, then the vectors $\left(\vec{a} + \vec{b}\right) \times \left(\vec{a} \times \vec{b}\right)$ is parallel to the vector

(a) $\vec{a} - \vec{b}$ (b) $\vec{a} + \vec{b}$ (c) $2\vec{a} - \vec{b}$ (d) $2\vec{a} + \vec{b}$

136) The area of the parallelogram having diagonals $\vec{a} = 3\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = \hat{i} - 3\hat{j} + 4\hat{k}$ is

(a) 4 (b) $2\sqrt{3}$ (c) $4\sqrt{3}$ (d) $5\sqrt{3}$

137) If $\vec{a}, \vec{b}$ and $\vec{c}$ are any three vectors, then $\vec{a} \times \left(\vec{b} \times \vec{c}\right) = \vec{a} \times \left(\vec{b} \times \vec{c}\right)$ if and only if

(a) $\vec{b}, \vec{c}$ are collinear (b) $\vec{a}$ and $\vec{c}$ are collinear (c) $\vec{a}$ and $\vec{b}$ are collinear (d) none

138) The volume of the parallelepiped whose sides are given by $\overrightarrow{OA} = 2\hat{i} - 3\hat{j}$, $\overrightarrow{OB} = \hat{i} + \hat{j} - \hat{k}$ and $\overrightarrow{OC} = 3\hat{i} - \hat{k}$ is

(a) $\frac{4}{13}$ (b) 4 (c) $\frac{2}{7}$ (d) $\frac{4}{9}$

139) If $\left|\vec{a}\right| = \left|\vec{b}\right| = 1$ such that $\vec{a} + 2\vec{b}$ and $5\vec{a} - \vec{b}$ are perpendicular to each other, then the angle between $\vec{a}$ and $\vec{b}$ is

(a) 45° (b) 60° (c) $\cos^{-1}\left(\frac{1}{3}\right)$ (d) $\cos^{-1}\left(\frac{2}{7}\right)$

140) The angle between the vector $3\hat{i} + 4\hat{j} + 5\hat{k}$ and the z-axis is

(a) 30° (b) 60° (c) 45° (d) 90°

141) The p.v, OP of a point P make angles 60° and 45° with X and Y axis respectively. The angle of inclination of $\overrightarrow{OP}$ with z-axis is

(a) 75° (b) 60° (c) 45° (d) 3

142) If $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = -\hat{i} + 2\hat{j} + \hat{k}$, $\vec{c} = 3\hat{i} + \hat{j}$ then $\vec{a} + \left(-\vec{b}\right)$ will be perpendiculur to $\vec{c}$ only when t =

(a) 5 (b) 4 (c) 3 (d) $\frac{7}{3}$

143) If θ is the angle between the vectors $\vec{a}$ and $\vec{b}$, then sinθ is

(a) $\frac{\vec{a}.\vec{b}}{\left|\vec{a}\right|\left|\vec{b}\right|}$ (b) $\frac{\left|\vec{a} \times \vec{b}\right|}{\vec{a}.\vec{b}}$ (c) $\sqrt{1 - \left(\frac{\vec{a}.\vec{b}}{\left|\vec{a}\right|\left|\vec{b}\right|}\right)^2}$ (d) 0

144) If the vector $\hat{i}+\hat{j}+2\hat{k}, -\hat{i}+2\hat{k}$ and $2\hat{i}+x\hat{j}-y\hat{k}$ are mutually orthogonal, then the values of x, y, z are

(a) (10, 4, 1) (b) (-10, 4, 1) (c) (-10, -4, $\frac{1}{2}$) (d) (-10, 4, $\frac{1}{2}$)

145) If $\vec{a} = |\vec{a}|\vec{e}$ then $\vec{e}.\vec{e}$

(a) 0 (b) e (c) 1 (d) $\vec{0}$

146) The value of $|\vec{a}+\vec{b}|^2 + |\vec{a}-\vec{b}|^2$ is

(a) $2\left(|\vec{a}|^2+|\vec{b}|^2\right)$ (b) $4\,\vec{a}.\vec{b}$ (c) $2\left(|\vec{a}|^2-|\vec{b}|^2\right)$ (d) $4\,|\vec{a}|^2|\vec{b}|^2$

147) If $\vec{p}\times\vec{q} = 2\hat{i}+3\hat{j}, \vec{r}\times\vec{s} = 3\hat{i}+2\hat{k}$ then $\vec{p}.\left(\vec{q}\left(\vec{r}\times\vec{s}\right)\right)$ is

(a) 9 (b) 6 (c) 2 (d) 5

148) If the work done by a force $\vec{F} = \hat{i}+m\hat{j}-\hat{k}$ in moving the point of application from(1, 1, 1) to (3, 3, 3) along a straight line is 12 units, then m is

(a) 5 (b) 2 (c) 3 (d) 6

149) The two planes 3x + 3y - 3z - 1 = 0 and x + y - z + 5 = 0 are

(a) mutually perpendicular (b) parallel (c) inclined at 45° (d) inclined at 30

150) The straight lines $\frac{x-3}{2} = \frac{y+5}{4} = \frac{z-1}{-13}$ and $\frac{x+1}{3} = \frac{y-4}{5} = \frac{z+2}{2}$ are

(a) parallel (b) perpendicular (c) inclined at 45° (d) none

151) For what value of $(\vec{a})$ will the straight lines $\frac{x+2}{a} = \frac{y}{3} = \frac{z-1}{4}$ and $\frac{x-3}{a} = \frac{y-1}{4} = \frac{z-7}{a}$ be perpendicular?

(a) 1 (b) 2 (c) 3 (d) -3

152) If $\left[\vec{a},\vec{b},\vec{c}\right]$=3 and $|\vec{c}| = 1$ then $\left|\left(\vec{b}\times\vec{c}\right)\times\left(\vec{c}\times\vec{a}\right)\right|$ is

(a) 1 (b) 3 (c) 6 (d) 9

153) If $\lambda\hat{i}+2\lambda\hat{j}+2\lambda\hat{k}$ is a unit vector, then the value of λ is

(a) $\pm\frac{1}{3}$ (b) $\pm\frac{1}{4}$ (c) $\pm\frac{1}{9}$ (d) $\frac{1}{2}$

154) For any three vectors $\vec{a}, \vec{b}$ and $\vec{c}$, $\left(\vec{a}+\vec{b}\right).\left(\vec{b}+\vec{c}\right)\times\left(\vec{c}+\vec{a}\right)$ is

(a) 0 (b) $\left[\vec{a},\vec{b},\vec{c}\right]$ (c) $2\left[\vec{a},\vec{b},\vec{c}\right]$ (d) $\left[\vec{a},\vec{b},\vec{c}\right]^2$

155) If the vectors $a\hat{i}+\hat{j}+\hat{k}, \hat{i}+b\hat{j}+\hat{k}$ and $\hat{i}+\hat{j}+c\hat{k}$ (a ≠ b ≠ c ≠ 1) are coplaner, then $\frac{1}{1-a}+\frac{1}{1-b}+\frac{1}{1-c} =$

(a) 0 (b) 1 (c) 2 (d) $\frac{abc}{(1-a)(1-b)(1-c)}$

156) The angle between the planes 2x + y - z = 9 and x + 2y + z = 7 is_____

(a) $\cos^{-1}(5/6)$ (b) $\cos^{-1}(5/36)$ (c) $\cos^{-1}(1/2)$ (d) $\cos^{-1}(1/12)$

157) The unit normal vector to the plane 2x + 3y + 4z = 5 is _____

(a) $\frac{2}{\sqrt{29}}\hat{i}+\frac{3}{\sqrt{29}}\hat{j}+\frac{4}{\sqrt{29}}\hat{k}$ (b) $\frac{2}{\sqrt{29}}\hat{i}-\frac{3}{\sqrt{29}}\hat{j}+\frac{4}{\sqrt{29}}\hat{k}$ (c) $\frac{2}{\sqrt{29}}\hat{i}-\frac{3}{\sqrt{29}}\hat{j}-\frac{4}{\sqrt{29}}\hat{k}$ (d) $\frac{2}{5}\hat{i}+\frac{3}{5}\hat{j}+\frac{4}{5}\hat{k}$

158) The work done by the force $\vec{F}=\hat{i}+\hat{j}+\hat{k}$ acting on a particle, if the particle is displaced from A(3, 3, 3) to the point B(4, 4, 4) is _____ units

(a) 2 (b) 3 (c) 4 (d) 7

159) The angle between the vectors $\hat{i}-\hat{j}$ and $\hat{j}-\hat{k}$ is _____

(a) $\frac{\pi}{3}$ (b) $\frac{-2\pi}{3}$ (c) $\frac{-\pi}{3}$ (d) $\frac{2\pi}{3}$

160) The unit normal vectors to the plane 2x - y + 2z = 5 are _____

(a) $2\hat{i}-\hat{j}+2\hat{k}$ (b) $\frac{1}{3}\left(2\hat{i}-\hat{j}+2\hat{k}\right)$ (c) $-\frac{1}{3}\left(2\hat{i}-\hat{j}+2\hat{k}\right)$ (d) $\pm\frac{1}{3}\left(2\hat{i}-\hat{j}+2\hat{k}\right)$

161) The distance from the origin to the plane $\vec{r}.\left(2\hat{i}-\hat{j}+5\hat{k}\right)=7$ is _____

(a) $\frac{7}{\sqrt{30}}$ (b) $\frac{\sqrt{30}}{7}$ (c) $\frac{30}{7}$ (d) $\frac{7}{30}$

162) If $\vec{a},\vec{b},\vec{c}$ are mutually $\perp^r$ unit vectors, then $\left|\vec{a}+\vec{b}+\vec{c}\right|$ is _____

(a) 3 (b) 9 (c) $3\sqrt{3}$ (d) $\sqrt{3}$

163) Let $\vec{u},\vec{v},\vec{w}$ be vectors such that $\vec{u}+\vec{v}+\vec{w}=\vec{0}$. If $|\vec{u}|=3$, $|\vec{v}|=4$, $|\vec{w}|=5$ then $\vec{u}.\vec{v}+\vec{v}.\vec{w}+\vec{w}.\vec{u}$ is _____

(a) 25 (b) -25 (c) 5 (d) $\sqrt{5}$

164) The length of the $\perp^r$ from the origin to plane $\vec{r}.\left(3\hat{i}+4\hat{j}+12\hat{k}\right)=26$ is _____

(a) 2 (b) $\frac{1}{2}$ (c) 26 (d) $\frac{26}{169}$

165) If $\left|\vec{a}\times\vec{b}\right|=\vec{a}.\vec{b}$, then the angle between the vector $\vec{a}$ and $\vec{b}$ is _____

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{6}$ (d) $\frac{\pi}{2}$

166) The value of $\left|\vec{a}+\vec{i}\right|^2+\left|\vec{a}+\vec{j}\right|^2+\left|\vec{a}+\vec{k}\right|^2$ if $|a|=1$ is _____

(a) 0 (b) 1 (c) 2 (d) 3

167) If $\vec{a},\vec{b},\vec{c}$ are three non - coplanar vectors, then $\frac{\vec{a}.\vec{b}\times\vec{c}}{\vec{c}\times\vec{a}.\vec{b}}+\frac{\vec{b}.\vec{a}\times\vec{c}}{\vec{c}.\vec{a}\times\vec{b}}=$ _____

(a) 0 (b) 1 (c) -1 (d) $\frac{\vec{a}.\vec{b}\times\vec{c}}{\vec{b}\times\vec{c}.\vec{c}}$

168) If $\vec{d} = \lambda\left(\vec{a} \times \vec{b}\right) + \mu\left(\vec{b} \times \vec{c}\right) + \omega\left(\vec{c} \times \vec{a}\right)$ and $\left|\vec{c} \times \vec{a}\right| = \frac{1}{8}$ then λ + μ + ω is

_____

(a) 0 (b) 1 (c) 8 (d) $8\vec{d}.\left(\vec{a} + \vec{b} + \vec{c}\right)$

169) The area of the parallelogram having diagonals $\vec{a} = 3\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = \hat{i} - 3\hat{j} + 4\hat{k}$ is

_____

(a) 4 (b) $2\sqrt{3}$ (c) $4\sqrt{3}$ (d) $5\sqrt{3}$

170) Let $\vec{a}, \vec{b},$ and $\vec{c}$ be three vectors having magnitudes 1, 1, 2 respectively.

If $\vec{a} \times \left(\vec{a} \times \vec{c}\right) + \vec{b} = 0$ then the acute angle between $\vec{a}$ and $\vec{c}$ is

_____

(a) 0 (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{6}$ (d) $\frac{2\pi}{3}$

171) If a particle moves in a straight line according to s = t^3-$6t^2$-15t, the time interval during which the velocity is negative and acceleration is positive is

(a) 2 < t < 5 (b) 2 ≤ t ≤ 5 (c) t ≥ 2 (d) t ≤ 2

172) The law of linear motion of a particle is given $\frac{1}{3}$ t^3-16t, the acceleration at the time when the velocity vanishes is

(a) 4 (b) 6 (c) 2 (d) 8

173) If the rate of increase of s =x^3-$5x^2$+5x+8 is twice the rate of increase of x, then one value of x is

(a) $\frac{3}{5}$ (b) $\frac{10}{3}$ (c) $\frac{3}{10}$ (d) $\frac{1}{3}$

174) The point on the curve y=x^2 is the tangent parallel to X-axis is

(a) (1,1) (b) (2,2) (c) (4,4) (d) (0,0)

175) The equation of the tangent to the curve y=x^2-4x+2 at (4,2) is

(a) x + 4y + 12 = 0 (b) 4x + y + 12 = 0 (c) 4x - y - 14 = 0 (d) x + 4y - 12 = 0

176) Equation of the normal to the curve y=$2x^2$+3 sin x at x=0 is

(a) x + y = 0 (b) 3y = 0 (c) x + 3y = 7 (d) x + 3y = 0

177) The least value of a when f f(x) =x^2+ax+1 is increasing on (1, 2) is

(a) -2 (b) 2 (c) 1 (d) -1

178) The value of $lim_{x \to \infty} e^{-x}$ is

(a) 0 (b) ∞ (c) e (d) $\frac{1}{e}$

179) The angle made by any tangent to the curve y = x^5 + 8x + 1 with the X-axis is a

(a) obtuse (b) right angle (c) acute angle (d) no angle

180) The critical points of the function f(x) = $(x-2)^{\frac{2}{3}}(2x+1)$ are

(a) -1, 2 (b) 1, $\frac{1}{2}$ (c) 1, 2 (d) none

181) The equation of the tangent to the curve x = t cost, y = t sin t at the origin is

(a) x = 0 (b) y = 0 (c) x +y = 0 (d) x + y = 7

182) In LMV theorem, we have f'(x_1) =$\frac{f(b)-f(a)}{b-a}$ then a < x_1 _____

(a) < b (b) ≤b (c) =b (d) ≠b

183) If the slope of the curve $2y^2$=ax^2+b at (1,-1) is - 1, then the values of a, b is

(a) 2, 0 (b) 0, 2 (c) 0, 0 (d) 2, 2

184) If the curves $y = 2e^x$ and $y = ae^{-x}$ intersect orthogonally, then a = _____

(a) $\frac{1}{2}$ (b) $-\frac{1}{2}$ (c) 2 (d) $2e^2$

185) The function -3x+12 is _____ function on R.

(a) decreasing (b) strictly decreasing (c) increasing (d) strictly increasing

186) The function/(x) = $x^9 + 3x^7 + 64$ is increasing on _____

(a) R (b) $(-\infty, 0)$ (c) $(0, \infty)$ (d) None of these

187) If x + y = 8, then the maximum value of xy is _____

(a) 8 (b) 16 (c) 20 (d) 24

188) The curve $y = e^x$ is _____

(a) convex (b) concave (c) convex upwards (d) concave upwards

189) $lim_{x \to 0} \frac{x}{tanx}$ is _____

(a) 1 (b) -1 (c) 0 (d) ∞

190) The statement " If f has a local extremum at c and if f'(c) exists then f'(c) = 0" is _____

(a) the extreme value theorem (b) Fermats' theorem (c) Law of mean (d) Rolle's theorem

191) If $y = x^4 - 10$ and if x changes from 2 to 1.99, the approximate change in y is

(a) -32 (b) -0.32 (c) - 10 (d) 10

192) If the radius of the sphere is measured as 9 em with an error of 0.03 cm, the approximate error in calculating its volume is

(a) 9.72 cm^3 (b) 0.972 cm^3 (c) 0.972π cm^3 (d) 9.72π cm^3

193) If $\log_e 4 = 1.3868$, then $\log_e 4.01$ =

(a) 1.3968 (b) 1.3898 (c) 1.3893 (d) none

194) If u = log $\sqrt{x^2+y^2}$, then $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$ is

(a) $\sqrt{x^2+y^2}$ (b) 0 (c) u (d) 2u

195) If $u = x^y + y^x$ then $u_x + u_y$ at x = y = 1 is

(a) 0 (b) 2 (c) 1 (d) ∞

196) If $u = (x-y)^4+(y-z)^4 +(z-x)^4$ then $\sum \frac{\partial u}{\partial x}$ =

(a) 4 (b) 1 (c) 0 (d) -4

197) If f(x, y, z) = sin (xy) + sin (yz) + sin (zx) then f_{xx} is

(a) -y sin (xy) + z^2 cos (xz) (b) y sin (xy) - z^2 cos (xz) (c) y sin (xy) + z^2 cos (xz) (d) -y sin (xy) - z^2 cos (xz)

198) If u = log $(x^3 + y^3 + z^3 - 3xyz)$ then $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$ =

(a) $\frac{3}{x+y+z}$ (b) x+y+z (c) $\frac{-9}{(x+y+z)^2}$ (d) $\frac{-9}{(x+y+z)^2}$

199) If f (x, y) = $x^3 + y^3 - 3xy^2$ then $\frac{\partial f}{\partial x}$ at x = 2,

(a) -15 (b) 15 (c) -9 (d) 16

200) If f(x,y) = $2x^2 - 3xy + 5y + 7$ then f(0, 0) and f(1, 1) is

(a) 7,11 (b) 11,7 (c) 0,7 (d) 1,0

201) The approximate value of $(627)^{\frac{1}{4}}$ is .....

(a) 5.002 (b) 5.003 (c) 5.005 (d) 5.004

202) The cube root of 127 is .....

(a) 5.026 (b) 5.26 (c) 5.028 (d) 5.075

203) If y = sin x and x changes from $\frac{\pi}{2}$ to π the approximate change in y is .....

(a) 0 (b) 1 (c) $\frac{\pi}{2}$ (d) $\frac{22}{14}$

204) If u = y^x then $\frac{\partial u}{\partial y}$ = ...

(a) xy^{x-1} (b) yx^{y-1} (c) 0 (d) 1

205) If u = sin-1 $\left(\frac{x^4+y^4}{x^2+y^2}\right)$ and f= sin u then f is a homogeneous function of degree .....

(a) 0 (b) 1 (c) 2 (d) 4

206) If x = r cos θ, y = r sin, then $\frac{\partial r}{\partial x}$ = ...

(a) sec θ (b) sin θ (c) cos θ (d) cosec θ

207) The percentage error in the 11th root of the number 28 is approximately ..... times the percentage error in 28.

(a) $\frac{1}{28}$ (b) $\frac{1}{11}$ (c) 11 (d) 28

208) If u = $(\frac{y}{x})$ then x $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}$ = ...

(a) 0 (b) 1 (c) 2u (d) u

209) If u = y sin x then $\frac{\partial^2 u}{\partial x \partial y}$ = ...

(a) cos x (b) cos y (c) sin x (d) 0

210) If is a homogeneous function of x and y of degree n, then $x\frac{\partial^2 u}{\partial x^2}+y\frac{\partial^2 u}{\partial x \partial y}$ = ... $\frac{\partial u}{\partial x}$

(a) n (b) 0 (c) 1 (d) n - 1

211) The value of $\int_0^{\frac{\pi}{2}} \frac{dx}{1+tanx}$

(a) π (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{4}$ (d) 0

212) The value of $\int_{\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{\frac{1-cos2x}{2x}}$ dx is

(a) $\frac{1}{2}$ (b) 2 (c) 0 (d) 1

213) $\int_1^{\sqrt{3}} \frac{dx}{1+x^2}$ is

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{6}$ (c) $\frac{\pi}{12}$ (d) $-\frac{\pi}{6}$

214) If $\int_0^{2a} f(x)dx = 2\int_0^{a} f(x)$ then

(a) f(2a -x) = - f(x) (b) f(2a - x) = f(x) (c) f(x) is odd (d) f(x) is even

215) The value of $\int_{-\pi}^{\pi} sin^3x\ cos^3x\ dx$ is

(a) 0 (b) π (c) 2π (d) 4π

216) The area enclosed by the curve y = $\frac{x^2}{2}$, the x - axis and the lines x = 1, x = 3 is

(a) 4 (c) 13

(b) $8\frac{2}{3}$ (d) $4\frac{1}{3}$

217) The area bounded by the parabola $y = x^2$ and the line $y = 2x$ is

(a) $\frac{4}{3}$ (b) $\frac{2}{3}$ (c) $\frac{51}{3}$ (d) $\frac{30}{3}$

218) The ratio of the volumes generated by revolving the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ about major and minor axes is

(a) 4:9 (b) 9:4 (c) 2:3 (d) 3:2

219) $\int_0^\infty e^{-mx} x^7$ dx is

(a) $\frac{\lfloor m}{7^m}$ (b) $\frac{\lfloor 7}{m^7}$ (c) $\frac{\lfloor m}{7^{m+1}}$ (d) $\frac{\lfloor 7}{m^8}$

220) If $\int_0^a f(x)dx + \int_0^a f(2a-x)dx =$

(a) $\int_0^a f(x)dx$ (b) $2\int_0^a f(x)dx$ (c) $\int_0^{2a} f(x)dx$ (d) $\int_0^{2a} f(a-x)dx$

221) $\int_{-1}^{1} x\, dx$ = ...

(a) -1 (b) 1 (c) 0 (d) 2

222) $\int_0^{2a} f(x)dx = 2\int_0^{2a} f(x)dx$

(a) f(2a -x) =f(x) (b) f(a -x) =f(x) (c) f(x) = - f(-x) (d) f(-x) =f(x)

223) The area enclosed by the curve $y^2 = 4x$, the x-axis and its latus rectum is ..... sq.units.

(a) $\frac{2}{3}$ (b) $\frac{4}{3}$ (c) $\frac{8}{3}$ (d) $\frac{16}{3}$

224) The area of the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$

(a) 6π (b) 36π (c) $6\pi^2$ (d) $36\pi^2$

225) The volume generated by the curve $y^2 = 16x$ from x = 2 to x = 3 rotating about x - axis ..... cu. units

(a) 72π (b) $\frac{256 \times 19}{3}\pi$ (c) 40π (d) 80π

226) $\int_a^b f(x)dx =$ .....

(a) $2\int_0^a f(x)dx$ (b) $\int_a^b f(a-x)dx$ (c) $\int_b^a f(b-x)dx$ (d) $\int_a^b f(a+b-x)dx$

227) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{sinx}{2+cosx} dx =$

(a) 0 (b) 2 (c) log 2 (d) log 4

228) $\int_0^{\frac{\pi}{2}} cos^3 2x\, dx =$

(a) $\frac{2}{3}$ (b) $\frac{1}{3}$ (c) 0 (d) $\frac{2\pi}{3}$

229) $\int_0^{\frac{\pi}{2}} \frac{sin\, x - cos\, x}{1 + sin\, x cos\, x} dx =$ .....

(a) $\frac{\pi}{2}$ (b) 0 (c) $\frac{\pi}{4}$ (d) π

230) The volume when $y = \sqrt{3+x^2}$ from x = 0 to x = 4 is rotated about x-axis is .....

(a) 100π (b) $\frac{100\pi}{9}$ (c) $\frac{100\pi}{3}$ (d) $\frac{100}{3}$

231) The order and degree of the differential equation $\left[\left(\frac{d^2y}{dx^2}\right)+\left(\frac{dy}{dx}\right)\right]^{\frac{1}{2}}=\frac{d^3y}{dx^3}$ are

(a) 1,2 (b) 2,1 (c) 3,2 (d) 2,3

232) The differential equation of the family of parabolas $y^2=4ax$ is

(a) $2y=x\left(\frac{dy}{dx}\right)$ (b) $y=2x\left(\frac{dy}{dx}\right)$ (c) $y=2x^2\left(\frac{dy}{dx}\right)$ (d) $y^2=2x\left(\frac{dy}{dx}\right)$

233) The solution of $\sec^2x \tan y\, dx+\sec^2y \tan x\, dy=0$ is

(a) tan x+tan y =c (b) sec x+sec y=c (c) tan x tan y=c (d) sec x-sec y =c

234) The solution of $(x^2-ay)dx=(ax-y^2)dy$ is

(a) $y=x^2+y^2-a(x+y)$ (b) $y=x^2+y^2-a(x+y)$ (c) $x^3+y^2=3ayx+c$ (d) $(x^2-ay)(ax-y^2)=0$

235) The transformation y=vx reduces $\frac{dy}{dx}=\frac{x+y}{3x}$

(a) $\frac{3av}{4v+1}=\frac{dx}{x}$ (b) $\frac{3dv}{v+1}=\frac{dx}{x}$ (c) $2x\frac{dv}{dx}=v$ (d) $\frac{3dv}{1-2v}==\frac{dx}{x}$

236) The I.F. of cosec x$\frac{dy}{dx}+y\sec^2x=0$ is

(a) $e^{\sec x}$ (b) $e^{\tan x}$ (c) $e^{\sec x \tan x}$ (d) $e^{\sec 2 x}$

237) The solution of $\frac{dy}{dx}+y$ cot x=sin 2x is

(a) y sin x=$\frac{2}{3}\sin_3x+c$ (b) y sec x=$\frac{x^2}{2}+c$ (c) y sin x =c+x (d) 2y sin x=sin x-$\frac{\sin 3x}{3}+c$

238) The general solution of $4\frac{d^2y}{dx^2}$+y=0 is

(a) $y=e^{\frac{x}{2}}\left[A\ \cos\frac{x}{2}+B\ \sin\frac{x}{2}\right]$ (b) $y=e^{\frac{x}{2}}\left[A\ \cos\frac{x}{2}-B\ \sin\frac{x}{2}\right]$ (c) $y=A\cos\frac{x}{2}+B\sin\frac{x}{2}$ (d) $t=Ae^{\frac{x}{2}}+Be^{\frac{-x}{2}}$

239) The I.F of y log y$\frac{dx}{dy}+x-\log y=0$ is

(a) log(log y) (b) log y (c) $\frac{1}{\log y}$ (d) $\frac{1}{\log(\log y)}$

240) The I.F of $\frac{dy}{dx}-y$ tan x=cos x is

(a) sec x (b) cos x (c) $e^{\tan x}$ (d) cot x

241) The order and degree of $y'+(y'')^2=(x+t'')^2$ are _____.

(a) 1,1 (b) 1,2 (c) 2,1 (d) 2,2

242) The differential equation corresponding to $xy=c^2$ where c is an arbitrary constant is _____.

(a) xy"+x=0 (b) y"=0 (c) xy'+y=0 (d) xy"-x=0

243) On finding the differential equation corresponding to $y=e^{mx}$ where m is the arbitrary constant, then m is _____.

(a) $\frac{y}{y^1}$ (b) $\frac{y^1}{y}$ (c) y' (d) y

244) The population p of a certain bacteria decreases at a rate proportional to the population p. The differential equation corresponding to the above statement is

_____.

(a) $\frac{dp}{dt} = \frac{k}{p}$ (b) $\frac{dp}{dt} = kt$ (c) $\frac{dp}{dt} = kp$ (d) $\frac{dp}{dt} = -kp$

245) The solution of log $\left(\frac{dy}{dx}\right)$=ax+by is_____.

(a) $\frac{e^{ax}}{a} + \frac{e^{-by}}{b} + c = 0$ (b) $ae^{ax}-be^{-by}+c=0$ (c) $ae^{x}+be^{y}=k$ (d) $be^{ax}+ae^{-by}=k$

246) The general solution of x $\frac{dy}{dx}$=y is _____.

(a) y=cx (b) $x^2+y^2=c$ (c) $x^2-y^2=c$ (d) $y=c^x$

247) The differential equation of $x^2y = k$ is _____.

(a) $x^2\frac{dy}{dx} = 0$ (b) $x^2\frac{dy}{dx} + y = 0$ (c) $x\frac{dy}{dx} + 2y = 0$ (d) $y\frac{dy}{dx} + 2x = 0$

248) Using y = vx, the differential equation $\frac{dy}{dx} = \frac{y}{x+\sqrt{xy}}$ is reduced to _____.

(a) $x(1+\sqrt{v})dv=v\sqrt{v}$ dx (b) $x(1-\sqrt{v})dv=v\sqrt{v}$ dx (c) $x(1+\sqrt{v})dv=-v\sqrt{v}$ dx (d) $v(1+\sqrt{v})dx-v\sqrt{v}$ dv=0

249) The I.F. of $(1+y^2)dx=(\tan^{-1}-t-x)dy$ is _____.

(a) $e^{\tan^{-1}}$ y (b) $e^{\tan^{-1}}$ x (c) $\tan^{-1}$ y (d) $\tan^{-1}x$

250) The differential equation associated with the family of concentric circles having their centres at the origin is _____.

(a) $\frac{dy}{dx} = \frac{-x}{y}$ (b) $\frac{dy}{dx} = \frac{-y}{x}$ (c) $\frac{dy}{dx} = \frac{x}{y}$ (d) $\frac{dy}{dx} = \frac{y}{x}$

251) The binary operation * defined on a set s is said to be commutative if

(a) a*b ∈ S ∀ a, b ∈ S (b) a*b = b*a ∀ a, b ∈ S (c) (a*b) * c = a*(b*c) ∀ a, b ∈ S (d) a*b = e ∀ a, b ∈ S

252) If * is defined by $a * b = a^2 + b^2 + ab + 1$, then (2 * 3) * 2 is

(a) 20 (b) 40 (c) 400 (d) 445

253) The number of binary operations that can be defined on a set of 3 elements is

(a) 3^2 (b) 3^3 (c) 3^9 (d) 3^1

254) The Identity element of $\left\{\begin{pmatrix} x & x \\ x & x \end{pmatrix}\right\}$ |x ∈ R, x≠0} under matrix multiplication is

(a) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (b) $\begin{pmatrix} \frac{1}{4x} & \frac{1}{4x} \\ \frac{1}{4x} & \frac{1}{4x} \end{pmatrix}$ (c) $\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$ (d) $\begin{pmatrix} \frac{1}{2x} & \frac{1}{2x} \\ \frac{1}{2x} & \frac{1}{2x} \end{pmatrix}$

255) Which one of the following is not a statement?

(a) 2 + 3 =5 (b) How beautiful is this flower? (c) Delhi is the capital of Tamil Nadu (d) A triangle has found angles.

256) Which of the following is a tautology?

(a) p v q (b) p ∧ q (c) q v ~ q (d) q ∧ ~ q

257) Which of the following is a contradiction?

(a) p v q (b) p ∧ q (c) q v ~ q (d) q ∧ ~ q

258) The identity element in the group {R - {1},x} where a * b = a + b - ab is

(a) 0 (b) 1 (c) $\frac{1}{a-1}$ (d) $\frac{a}{a-1}$

259) Define * on Z by a*b = a+b+1 ∀ a,b ∈ Z. Then the identity element of z is
(a) 1 (b) 0 (c) 1 (d) -1

260) A binary operation * is defined on the set of positive rational numbers Q^+ by a*b = $\frac{ab}{4}$. Then 3 * $\left(\frac{1}{5} * \frac{1}{2}\right)$ is
(a) $\frac{3}{160}$ (b) $\frac{5}{160}$ (c) $\frac{3}{10}$ (d) $\frac{3}{40}$

261) If a * b = a^2b^2 - ab then 3 * (1 * 1)
(a) 0 (b) 1 (c) 2 (d) 4

262) The number whose multiplication universe does not exist in C.
(a) 0 (b) 1 (c) 0 (d) 1

263) Let p: Kamala is going to school
q: There are 20 students in the class. Then Kamala is not going to school or there are 20 students in the class is represented by
(a) p v q (b) p ∧ q (c) ~ p (d) ~ p v q

264) If p is true and q is unknown, then _____
(a) ~ p is true (b) p v (~p) is false (c) p ∧ (~p) is true (d) p v q is true

265) '+' is not a binary operation on
(a) ~ (b) z (c) c (d) Q- {0}

266) '-' is a binary operation on
(a) ~ (b) Q-{0} (c) R-{0} (d) Z

267) Which of the following is a statement?
(a) 7+2< 10 (b) Wish you all success (c) All the best (d) How old are you?

268) In (N, *), x * y = max(x, y), x, y ∈ N then 7 * (-7)
(a) 7 (b) -7 (c) 0 (d) -49

269) In (S, *), is defined by x * y = x where x, y ∈ S, then
(a) associative (b) Commutative (c) associative and commutative (d) neither associative nor commutative

270) The number of commutative binary operations which can be defined on a set containing n elements is
(a) $n^{\frac{n(n+1)}{2}}$ (b) n^{n^2} (c) $n^{\frac{n}{2}}$ (d) n^2

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RAVI MATHS TUITION CENTER, NEAR VILLIVAKKAM RLY STATION, CHENNAI – 82.
WHATSAPP - 8056206308

MCQ CREATIVE 2

12th Standard

Maths

Exam Time : 02:40:00 Hrs

Total Marks : 160
160 x 1 = 160

1) Trivial solution of AX=0
2) Non - Trivial solution of AX=0
3) $\rho(A) = \rho[(A/0)] < n$
4) $\rho(A) = \rho[(A/0)] = n$
5) $\rho(A) = \rho[(A|B]) = 3$ = number of unknowns
6) $\rho(A) = \rho[(A|B]) = 2 <$ number of unknowns
7) $\rho(A) = \rho[(A|B]) = 1 <$ number of unknowns
8) $\rho(A) \neq \rho[(A|B])$
The values given to the arbitrary constraints in the general stores
9) [adj A]
10) $(\text{adj } A)^T$
11) adj (adj A)
12) |adj (adj A)|
Consistent with two parameter family of solution
13) $(\text{adj } A)^{-1}$
14) $(\lambda A)^{-1}$
15) adj (AB)
16) $(A^T)^{-1}$
17) A(adj A)
18) $(AB)^{-1}$
19) $(A^{-1})^{-1}$
20) adj (λA)
21) Re(z)
22) Im(z)
23) z is real
24) z is imaginary
25) |z|
26) $|z_1 + z_2|$
27) $|z_1 - z_2|$

(1) (a, 0)
(2) $\frac{1}{\lambda} A^{-1}$
(3) $\frac{\pi}{4}$
(4) Non - trivial solution
(5) Vieta's formula
(6) Distributive law
(7) (0, 1)
(8)
(9) $\frac{\text{Absolute error}}{\text{Actual error}}$
(10) $-\operatorname{cosec}^2(x^2)(2x)\,dx$
(11) ~q
(12)
(13) $\frac{n-1}{n} I_{n-2}$ $n \geq 2$
(14) $\frac{1}{30}$
(15) $\text{adj}(A^{-1})$
(16) $\frac{\pi}{2}$
(17) $\frac{\pi}{4a}$
(18) Idempotent law
(19) velocity
(20) $2n\pi - \frac{\pi}{2}$
(21) Reciprocal equation of type I
(22) Inverse element
(23) 8
(24) $\frac{-\pi}{6}$
(25) $\frac{\pi}{2a}$
(26) Complement law
(27) Involution law

28) $|z_1z_2|$
29) arg (0)
30) arg (i)
31) arg (-1)
32) arg (-i)
33) arg (z_1z_2)
34) $\arg\left(\frac{z_1}{z_2}\right)$
35) arg (z^n)
36) arg (z)
37) $\left|\sqrt{3}+i\right|$
38) $\left|-\sqrt{3}+i\right|$
39) |1+i|
40) |2+2i|
41) $6x^6 - 35x^5 + 56x^4 - 56x^2 + 35x - 6 = 0$
42) $p(x)=x^n.p\left(\frac{1}{x}\right)$
43) $2x^7-3x^6-4x^5+5x^4+6x^3-7x+8 = 0$
44) $9x^9+2x^5-x^4-7x^2+2=0$
45) $x^9-5x^8-14x^7=0$
46) $x^9+9x^7+3x+7x^5+5x^3$
47) imaginary roots occur in pairs
48) A polynomial of degree n ≥ 1 has at least one root in C
49) $\Sigma\propto\beta=\frac{c}{a}$ in $ax^3+bx^2+cx+d=0$
50) $x^3-3x^2-33x+35=0$
51) $7x^3 - 43x^2 - 43x + 7 = 0$
52) $x^4 - 10x^3 + 26x^2 - 10x + 1= 0$
53) Period of sine function
54) Period of tangent function
55) Distance required to complete one full cycle
56) Amplitude of sine function
57) $\sin^{-1}(3x-4x^3)$
58) $\cos^{-1}(4x^3-3x)$
59) $\sin^{-1}\left(\frac{1}{x}\right)$
60) $\cos^{-1}(-x)$
61)

(28) $y^{11}-4y=0$
(29) $\frac{1}{1+x^2}$
(30) local minimum at c
(31) acceleration
(32) f is an identity function
(33) f is a constant function
(34) 2,2
(35) zero sum of all co-efficients
(36) $|A|^{n-2}.A$
(37) $xy^1=y\log y$
(38) Demorgan's law
(39) 2
(40) f'(x)
(41) tan(x+y)-sec(x+y)=x+c
(42) x = -1 is one root
(43) Steepness of a hill side
(44) (adj B) (adj A)
(45) $\arg(z_1) - \arg(z_2)$
(46) fdg +gdf
(47) Integral power of the highest order derivative.
(48) $f^v(0)$
(49) $\leq |z_1| + |z_2|$
(50) $\frac{-\pi}{4}$
(51) $\frac{z+z}{2}$
(52) $\frac{\pi}{4}$
(53) $y + \frac{\sin 2y}{2} + \frac{\cos 7x}{7} + \frac{\cos 3x}{3} = c$
(54) f'(x) dx
(55) $2\sqrt{2}$
(56) Highest derivative in the differential equation
(57) 1,1
(58) Trivial solution
(59) Maclaurin's series
(60) $\frac{n-1}{m+n}I_{m,n-2}$ n ≥ 2
(61) $|z_1|\ |z_2|$

$sec^{-1}\left(\frac{2}{3}\right) + cosec^{-1}\left(\frac{2}{3}\right)$

62) $cos\left(tan^{-1}\left(\frac{3}{4}\right)\right)$ (62) adj (A^T)

63) $sin^{-1}\left(cos^{-1}\left(\frac{3}{5}\right)\right)$ (63) $|A|^{n-1}$

64) $cos^{-1}(-1)$ (64) $\frac{\pi}{a}$

65) $sec^{-1}(2)$ (65) $(x^2+1)^2$

66) $sin^{-1}\left(\frac{-1}{2}\right)$ (66) 2π

67) $sin^{-1}\left(sin\left(\frac{-\pi}{3}\right)\right)$ (67) |z |

68) $sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (68) F(t) - F (a)

69) $sin^{-1}\left(sin\frac{2\pi}{3}\right)$ (69) $\frac{\pi}{12}$

70) $tan^{-1}\left(tan\left(\frac{3\pi}{3}\right)\right)$ (70) 2,1

71) $tan^{-1}\left(tan\left(\frac{7\pi}{6}\right)\right)$ (71) A

72) $tan^{-1}\left(cos\left(\frac{13\pi}{6}\right)\right)$ (72) Taylor's series

73) Focus of $(y + 1)^2 = 8(x - 3)$ (73) 1

74) Vertex of $(y -1)^2 = 8x$ (74) 2nπ

75) LR of $(y + 1)^2 = 8(x - 3)$ (75) $xy^1+y=0$

76) LR of $(x - 4)^2 = 4(x + 2)$ (76) complex conjugate root theorem

77) f'(x) changes from + ve to - ve at c (77) e^x

78) f'(x) changes from - ve to + ve at c (78) 2

79) f'(x) +ve on both sides of c. (79) $3\cos^{-1}x$

80) f'(x) - ve on both sides of c (80) $ex(x^2-2x+2)+\log y=c$

81) f'(c) = $\frac{f(b)-f(a)}{b-a}$ (81) $\frac{2}{3}$

82) f'(c) =0 (82) fundamental theorem of Algebra

83) $\sum_{n=0}^{\infty}\frac{f^n(a)}{n!}$ $(x-a)^n$ (83) e - 2

84) $\sum_{n=0}^{\infty}\frac{f^n(0)}{n!}$ $.x^n$ (84)

2 is the only even prime or sin x is an even function

85) Rate of change In vertical length with respect to horizontal length (85) $\pi - \cos^{-1}x$

(86) 3,3

86) Rate of displacement with respect to time

87) Rate of change In velocity with respect to time

(87) $1+x^2$

88) Rate of change In elevation with respect to linear distance

(88) $y=2xy^1$

89) $\sec^2 x$

(89) period

90) $2 \sec^2 x \tan x$

(90) $|A| \neq 0$

91) 2

(91) $\frac{2x}{x^2+1}$

92) 16

(92)

Consistent with one parameter family of solution

93) Absolute error

(93) 2 is not the only even prime

94) Relative error

(94) $\arg z_1 + \arg z_2$

95) Percentage error

(95) $|A| = 0$

96) $\frac{Absolute\ error}{Actual\ error}$ x 100

(96) $z=-z$

97) df

(97) 4 Change of sign

98) Δf

(98) $e^{\frac{x}{x}}$

99) $\frac{df}{dx}$

(99) $-\sin(x^2+7x)(2x+7)dx$

100) f(x)

(100) f'''(0)

101) df = 0

(101) $\frac{\pi}{3}$

102) df = dx

(102) f''(0)

103) $dh = e^{f(x)} f'(x)dx$

(103) λ^{n-1} adj(A)

104) d(fg)

(104) $(A^{-1})^T$

105) $\log(x^2 + 1)$

(105) (G, *) is closed

106) $\cos(x^2+ 7x)$

(106)

neither a local minimum nor a local maximum

107) $\tan^{-1} x$

(107) Percentage error

108) $\cot(x^2)$

(108) at least 6 imaginary roots

109) $\int_0^1 x^2 e^x dx$

(109) $2n\pi + \pi$

110) $\int_0^{\frac{\pi}{2}} sin2x\ cos\ dx$

(110) $|A|. I_n$

111) $\int_0^1 \frac{dx}{\sqrt{4-x^2}}$

(111) In consistent and has no solution

112) $\int_0^{\frac{\pi}{2}} sin^2 x dx$

(112) π

113) $\int_0^1 log\left(\frac{1}{x} - 1\right)dx$

(113) $f(x_0) + f'(x_0)(x - x_0) \forall x \in (a, b)$

114) $\int_0^1 x(1-x)^{10} dx$

(114) Lagranges mean value theorem

115) $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{dx}{1+\sqrt{tanx}}$

(115) z=z

116) $\int_0^1 x(1-x)^4 dx$

(116)

It contains many arbitrary constants as the order of the differential equation.

117) $\int_0^t f(x)dx$

118) $\int_0^\infty \frac{dx}{a^2+x^2}$

119) $\int_a^\infty \frac{dx}{a^2+x^2}$

120) $\int_{-\infty}^\infty \frac{dx}{a^2+x^2}$

121) $\int_0^\infty e^{-x}.x^n dx$

122) $\int_0^{\frac{\pi}{2}} sin^n x dx$

123) $\int_0^1 x^m(1-x)^n dx$

124) $\int_0^{\frac{\pi}{2}} sin^m x cos^n x dx$

125) $\frac{dy}{dx}+y=x^2$

126) $\frac{d^2y}{dx^2}+x=\sqrt{y+\frac{dy}{dx}}$

127) $\frac{d^2y}{dx^2}-y+\left(\frac{dy}{dx}+\frac{d^2y}{dx^3}\right)^{\frac{3}{2}}$

128) $y'+(y'')^2=(x+y'')^2$

129) $y^2=4ax$

130) $xy=c^2$

131) $y=e^{mx}$

132) $y=Ae^{2x}+Be^{-2x}$

133) $\frac{dy}{dx}-sin(x+y)$

134) $yx^2dx+e^{-y}dy=0$

135) $cos^2xdy+y\ e^{tanx}\ dx=0$

136) $sec2x\ dy-sin5x\ sec^2ydx=0$

137) $\frac{dy}{dx}+y=x$

138) $\frac{dy}{dx}+\frac{4x}{x^2+1}y=\frac{1}{(x^2+1)^2}$

139)

$(1+x^2)\frac{dy}{dx}+2xy=cosx$

140) $\frac{dy}{dx}+xy=x$

141) Order

142) Degree

(117) $\frac{\Pi}{m+n+1}$, m, n − 1, n ≥ 1

(118) $\frac{4}{5}$

(119) slope

(120) $\frac{-\pi}{3}$

(121) $B^{-1}A^{-1}$

(122) arg z + 2nπ, n∈ Z

(123) Unique solution

(124) 4

(125) $\frac{\Pi}{6}$

(126) $|A|^{n-2}.A$

(127) Absorption law

(128) Identity law

(129) $\frac{\Pi}{2}$

(130) n!

(131) Relative error x 100

(132) $\frac{\pi}{6}$

(133) 4 Change of sign

(134) $cosec^{-1}x$

(135) n arg (z)

(136) $\frac{z-\bar{z}}{2}$

(137) does not exist

(138) $e^{f(x)}$

(139) no positive and no negative root

(140) 0

(141) $\sqrt{2}$

(142)

2 is the only even prime if and only if sin x is an even function

143) General solution

144) Particular solution

145) p v p ≡ p, p ∧ p ≡ p

146) p v (q v r) = (p v p) v r

(143) even degree reciprocal equation of type II

(144) local maximum at c

(145) f(x + Δx) - f(x)

(146) $\lim_{\triangle\to 0}\frac{\triangle f}{\triangle x}$

147) p v (q v r) = (p ∧ p) v (p ∧ r)

148) p v II = II

149) ~ II = F

150) ~(~p)

151) p v (p ∧ p) ≡ p

152) ~(p v p) = ~p v ~ q

153) p v q

154) ~ p

155) sin x is an odd function

156) p ↔ q

157) $\begin{pmatrix} ab & 0 \\ 0 & 0 \end{pmatrix} \in G \forall a, b \in R - \{0\}$

158) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \in G$

159) $\begin{pmatrix} abc & 0 \\ 0 & 0 \end{pmatrix} \in G$

160) $\begin{pmatrix} \frac{1}{a} & 0 \\ 0 & 0 \end{pmatrix} \in G \forall a \in G$

(147) Actual value - approximate value

(148) $\frac{\pi}{6}$

(149) $3\sin^{-1}x$

(150) π

(151) neither a local maximum nor a local minimum

(152) Identify element

(153) $\frac{4}{5}$

(154) Associative law

(155) $\geq |z_1| + |z_2|$

(156) Type I even degree reciprocal equation

(157) (G, *) is associative

(158) Rolles' theorem

(159) tan(c+y)-sec(x+y)=x+c

(160) $\frac{1}{132}$

46 x 2 = 92

161) The rank of any 3 x 4 matrix is
   (1) May be 1
   (2) May be 2
   (3) May be 3
   (4) Maybe 4

162) If A is symmetric then
   (1) $A^T = A$
   (2) adj A is symmetric
   (3) adj (A^T)= $(\text{adj } A)^T$
   (4) A is orthogonal

163) If A is a non-singular matrix of odd order them
   1) Order of A is 2m + 1
   (2) Order of A is 2m + 2
   (3) |adj AI is positive
   (4) IAI ≠ 0

164) If A is a orthogonal matrix, then
   (1) $AA^T = A^TA = I$
   (2) A is non-singular
   (3) IAI= 0
   (4) $A^{-1} = A^T$

165) A matrix which is obtained from an identity matrix by applying only one elementary transformation is
(1) Identity matrix
(2) Elementary matrix
(3) Square matrix
(4) Equivalent to identify matrix

166) i^{-1} =
(i) $\frac{1}{i}$
(ii) i
(iii) -i
(4) $\frac{1}{i^2}$

167) When z =x+iy, then iz is
(1) x-iy
(2) i(x+iy)
(3) -y+ix
(4) Rotation ofz by 90° in the counter clockwise direction

168) (1+3i) (1-3i)
(i) $(1)^2 - (3i)^2$
(2) 1 + 9
(3) 10
(4) -8

169) If z=x+iy, then z $\bar{z}$ =
(i) (x+iy) (x-iy)
(2) $|z|^2$
(3) $x^2 + y^2$
(4) |z|

170) The princlple argumentofa complex number.
(1) θ = α
(2) θ = -α
(3) $\frac{\pi}{2} - \alpha$
(4) θ = α - π

171) Application of De Moivre's theorem.
(1) $(\sin\theta + i\cos\theta)^n = \sin n\theta + i\cos n\theta$
(2) $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$
(3) $(\cos\theta + i\sin\theta)^{-n} = \cos n\theta - i\sin n\theta$
(4) $(\cos\theta - i\sin\theta)^{-n} = \cos n\theta + i\sin n\theta$

172) 1) $x + \frac{1}{x} = 2$
2) $ax^2+bx+c=0$
3) $\sqrt{x} + \frac{1}{\sqrt{x}} = 4$
4) $ax^2 + \frac{b}{x} + c = 0$

173) (1) $2x^2 + 7x - 2x + 7 = 0$
(2) $6x^2 - 6x^3 + 5 = 0$
(3) $-5 + 6x + 5x^2 - 6x^3 = 0$
(4) $9x^4 - 5x^3 + 5x^2 - 9 = 0$

174) (1) $\left(\frac{3}{5}\right)^x = x - x^2 - 9$
(2) sin x = 4
(3) tan x= 1
(4) cos x=7

175) If ≠ 0, then $\frac{p}{2x} = \frac{a}{a+c} + \frac{b}{x-c}$ has two equal roots then p = _____
(1) $p = (\sqrt{a} - \sqrt{b})^2$
(2) $(\sqrt{a} + \sqrt{b})^2$
(3) $(\sqrt{a} \pm \sqrt{b})^2$
(4) 0

176) Ifax + by = 1, $Cx^2 + dy^2 = 1$ have only one solution, then
(1) $\frac{a^2}{c} + \frac{b^2}{d} = 1$
(2) $x = \frac{a}{c}$
(3) $x = \frac{c}{a}$
(4) $x = \frac{b}{d}$

177) (1) Domain is $(-\infty, -1] \cup [1, \infty)$
(2) Range is $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right) - \{0\}$
(3) Odd function
(4) Periodic function

178) (1 tan(tan-1x)= x if $x \epsilon R$
(2) $sin^{-1}\left(\frac{1}{x}\right) = cosec \quad x \quad if \quad x \epsilon R(-4/1)$
(3) $cos^{-1}\left(\frac{1}{x}\right) = sec \quad x \quad if \quad x \epsilon R(-1/1)$
(4)
$$\tan^{-1}\left(\frac{1}{x}\right) = \begin{cases} \cot^{-1} x, & \text{if} \quad x > 0 \\ -A + \cot^{-1} & \text{if} \quad x < 0 \end{cases}$$

179)
(1) The x-intercept is 1 and the y-intercept is $\frac{\pi}{2}$
(2) It is an even function
(3) Not symmetric with respect to origin
(4) Not symmetric with respect to y-axis

180) (1) $sin^{-1}x = cos^{-1}\sqrt{1-x^2}$ if $0 \le x \le 1$
(2) $sin^{-1}x = -cos^{-1}\sqrt{1-x^2}$ if $-1 \le x \le 0$
(3) $cos^{-1}x = sin^{-1}\sqrt{1+x^2}$ if $0 \le x \le 1$
(4) $cos^{-1}x = sin^{-1}\sqrt{1-x^2}$ if $0 \le x \le 1$

181) (1) $\cot(\cot^{-1}(+600)) = -600$
(2) $\cot(\cot^{-1}(1782)) = 1782$

(3) $cot\left(cot^{-1}\left(\frac{-17}{9}\right)\right) = \frac{-17}{9}$

(4) $cot(cot^{-1}\left(\sqrt{3}\right) = \sqrt{3}$

182) (1) x = a cos θ, y = a sin θ

(2) θ

(3) 0 ≤ θ ≤ 2π

(4) (a cos θ, b sin θ)

183) (1) $y^2 = 4ax$

(2) c = $\frac{a}{m}$

(3) $c^2 = a^2(1+m^2)$

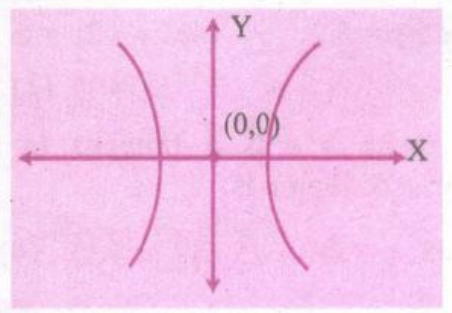
(4) $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$

184) (1) Transverse axis is parallel to x-axis

(2) Direction are x = ± $\frac{a}{e}$

(3) Centre is (0, 0)

(4) Transverse axis parallel to y-axis

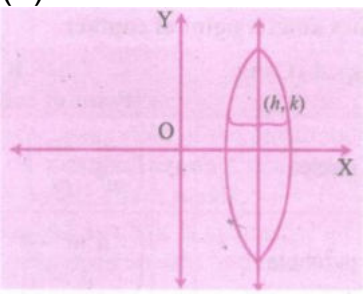


185) (1) Major axi parallel to x-axis

(2) $c^2 \equiv a^2 - b^2$

(3) Foci are a units right and a units left of centre

(4) $c^2 = a^2 + b^2$

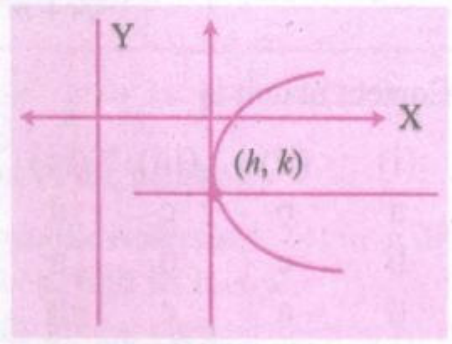


186) (1) Vertex (h, k)

(2) Equation of directrix is x = h + a

(3) Axis of symmetry is y = k

(4) Length of latus rectum = 4a



187) (1) displacement
(2) length
(3) weight
(4) velocity

188) For any non-zero vectors $\vec{a}$ and $\vec{b}$ $\vec{a} \times \vec{b}$ is
(1) cross product of $\vec{a}$ and $\vec{b}$
(2) $|\vec{a}|$ $\left|\vec{b}\right|$ sinθ
(3) $|\vec{a}|$ $\left|\vec{b}\right|$ sinθ $\hat{n}$
(4) -$\left(\vec{b} \times \vec{a}\right)$

189) For any non-zero vectors $\vec{a}, \vec{b}$ and $\vec{c}$, $\left(\vec{a} \times \vec{b}\right) . \vec{c}$ is _____
(1) $\vec{a}. \left(\vec{b} \times \vec{c}\right)$
(2) $\left(\vec{b} \times \vec{a}\right) . \vec{c}$
(3) $\left(\vec{b} \times \vec{c}\right) . \vec{a}$
(4) $(\vec{c} \times \vec{a}) . \vec{b}$

190) $\vec{a}, \vec{b}$ and $\vec{c}$ are said to be coplanar if
(1) $\left[\vec{a}, \vec{b}, \vec{c}\right]$ =0
(2) $\vec{a}, \vec{b}, \vec{c}$ lie on the same plane
(3) They are either parallel or intersecting
(4) Skew lines

191) The equation of the plane at a distance p from the origin and perpendicular to the unit normal vector $\hat{d}$ is
(1) $\vec{r}. \vec{d} = p$
(2) $\vec{r}. \hat{d} = p$
(3) $\vec{r}. \vec{d} = q$ where $q = p\left|\vec{d}\right|$
(4) $\vec{r}. \frac{\vec{d}}{|d|} = p$

192) The approximate value of $\frac{1}{10.1}$
(1) 0.099
(2) $\frac{1}{10} - 0.001$
(3) f(10) - 0.001
(4) 0.99

193) The differential of $\frac{x-2}{2x+3}$ is
(1) $\frac{7}{(2x+3)^2}$ dx
(2) $\frac{7}{(2x+3)^2}$

(3) $\frac{(2x+3-2x+4)}{(2x+3)^2}$

(4) $\frac{(2x+3)(1)-(x-2)(2)}{(2x+3)^2}$ dx

194) If u = $\frac{x}{y^2} - \frac{y}{x^2}$ then

(1) $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$

(2) $\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial x \partial y}$

(3) $\frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 f}{\partial y^2}$

(4) $\frac{\partial^2 u}{\partial x \partial y} = \frac{-2}{y^3} + \frac{2}{x^3}$

195) If u = log $\left(\frac{x^2+y^2}{xy}\right)$ then

(1) u is a homogeneous function

(2) $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ = 0

(3) $\frac{x^2+y^2}{xy}$ is a homogeneous function

(4) $\frac{x^2+y^2}{xy}$ is a homogeneous function of degree 0.

196) Suppose that A = {(x, y) / a < x < b, c < y < d} ⊂ R^2, F: A ⟶ R. F is continuous at (u, v) if

(1) F is defined at (u, v)

(2) $\lim_{(x,y)\to(u,v)} F(x,y)$ =L exists

(3) L= F (u, v)

(4) either (a) or (b) or (c) holds true

197) $\int_0^{\frac{\pi}{2}} \frac{cos^{\frac{5}{3}} x}{cos^{\frac{5}{3}} x + sin^{\frac{5}{3}} x} dx$ =

(1) $\int_0^{\frac{\pi}{2}} \frac{cos^{\frac{5}{3}} (\frac{\pi}{2} - x)}{cos^{\frac{5}{3}} (\frac{\pi}{2} - x) + sin^{\frac{5}{3}} (\frac{\pi}{2} - x)}$ dx

(2) $\frac{\pi}{2}$

(3) $\int_0^{\frac{\pi}{2}} \frac{sin^{\frac{5}{3}}}{sin^{\frac{5}{3}} + sin^{\frac{5}{3}}} dx$

(4) $\frac{\pi}{4}$

198) The area bounded by the parabola y^2 = x and its latus rectum is

(1) $2\int_0^{\frac{1}{4}} y dx$

(2) $2\int_0^{\frac{1}{4}} \sqrt{x} dx$

(3) $\frac{1}{6}$

(4) $\frac{2}{3}$

199) The area of the region bounded by the graph of y = sin x and y = cos x between x = 0 and x = $\frac{\pi}{4}$

(1) $\int_0^{\frac{\pi}{4}} (cos\ x - sin\ x) dx$

(2) $[sinx + cosx]_0^{\frac{\pi}{4}}$

(3) $\int_0^{\frac{\pi}{4}} (sinx - cosx)dx$

(4) $\sqrt{2} - 1$

200) $\int_a^b f(x)dx =$

(1) $\int_a^b f(y)dy$

(2) $-\int_a^b f(x)dx$

(3) $\int_a^b f(a+b-x)dx$

(4) $\int_0^a f(a-x)dx$

201) The volume generated by rotating the triangle with vertices at (0,0) (3, 0) and (3,3) about x - axis is

(1) 36π

(2) 9π

(3) $\frac{1}{3}\pi 3^2 \times 3$

(4) Volume of the cone with r = 3, h = 3

202) The differential equation obtained by eliminating a and b from $y=ae^{3x}+be^{-3x}$ is

1) $\frac{d^2y}{dx^2} - 9y$

2) $\frac{d^2y}{dx^2} + 9y$

3) $y^{11}-9y=0$

4) $y^1=3ae^{3x}-3be^{-3x}$

203) If $\frac{dy}{dx} = \frac{x-y}{x+y}$ then

1) xdy+y dx=x dx+y dy

2) $\int d(xy) = \int xdx + \int ydy$

3) $x^2-y^2+2xy=c$

4) $x^2-y^2-2xy=c$

204) $y=cx=c^2$ is the general solution of the differential equation.

1) y'=c

2) $y=y'x-(y')^2$

3) y"=0

4) $(y')^2-xy'+y=0$

205) If cos x is an I.F of $\frac{dy}{dx} + py = Q,$ then p

1) tan x

2) -tan x

3) $p = \frac{f^1(x)}{f(x)}$ where f(x) =cos x

4) $\frac{\frac{d}{dx}(cos\ x)}{cos\ x}$

206) Solution of $\frac{dy}{dx}$ +mx=0 where m< 0 is

1) $\frac{dy}{dx}$ =-m dy

2) $y=ce^{mx}$

3) log x=-my+log x

4) $x=ce^{-my}$

8 x 2 = 16

207) Which of the following statement is incorrect?
(1) Initial velocity means velocity at t = 0.
(2) Initial acceleration means, acceleration at t=0.
(3) If the motion is upward, at the maximum height the velocity is not zero.
(4) If the motion is horizontal, u = 0 when the particle comes to rest.

208) Identify the incorrect statement.
(1) Every constant function is an increasing function.
(2) Every constant function is a decreasing function.
(3) Every identity function is an increasing function.
(4) Every polynomial function is continuous.

209) Identify the false statement.
(1) All the stationary numbers are critical numbers.
(2) At the stationary point, the first derivative is zero.
(3) At critical numbers, the first derivative does not exist.
(4) All the critical numbers are stationary numbers.

210) Which of the following are not statements?
(i) 3 + 4 = 8
(ii) The sun is a planet
(iii) Switch on the light
(iv) Where are you going?
(1) (i), (ii)
(2) (ii), (iii)
(3) (iii)
(iv) (4) (iv) only

211) If p is true and q is false, then which of the following is not true?
(1) p ⟶ q is F
(2) p v q is T
(3) p ∧ q is F
(4) p ⇔ q is F

212) p v q is false when
(a) p is T, q is F
(c) p is T, q is T
(b) p is F, q is T
(d) p is F, q is F

213) Which of the following is not a binary operation on R?
(1) a * b = ab
(2) a * b = a - b
(3) a * b =$\sqrt{ab}$
(4) a * b = $\sqrt{a^2+b^2}$

214) Which of the following is not a contradiction?
(1) p v q
(2) p ∧ q
(3) p v ~q
(4) p ∧ ~p

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2 MARKS VOLUME 1

12th Standard

Maths

216 x 2 = 432

1) If A = $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is non-singular, find A^{-1}.

2) If A is a non-singular matrix of odd order, prove that |adj A| is positive

3) Find a matrix A if adj(A) = $\begin{bmatrix} 7 & 7 & -7 \\ -1 & 11 & 7 \\ 11 & 5 & 7 \end{bmatrix}$.

4) If adj A = $\begin{bmatrix} -1 & 2 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, find A^{-1}.

5) If A is symmetric, prove that then adj Ais also symmetric.

6) Prove that $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ is orthogonal

7) Find the adjoint of the following:

$\begin{bmatrix} -3 & 4 \\ 6 & 2 \end{bmatrix}$

8) Find the inverse (if it exists) of the following:

$\begin{bmatrix} -2 & 4 \\ 1 & -3 \end{bmatrix}$

9) If adj(A) = $\begin{bmatrix} 0 & -2 & 0 \\ 6 & 2 & -6 \\ -3 & 0 & 6 \end{bmatrix}$, find A^{-1}.

10) Reduce the matrix $\begin{bmatrix} 3 & -1 & 2 \\ -6 & 2 & 4 \\ -3 & 1 & 2 \end{bmatrix}$ to a row-echelon form.

11) Find the rank of each of the following matrices:

$$\begin{bmatrix} 3 & 2 & 5 \\ 1 & 1 & 1 \\ 3 & 3 & 6 \end{bmatrix}$$

12) Find the rank of the following matrices which are in row-echelon form :

$$\begin{bmatrix} 2 & 0 & -7 \\ 0 & 3 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

13)

Find the rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ 3 & 0 & 5 \end{bmatrix}$ by reducing it to a row-echelon form.

14) Find the rank of the following matrices by minor method:

$$\begin{bmatrix} 2 & -4 \\ -1 & 2 \end{bmatrix}$$

15) Find the rank of the following matrices by minor method:

$$\begin{bmatrix} -1 & 3 \\ 4 & -7 \\ 3 & -4 \end{bmatrix}$$

16) Find the rank of the following matrices by minor method:

$$\begin{bmatrix} 1 & -2 & -1 & 0 \\ 3 & -6 & -3 & 1 \end{bmatrix}$$

17) Find the rank of the following matrices by minor method:

$$\begin{bmatrix} 1 & -2 & 3 \\ 2 & 4 & -6 \\ 5 & 1 & -1 \end{bmatrix}$$

18) Find the rank of the following matrices by minor method:

$$\begin{bmatrix} 0 & 1 & 2 & 1 \\ 0 & 2 & 4 & 3 \\ 8 & 1 & 0 & 2 \end{bmatrix}$$

19) Solve the following system of homogenous equations.

$2x + 3y - z = 0$, $x - y - 2z = 0$, $3x + y + 3z = 0$

20) Find the rank of each of the following matrices:

$$\begin{bmatrix} 4 & 3 & 1 & -2 \\ -3 & -1 & -2 & 4 \\ 6 & 7 & -1 & 2 \end{bmatrix}$$

21) Find the rank of the following matrices which are in row-echelon form :

$$\begin{bmatrix} -2 & 2 & -1 \\ 0 & 5 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

22) Find the rank of the following matrices which are in row-echelon form :

$$\begin{bmatrix} 6 & 0 & -9 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

23) Simplify the following

$i^{1947}+i^{1950}$

24) Evaluate the following if z=5−2i and w= −1+3i

z+w

25) If z_1=1-3i,z_2=4i, and z_3 = 5 , show that $(z_1+z_2)+z_3=z_1+(z_2+z_3)$

26) Write $\frac{3+4i}{5-12i}$ in the x+iy form, hence find its real and imaginary parts.

27) Simplify $\left(\frac{1+i}{1-i}\right)^3-\left(\frac{1-i}{1+i}\right)^3$

28) If z_1=3-2i and z_2=6+4i, find $\frac{z_1}{z_2}$

29) Find z^{-1}, if z=(2+3i)(1− i).

30) Write the following in the rectangular form:

$(5+9i)+(2-4i)$

31) If z=x+iy, find the following in rectangular form.

$Re\left(\frac{1}{z}\right)$

32) Find the following $\left|\frac{2+i}{-1+2i}\right|$

33) Which one of the points i,−2+i , and 3 is farthest from the origin?
34) Find the square root of 6−8i .
35) Find the modulus of the following complex numbers

$$\frac{2i}{3+4i}$$

36) If the area of the triangle formed by the vertices z,iz and z+iz is 50 square units, find the value of |z|
37) Find the square roots of 4+3i
38) Obtain the Cartesian form of the locus of z=x+iy in
$[Re(iz)]^2 = 3$
39) Show that the following equations represent a circle, and, find its centre and radius|
$|z-2-i| = 3$
40) Obtain the Cartesian equation for the locus of z=x+iy in
|z-4|=16
41) Write in polar form of the following complex numbers
$2 + i2\sqrt{3}$
42)
If $\omega \neq 1$ is a cube root of unity, then the show that $\frac{a+b\omega+c\omega^2}{b+c\omega+a\omega^2} + \frac{a+b\omega+c\omega^2}{c+a\omega+b\omega^2} = 1$
43)
Find the value of $\sum_{k=1}^{8}\left(cos\frac{2k\pi}{9} + isin\frac{2k\pi}{9}\right)$.
44) Show that |3z−5+i|= 4 represents a circle, and, find its centre and radius.
45) Show that |z+2−i|< 2 represents interior points of a circle. Find its centre and radius.
46) Find the modulus and principal argument of the following complex numbers.
$\sqrt{3} + i.$
47) Represent the complex number −1−i
48)
Find the principal argument Arg z , when z = $\frac{-2}{1+i\sqrt{3}}$
49) Simplify the following
$i^{1948}-i^{-1869}$
50) Simplify the following
$\sum_{n=1}^{12} i^n$
51) Simplify the following

$$i^{59} + \frac{1}{i^{59}}$$

52) Simplify the following
$i\, i^{2} i^{3} \ldots i^{2000}$

53) Simplify the following

$\sum_{n=1}^{10} i^{n+50}$.

54) Evaluate the following if z=5−2i and w= −1+3i

z−iw

55) Evaluate the following if z=5−2i and w= −1+3i

2z+3w

56) Evaluate the following if z=5−2i and w= −1+3i

z w

57) If z_1=1-3i,z_2=4i, and z_3 = 5 ,show that $(z_1+z_2)z_3=z_1(z_2z_3)$

58) Write the following in the rectangular form:

$$\frac{10-5i}{6+2i}$$

59) Write the following in the rectangular form:

$$\overline{3i+\frac{1}{2-i}}.$$

60) If z=x+iy , find the following in rectangular form.

Re(i z)

61) If z=x+iy , find the following in rectangular form.

Im(3z+4z−4i)

62) Find the modulus of the following complex number $\frac{2-i}{1+i}+\frac{1-2i}{1-i}$

63) Find the modulus of the following complex numbers

$(1-i)^{10}$

64) Find the modulus of the following complex numbers

2i(3−4i)(4−3i).

65) Find the square roots of −6+8i

66) Find the square roots of

−5−12i .

67) Obtain the Cartesian form of the locus of z=x+iy in

$$\overline{z} = 2^{-1}$$

68) Show that the following equations represent a circle, and, find its centre and radius

$|2z+2-4i| = 2$

69) Show that the following equations represent a circle, and, find its centre and radius

|3z-6+12i|=8

70) Write in polar form of the following complex numbers

$3-i\sqrt{3}$

71) Write in polar form of the following complex numbers

-2-i2

72) Simplify the following:

$i^{-1924}+i^{2018}$

73) Simplify the following:
$\sum_{n=1}^{102} i^n$

74) Simplify the following:
i $i^2 i^3 \ldots i^{40}$

75)
Find the following $\left|\overline{(1+i)(2+3i)(4i-3}\right|$

76)
Find the following $\left|\frac{i(2+i)^3}{(1+i)^2}\right|$

77) Find the modulus and principal argument of the following complex numbers.
$-\sqrt{3}+i$

78) Find the modulus and principal argument of the following complex numbers:
$-\sqrt{3}-i$

79) Find the modulus and principal argument of the following complex numbers.
$\sqrt{3}$-i

80) Represent the complex numbe $1+i\sqrt{3}$ in polar form.

81) If α, β, and γ are the roots of the equatio $x^3+pz^2+qx+r=0$, find the value of $\Sigma\frac{1}{\beta\gamma}$ in terms of the coefficients.

82) Construct a cubic equation with roots 1,2, and 3

83) If α, β, γ and δ are the roots of the polynomial equation $2x^4+5x^3-7x^2+8=0$, find a quadratic equation with integer coefficients whose roots are α + β + γ + δ and αβ$\gamma\delta$.

84) Formalate into a mathematical problem to find a number such that when its cube root is added to it, the result is 6.

85) A 12 metre tall tree was broken into two parts. It was found that the height of the part which was left standing was the cube root of the length of the part that was cut away. Formulate this into a mathematical problem to find the height of the part which was cut away.

86) Find the monic polynomial equation of minimum degree with real coefficients having 2-$\sqrt{3}$i as a root.

87) Find a polynomial equation of minimum degree with rational coefficients, having 2-$\sqrt{3}$i as a root.

88) Show that the equation $2x^2-6x+7=0$ cannot be satisfied by any real values of x.

89) If $x^2+2(k+2)x+9k=0$ has equal roots, find k.

90) Show that, if p,q,r are rational, the roots of the equation $x^2-2px+p^2-q^2+2qr-r^2=0$ are rational.

91) Find a polynomial equation of minimum degree with rational coefficients, having 2+$\sqrt{3}$ i as a root.

92) Find a polynomial equation of minimum degree with rational coefficients, having 2i+3 as a root.

93) Obtain the condition that the roots of $x^3+px^2+qx+r=0$ are in A.P.

94) Solve: (2x-1)(x+3)(x-2)(2x+3)+20=0

95) Solve the equation : $x^4-14x^2+45=0$

96) Solve: $8x^{\frac{3}{2x}} - 8x^{\frac{-3}{2x}}=63$

97) Show that the polynomial $9x^9+2x^5-x^4-7x^2+2$ has at least six imaginary roots.

98) Discuss the nature of the roots of the following polynomials:
$x^{2018}+1947x^{1950}+15x^8+26x^6+2019$

99) Solve: $2\sqrt{\frac{x}{a}}+3\sqrt{\frac{a}{x}}=\frac{b}{a}+\frac{6a}{b}$

100) Solve the equations:
$6x^4-35x^3+62x^2-35x+6=0$

101) Discuss the maximum possible number of positive and negative roots of the polynomial equation $9x^9-4x^8+4x^7-3x^6+2x^5+x^3+7x^2+7x+2=0$

102) Show that the equation $x^9-5x^5+4x^4+2x^2+1=0$ has atleast 6 imaginary solutions.

103) Determine the number of positive and negative roots of the equation $x^9-5x^4-14x^2=0$.

104) Find the exact number of real roots and imaginary of the equation
$x^9+9x^7+7x^5+5x^3+3x$.

105) Construct a cubic equation with roots 1,1, and −2

106) Construct a cubic equation with roots 2,−2, and 4.

107) Examine for the rational roots of $x^8-3x+1=0$

108) Discuss the nature of the roots of the following polynomials:
$x^5-19x^4+2x^3+5x^2+11$

109) Find the principal value of $\sin^{-1}\left(-\frac{1}{2}\right)$(in radians and degrees).

110) Find the principal value of $\sin^{-1}(2)$, if it exists.

111) Find the principal value of

$$Sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$$

112) Find all the values of x such that
$-10\pi \le x \le 10\pi$ and sin x=0

113) Find the period and amplitude of
y=sin 7x

114) For what value of x does $\sin x=\sin^{-1}x$?

115) Find the value of $\sin^{-1}\left(sin\frac{5\pi}{9}cos\frac{\pi}{9}+cos\frac{5\pi}{9}sin\frac{\pi}{9}\right)$.

116) Find all values of x such that
$6\pi \le x \le 6\pi$ and cos x -0

117) State the reason for $\cos^{-1}[cos(-\frac{\pi}{6})] \neq \frac{\pi}{6}$.

118) Is $\cos^{-1}(-x)=\pi-\cos^{-1}(x)$ true? Justify your answer.

119) Find the principal value of cos-1$(\frac{1}{2})$.

120) Find the value of

$$2cos^{-1}\left(\frac{1}{2}\right)+sin^{-1}\left(\frac{1}{2}\right)$$

121) Find the value of

$$tan^{-1}(tan\frac{5\pi}{4})$$

122) Find the value of

$$tan(tan^{-1}(\frac{7\pi}{4}))$$

123) Find the value of $sec^{-1}\left(-\frac{2\sqrt{3}}{3}\right)$

124) If $cot^{-1}\frac{1}{7}=\theta$, find the value of $cos\theta$.

125) Show that $cot^{-1}\left(\frac{1}{\sqrt{x^2-1}}\right)=sec^{-1}x,\ |x|>1$

126) Find the principal value of

$sec^{-1}(\frac{2}{\sqrt{3}})$

127) Find the value of

$$tan^{-1}\left(\sqrt{3}\right)-sec^{-1}(-2)$$

128) Prove that $\frac{\pi}{2}\leq sin^{-1}x+2cos^{-1}x\leq\frac{3\pi}{2}$.

129) Simplify

$$cos^{-1}\left(cos\left(\frac{13\pi}{3}\right)\right)$$

130) Find the value of

i) $sin\left[\frac{\pi}{3}-sin^2\left(-\frac{1}{2}\right)\right]$

131) Prove that $tan\left(sin^{-1}x\right)=\frac{x}{\sqrt{1-x^2}}for\,|x|<1$

132) Solve $tan^{-1}\left(\frac{1-x}{1+x}\right)=\frac{1}{2}tan^{-1}x$ for $x>0$

133) Show that $cot(sin^{-1}x)=\frac{\sqrt{1-x^2}}{x}$ $-1\leq x\leq$ and $x\neq 0$

134) Find the value, if it exists. If not, give the reason for non-existence.

$sin^{-1}(cos\pi)$

135) Prove that

i$tan^{-1}(\frac{2}{11}) + tan^{-1}(\frac{7}{24}) = tan^{-1}(\frac{1}{2})$

136)

Prove that $\tan^{-1} x+\tan^{-1} z=\tan^{-1}\left[\frac{x+y+z-xyz}{1-xy-yz-zx}\right]$

137) If $\tan^{-1} x+\tan^{-1}y+\tan^{-1}z=\pi$, show that x+y+z+=xyz

138) Find all the values of x such that

$-8\pi \leq x \leq -8\pi$ and sin x=-1

139) Find the period and amplitude of

y=-sin($\frac{1}{3}x$)

140) Find the period and amplitude of

y=4sin(−2x)

141) Find the value of

i) $sin^{-1}\left(sin\left(\frac{5\pi}{4}\right)\right)$

ii) $sin^{-1}\left(sin\left(\frac{2\pi}{3}\right)\right)$

142) Find all values of x such that

$-5\pi \leq x \leq 5\pi$ and cos x -1

143) Find the value of

$cos^{-1}\left(\frac{1}{2}\right) + sin^{-1}(-1)$

144) Find the value of

$cos^{-1}\left(cos\frac{\pi}{7}cos\frac{\pi}{17} - sin\frac{\pi}{17}sin\frac{\pi}{17}\right).$

145) Find the value of

$\tan^{-1}(tan(-\frac{\pi}{6}))$

146) Find the value of

tan ($\tan^{-1}$(-0.2021))

147) Find the value of

tan($\tan^{-1}$(-0.2021)).

148) Find the principal value of

$\cot^{-1}(\sqrt{3})$

149) Find the principal value of

$\text{cosec}^{-1}(-\sqrt{2})$

150) Find the value, if it exists. If not, give the reason for non-existence

$tan^{-1}\left(sin\left(-\frac{5\pi}{2}\right)\right)$

151) Find the value, if it exists. If not, give the reason for non-existence.
$\sin^{-1}[\sin 5]$

152) Find the principal value of

$$Sin^{-1}\left(sin\left(-\frac{\pi}{3}\right)\right)$$

153) Find the principal value of

$$Sin^{-1}\left(sin\left(\frac{5\pi}{6}\right)\right)$$

154) Simplify

$$tan^{-1}\left(tan\left(\frac{3\pi}{4}\right)\right)$$

155) Simplify

$$sec^{-1}\left(sec\left(\frac{5\pi}{3}\right)\right)$$

156) Simplify
$\sin^{-1}[\sin 10]$

157) Find the value of

$$cos\left[\frac{1}{2}cos^{-1}\left(\frac{1}{8}\right)\right]$$

158) Find the general equation of a circle with centre(-3,-4) and radius 3 units.

159) Determine whether $x+y-1=0$ is the equation of a diameter of the circle $x^2+y^2-6x+4y+c = 0$ for all possible values of c .

160) Find the general equation of the circle whose diameter is the line segment joining the points $(-4,-2)$and $(1,1)$.

161) Examine the position of the point (2,3) with respect to the circle $x^2+y^2-6x-8y+12=0$.

162) The line $3x+4y-12 = 0$ meets the coordinate axes at A and B . Find the equation of the circle drawn on AB as diameter.

163) If $y=4x+c$ is a tangent to the circle $x^2+y^2=9$, find c .

164) Obtain the equation of the circles with radius 5 cm and touching x-axis at the origin in general form.

165) Find the equation of the circle with centre (2,-1) and passing through the point (3,6) in standard form.

166) Obtain the equation of the circle for which (3,4) and (2,-7) are the ends of a diameter.

167) If $y=2\sqrt{2}x +c$ is a tangent to the circle$x^2+y^2=16$, find the value of c.

168) Find centre and radius of the following circles.
$x_2+ (y+2^{)2} =0$

169) Find the vertices, foci for the hyperbola $9x^2-16y^2=144$.

170) Find the equation of the parabola in each of the cases given below:
(i) focus (4,0) and directrix $x = -4$.

(ii) passes through (2,-3) and symmetric about y -axis.
(iii) vertex(1,-2) and focus(4,-2) .
(iv) end points of latus rectum(4,-8) and(4,8) .

171) $2x^2-y^2=7$

172) $3x^2+3y^2-4x+3y+10 = 0$

173) $3x^2+2y^2=14$

174) $x^2+y^2+x-y=0$

175) $11x^2-25y^2-44x+50y-256 = 0$

176) $y^2+4x+3y+4=0$

177) Find centre and radius of the following circles.
$x^2+y^2+6x-4y+4=0$

178) Find centre and radius of the following circles.
$x^2+y^2-x+2y-3= 0$

179) Find centre and radius of the following circles.
$2x^2+2y^2-6x+4y+2=0$

180) If $\hat{a} = -3\hat{i} - \hat{j} + 5\hat{k}$, $\hat{b} = \hat{i} - 2\hat{j} + \hat{k}$, $\hat{c} = 4\hat{i} - 4\hat{k}$ and $\hat{a}.(\hat{b} \times \hat{c})$

181) Find the volume of the parallelepiped whose coterminus edges are given by the vectors $2\hat{i} - 3\hat{j} + 4\hat{k}$, $\hat{i} - 2\hat{j} + 4\hat{k}$ and $3\hat{i} - \hat{j} + 2\hat{k}$

182) Show that the vectors $\hat{i} + 2\hat{j} - 3\hat{k}$, $\hat{i} + 2\hat{j} - 3\hat{k}$ and $3\hat{i} + \hat{j} - \hat{k}$

183) If $2\hat{i} - \hat{j} + 3\hat{k}$, $3\hat{i} + 2\hat{j} + \hat{k}$, $\hat{i} + m\hat{j} + 4\hat{k}$ are coplanar, find the value of m.

184) If $\vec{a}, \vec{b}, \vec{c}$ are three vectors, prove that $[\vec{a} + \vec{c}, \vec{a} + \vec{b}, \vec{a} + \vec{b} + \vec{c}] = [\vec{a}, \vec{b}, \vec{c}]$

185) If $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{6} = 2\hat{i} + \hat{j} - 2\hat{k}$, $\vec{c} = 3\hat{i} + 2\hat{j} + \hat{k}$ find $\vec{a}.(\vec{b} \times \vec{c})$.

186) Find the volume of the parallelepiped whose coterminous edges are represented by the vectors $-6\hat{i} + 14\hat{j} + 10\hat{k}$, $14\hat{i} - 10\hat{j} - 6\hat{k}$ and $2\hat{i} + 4\hat{j} - 2\hat{k}$

187) The volume of the parallelepiped whose coterminus edges are $7\hat{i} + \lambda\hat{j} - 3\hat{k}$, $\hat{i} + 2\hat{j} - \hat{k}$, $-3\hat{i} + 7\hat{j} + 5\hat{k}$ is 90 cubic units. Find the value of λ.

188) Determine whether the three vectors $2\hat{i} + 3\hat{j} + \hat{k}$, $\hat{i} - 2\hat{j} + 2\hat{k}$ and $3\hat{i} + \hat{j} + 2\hat{k}$ are coplanar.

189) Prove that $(\vec{a}.(\vec{b} \times \vec{c}))\vec{a} = (\vec{a} \times \vec{b}) \times (\vec{a} \times \vec{c})$

190) If $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$, $\vec{b} = 3\hat{i} + 5\hat{j} + 2\hat{k}$, $\vec{c} = -\hat{i} - 2\hat{j} + 3\hat{k}$, verify that

(i) $(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a}.\vec{c}) \times \vec{b} - (\vec{b}.\vec{c})\vec{a}$

(ii) $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a}.\vec{c}) \times \vec{b} - (\vec{a}.\vec{b})\vec{c}$

191) Show that the straight line passing through the points (6,7,5)A and (8,10,6)B is perpendicular to the straight line passing through the points C(10, 2, -5) and D(8, 3, -4)

192) Show that the lines $\frac{x-1}{4} = \frac{2-y}{6} = \frac{z-4}{12}$ and $\frac{x-1}{4} = \frac{2-y}{6} = \frac{z-4}{12}$ are parallel.

193) Find the non-parametric form of vector equation and Cartesian equations of the straight line passing through the point with position vector $4\hat{i} + 3\hat{j} - 7\hat{k}$ and parallel to the vector $2\hat{i} - 6\hat{j} + 7\hat{k}$.

194) Find the parametric form of vector equation and Cartesian equations of the straight line passing through the point (−2,3, 4) and parallel to the straight line $\frac{x-1}{-4} = \frac{y-3}{5} = \frac{z-8}{6}$

195) Find the angle between the following lines.

$\vec{r} = (4\hat{i} - \hat{j}) + t(\hat{i} + 2\hat{j} - 2\hat{k}),\ \hat{r} = (\hat{i} + 2\hat{j} - 2\hat{k}) + s(-\hat{i} - 2\hat{j} + 2\hat{k})$

196) Show that the points (2, 3, 4),(−1, 4, 5) and (8,1, 2) are collinear.

197) Find the vector and Cartesian form of the equations of a plane which is at a distance of 12 units from the origin and perpendicular to $6\hat{i} + 2\hat{j} - 3\hat{k}$

198) Find the direction cosines and length of the perpendicular from the origin to the plane $\vec{r}.(3\hat{i} - 4\hat{j} + 12\hat{k}) = 5$

199) Find the vector and Cartesian equations of the plane passing through the point with position vector $4\hat{i} + 2\hat{j} - 3\hat{k}$ and normal to vector $2\hat{i} - \hat{j} + \hat{k}$

200) A variable plane moves in such a way that the sum of the reciprocals of its intercepts on the coordinate axes is a constant. Show that the plane passes through a fixed point

201) Find a parametric form of vector equation of a plane which is at a distance of 7 units from the origin having 3, −4,5 as direction ratios of a normal to it.

202) Find the direction cosines of the normal to the plane 12x + 3y − 4z = 65 . Also, find the non-parametric form of vector equation of a plane and the length of the perpendicular to the plane from the origin.

203) Find the vector and Cartesian equations of the plane passing through the point with position vector $2\hat{i} + 6\hat{j} + 3\hat{k}$ and normal to the vector $\hat{i} + 3\hat{j} + 5\hat{k}$

204) A plane passes through the point (−1,1, 2) and the normal to the plane of magnitude $3\sqrt{3}$ makes equal acute angles with the coordinate axes. Find the equation of the plane.

205) Find the intercepts cut off by the plane $\vec{r}.(6\hat{i} + 4\hat{j} - 3\hat{k})$=12 on the coordinate axes.

206) Verify whether the line $\frac{x-3}{-4} = \frac{y-4}{-7} = \frac{z+3}{12}$ lies in the plane 5x-y+z=8.

207) Find the angle between the straight line $\vec{r} = (2\hat{i} + \hat{j} + \hat{k}) + t(\hat{i} - \hat{j} + \hat{k})$ and the plane 2x-y+z=5

208) Find the angle between the line $\vec{r} = (2\hat{i} - \hat{j} + \hat{k}) + t(\hat{i} + 2\hat{j} - 2\hat{k})$ and the plane $\vec{r} = (6\hat{i} + 3\hat{j} + 2\hat{k}) = 8$

209) Find the angle between the planes $\vec{r}.(\hat{i} + \hat{j} - 2\hat{k})$ = 3 and 2x - 2y + z =2

210) Find the length of the perpendicular from the point (1, -2, 3) to the plane x - y + z =5.

211) Find the acute angle between the planes $\vec{r}.(2\hat{i} + 2\hat{j} + 2\hat{k})$ and 4x-2y+2z=15.

212) Find the distance of a point (2,5, −3) from the plane $\vec{r}.(6\hat{i} - 3\hat{j} + 2\hat{k})$=5

213) Find the distance between the parallel planes x+2y-2z=0 and 2x+4y-4z+5=0

214) Find the distance between the planes $\vec{r}.(2\hat{i}-\hat{j}-\hat{k})$=6 and $\vec{r}.(6\hat{i}-\hat{j}-2\hat{k})$= 27

215) Find the angle between the following lines.

$\frac{x+4}{3}=\frac{y-7}{4}=\frac{z+5}{5}$, $\vec{r}=4\hat{k}+t(2\hat{i}+\hat{j}+\hat{k})$

216) Find the angle between the following lines.

2x = 3y = −z and 6x = − y = −4z.

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2 MARKS VOLUME 1 CREATIVE

12th Standard

Maths

55 x 2 = 110

1) For any 2 x 2 matrix, if A (adj A) = $\begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}$ then find |A|.
2) For the matrix A, if A^3 = I, then find A^{-1}.
3) If A is a square matrix such that A^3 = I, then prove that A is non-singular.
4) Show that the system of equations is inconsistent. 2x + 5y= 7, 6x + 15y = 13.
5) Flod the rank of the matrix $\begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$.
6) Find the rank of the matrix A = $\begin{bmatrix} 4 & 5 & -6 & 1 \\ 7 & -3 & 0 & 8 \end{bmatrix}$.
7) Show that the equations 3x + y + 9z = 0, 3x + 2y + 12z = 0 and 2x + y + 7z = 0 have nontrivial solutions also.
8) Find k if the equations x + 2y + 2z = 0, x - 3y - 3z = 0, 2x + y + kz = 0 have only the trivial solution.
9) Solve : 2x - y = 3, 5x + y = 4 using matrices.
10) Solve 6x - 7y = 16, 9x - 5y = 35 using (Cramer's rule).
11) It z_1 and z_2 are two complex numbers, such that $|z_1| = |z_2|$, then is it necessary that $z_1 = z_2$?
12) Find Re (z) and im (z) if z = $5i^{11} + 7i^3$
13) If $(\cos\theta + i \sin\theta)^2 = x + iy$, then show that $x^2+y^2 =1$
14) If z_1 and z_2 are 1-i, -2+4i then find Im$\left(\frac{z_1 z_2}{\bar{z}_1}\right)$.
15) If z=$\left(\frac{\sqrt{3}}{2}+\frac{i}{2}\right)^{107}+\left(\frac{\sqrt{3}}{2}-\frac{i}{2}\right)^{107}$, then show that Im (z) =0
16) Find the modulus of the complex number i^{25}.
17) If 1, ω, ω^2 are the cube roots of unity show that $(1+\omega^2)^3 - (1+\omega)^3 =0$
18) Find the argument of -2
19) Find the modules of $(1+ 3i)^3$
20) Find the values of the real number x and y if 3x + (2x - 3y) i = 6 + $3i^9$.
21) If sin ∝, cos ∝ are the roots of the equation $ax^2 + bx + c$-0 (c ≠ 0), then prove that $(n + c)^2 - b^2 + c^2$
22) Find value of a for which the sum of the squares of the equation x^2 - (a- 2) x - a-1=0 assumes the least value.
23) Find th Int rval for a for which $3x^2+2(a^2+1)$ x+(a^2-3n+2) possesses roots of opposite sign.
24) Find x If $x=\sqrt{2+\sqrt{2+\sqrt{2+....+upto\infty}}}$
25) Find the number of positive and negative roots of the equation $x^7 - 6x^6 + 7x^5 + 5x^2+2x+2$

26) Find the principal value of $tan^{-1}\left(\frac{-1}{\sqrt{3}}\right)$

27) Find the principal value of $\sin^{-1}(-l)$.

28) Find the principal value of $cos^{-1}\left(\frac{-1}{2}\right)$

29) If $cot^{-1}\left(\frac{1}{7}\right) = \theta$ find the value of cos θ

30) If $sin^{-1}\left(\frac{1}{2}\right) = tan^{-1}x$ then find the value of x,

31) Prove that $tan^{-1}\left(\frac{1}{7}\right) + tan^{-1}\left(\frac{1}{13}\right) = tan^{-1}\left(\frac{2}{9}\right)$

32) Ecalute $sin\left(cos^{-1}\left(\frac{3}{5}\right)\right)$

33) Evaluate $sin\left(\frac{1}{2}cos^{-1}\frac{4}{5}\right)$

34) Prove that $2tan^{-1}\left(\frac{2}{3}\right) = tan^{-1}\left(\frac{12}{5}\right)$

35) Evaluate $sin\left(cos^{-1}\left(\frac{1}{2}\right)\right)$

36) Find the equation of tangent to the circle $x^2 + y^2 + 2x - 3y - 8 = 0$ at (2, 3).

37) Find the length of the tangent from (2, -3) to the circle $x^2 + y^2 - 8x - 9y + 12 = 0$.

38) Find the equation of the parabola with vertex at the origin, passing through (2, -3) and symmetric about x-axis

39) If a parabolic reflector is 24 cm in diameter and 6 cm deep, find its locus.

40) If the line y = 3x + 1, touches the parabola $y^2 = 4ax$, find the length of the latus rectum?

41) Find the locus of a point which divides so that the sum of its.distances from (-4, 0) and (4, 0) is 10 units.

42) For the ellipse $x^2 + 3y^2 = a^2$, find the length of major and minor axis.

43) Find the eccentricity of the ellipse with foci on x-axis if its latus rectum be equal to one half of its major axis.

44) Find the eccentricity of the hyperbola. with foci on the x-axis if the length of its conjugate axis is $\left(\frac{3}{4}\right)^{th}$ of the length of its tranverse axis.

45) Find the equation of the hyperbola whose vertices are (0, ±7) and e = $\frac{4}{3}$

46) If $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = -\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{c} = 3\hat{i} + \hat{j}$ find $\frac{\lambda}{c}$ such that $\vec{a} + \lambda\vec{b}$ is perpendicular to $\vec{c}$

47) A force of magnitude 6 units acting parallel to $2\hat{i} - 2\hat{j} + \hat{k}$ displaces the point of application from (1, 2, 3) to (5, 3, 7). Find the work done.

48) Find the area of the triangle whose vertices are A(3, -1, 2) B(I, -1, -3) and C(4, -3,1)

49) Forces $2\hat{i} + 7\hat{j}$, $2\hat{i} - 5\hat{j} + 6\hat{k}$, $-\hat{i} + 2\hat{j} - \hat{k}$ act at a point P whose position vector is $4\hat{i} - 3\hat{j} - 2\hat{k}$ .Find the vector moment of the resultant of these forces acting at P about this point Q whose position vector is $6\hat{i} + \hat{j} - 3\hat{k}$

50) Find the Cartesian equation of a.line passing through the pointsA(2, -1, 3) and B(4, 2, 1)

51) Find the parametric form of vector equation of a line passing through a point (2, -1, 3) and parallel to line $\vec{r} = \left(\hat{i} + \hat{j}\right) + t\left(2\hat{i} + \hat{j} - 2\hat{k}\right)$

52) Find the parametric form of vector equation of the plane passing through the point (1, -1, 2) having 2, 3, 3.as direction ratios of normal to the plane.

53) If the planes $\vec{r}.\left(\hat{i} + 2\hat{j} + 3\hat{k}\right) = 7$ =and

$\vec{r}.\left(\lambda\hat{i} + 2\hat{j} - 7\hat{k}\right) = 26$ are perpendicular. Find the value of λ.

54) FInd the equation of the plane containing the line of intersection of the planes x + y + Z - 6 = 0 and
2x + 3y + 4z + 5 = 0 and passing through the point (1, 1, 1)

55) Let $\vec{a}, \vec{b}, \vec{c}$ be unit vectors such $\vec{a}.\vec{b} = \vec{a}.\vec{c} = 0$ and the angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{6}$. Prove that $\vec{a} = \pm 2\left(\vec{b} \times \vec{c}\right)$

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3 MARKS V1 CRE

12th Standard

Maths

Exam Time : 01:40:00 Hrs

Total Marks : 100

55 x 3 = 165

1) Solve: 2x + 3y = 10, x + 6y = 4 using Cramer's rule.
2) For what value of t will the system tx +3y - z = 1, x + 2y + z = 2, -tx + y + 2z = -1 fail to have unique solution?
3) Solve: 3x+ay =4, 2x+ay=2, a≠0 by Cramer's rule.
4) Verify $(AB)^{-1} = B^{-1} A^{-1}$ for A= $\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$ and B= $\begin{bmatrix} 4 & 5 \\ 3 & 4 \end{bmatrix}$.
5) Under what co.nditions will the rank of the matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & h-2 & 2 \\ 0 & 0 & h+2 \\ 0 & 0 & 3 \end{bmatrix}$ be less than 3?
6) Find,the rank of the matrix math $\begin{bmatrix} 4 & 4 & 0 & 3 \\ -2 & 3 & -1 & 5 \\ 1 & 4 & 8 & 7 \end{bmatrix}$.
7) Verify that $(A^{-1})^T = (A^T)^{-1}$ for A= $\begin{bmatrix} -2 & -3 \\ 5 & -6 \end{bmatrix}$.
8) Solve 2x - 3y = 7, 4x - 6y = 14 by Gaussian Jordan method.
9) Solve: x + y + 3z = 4, 2x + 2y + 6z = 7, 2x + y + z = 10.
10) If the rank of the matrix $\begin{bmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ -1 & 0 & \lambda \end{bmatrix}$ is 2, then find λ.
11) Explain the falacy:
12) Find the circle roots of -27.
13) Find the principal value of -2i.
14) Show that $\left|\frac{z-3}{z+3}\right|$ = 2 represent a circle.
15) Show that the complex numbers 3 + 2i, 5i, -3 + 2i and -i form a square.
16) If the complex number 2 + i and 1-2i are equidistant from x + iy then show that x+3y = 0.

17) Find the locus of z if Re$\left(\frac{z+1}{z-i}\right)$ =0 where z=x+iy.

18) Find the locus of Z if |3z - 5| = 3 |z + 1| where z=x+iy.

19) Find the locus of z if Re$\left(\frac{\bar{z}+1}{\bar{z}-i}\right)$ =0.

20) If $\frac{(a+i)^2}{2a-i}$ =p+iq, show that $p^2+q^2 = \frac{(a^2+i)^2}{4a^2+1}$.

21) Find the number .of real solu,tlons of sin (e^x) $-5^x + 5^{-x}$

22) Find the number of positive integral solutions of (pairs of positive integers satisfying) $x^2 - y^2 = 353702$.

23) Solve: $2x+2^{x-1}+2^{x-2}=7x+7^{x-1}+7^{x-2}$

24) Solve:$(x-1)^4+(x-5)^4=82$

25) Solve: $(5+2\sqrt{6})^{x^2-3}+(5-2\sqrt{6})^{x^2-3}=10$

26) Prove that $cos^{-1}\left(\frac{4}{5}\right)+tan^{-1}\left(\frac{3}{5}\right)=tan^{-1}\left(\frac{27}{11}\right)$

27) Evaluate $cos\left[sin^{-1}\frac{3}{5}+sin^{-1}\frac{5}{13}\right]$

28) Prove that $tan^{-1}\left(\frac{m}{n}\right)-tan^{-1}\left(\frac{m-n}{m+n}\right)=\frac{\pi}{4}$

29) Solve: $tan^{-1}\left(\frac{x-1}{x-2}\right)+tan^{-1}\left(\frac{x+1}{x+2}\right)=\frac{\pi}{4}$

30) Solve $tan^{-1}\left(\frac{2x}{1-x^2}\right)+cot^{-1}\left(\frac{1-x^2}{2x}\right)=\frac{\pi}{3}, x>0$

31) If $sin\left(sin^{-1}\frac{1}{5}+cos^{-1}x\right)=1$ then find the value ofx.

32) Prove that $tan^{-1}\sqrt{x}=\frac{1}{2}cos^{-1}=\frac{1}{2}cos^{-1}\left(\frac{1-x}{1+x}\right), x\epsilon|0, 1|$

33) Evaluate $cos\left[cos^{-1}\left(\frac{-\sqrt{3}}{2}+\frac{\pi}{6}\right)\right]$

34) Find the real solutions of the equation

$$tan^{-1}\sqrt{x(x+1)} + sin^{-1}\sqrt{x^2+x+1} = \frac{\pi}{2}$$

35) Solve: cos(tan^{-1}x) = $sin\left(cot^{-1}\frac{3}{4}\right)$

36) Find the circumference and area of the circle $x^2 + y^2 - 2x + 5y + 7 = 0$

37) Find the value of p so that 3x + 4y - p = 0 is a tangent to the circle $x^2 + y^2 - 64 = 0$.

38) Find the condition for the line lx + my + n = 0 is Rtangent to the circle $x^2 + y^2 = a^2$

39) Find the area of th triangle found by the Unel Joining the vertex of the parabola x^2 = -36y to the ends of the latus rectum.

40) Find the equatlon of the ellipse whose e = $\frac{3}{4}$, foci ony-axl ,centre at origin and passing through (6,4).

41) Find the equation of the ellipse whose latus rectum is 5 and e= $\frac{2}{3}$

42) Find the equation of the hyperbola whose conjugate axis is 5 and the distance between the foci is 13.

43) For the hyperbola $3x^2 - 6y^2 = -18$, find the length of transverse and conjugate axes and eccentricity.

44) Find the value of c if y = x + c is a tangent to the hyperbola $9x^2 - 16y^2 = 144$.

45) Show that the line x + y + 1 = 0 touches the hyperbola $\frac{x^2}{16} - \frac{y^2}{15} = 1$ and find the co-ordinates of the point of contact

46) Dot product of a vector with vector $3\hat{i} - 5\hat{k}$, $2\hat{i} + 7\hat{j}$ and $\hat{i} + \hat{j} + \hat{k}$ are respectively -1, 6 and 5. Find the vector.

47) Find the Cartesian form of the equation of the plane $\vec{r} = (s - 2t)\hat{i} + (3 - t)\hat{j} + (2s + t)\hat{k}$

48) Find the equation of the plane through the intersection of the planes lx-3y+ z-4 -0 and x - y + Z + 1 - 0 and perpendicular to the plane x + 2y - 3z + 6 = 0

49) Find the angle between the line $\frac{x-2}{3} = \frac{y-1}{-1} = \frac{z-3}{2}$ and the plane 3x + 4y + z + 5 = 0

50) If $\vec{a} = \hat{i} - \hat{j}, \vec{b} = \hat{j} - \hat{k}, \vec{c} = \hat{k} - \hat{i}$ then find $\left[\vec{a} - \vec{b}, \vec{b} - \vec{c}, \vec{c} - \vec{a}\right]$

51) Prove that $\left[\vec{a} + \vec{b} + \vec{c}, \vec{b} + \vec{c}, \vec{c}\right] = \left[\vec{a}\vec{b}\vec{c}\right]$

52) Prove by vector method, that in a right angled triangle the square of the hypotenuse is equal to the sum of the square of the other two sides.

53) If $\vec{a} + \vec{b} + \vec{c} = 0$ then show that $\vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$

54) Show that the four points whose position vectors are $6\hat{i} - 7\hat{j}, 16\hat{i} - 29\hat{j} - 4\hat{k}, 3\hat{i} - 6\hat{j}$ are co-planar

55) Show that the lines $\frac{x-1}{3} = \frac{y+1}{2} = \frac{z-1}{5}$ and $\frac{x+2}{4} = \frac{y-1}{3} = \frac{z+1}{-2}$ do not intersect

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3 MARKS BB

12th Standard

Maths

289 x 3 = 867

1) Find the inverse of the matrix $\begin{bmatrix} 2 & -1 & 3 \\ -5 & 3 & 1 \\ -3 & 2 & 3 \end{bmatrix}$.

2) Verify the property $(A^T)^{-1} = (A^{-1})$ with A = $\begin{bmatrix} 2 & 9 \\ 1 & 7 \end{bmatrix}$.

3) Verify $(AB)^{-1} = B^{-1}A^{-1}$ with A = $\begin{bmatrix} 0 & -3 \\ 1 & 4 \end{bmatrix}$, B = $\begin{bmatrix} -2 & -3 \\ 0 & -1 \end{bmatrix}$.

4) If A = $\frac{1}{9}\begin{bmatrix} -8 & 1 & 4 \\ 4 & 4 & 7 \\ 1 & -8 & 4 \end{bmatrix}$, prove that $A^{-1} = A^T$.

5) If A = $\begin{bmatrix} 8 & -4 \\ -5 & 3 \end{bmatrix}$, verify that A(adj A) = $|A|I_2$.

6) If A = $\begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix}$ and B = $\begin{bmatrix} -1 & -3 \\ 5 & 2 \end{bmatrix}$, verify that $(AB)^{-1} = B^{-1}A^{-1}$

7) If adj(A) = $\begin{bmatrix} 2 & -4 & 2 \\ -3 & 12 & -7 \\ -2 & 0 & 2 \end{bmatrix}$, find A.

8) Find adj(adj A) if adj A = $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ -1 & 0 & 1 \end{bmatrix}$.

9) A = $\begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$, show that $A^TA^{-1} = \begin{bmatrix} \cos 2x & -\sin 2x \\ \sin 2x & \cos 2x \end{bmatrix}$

10) Find the matrix A for which $A\begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} 14 & 7 \\ 7 & 7 \end{bmatrix}$.

11) Given A = $\begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}$, B = $\begin{bmatrix} 3 & -2 \\ 1 & 1 \end{bmatrix}$ and C = $\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$, find a matrix X such that AXB = C.

12) If A = $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$, show that $A^{-1} = \frac{1}{2}(A^2 - 3I)$.

13) Decrypt the received encoded message $\begin{bmatrix} 2 & -3 \end{bmatrix}\begin{bmatrix} 20 & 4 \end{bmatrix}$ with the encryption matrix

$\begin{bmatrix} -1 & -1 \\ 2 & 1 \end{bmatrix}$

and the decryption matrix as its inverse, where the system of codes are described by the numbers 1 - 26 to the letters A - Z respectively, and the number 0 to a blank space.

14) Reduce the matrix $\begin{bmatrix} 0 & 3 & 1 & 6 \\ -1 & 0 & 2 & 5 \\ 4 & 2 & 0 & 0 \end{bmatrix}$ to row-echelon form.

15) Find the rank of the matrix $\begin{bmatrix} 2 & -2 & 4 & 3 \\ -3 & 4 & -2 & -1 \\ 6 & 2 & -1 & 7 \end{bmatrix}$ by reducing it to an echelon form.

16) Find the inverse of the non-singular matrix A = $\begin{bmatrix} 0 & 5 \\ -1 & 6 \end{bmatrix}$, by Gauss-Jordan method.

17) Find the rank of the following matrices by row reduction method:

$\begin{bmatrix} 1 & 1 & 1 & 3 \\ 2 & -1 & 3 & 4 \\ 5 & -1 & 7 & 11 \end{bmatrix}$

18) Find the inverse of each of the following by Gauss-Jordan method:

$\begin{bmatrix} 2 & -1 \\ 5 & -2 \end{bmatrix}$

19) Solve the following system of linear equations, using matrix inversion method:
5x + 2y = 3, 3x + 2y = 5.

20) Solve the following system of linear equations by matrix inversion method:
2x + 5y = −2, x + 2y = −3

21) A man is appointed in a job with a monthly salary of certain amount and a fixed amount of annual increment. If his salary was Rs.19,800 per month at the end of the first month after 3 years of service and Rs.23,400 per month at the end of the first

month after 9 years of service, find his starting salary and his annual increment. (Use matrix inversion method to solve the problem.)

22) 4 men and 4 women can finish a piece of work jointly in 3 days while 2 men and 5 women can finish the same work jointly in 4 days. Find the time taken by one man alone and that of one woman alone to finish the same work by using matrix inversion method.

23) Solve the following systems of linear equations by Cramer's rule:
$\frac{3}{x} + 2y = 12$, $\frac{2}{x} + 3y = 13$

24) In a competitive examination, one mark is awarded for every correct answer while $\frac{1}{4}$ mark is deducted for every wrong answer. A student answered 100 questions and got 80 marks. How many questions did he answer correctly ? (Use Cramer's rule to solve the problem).

25) A chemist has one solution which is 50% acid and another solution which is 25% acid. How much each should be mixed to make 10 litres of a 40% acid solution ? (Use Cramer's rule to solve the problem).

26) A fish tank can be filled in 10 minutes using both pumps A and B simultaneously. However, pump B can pump water in or out at the same rate. If pump B is inadvertently run in reverse, then the tank will be filled in 30 minutes. How long would it take each pump to fill the tank by itself ? (Use Cramer's rule to solve the problem).

27) Solve the system of linear equations, by Gaussian elimination method $4x + 3y + 6z = 25$, $x + 5y + 7z = 13$, $2x + 9y + z = 1$.

28) Test for consistency of the following system of linear equations and if possible solve:
$x - y + z = -9$, $2x - 2y + 2z = -18$, $3x - 3y + 3z + 27 = 0$.

29) Solve the following system:
$x + 2y + 3z = 0$, $3x + 4y + 4z = 0$, $7x + 10y + 12z = 0$.

30) Find the adjoint of the following:

$$\begin{bmatrix} 2 & 3 & 1 \\ 3 & 4 & 1 \\ 3 & 7 & 2 \end{bmatrix}$$

31) Find the adjoint of the following:

$$\frac{1}{3}\begin{bmatrix} 2 & 2 & 1 \\ -2 & 1 & 2 \\ 1 & -2 & 2 \end{bmatrix}$$

32) Find the inverse (if it exists) of the following:

$$\begin{bmatrix} 5 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 5 \end{bmatrix}$$

33) Find the inverse (if it exists) of the following:

$$\begin{bmatrix} 2 & 3 & 1 \\ 3 & 4 & 1 \\ 3 & 7 & 2 \end{bmatrix}$$

34) Find the rank of the following matrices by row reduction method:

$$\begin{bmatrix} 1 & 2 & -1 \\ 3 & -1 & 2 \\ 1 & -2 & 3 \\ 1 & -1 & 1 \end{bmatrix}$$

35) Find the rank of the following matrices by row reduction method:

$$\begin{bmatrix} 3 & -8 & 5 & 2 \\ 2 & -5 & 1 & 4 \\ -1 & 2 & 3 & -2 \end{bmatrix}$$

36) Solve the following system of linear equations by matrix inversion method:
$2x - y = 8$, $3x + 2y = -2$

37) Solve the following systems of linear equations by Cramer's rule:
$\frac{3}{x} + 2y = 12$, $\frac{2}{x} + 3y = 13$

38) Simplify the following i^7

39) Find the value of the real numbers x and y, if the complex number $(2+i)x+(1-i)y+2i-3$ and $x+(-1+2i)y+1+i$are equal

40) Given the complex number $z=2+3i$, represent the complex numbers in Argand diagram
z, iz , and $z+iz$

41) Find the values of the real numbers x and y, if the complex numbers $(3-i)x-(2-i)y+2i+5$ and $2x+(-1+2i)y+3+2i$ are equal.

42) If $z_1=3, z_2=-7i$, and $z_3=5+4i$, show that $z_1(z_2+z_3)=z_1z_2+z_1z_3$

43) If $z_1=2+5i$, $z_2=-3-4i$, and $z_3=1+i$, find the additive and multiplicate inverse of z_1, z_2 and z_3

44) If $\frac{z+3}{z-5i} = \frac{1+4i}{2}$, find the complex number z

45) If $z_i = 2-i$ and $z_2=-4+3i$, find the inverse of z_1z_2 and $\frac{z_1}{z_2}$

46) The complex numbers u,v, and w are related by $\frac{1}{u} = \frac{1}{v} + \frac{1}{w}$ If $v=3-4i$ and $w=4+3i$, find u in rectangular form.

47) Prove the following properties z is real if and only if $z = \bar{z}$

48) Find the least value of the positive integer n for which $\left(\sqrt{3}+i\right)^n$ real

49) Show that $\left(2+i\sqrt{3}\right)^{10}-\left(2-i\sqrt{3}\right)^{10}$ is purely imaginary

50) If z_1=3+4i,z_2=5-12i, and z_3 =6+8 , find $|z_1|$,$|z_2|$,$|z_3|$,$|z_1+z_2|$,$|z_2-z_3|$,and$|z_1+z_3|$

51) If z_1,z_2 and z_3 ,are complex numbers such that $|z_1|=|z_2|=|z_3|=|z_1+z_2+z_3|=1$ find the value of $\left|\frac{1}{z_1}+\frac{1}{z_2}+\frac{1}{z_3}\right|$

52) If |z|=2 show that $3 \le |z+3+4i| \le 7$

53) Show that the points 1, $\frac{-1}{2}+i\frac{\sqrt{3}}{2}$, and $\frac{-1}{2}-i\frac{\sqrt{3}}{2}$ are the vertices of an equilateral triangle.

54) Show that the equation $z^2=\bar{z}$ has four solutions.

55) For any two complex number z_1 and z_2 such that $|z_1|=|z_2|=1$ and z1z2 $\neq$, then show that $\frac{z_1+z_2}{1+z_1z_2}$ is real number.

56) Which one of the points10 − 8i , 11+ 6i is closest to1+ i .

57) If |z|= 3, show that $7 \le |z+6-8i| \le 13$.

58) If |z| =1, show that $2 \le \left|z^2-3\right| \le 4$

59) If $\left|z-\frac{2}{z}\right|=2$ show that the greatest and least value of |z| are $\sqrt{3}+1$ and $\sqrt{3}-1$ respectively.

60) Show that the equation $z^3+2\bar{z}=0$ has five solutions

61) If z=x+iy is a complex number such that $\left|\frac{z-4i}{z+4i}\right|=1$ show that the locus of z is real axis.

62) Find the rectangular form of the complex numbers

$$\left(cos\frac{\pi}{6}+isin\frac{\pi}{6}\right)\left(cos\frac{\pi}{12}+isin\frac{\pi}{12}\right)$$

63) If $(x_1+iy_1)(x_2+iy_2)(x_3+iy_3)...(x_n+iy_n)$ =a+ib, show that

i) $(x_1^2+y_1^2)(x_2^2+y_2^2)(x_3^2+y_3^2)...(x_n^2+y_n^2)=a^2+b^2$

ii) $\sum_{r=1}^{n} tan^{-1}\left(\frac{y_r}{x_r}\right)=tan^{-1}\left(\frac{b}{a}\right)+2k\pi, k\epsilon Z$

64) If $\frac{1+z}{1-z} = cos2\theta + isin2\theta$, show that z=itan$\theta$

65) If $cos\alpha + cos\beta + cos\gamma = sin\alpha + sin\beta + sin\gamma = 0$ then show that
(i) $cos3\alpha + cos3\beta + cos3\gamma = 3cos(\alpha + \beta + \gamma)$
(ii) $sin3\alpha + sin3\beta + sin3\gamma + sin3\gamma = 3sin(\alpha + \beta + \gamma)$

66) Find the value of $\left(\frac{1 + sin\frac{\pi}{10} + icos\frac{\pi}{10}}{1 + sin\frac{\pi}{10} - icos\frac{\pi}{10}}\right)^{10}$

67) If $\omega \neq 1$ is a cube root of unity, show that the roots of the equation $(z-1)^3 + 8 = 0$ are $-1, 1 - 2\omega, 1 - 2\omega^2$.

68) If $\omega \neq 1$ is a cube root of unity, show that $\left(1 - \omega + \omega^2\right)^6 + \left(1 + \omega - \omega^2\right)^6 = 128$

69) If z=2−2i, find the rotation of z by θ radians in the counter clockwise direction about the origin when $\theta = \frac{\pi}{3}$.

70) Obtain the Cartesian form of the locus of z in
|z|=|z-i|

71) Find the product $\frac{3}{2}\left(cos\frac{\pi}{3} + isin\frac{\pi}{3}\right).6\left(cos\frac{5\pi}{6} + isin\frac{5\pi}{6}\right)$ in rectangular from

72) Find the quotient $\frac{2\left(cos\frac{9\pi}{4} + isin\frac{9\pi}{4}\right)}{4\left(cos\left(\frac{-3\pi}{2} + \right)isin\left(\frac{-3\pi}{2}\right)\right)}$ in rectangular form

73) Find the fourth roots of unity.

74) Find the cube roots of unity.

75) Simplify $\left(\frac{1 + cos2\theta + isin2\theta}{1 + cos2\theta - isin2\theta}\right)^{30}$

76) If $z = (cos\theta + isin\theta)$, show that $z^n + \frac{1}{z^n} = 2cosn\theta$ and $z^n - \frac{1}{z^n} = 2isinn\theta$

77) Simplify $\left(sin\frac{\pi}{6} + icos\frac{\pi}{6}\right)^{18}$

78) Evaluate the following if z=5−2i and w= −1+3i
$z^2+2zw+w^2$

79) Evaluate the following if z=5−2i and w= −1+3i
$(z+w)^2$

80) Given the complex number z=2+3i, represent the complex numbers in Argand diagram
z, −iz , and z−iz

81) If $z_1=3, z_2=-7i$, and $z_3=5+4i$, show that
$(z_1+z_2)z_3=z_1z_3+z_2+z_3$

82) Prove the following properties

$Ro(z) = \frac{z+\bar{z}}{2}$ and $\text{Im}(z) = \frac{z-\bar{z}}{2i}$

83) Find the least value of the positive integer n for which $\left(\sqrt{3}+i\right)^n$ purely imaginary

84) Show that $\left(\frac{19-7i}{9+i}\right)^{12} + \left(\frac{20-5i}{7-6i}\right)^{12}$ is real

85) Obtain the Cartesian form of the locus of z=x+iy in
Im[(1−i)z+1]= 0

86) Obtain the Cartesian form of the locus of z=x+iy in
|z+i|=|z-1|

87) Obtain the Cartesian equation for the locus of z=x+iy in
$|z-4|^2-|z-1|^2=16$

88) Write in polar form of the following complex numbers

$$\frac{i-1}{\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}}$$

89) Find the rectangular form of the complex numbers

$$\frac{\cos\frac{\pi}{6}-i\sin\frac{\pi}{6}}{2\left(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}\right)}$$

90) If $\omega \neq 1$ is a cube root of unity, show that

$$(1+\omega)\left(1+\omega^2\right)\left(1+\omega^4\right)\left(1+\omega^8\right)\ldots\left(1+\omega^{2^{11}}\right) = 11.$$

91) If z=2−2i, find the rotation of z by θ radians in the counter clockwise direction about the origin when $\theta = \frac{2\pi}{3}$.

92) If z=2−2i, find the rotation of z by θ radians in the counter clockwise direction about the origin when $\theta = \frac{3\pi}{2}$.

93) Simplify the following:
i^{1729}

94) Obtain the Cartesian form of the locus of z in
$|2z-3-i|=3$

95) Given the complex number z=3+2i, represent the complex numbers z,iz, and z+ iz in one Argand diagram. Show that these complex numbers form the vertices of an isosceles right triangle.

96) If α and β are the roots of the quadratic equation $17x^2+43x-73 = 0$, construct a quadratic equation whose roots are α + 2 and β + 2.

97) If α and β are the roots of the quadratic equation $2x^2-7x+13 = 0$, construct a quadratic equation whose roots are α^2 and β^2.

98) If the sides of a cubic box are increased by 1, 2, 3 units respectively to form a cuboid, then the volume is increased by 52 cubic units. Find the volume of the cuboid.

99) If α, β and γ are the roots of the cubic equation $x^3+2x^2+3x+4=0$, form a cubic equation whose roots are, 2α, 2β, 2γ

100) Find the sum of squares of roots of the equation $2x^4-8x+6x^2-3=0$.

101) Find the sum of the squares of the roots of $ax^4+bx^3+cx^2+dx+e = 0$.

102) Find the condition that the roots of $x^3+ax^2+bx+c = 0$ are in the ratio p:q:r.

103) If p is real, discuss the nature of the roots of the equation $4x^2+4px+p+2=0$ in terms of p.

104) If α, β, and γ are the roots of the polynomial equation $ax^3+bx^2+cx+d=0$, find the value of $\Sigma\frac{\alpha}{\beta\gamma}$ in terms of the coefficients.

105) If p and q are the roots of the equation $lx^2+nx+n = 0$, show that $\sqrt{\frac{p}{q}}+\sqrt{\frac{q}{p}}+\sqrt{\frac{n}{l}}=0$.

106) If the equations $x^2+px+q= 0$ and $x^2+p'x+q'= 0$ have a common root, show that it must be equal to $\frac{pq'-p'q}{q-q'}$ or $\frac{q-q'}{p'-p}$.

107) Form a polynomial equation with integer coefficients with $\sqrt{\frac{\sqrt{2}}{\sqrt{3}}}$ as a root.

108) Prove that a line cannot intersect a circle at more than two points.

109) If k is real, discuss the nature of the roots of the polynomial equation $2x^2+kx+k=0$, in terms of k.

110) Prove that a straight line and parabola cannot intersect at more than two points.

111) Solve the equation $x^4-9x^2+20=0$.

112) Solve the equation $x^3-3x^2-33x+35 = 0$.

113) Solve the equation $2x^3+11x^2-9x-18=0$.

114) Find the condition that the roots of $ax^3+bx^2+cx+d=0$ are in geometric progression. Assume a,b,c,d ≠0.

115) If the roots of $x^3+px^2+qx+r=0$ are in H.P. prove that $9pqr = 27r^3+2p$.

116) It is known that the roots of the equation $x^3-6x^2-4x+24=0$ are in arithmetic progression. Find its roots.

117) Solve the cubic equation : $2x^3-x^2-18x+9=0$ if sum of two of its roots vanishes.

118) Solve the equation $9x-36x^2+44x-16=0$ if the roots form an arithmetic progression.

119) Solve the equation $3x^3-26x^2+52x-24=0$ if its roots form a geometric progression.

120) Solve the cubic equations:
$8x^3-2x^2-10x+3=0$

121) Solve the equation $x^3-5x^2-4x+20=0$

122) Find the roots of $2x^3+3x^2+2x+3$

123) Solve the equation $7x^3-43x^2=43x-7$

124) Find solution, if any, of the equation $2\cos^2x-9\cos x+4=0$

125) Solve the following equations,
$\sin^2x-5\sin x+4=0$

126) Examine for the rational roots of $2x^2-x^2-1=0$

127) Find all real numbers satisfying $4^x-3(2^{x+2})+2^5=0$

128) If α, β and γ are the roots of the cubic equation $x^3+2x^2+3x+4=0$, form a cubic equation whose roots are
$\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$

129) If α, β and γ are the roots of the cubic equation $x^3+2x^2+3x+4=0$, form a cubic equation whose roots are
$-\alpha, \beta, -\gamma$

130) Solve the cubic equations:
$8x^3-2x^2-7x+3=0$

131) Solve the following equations,
$12x+8x=29x^2-4$

132) Solve the equations
$x^4+3x^3-3x-1=0$

133) Find the domain of $\sin^{-1}(2-3x^2)$

134) Sketch the graph of $y=\sin(\frac{1}{3}x)$ for $0 \le x < 6\pi$.

135) Find the domain of the following

$$\sin^{-1}\left(\frac{x^2+1}{2x}\right)$$

136) Find the domain of $\cos^{-1}(\frac{2+\sin x}{3})$

137) Find the domain of
$f(x)=\sin^{-1}(\frac{|x|-2}{3}) + \cos^{-1}(\frac{1-|x|}{4})$

138) For what value of x , the inequality $\frac{\pi}{2} < \cos^{-1}(3x-1) < \pi$

139) Find the value of

$$\cos\left(\cos^{-1}\left(\frac{4}{5}\right) + \sin^{-1}\left(\frac{4}{5}\right)\right)$$

140) Find the principal value of $\tan^{-1}(\sqrt{3})$

141) Find

i) $\tan^{-1}(-\sqrt{3})$

ii) $\tan^{-1}(tan\frac{3\pi}{5})$

iii) $\tan(\tan^{-1}-(2019))$

142) Find the value of $\tan^{-1}(-1)+\cos^{-1}(\frac{1}{2}) + sin^{-}1(-\frac{1}{2})$

143) Prove that $\tan(\sin^{-1}x)=\frac{x}{\sqrt{1-x^2}}, -1 < x < 1.$

144) Find the domain of the following functions :

$tan^{-1}(\sqrt{9-x^2})$

145) Find the value of

$$tan\left(cos^{-1}\left(\frac{1}{2}\right) - sin^{-1}\left(-\frac{1}{2}\right)\right)$$

146) Evaluate $sin\left[sin^{-1}\left(\frac{3}{5}\right) + sec^{-1}\left(\frac{5}{4}\right)\right]$

147) Prove that

$\tan^{-1}\frac{1}{2} + tan^{-1}\frac{1}{3} = \frac{\pi}{4}$

148) If $\cos^{-1}x+\cos^{-1}y+\cos^{-1} z$ = and 0< x,y,z< 1, show that $x^2+y^2+z^2+2xyz=1$

149) Solve sin-1 x > $\cos^{-1}x$

150) Solve $\tan^{-1} 2x+\tan^{-1} 3x=\frac{\pi}{4}$, if $6x^2< 1$

151) Find the value of the expression in terms of x, with the help of a reference triangle.

$\sin(\cos^{-1}(1-x))$

152) Find the value of

$$sin^{-1}\left(cos\left(sin^{-1}\left(\frac{\sqrt{3}}{2}\right)\right)\right)$$

153) Prove that $tan^{-1}x + tan^{-1}\frac{2x}{1-x^2} = tan^{-1}\frac{3x-x^2}{1-3x^2}, |x| < \frac{1}{\sqrt{3}}$

154) Simplify: $tan^{-1}\frac{x}{y} - tan^{-1}\frac{x-y}{x+y}$

155) Solve

$sin^{-1}\frac{5}{x} + sin^{-1}\frac{12}{x} = \frac{\pi}{2}$

156) Find the number of solution of the equation $\tan^{-1}(x-1) \tan^{-1}x+\tan^{-1}(x+1)\tan^{-1}(3x)$

157) Find the domain of the following

$g(x)=2\sin^{-1}(2x-1)-\frac{\pi}{4}$

158) Find the domain of

$g(x)=\sin^{-1}x+\cos^{-1}x$

159) Find the value of

$$cos^{-1}\left(cos\left(\frac{4\pi}{3}\right)\right)+cos^{-1}\left(cos\left(\frac{5\pi}{4}\right)\right).$$

160) Find the domain of the following functions

$$\frac{1}{2}tan^{-1}(1-x^2)-\frac{\pi}{4}$$

161) Find the value of

$$sin\left(tan^{-1}\left(\frac{1}{2}\right)-cos^{-1}\left(\frac{4}{5}\right)\right)$$

162) Find the value of

$$cos\left(sin^{-1}\left(\frac{4}{5}\right)-tan^{-1}\left(\frac{3}{4}\right)\right)$$

163) Find the value of

$$sin^{-1}(-1)+cos^{-1}\left(\frac{1}{2}\right)+cot^{-1}(2)$$

164) Find the value of

$$cot^{-1}(1)+sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)-sec^{-1}(-\sqrt{2})$$

165) Find the value of the expression in terms of x, with the help of a reference triangle.
cos (tan^{-1}(3x-1))

166) Find the value of the expression in terms of x, with the help of a reference triangle.

$$\tan\left(sin^{-1}\left(x+\frac{1}{2}\right)\right)$$

167) Find the value of

$$cot\left(sin^{-1}\frac{3}{5}+sin^{-1}\frac{4}{5}\right)$$

168) Find the value of

$$tan\left(sin^{-1}\frac{3}{5}+cot^{-1}\frac{3}{2}\right)$$

169) Prove that

$$sin^{-1}(\frac{3}{5})-cos^{-1}(\frac{12}{13})=sin^{-1}(\frac{16}{65})$$

170) Solve:

$$2tan^{-1}x=cos^{-1}\frac{1-a^2}{1+a^2}-cos^{-1}\frac{1-b^2}{1+b^2}, a>0, b>0$$

171) Solve:

$$2tan^{-1}(cosx)=tan^{-1}(2cosec\quad x)$$

172) Solve

$$cot^{-1}x - xot^{-1}(x+2) = \frac{\pi}{12}, x > 0$$

173) Find the value of

$$tan\left[\frac{1}{2}sin^{-1}\left(\frac{2a}{1+a^2}\right) + \frac{1}{2}cos^{-1}\left(\frac{1-a^2}{1+a^2}\right)\right]$$

174) Prove that

$$2\tan^{-1}\frac{1}{2} + tan^{-1}\frac{1}{7} = tan^{-1}\frac{31}{17}$$

175) Find the equation of the circle described on the chord 3x+y+5= 0 of the circle $x^2+y^2=16$ as diameter.

176) A line 3x+4y+10 = 0 cuts a chord of length 6 units on a circle with centre of the circle (2,1) . Find the equation of the circle in general form.

177) A circle of radius 3 units touches both the axes. Find the equations of all possible circles formed in the general form.

178) Find the centre and radius of the circle$3x^2+(a+1)y^2+6x-9y+a+4=0$.

179) Find the equations of the tangent and normal to the circle $x^2+y^2=25$ at P(-3,4).

180) Find the equation of circles that touch both the axes and pass through (-4,-2) in general form.

181) Find the equation of the circle with centre (2,3) and passing through the intersection of the lines 3x−2y−1=0and 4x+y−27=0.

182) A circle of area 9π square units has two of its diameters along the lines x+y=5 and x−y=1.
Find the equation of the circle.

183) Find the equation of the tangent and normal to the circle $x^2+y^2-6x+6y-8=0$ at (2,2) .

184) Determine whether the points(-2,1),(0,0) and (-4,-3) lie outside, on or inside the circle
$x^2+y^2-5x+2y-5=0$.

185) If the equation$3x^2+(3-p)xy+qy^2-2px=8pq$ represents a circle, find p and q . Also determine the centre and radius of the circle

186) Find the length of Latus rectum of the parabola $y^2 = 4ax$.

187) Find the length of Latus rectum of the ellipse$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

188) Find the equation of the parabola with focus $\left(-\sqrt{2}, 0\right)$ and directrix $x=\sqrt{2}$.

189) Find the equation of the parabola whose vertex is (5,-2) and focus(2,-2) .

190) Find the equation of the parabola with vertex (-1,-2) , axis parallel to y -axis and passing through (3,6)

191) Find the equation of the ellipse with foci (±2,0) , vertices (±3,0) .

192) Find the equation of the hyperbola with vertices (0,±4) and foci(0,±6).

193) The orbit of Halley's Comet is an ellipse 36.18 astronomical units long and by 9.12 astronomical units wide. Find its eccentricity.

194) Find the equation of the ellipse in each of the cases given below:

(i) foci(±3 0),e =$\frac{1}{2}$

(ii) foci (0,±4)and end points of major axis are(0,±5).

(iii) length of latus rectum 8, eccentricity =$\frac{3}{5}$ and major axis on x -axis.

(iv) length of latus rectum 4 , distance between foci4 $\sqrt{2}$ and major axis as y - axis.

195) Find the equation of the hyperbola in each of the cases given below:

(i) foci(±2,0), eccentricity =$\frac{3}{2}$

(ii) Centre (2,1) , one of the foci (8,1) and corresponding directrix x = 4.

(iii) passing through(5,−2)and length of the transverse axis along x axis and of length 8 units.

196) Find the vertex, focus, equation of directrix and length of the latus rectum of the following:

$y^2 = 16x$

197) Identify the type of conic and find centre, foci, vertices, and directrices of each of the following:

$\frac{x^2}{25} + \frac{y^2}{9} = 1$

198) Prove that the length of the latus rectum of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $\frac{2b^2}{a}$.

199) Show that the absolute value of difference of the focal distances of any point P on the hyperbola is the length of its transverse axis.

200) Identify the type of conic and find centre, foci, vertices, and directrices of each of the following :

$\frac{(x-3)^2}{225} + \frac{(y-4)^2}{289} = 1$

201) Identify the type of the conic for the following equations:

(1) $16y^2=-4x^2+64$

(2) $x^2+y^2=-4x-y+4$

(3) $x^2-2y=x+3$

(4) $4x^2-9y^2-16x+18y-29 = 0$

202) Find the equations of tangent and normal to the parabola $x^2+6x+4y+5=0$ at (1,-3) .

203) Find the equations of tangent and normal to the ellipse $x^2+4y^2=32$ when $\theta = \frac{\pi}{4}$

204) Find the equation of the tangent to the parabola $y^2 = 16x$perpendicular to $2x+2y+3=0$.

205) Find the equation of the tangent at t = 2 to the parabola$y^2 = 8x$. (Hint: use parametric form)

206) Find the equations of the tangent and normal to hyperbola $12x^2-9y^2=108$ at $\theta = \frac{\pi}{3}$ (Hint: use parametric form)

207) Prove that the point of intersection of the tangents at ‘ t_1 ’ and ‘ t_2 ’on the parabola $y^2 = 4ax$ is $\left[at_1t_2, a(t_1+t_2)\right]$.

208) If the normal at the point ‘ t_1 ’ on the parabola $y^2 = 4ax$ meets the parabola again at the point‘ t_2 ’, then prove that $t_2=-\left(t_1\frac{2}{t_1}\right)$

209) A semielliptical archway over a one-way road has a height of 3m and a width of 12m. The truck has a width of 3m and a height of 2.7m. Will the truck clear the opening of the archway?

210) The parabolic communication antenna has a focus at 2m distance from the vertex of the antenna. Find the width of the antenna 3m from the vertex.

211) The equation $y=\frac{1}{32}x^2$ models cross sections of parabolic mirrors that are used for solar energy. There is a heating tube located at the focus of each parabola; how high is this tube located above the vertex of the parabola?

212) A search light has a parabolic reflector (has a cross-section that forms a ‘bowl’). The parabolic bowl is 40cm wide from rim to rim and 30cm deep. The bulb is located at the focus.
(1) What is the equation of the parabola used for reflector?
(2) How far from the vertex is the bulb to be placed so that the maximum distance covered?

213) An equation of the elliptical part of an optical lens system is $\frac{x^2}{16}+\frac{y^2}{9}=1$. The parabolic part of the system has a focus in common with the right focus of the ellipse .The vertex of the parabola is at the origin and the parabola opens to the right. Determine the equation of the parabola.

214) A room 34m long is constructed to be a whispering gallery. The room has an elliptical ceiling, as shown in Fig. 5.64. If the maximum height of the ceiling is 8m, determine where the foci are located.

215) The equation of the ellipse is $\frac{(x-11)^2}{484}+\frac{y^2}{64}=1.$ (x and y are measured in centimeters) where to the nearest centimeter, should the patient’s kidney stone be placed so that the reflected sound hits the kidney stone?

216) Find the vertex, focus, equation of directrix and length of the latus rectum of the following:
$x^2 = 24y$

217) Find the vertex, focus, equation of directrix and length of the latus rectum of the following:
$y^2 = -8x$

218) Identify the type of conic and find centre, foci, vertices, and directrices of each of the following:
$\frac{x^2}{3}+\frac{y^2}{10}=1$

219) Identify the type of conic and find centre, foci, vertices, and directrices of each of the following:
$\frac{x^2}{25}+\frac{y^2}{144}=1$

220) Identify the type of conic and find centre, foci, vertices, and directrices of each of the following:

$$\frac{x^2}{16}+\frac{y^2}{9}=1$$

221) Identify the type of conic and find centre, foci, vertices, and directrices of each of the following :

$$\frac{(x+1)^2}{100}+\frac{(y-2)^2}{64}=1$$

222) Identify the type of conic and find centre, foci, vertices, and directrices of each of the following :

$$\frac{(x+3)^2}{225}-\frac{(y-4)^2}{64}=1$$

223) Identify the type of conic and find centre, foci, vertices, and directrices of each of the following :

$$\frac{(y-2)^2}{25}\frac{(x+1)^2}{16}=1$$

224) With usual notations, in any triangle ABC, prove the following by vector method.
(i) $a^2=b^2+c^2-2bc\cos A$
(ii) $b^2=c^2+a^2-2ca\cos B$
(iii) $c^2=a^2+b^2-2ab\cos C$

225) With usual notations, in any triangle ABC, prove the following by vector method.
(i) $a=b\cos C+c\cos B$
(ii) $b=c\cos A+a\cos C$
(iii) $c=a\cos B+b\cos A$

226) If D is the midpoint of the side BC of a triangle ABC, then show by vector method that $\left|\overrightarrow{AB}\right|^2+\left|\overrightarrow{AC}\right|^2=2(\left|\overrightarrow{AD}\right|^2+\left|\overrightarrow{BD}\right|^2)$

227) In triangle, ABC the points, D, E, F are the midpoints of the sides, BC, CA and AB respectively. Using vector method, show that the area of ΔDEF is equal to $\frac{1}{2}$(area of ΔABC)

228) A particle acted upon by constant forces $2\hat{j}+5\hat{j}+6\hat{k}$ and $-\hat{i}-2\hat{j}-\hat{k}$ is displaced from the point
(4, −3, −2) to the point (6, 1, −3) . Find the total work done by the forces.

229) A particle is acted upon by the forces $(3\hat{i}-2\hat{j}+2\hat{k})$ and $(2\hat{i}+\hat{j}-\hat{k})$ is displaced from the point (1, 3, -1) to the point (4,1, -λ). If the work done by the forces is 16 units, find the value of λ.

230) Find the magnitude and the direction cosines of the torque about the point (2, 0, -1) of a force $(2\hat{i}+\hat{j}-\hat{k})$, whose line of action passes through the origin

231) Prove by vector method that if a line is drawn from the centre of a circle to the midpoint of a chord, then the line is perpendicular to the chord.

232) Prove by vector method that the median to the base of an isosceles triangle is perpendicular to the base.

233) Prove by vector method that an angle in a semi-circle is a right angle.

234) Prove by vector method that the diagonals of a rhombus bisect each other at right angles.

235) Using vector method, prove that if the diagonals of a parallelogram are equal, then it is a rectangle

236) Prove by vector method that the area of the quadrilateral ABCD having diagonals AC and BD is $\frac{1}{2}\left|\vec{AC} \times \vec{BC}\right|$.

237) Prove by vector method that the parallelograms on the same base and between the same parallels are equal in area.

238) If G is the centroid of a ΔABC, prove that (area of ΔGAB) = (area of ΔGBC) = (area of ΔGCA) = $\frac{1}{3}$ (area of ΔABC)

239) A particle acted on by constant forces $8\vec{i} + 2\vec{j} - 6\vec{k}$ and $6\vec{i} + 2\vec{j} - 2\vec{k}$ is displaced from the point (1, 2, 3) to the point (5, 4, 1). Find the total work done by the forces.

240) Forces of magnit $5\sqrt{2}$ and $5\sqrt{2}$ units acting in the directions $3\hat{i} + 4\hat{j} + 5\hat{k}$ and $10\hat{i} + 6\hat{j} - 8\hat{k}$ respectively, act on a particle which is displaced from the point with position vector $4\hat{i} + 3\hat{j} - 2\hat{k}$ to the point with position vector $6\hat{i} + \hat{j} - 3\hat{k}$. Find the work done by the forces.

241) Find the magnitude and direction cosines of the torque of a force represented by $3\hat{i} + 4\hat{j} - 5\hat{k}$ about the point with position vector $2\hat{i} - 3\hat{j} + 4\hat{k}$ acting through a point whose position vector is $4\hat{i} + 2\hat{j} - 3\hat{k}$.

242) Find the torque of the resultant of the three forces represented by $-3\hat{i} + 6\hat{j} + 3\hat{k}$, $4\hat{i} - 10\hat{j} + 12\hat{k}$ and $4\hat{i} + 7\hat{j}$ acting at the point with position vector $8\hat{i} - 6\hat{j} - 4\hat{k}$, about the point with position vector $18\hat{i} + 3\hat{j} - 9\hat{k}$

243) Show that the four points (6, -7, 0), (16, -19, -4), (0, 3, -6), (2, -5, 10) lie on a same plane.

244) If the vectors $\vec{a}, \vec{b}, \vec{c}$ are coplanar, then prove that the vectors $\vec{a} + \vec{b}, \vec{b} + \vec{c}, \vec{c} + \vec{a}$ are also coplanar

245) If $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar vectors represented by concurrent edges of a parallelepiped of volume 4 cubic units, find the value of

$(\vec{a} + \vec{b}).(\vec{b} \times \vec{c}) + (\vec{b} + \vec{c}).(\vec{c} \times \vec{a}) + (\vec{c} + \vec{a})(\vec{a} \times \vec{b})$

246) Find the altitude of a parallelepiped determined by the vectors $\vec{a} = 2\hat{i} + 5\hat{j} + 3\hat{k}$, $\hat{b} = \hat{i} + 3\hat{j} - 2\hat{k}$ and $\vec{c} = -3\vec{i} + \vec{j} + 4\vec{k}$ if the base is taken as the parallelogram determined by $\vec{b}$ and $\vec{c}$

247) Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i}$ and $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$. If $c_1 = 1$ and $c_2 = 2$, find c_3 such that $\vec{a}, \vec{b}$ and $\vec{c}$ are coplanar.

248) If $\vec{a} = \hat{i} - \hat{k}, \vec{b} = x\hat{i} + \hat{j} + (1-x)\hat{k}, \vec{c} = y\hat{i} + x\hat{j} + (1+x+y)\hat{k}$show that $[\vec{a}, \vec{b}, \vec{c}]$ depends on neither x nor y.

249) If the vectors $a\hat{i} + a\hat{j} + c\hat{k}, \hat{i} + \hat{k}$ and $c\hat{i} + c\hat{j} + b\hat{k}$ are coplanar, prove that c is the geometric mean of a and b.

250) Let $\vec{a}, \vec{b}, \vec{c}$ be three non-zero vectors such that $\vec{c}$ is a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$. If the angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{6}$, show that $[\vec{a}, \vec{b}, \vec{c}]^2 = \frac{1}{4}|\vec{a}|^2|\vec{b}|^2$

251) Prove that $[\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}] = [\vec{a}, \vec{b}, \vec{c}]^2$

252) For any four vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ we have

$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = [\vec{a}, \vec{b}, \vec{d}]\vec{c} - [\vec{a}, \vec{b}, \vec{c}]\vec{d} = [\vec{a}, \vec{c}, \vec{d}]\vec{b} - [\vec{b}, \vec{c}, \vec{d}]\vec{a}$

253) If $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}, \vec{b} = 2\hat{i} + \hat{j} + \hat{k}, \vec{c} = 3\hat{i} + 2\hat{j} + \hat{k}$ find

(i) $(\vec{a} \times \vec{b}) \times \vec{c}$

(ii) $\vec{a} \times (\vec{b} \times \vec{c})$

254) For any vector $\vec{a}$, prove that $\hat{i} \times (\vec{a} \times \hat{i}) + \hat{j} \times (\vec{a} \times \hat{j}) + \hat{k} \times \vec{a} \times \hat{k} = 2\vec{a}$.

255) Prove that $[\vec{a} - \vec{b}, \vec{b} - \vec{c}, \vec{c} - \vec{a}] = 0$

256) If $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}, \vec{b} = -\hat{i} + 2\hat{j} - 4\hat{k}, \vec{c} = \hat{i} + \hat{j} + \hat{k}$ then find the value of $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d})$.

257) If $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are coplanar vectors, then show that $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \vec{0}$.

258) If $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}, \vec{b} = 2\hat{i} - \hat{j} + \hat{k}, \vec{c} = 3\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{a} \times (\vec{b} \times \vec{c}) = l\vec{a} + m\vec{b} + n\vec{c}$, find the values of l, m, n.

259) If $\hat{a}, \hat{b}, \hat{c}$ are three unit vectors such that $\hat{b}$ and $\hat{c}$ are non-parallel and $\hat{a} \times \hat{b} \times \hat{c} = \frac{1}{2}\hat{b}$ the angle between $\hat{a}$ and $\vec{c}$.

260) A straight line passes through the point (1, 2, −3) and parallel to $4\hat{i} + 5\hat{j} - 7\hat{k}$. Find
(i) vector equation in parametric form
(ii) vector equation in non-parametric form
(iii) Cartesian equations of the straight line.

261) The vector equation in parametric form of a line is $\vec{r} = (3\hat{i} - 2\hat{j} + 6\hat{k}) + t(2\hat{i} - \hat{j} + 3\hat{k})$. Find

(i) the direction cosines of the straight line
(ii) vector equation in non-parametric form of the line
(iii) Cartesian equations of the line.

262) Find the vector equation in parametric form and Cartesian equations of the line passing through (-4, 2, -3) and is parallel to the line $\frac{-x-2}{4} = \frac{y+3}{-2} = \frac{2z-6}{3}$

263) Find the angle between the straight line $\frac{x+3}{2} = \frac{y-1}{2} = -z$ with coordinate axes.

264) Find the angle between the lines $\vec{r} = (\hat{i} + 2\hat{j} + 4\hat{k}) + t(2\hat{i} + 2\hat{j} + \hat{k})$ and the straight line passing through the points (5, 1, 4) and (9, 2, 12)

265) Find the angle between the straight lines $\frac{x-4}{2} = \frac{y}{1} = \frac{z-1}{-2}$ and $\frac{x-4}{2} = \frac{y}{1} = \frac{z-1}{-2}$ and state whether they are parallel or perpendicular.

266) Find the direction cosines of the straight line passing through the points (5,6,7) and (7,9,13) . Also, find the parametric form of vector equation and Cartesian equations of the straight line passing through two given points.

267) The vertices of ΔABC are A(7, 2, 1),B(6,0,3) , and C(4, 2, 4) . Find ∠ABC .

268) If the straight line joining the points (2, 1, 4) and (a−1, 4, −1) is parallel to the line joining the points (0, 2, b −1) and (5, 3, −2) , find the values of a and b.

269) If the straight lines $\frac{x-5}{5m+2} = \frac{2-y}{5} = \frac{1-z}{-1}$ and $x = \frac{2y+1}{4m} = \frac{1-z}{-3}$ are perpendicular to each other, find the value of m.

270) Find the shortest distance between the two given straight lines
$\vec{r} = (2\hat{i} + 3\hat{j} + 4\hat{k}) + t(-2\hat{i} + \hat{j} - 2\hat{k})$ and $\frac{x-3}{2} = \frac{y}{-1} = \frac{z+2}{2}$

271) Find the coordinates of the foot of the perpendicular drawn from the point (-1, 2, 3) to the straight line $\vec{r} = (\hat{i} - 4\hat{j} + 3\hat{k}) + t(2\hat{i} + 3\hat{j} + \hat{k})$. Also, find the shortest distance from the point to the straight line.

272) Find the parametric form of vector equation and Cartesian equations of a straight line passing through (5, 2,8) and is perpendicular to the straight lines
$\vec{r} = (\hat{i} + \hat{j} - \hat{k}) + s(2\hat{i} - 2\hat{j} + \hat{k})$
^
$\vec{r} = (2i - \hat{j} - 3\hat{k}) + t(\hat{i} + 2\hat{j} + 2\hat{k})$.

273) Show that the lines $\vec{r} = (6\hat{i} + \hat{j} + 2\hat{k}) + s(\hat{i} + 2\hat{j} - 3\hat{k})$ and
$\vec{r} = (3\hat{i} + 2\hat{j} + 2\hat{k}) + t(2\hat{i} + 4\hat{j} - 5\hat{k})$ are skew lines and hence find the shortest distance between them.

274) If the two lines $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4}$ and $\frac{x-3}{1} = \frac{y-m}{2} = z$ intersect at a point, find the value
of m

275) Show that the straight lines x + 1= 2y = −12z and x = y + 2 = 6z − 6 are skew and hence find the shortest distance between them.

276) Find the parametric form of vector equation of the straight line passing through (−1, 2,1) and parallel to the straight line $\vec{r} = (2\hat{i} + 3\hat{j} - \hat{k}) + t(\hat{i} - 2\hat{j} + \hat{k})$ and hence find the

shortest distance
between the lines.

277) Find the foot of the perpendicular drawn from the point (5, 4, 2) to the line $\frac{x+1}{2} = \frac{y-3}{3} = \frac{z-1}{-1}$. Also, find the equation of the perpendicular

278) If a plane meets the coordinate axes at A,B,C such that the centriod of the triangle ABC is the point (u,v,w), find the equation of the plane.

279) Find the non-parametric form of vector equation, and Cartesian equation of the plane passing through the point (0, 1, -5) and parallel to the straight lines $\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + s(\hat{i} + 3\hat{j} + 6\hat{k})$ and $\hat{r} = (\hat{i} - 3\hat{j} + 5\hat{k}) + t(\hat{i} + \hat{j} - \hat{k})$

280) Find the non-parametric form of vector equation, and Cartesian equations of the plane $\vec{r} = (6\hat{i} - \hat{j} + \hat{k}) + s(-\hat{i} + 2\hat{j} + \hat{k}) + (-5\hat{i} - 4\hat{j} - 5\hat{k})$

281) If the straight lines $\frac{x-1}{1} = \frac{y-2}{1} = \frac{z-3}{m^2}$ and $\frac{x-3}{1} = \frac{y-2}{m^2} = \frac{z-1}{2}$ are coplanar, find the distinct real values of m.

282) Find the equation of the plane passing through the line of intersection of the planes $\vec{r}.(2\hat{i} - 7\hat{j} + 4\hat{k}) = 3$ and 3x - 5y + 11 = 0, and the point (-2, 1, 3)

283) Find the equation of the plane passing through the line of intersection of the planes x + 2y + 3z = 2 and x - y - z + 11 = 3 and at a distance $\frac{2}{\sqrt{3}}$ from the point (3, 1, -1)

284) Find the equation of the plane which passes through the point (3, 4, -1) and is parallel to the plane 2x - 3y + 5z = 0. Also, find the distance between the two planes.

285) Find the point of intersection of the line x - 1 = $\frac{y}{2}$ = z + 1 with the plane 2x - y + 2z = 2. Also, find the angle between the line and the plane.

286) Find the distance of the point (5,-5,-10) from the point of intersection of a straight line passing through the points A(4,1,2) and B(7,5,4) with the plane x-y+z=5

287) Find the equation of the plane passing through the intersection of the planes $\vec{r}.(\hat{i} + \hat{j} + \hat{k}) + 1 = 0$ and $\vec{r}.(2\hat{i} - 3\hat{j} + 5\hat{k}) = 2$ and the point (-1, 2, 1)

288) Find the equation of the plane passing through the intersection of the planes 2x+3y−z+7=0 and andx+y−2z+5=0 and is perpendicular to the planex+y−3z−5=0.

289) Find the coordinates of the point where the straight line $\vec{r} = (2\hat{i} - \hat{j} + 2\hat{k}) + t(3\hat{i} + 4\hat{j} + 2\hat{k})$ intersects the plane x−y+z−5=0.

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5 MARKS BB V1

12th Standard

Maths

Exam Time : 01:40:00 Hrs

Total Marks : 100
134 x 5 = 670

1) If A = $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & 4 \\ 2 & -4 & 3 \end{bmatrix}$, verify thatA(adj A)=(adj A)A = |A| I_3.

2) If A = $\begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}$, find x and y such that $A^2 + xA + yI_2 = O_2$. Hence, find A^{-1}.

3) If A = $\frac{1}{7}\begin{bmatrix} 6 & -3 & a \\ b & -2 & 6 \\ 2 & c & 3 \end{bmatrix}$ is orthogonal, find a, b and c , and hence A^{-1}.

4) If F(α) = $\begin{bmatrix} \cos\alpha & 0 & \sin\alpha \\ 0 & 1 & 0 \\ -\sin\alpha & 0 & \cos\alpha \end{bmatrix}$, show that $[F(\alpha)]^{-1} = F(-\alpha)$.

5) If A = $\begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix}$, show that $A^2 - 3A - 7I_2 = O_2$. Hence find A^{-1}.

6) Show that the matrix $\begin{bmatrix} 3 & 1 & 4 \\ 2 & 0 & -1 \\ 5 & 2 & 1 \end{bmatrix}$ is non-singular and reduce it to the identity matrix by elementary row transformations.

7) Find the inverse of A = $\begin{bmatrix} 2 & 1 & 1 \\ 3 & 2 & 1 \\ 2 & 1 & 2 \end{bmatrix}$ by Gauss-Jordan method.

8) Solve the following system of equations, using matrix inversion method:
$2x_1 + 3x_2 + 3x_3 = 5$, $x_1 - 2x_2 + x_3 = -4$, $3x_1 - x_2 - 2x_3 = 3$.

9) If A = $\begin{bmatrix} -4 & 4 & 4 \\ -7 & 1 & 3 \\ 5 & -3 & -1 \end{bmatrix}$ and B = $\begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & -2 \\ 2 & 1 & 3 \end{bmatrix}$, find the productsAB and BAand hence solve the system of equations x - y + z = 4, x - 2y - 2z = 9, 2x + y + 3z = 1.

10) If $A = \begin{bmatrix} -5 & 1 & 3 \\ 7 & 1 & -5 \\ 1 & -1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix}$, find the products AB and BA and hence solve the system of equations $x + y + 2z = 1$, $3x + 2y + z = 7$, $2x + y + 3z = 2$.

11) The prices of three commodities A, B and C are Rs.x, y and z per units respectively. A person P purchases 4 units of B and sells two units of A and 5 units of C . Person Q purchases 2 units of C and sells 3 units of A and one unit of B . Person R purchases one unit of A and sells 3 unit of B and one unit of C . In the process, P, Q and R earn Rs.15,000, Rs.1,000 and Rs.4,000 respectively. Find the prices per unit of A, B and C . (Use matrix inversion method to solve the problem.)

12) Solve, by Cramer's rule, the system of equations
$x_1 - x_2 = 3$, $2x_1 + 3x_2 + 4x_3 = 17$, $x_2 + 2x_3 = 7$.

13) In a T20 match, Chennai Super Kings needed just 6 runs to win with 1 ball left to go in the last over. The last ball was bowled and the batsman at the crease hit it high up. The ball traversed along a path in a vertical plane and the equation of the path is $y = ax^2 + bx + c$ with respect to a xy-coordinate system in the vertical plane and the ball traversed through the points (10, 8), (20, 16) (30, 18) can you conclude that Chennai Super Kings won the match?
Justify your answer. (All distances are measured in metres and the meeting point of the plane of the path with the farthest boundary line is (70, 0).)

14) A family of 3 people went out for dinner in a restaurant. The cost of two dosai, three idlies and two vadais is Rs.150. The cost of the two dosai, two idlies and four vadais is Rs.200. The cost of five dosai, four idlies and two vadais is Rs.250. The family has Rs.350 in hand and they ate 3 dosai and six idlies and six vadais. Will they be able to manage to pay the bill within the amount they had ?

15) The upward speed v(t)of a rocket at time t is approximated by $v(t) = at^2 + bt + c \le t \le 100$ where a, b and c are constants. It has been found that the speed at times t = 3, t = 6, and t = 9 seconds are respectively, 64, 133, and 208 miles per second respectively. Find the speed at time t = 15 seconds. (Use Gaussian elimination method.)

16) Solve the following systems of linear equations by Gaussian elimination method:
(i) $2x - 2y + 3z = 2$, $x + 2y - z = 3$, $3x - y + 2z = 1$
(ii) $2x + 4y + 6z = 22$, $3x + 8y + 5z = 27$, $-x + y + 2z = 2$

17) If $ax^2 + bx + c$ is divided by $x + 3$, $x - 5$, and $x - 1$, the remainders are 21,61 and 9 respectively. Find a,b and c. (Use Gaussian elimination method.)

18) An amount of Rs.65,000 is invested in three bonds at the rates of 6%,8% and 9% per annum respectively. The total annual income is Rs.4,800. The income from the third bond is Rs.600 more than that from the second bond. Determine the price of each bond. (Use Gaussian elimination method.)

19) A boy is walking along the path $y = ax^2 + bx + c$ through the points (−6, 8),(−2, −12) , and (3, 8) . He wants to meet his friend at P(7,60) . Will he meet his friend? (Use Gaussian elimination method.)

20) Test for consistency of the following system of linear equations and if possible solve:
$x + 2y - z = 3$, $3x - y + 2z = 1$, $x - 2y + 3z = 3$, $x - y + z + 1 = 0$

21) Test for consistency of the following system of linear equations and if possible solve:
$4x - 2y + 6z = 8$, $x + y - 3z = -1$, $15x - 3y + 9z = 21$.

22) Test the consistency of the following system of linear equations
$x - y + z = -9$, $2x - y + z = 4$, $3x - y + z = 6$, $4x - y + 2z = 7$.

23) Find the condition on a, b and c so that the following system of linear equations has one parameter family of solutions: $x + y + z = a$, $x + 2y + 3z = b$, $3x + 5y + 7z = c$.

24) Investigate for what values of λ and μ the system of linear equations
$x + 2y + z = 7$, $x + y + \lambda z = \mu$, $x + 3y - 5z = 5$ has
(i) no solution
(ii) a unique solution
(iii) an infinite number of solutions

25) Test for consistency and if possible, solve the following systems of equations by rank method.
i) $x - y + 2z = 2$, $2x + y + 4z = 7$, $4x - y + z = 4$
ii) $3x + y + z = 2$, $x - 3y + 2z = 1$, $7x - y + 4z = 5$
iii) $2x + 2y + z = 5$, $x - y + z = 1$, $3x + y + 2z = 4$
iv) $2x - y + z = 2$, $6x - 3y + 3z = 6$, $4x - 2y + 2z = 4$

26) Find the value of k for which the equations $kx - 2y + z = 1$, $x - 2ky + z = -2$, $x - 2y + kz = 1$ have
(i) no solution
(ii) unique solution
(iii) infinitely many solution

27) Investigate the values of λ and μ the system of linear equations $2x + 3y + 5z = 9$, $7x + 3y - 5z = 8$, $2x + 3y + \lambda z = \mu$, have
(i) no solution
(ii) a unique solution
(iii) an infinite number of solutions.

28) Solve the system: $x + 3y - 2z = 0$, $2x - y + 4z = 0$, $x - 11y + 14z = 0$

29) Solve the system: $x + y - 2z = 0$, $2x - 3y + z = 0$, $3x - 7y + 10z = 0$, $6x - 9y + 10z = 0$.

30) Determine the values of λ for which the following system of equations $(3\lambda - 8)x + 3y + 3z = 0$, $3x + (3\lambda - 8)y + 3z = 0$, $3x + 3y + (3\lambda - 8)z = 0$. has a non-trivial solution.

31) By using Gaussian elimination method, balance the chemical reaction equation:
$C_5H_8 + O_2 \longrightarrow CO_2 + H_2O$. (The above is the reaction that is taking place in the burning of organic compound called isoprene.)

32) If the system of equations $px + by + cz = 0$, $ax + qy + cz = 0$, $ax + by + rz = 0$ has a non-trivial solution and $p \neq a, q \neq b, r \neq c$, prove that $\frac{p}{p-a} + \frac{q}{q-b} + \frac{r}{r-c} = 2$.

33) Solve the following system of homogenous equations.
$3x + 2y + 7z = 0$, $4x - 3y - 2z = 0$, $5x + 9y + 23z = 0$

34) Determine the values of λ for which the following system of equations $x + y + 3z = 0$, $4x + 3y + \lambda z = 0$, $2x + y + 2z = 0$ has
(i) a unique solution
(ii) a non-trivial solution

35) By using Gaussian elimination method, balance the chemical reaction equation:
$C_2H_5 + O_2 \rightarrow H_2O + CO_2$

36) Find the inverse of each of the following by Gauss-Jordan method:

$$\begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 6 & -2 & -3 \end{bmatrix}$$

37) Find the inverse of each of the following by Gauss-Jordan method:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$$

38) Solve the following system of linear equations by matrix inversion method:
$2x + 3y - z = 9$, $x + y + z = 9$, $3x - y - z = -1$

39) Solve the following system of linear equations by matrix inversion method:
$x + y + z - 2 = 0$, $6x - 4y + 5z - 31 = 0$, $5x + 2y + 2z = 13$.

40) Solve the following systems of linear equations by Cramer's rule:
$3x + 3y - z = 11$, $2x - y + 2z = 9$, $4x + 3y + 2z = 25$.

41) Solve the following systems of linear equations by Cramer's rule:
$\frac{3}{x} - \frac{4}{y} - \frac{2}{z} - 0$, $\frac{1}{x} + \frac{2}{y} + \frac{1}{z} - 2 = 0$, $\frac{2}{x} - \frac{5}{y} - \frac{4}{z} + 1 = 0$

42) Show that $\left(2 + i\sqrt{3}\right)^{10} + \left(2 - i\sqrt{3}\right)^{10}$ is real ii) $\left(\frac{19 + 9i}{5 - 3i}\right)^{15} - \left(\frac{8 + i}{I + 2i}\right)^{15}$ is purely imaginary.

43) Let z_1,z_2, and z_3 be complex numbers such that $\left|z_1\right\| = \left|z_2\right| = \left|z_3\right| = r > 0$ and $z_1+z_2+z_3 \neq 0$ prove that $\left|\frac{z_1z_2 + z_2z_3 + z_3z_1}{z_1 + z_2 + z_3}\right| = r$

44) If z_1,z_2, and z_3 are three complex numbers such that $|z_1|=1$,$|z_2|=2|z_3|=3$ and $|z_1+z_2+z_3|=1$,show that $|9z_1z_2+4z_1z_2+z_2z_3|=6$

45) If z=x+iy is a complex number such that Im $\left(\frac{2z + 1}{iz + 1}\right) = 0$ show that the locus of z is $2x^2+2y^2+x-2y=0$

46) If z=x+iy and arg $\left(\frac{z - i}{z + 2}\right) = \frac{\pi}{4}$, then show that $x^2+y^2+3x-3y+2=0$

47) Show that $\left(\frac{\sqrt{3}}{2} + \frac{i}{2}\right)^5 + \left(\frac{\sqrt{3}}{2} - \frac{i}{2}\right)^5 = -\sqrt{3}$

48) If $2cosa = x + \frac{1}{x}$ and $2cos\beta = y + \frac{1}{y}$, show that

i) $\frac{x}{y} + \frac{y}{x} = 2cos(\alpha - \beta)$.

ii) $xy - \frac{1}{xy} = 2isin(\alpha + \beta)$

iii) $\frac{x^m}{y^n} - \frac{y^n}{x^m} = 2isin(m\alpha - n\beta)$

iv) $x^m y^n + \frac{1}{x^m y^n} = 2cos(m\alpha + n\beta)$

49) Solve the equation $z^3+27=0$.

50) Prove that the values of $\sqrt[4]{-1}arr \quad \pm\frac{1}{\sqrt{2}}(1 \pm i)$.Let z=(-1)

51) If z=x+iy and $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{2}$,then show that $x^2+y^2=1$.

52) Suppose z1,Z2, and z3 are the vertices of an equilateral triangle inscribed in the circle |z|= 2. If $z_1 = 1+i\sqrt{3}$ then find z_2 and z_3

53) Find all cube roots of $\sqrt{3} + i$

54) Solve the equation $z^3+8i=0$,where $z \in \mathbb{C}$

55) Simplify: $(1+i)^{18}$

56) Simplify: $\left(-\sqrt{3} + 3i\right)^{31}$

57) Solve the equation $3x^2-16x^2+23x-6=0$ if the product of two roots is 1.

58) Form the equation whose roots are the squares of the roots of the cubic equation $x^3+ax^2+bx+c = 0$.

59) Solve the equation $x^3-9x^2+14x+24=0$ if it is given that two of its roots are in the ratio 3:2.

60) Find a polynomial equation of minimum degree with rational coefficients, having $\sqrt{5}-\sqrt{3}$ as a root.

61) If 2+i and $3-\sqrt{2}$ are roots of the equation $x^6-13x^5+62x^4-126x^3+65x^2+127x-140=0$, find all roots.

62) Solve the equation (x-2)(x-7)(x-3)(x+2)+19=0

63) Solve the equation (2x-)(6x-1)(3x-2)(x-12)-7=0

64) Solve:
(x-5)(x-7)(x+6)(x+4)=504

65) Determine k and solve the equation $2x^3-6x^2+3x+k=0$ if one of its roots is twice the sum of the other two roots.

66) Find all zeros of the polynomial $x^6-3x^5-5x^4+22x^3-39x^2-39x+135$, if it is known that 1+2i and $\sqrt{3}$ are two of its zeros.

67) Solve the following equation: $x^4-10x^3+26x^2-10x+1=0$

68) Solve the equation $6x^4-5x^3-38x^2-5x+6=0$ if it is known that $\frac{1}{3}$ is a solution.

69) Discuss the maximum possible number of positive and negative roots of the polynomial equations x^2-5x+6 and $x^2-5x+16$. Also draw rough sketch of the graphs.

70) Solve:
(x-4)(x-7)(x-2)(x+1)=16

71) Find the principal value of $\cos^{-1}\left(\frac{\sqrt{3}}{3}\right)$

72) Find (i) $\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)$

ii) $\cos^{-1}\left(\cos\left(-\frac{\pi}{3}\right)\right)$

iii) $\cos^{-1}\left(\cos\left(-\frac{7\pi}{6}\right)\right)$

73) Find the principal value of
$\operatorname{cosec}^{-1}(-1)$

74) If a_1, a_2, a_3, ... an is an arithmetic progression with common difference d, prove that tan

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a} \quad \left[\tan^{-1}\left(\frac{d}{1+a_1a_2}\right) + \tan^{-1}\left(\frac{d}{1+a_2a_3}\right) + \ldots \tan^{-1}\left(\frac{d}{1+a_na_{n-1}}\right)\right] = \frac{a_n-a_1}{1+a_1a_n}$$

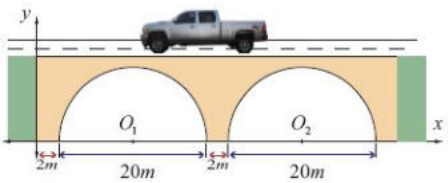
75) Solve $\tan^{-1}\left(\frac{x-1}{x-2}\right) + \tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$

76) Solve $\cos\left(\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)\right) = \sin\left\{\cot^{-1}\left(\frac{3}{4}\right)\right\}$

77) Find the principal value of
$\sec^{-1}(-2)$.

78) Find the equation of the circle passing through the points(1,1), (2,-1) , and(3,2) .

79) A road bridge over an irrigation canal have two semi circular vents each with a span of 20m and the supporting pillars of width 2m. Use Fig.5.16 to write the equations that model the arches.



80) Find the equation of the circle through the points (1,0),(-1,0) , and (0,1) .

81) Find the vertex, focus, directrix, and length of the latus rectum of the parabola $x^2-4x-5y-1=0$.

82) Find the equation of the ellipse whose eccentricity is $\frac{1}{2}$, one of the foci is(2,3) and a directrix is x = 7 . Also find the length of the major and minor axes of the ellipse.

83) Find the foci, vertices and length of major and minor axis of the conic $4x^2+36y^2+40x-288y+532 = 0$.

84) For the ellipse$4x^2+y^2+24x-2y+21 = 0$, find the centre, vertices, and the foci. Also prove that the length of latus rectum is 2 .

85) Find the centre, foci, and eccentricity of the hyperbola $11x^2-25y^2-44x+50y-256 = 0$

86) Find the equations of the two tangents that can be drawn from (5,2) to the ellipse $2x^2+7y^2=14$.

87) Find the equations of tangents to the hyperbola$\frac{x^2}{16}-\frac{y^2}{64}$ =1which are parallel to$10x-3y+9= 0$.

88) Show that the line$x-y+4=0$ is a tangent to the ellipse $x^2+3y^2=12$. Also find the coordinates
of the point of contact.

89) The maximum and minimum distances of the Earth from the Sun respectively are 152×106 km and 94.5×106 km. The Sun is at one focus of the elliptical orbit. Find the distance from the Sun to the other focus.

90) A concrete bridge is designed as a parabolic arch. The road over bridge is 40m long and the maximum height of the arch is 15m. Write the equation of the parabolic arch.

91) Two coast guard stations are located 600 km apart at points A(0,0) and B(0,600). A distress signal from a ship at P is received at slightly different times by two stations. It is determined that the ship is 200 km farther from station A than it is from station B. Determine the equation of hyperbola that passes through the location of the ship.

92) Certain telescopes contain both parabolic mirror and a hyperbolic mirror. In the telescope shown in figure the parabola and hyperbola share focus F_1 which is 14mabove the vertex of the parabola. The hyperbola's second focus F_2 is 2m above the parabola's vertex. The vertex of the hyperbolic mirror is 1m below F_1. Position a coordinate system with the origin at the centre of the hyperbola and with the foci on the y-axis. Then find the equation of the hyperbola.

93) A bridge has a parabolic arch that is 10m high in the centre and 30m wide at the bottom. Find the height of the arch 6m from the centre, on either sides.

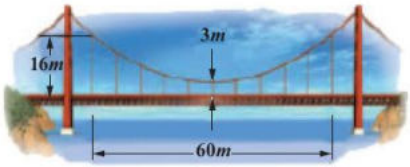
94) A tunnel through a mountain for a four lane highway is to have a elliptical opening. The total width of the highway (not the opening) is to be 16m, and the height at the edge of the road must be sufficient for a truck 4m high to clear if the highest point of the opening is to be 5m approximately . How wide must the opening be?

95) At a water fountain, water attains a maximum height of 4m at horizontal distance of 0 5 . m from its origin. If the path of water is a parabola, find the height of water at a horizontal distance of 0.75m from the point of origin.

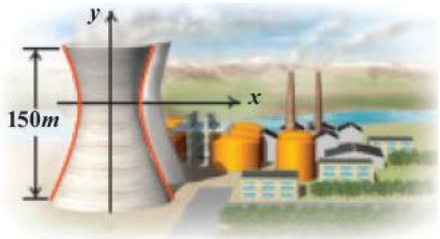
96) An engineer designs a satellite dish with a parabolic cross section. The dish is 5m wide at the opening, and the focus is placed 1 2 . m from the vertex
(a) Position a coordinate system with the origin at the vertex and the x -axis on the

parabola's axis of symmetry and find an equation of the parabola.
(b) Find the depth of the satellite dish at the vertex.

97) Parabolic cable of a 60m portion of the roadbed of a suspension bridge are positioned as shown below. Vertical Cables are to be spaced every 6m along this portion of the roadbed. Calculate the lengths of first two of these vertical cables from the vertex.



98) Cross section of a Nuclear cooling tower is in the shape of a hyperbola with equation $\frac{x^2}{30^2} - \frac{y^2}{44^2} = 1$ The tower is 150m tall and the distance from the top of the tower to the centre of the hyperbola is half the distance from the base of the tower to the centre of the hyperbola. Find the diameter of the top and base of the tower.



99) A rod of length 1 2 . m moves with its ends always touching the coordinate axes. The locus of a point P on the rod, which is 0 3 . m from the end in contact with x -axis is an ellipse. Find the eccentricity.

100) Assume that water issuing from the end of a horizontal pipe, 7 5 . m above the ground, describes a parabolic path. The vertex of the parabolic path is at the end of the pipe. At a position 2 5 . m below the line of the pipe, the flow of water has curved outward 3m beyond the vertical line through the end of the pipe. How far beyond this vertical line will the water strike the ground?

101) On lighting a rocket cracker it gets projected in a parabolic path and reaches a maximum height of 4m when it is 6m away from the point of projection. Finally it reaches the ground 12m away from the starting point. Find the angle of projection.

102) Points A and B are 10km apart and it is determined from the sound of an explosion heard at those points at different times that the location of the explosion is 6 km closer to A than B . Show that the location of the explosion is restricted to a particular curve and find an equation of it.

103) Find the vertex, focus, equation of directrix and length of the latus rectum of the following:
$x^2-2x+8y+17=0$

104) Find the vertex, focus, equation of directrix and length of the latus rectum of the following: $y^2-4y-8x+12=0$

105) Identify the type of conic and find centre, foci, vertices, and directrices of each of the following :
$18x^2+12y^2-144x+48y+120 = 0$

106) Identify the type of conic and find centre, foci, vertices, and directrices of each of the following :
$9x^2-y^2-36x-6y+18 = 0$

107) By vector method, prove that $\cos(\alpha + \beta) = \cos\alpha \cos\beta - \sin\alpha \sin\beta$

108) With usual notations, in any triangle ABC, prove by vector method that $\frac{a}{sinA} = \frac{b}{sinB} = \frac{c}{sinc}$

109) Prove by vector method that sin(α −β)=sinα cosβ −cosα sinβ

110) Prove by vector method that the perpendiculars (attitudes) from the vertices to the opposite sides of a triangle are concurrent.

111) Using vector method, prove that cos(α − β)=cos α cos β +sin α sin β

112) Prove by vector method that sin(α + β)=sin α cos β + cos α sin β

113) If $\vec{a} = -2\hat{i} + 3\hat{j} - 2\hat{k}, \vec{b} = 3\hat{i} - \hat{j} + 3\hat{k}, \vec{c} = 2\hat{i} - 5\hat{j} + \hat{k}$ find $(\vec{a} \times \vec{b}) \times \vec{c}$ and $(\vec{a} \times \vec{b}) \times \vec{c}$. State whether they are equal.

114) If $\vec{a} = \vec{i} - \vec{j}, \vec{b} = \hat{i} - \hat{j} - 4\hat{k}, \vec{c} = 3\hat{j} - \hat{k}$ and $\vec{d} = 2\hat{i} + 5\hat{j} + \hat{k}$

(i) $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = [\vec{a}, \vec{b}, \vec{d}]\vec{c} - [\vec{a}, \vec{b}, \vec{c}]\vec{d}$

(ii) $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = [\vec{a}, \vec{c}, \vec{d}]\vec{b} - [\vec{b}, \vec{c}, \vec{d}]\vec{a}$

115) Find the vector equation in parametric form and Cartesian equations of a straight passing through the points (-5, 7, 14) and (13, -5, 2). Find the point where the straight line crosses the xy - plane.

116) Find the points where the straight line passes through (6,7, 4) and (8, 4,9) cuts the xz and yz planes.

117) Find the point of intersection of the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-4}{5} = \frac{y-1}{2} = z$

118) Find the equation of a straight line passing through the point of intersection of the straight lines $\vec{r} = (\hat{i} + 3\hat{j} - \hat{k}) + t(2\hat{i} + 3\hat{j} + 2\hat{k})$ and $\frac{x-2}{1} = \frac{y-4}{2} = \frac{z+3}{4}$ and perpendicular to both straight lines.

119) Determine whether the pair of straight lines $\vec{r}(2\hat{i} + 3\hat{j} - \hat{k}) + t(2\hat{i} + 3\hat{j} + 2\hat{k})$, $\vec{r} = (2\hat{j} - 3\hat{k}) + s(\hat{i} + 2\hat{j} + 3\hat{k})$ are parallel. Find the shortest distance between them.

120) Show that the lines $\frac{x-3}{3} = \frac{y-3}{-1} = z - 1$ and $\frac{x-6}{2} = \frac{z-1}{3}, y - 2 = 0$ intersect. Also find the point of intersection

121) If the Cartesian equation of a plane is 3x - 4y + 3z = -8, find the vector equation of the plane in the standard form.

122) Find the vector parametric, vector non-parametric and Cartesian form of the equation of the plane passing through the points (-1, 2, 0), (2, 2, -1)and parallel to the straight line $\frac{x-1}{1} = \frac{2y+1}{2} = \frac{z+1}{-1}$

123) Find the non-parametric form of vector equation, and Cartesian equation of the plane passing through the point (2,3,6) and parallel to the straight lines $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-3}{1}$ and $\frac{x+3}{2} = \frac{y-3}{-5} = \frac{z+1}{-3}$

124) Find the parametric form of vector equation, and Cartesian equations of the plane passing through the points (2, 2,1), (9,3,6) and perpendicular to the plane 2x + 6y + 6z = 9

125) Find parametric form of vector equation and Cartesian equations of the plane passing through the points (2, 2,1), (1, −2,3) and parallel to the straight line passing through the points (2,1,−3) and (−1,5,−8)

126) Find the non-parametric form of vector equation of the plane passing through the point (1, −2, 4) and perpendicular to the plane x + 2y −3z = 11 and parallel to the line $\frac{x+7}{3} = \frac{y+3}{-1} = \frac{z}{1}$

127) Find the parametric form of vector equation, and Cartesian equations of the plane containing the line $\vec{r} = (\hat{i} - \hat{j} + 3\hat{k}) + t(2\hat{i} - \hat{j} + 4\hat{k})$ and perpendicular to plane $\vec{r}.(\hat{i} + 2\hat{j} + \hat{k}) = 8$

128) Find the parametric vector, non-parametric vector and Cartesian form of the equations of the plane passing through the points (3,6,−2), (−1,−2,6) , and (6,−4,−2).

129) Show that the straight lines $\vec{r} = (5\hat{i} + 7\hat{j} - 3\hat{k}) + s(-4\hat{i} + 4\hat{j} - 5\hat{k})$ and $\vec{r} = (8\hat{i} + 4\hat{j} + 5\hat{k}) + t(7\hat{i} + \hat{j} + 3\hat{k})$ are coplanar. Find the vector equation of the plane in which they lie.

130) Show that the lines $\frac{x-2}{1} = \frac{y-3}{1} = \frac{z-4}{3}$ and $\frac{x-2}{1} = \frac{y-3}{1} = \frac{z-4}{3}$ coplanar.Also, find the plane containing these lines.

131) If the straight lines $\frac{x-1}{2} = \frac{y+1}{\lambda} = \frac{z}{2}$ and $\frac{x-1}{2} = \frac{y+1}{\lambda} = \frac{z}{\lambda}$ are coplanar, find λ and equations of the planes containing these two lines.

132) Show that the lines $\vec{r} = (\hat{i} - 3\hat{j} - 5\hat{k}) + s(3\hat{i} + 5\hat{j} + 7\hat{k})$ and $\vec{r} = (2\hat{i} + 4\hat{j} + 6\hat{k}) + t(\hat{i} + 4\hat{j} + 7\hat{k})$ are coplanar. Also,find the non-parametric form of vector equation of the plane containing these lines

133) Find the coordinates of the foot of the perpendicular and length of the perpendicular from the point (4,3,2) to the plane x + 2y + 3z = 2

134) Find the image of the point whose position vector is $\hat{i} + 2\hat{j} + 3\hat{k}$ in the plane $\vec{r}.(\hat{i} + 2\hat{j} + 4\hat{k})$ = 38

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5 MARKS V1 CREATIVE

12th Standard

Maths

Exam Time : 01:40:00 Hrs | Total Marks : 100

32 x 5 = 160

1) Using determinants; find the quadratic defined by f(x) =ax^2 + bx + c, if f(1) =0, f(2) =-2 and f(3) = -6.
2) Solve: $\frac{2}{x}+\frac{3}{y}+\frac{10}{z}=4, \frac{4}{x}-\frac{6}{y}+\frac{5}{z}=1, \frac{6}{x}+\frac{9}{y}-\frac{20}{z}$ =2
3) The sum of three numbers is 20. If we multiply the third number by 2 and add the first number to the result we get 23. By adding second and third numbers to 3 times the first number we get 46. Find the numbers using Cramer's rule.
4) For what value of λ, the system of equations x+y+z=1, x+2y+4z=λ, x+4y+10z=λ^2 is consistent.
5) Show that the equations -2x + y + z = a, x - 2y + z = b, x + y -2z = c are consistent only if a + b + c =0.
6) Using Gaussian Jordan method, find the values of λ and μ so that the system of equations 2x - 3y + 5z = 12, 3x + y + λz =μ, x - 7y + 8z = 17 has (i) unique solution (ii) infinite solutions and (iii) no solution.
7) If 1, ω, ω^2 are the cube roots of unity then show that $(1+5\omega^2+\omega^4)(1+5\omega+\omega^2)(5+\omega+\omega^5)=64$
8) Show that $\left(\frac{i+\sqrt{3}}{-i+\sqrt{3}}\right)^{2\omega}+\left(\frac{i-\sqrt{3}}{i+\sqrt{3}}\right)^{2\omega}$=-1
9) Verify that 2 arg(-1) ≠ $\arg(-1)^2$
10) Verify that arg(1+i) + arg(1-i) = arg[(1+i) (1-i)]
11) Find all the roots $(2-2i)^{\frac{1}{3}}$ and also find the product of its roots.
12) Find the radius and centre of the circle $z\bar{z}$-(2+3i)z-(2-3i)$\bar{z}$+9 =0 where z is a complex number.
13) If the sum of the roots of the quadratic equation ax^2+ bx + c = 0 (abe≠ 0) is equal to the sum of the squares of their reciprocals, then $\frac{a}{c}, \frac{b}{a}, \frac{c}{b}$ are H.P.
14) If a, b, c, d and p are distinct non-zero real numbers such that $(a^2+b^2+c^2)p^2-2(ab+bc+cd)p+(b^2+c^2+d^2)\leq 0$ the n. Prove that a,b,c,d are in G.P and ad=bc
15) If c ≠ 0 and $\frac{p}{2x}=\frac{a}{x+x}+\frac{b}{x-c}$ has two equal roots, then find p.
16) If the equation x^2 + bx + ca = 0 and x^2 + cx + ab = 0 have a comnion root and b≠c, then prove that their roots will satisfy the equation x^2 + ax + bc =0.
17) Solve: $(2x^2 - 3x + 1)(2x^2 + 5x + 1) = 9x^2$.
18) Find the domain of the following functions
    (i) f(x) = $\sin^{-1}(2x - 3)$
    (ii) f(x) = $\sin^{-1}x$ + cos x
19) Write thefunction $f(x)=tan^{-1}\sqrt{\frac{a-x}{a+x}}-a<x<a$ in the simplest form

20) Simplify $sin^{-1}\left(\frac{sinx+cosx}{\sqrt{2}}\right), \frac{\pi}{4}<x<\frac{\pi}{4}$

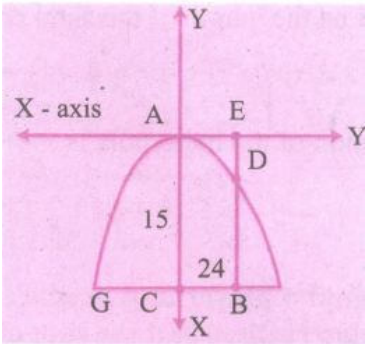
21) If $tan^{-1}\left(\frac{\sqrt{1+x^2}-\sqrt{1-x^2}}{\sqrt{1+x^2}+\sqrt{1-x^2}}\right)=a$ than prove that x^2=sin2a

22) Provethat $tan^{-1}\left(\frac{1-x}{1+x}\right)-tan^{-1}\left(\frac{1-y}{1+y}\right)=sin^{-1}\left(\frac{y-x}{\sqrt{1+x^2}.\sqrt{1+y^2}}\right)$

23) An arch is in the form of a parabola with its axis vertical. The arch is 10 m high and 5 m wide at the base. How wide is it 2 m from the vertex of the parabola?

24) An equilateral triangle is inscribed in the parabola y^2 = 4ax whose vertex is at the vertex of the parabola. Find the length of its side.

25) The guides of a railway bridge is a parabola with its vertex at the highest point 15 m above the ends. If the span is 120 m, find the height of the bridge at 24 m from the middle point.



26) The foci of a hyperbola coincides with the foci of the ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$. Find the equation of the hyperbola if its eccentricity is 2.

27) A kho-kho player In a practice Ion while running realises that the sum of tne distances from the two kho-kho poles from him is always 8m. Find the equation of the path traced by him of the distance between the poles is 6m.

28) Show that the points A, B, C with position vector $2\hat{i}-\hat{j}+\hat{k}, \hat{i}-3\hat{j}-5\hat{k}$ and $3\hat{i}-4\hat{j}+4\hat{k}$ respectively are the vector of a right angled, triangle. Also, find the remaining angles of the triangle.

29) ABCD is a quadrilateral with $\overrightarrow{AB}=\vec{\alpha}$ and $\overrightarrow{AD}=\vec{\beta}$ and $\overrightarrow{AC}=2\vec{\alpha}+3\vec{\beta}$. If. the area of the quadrilateral is λ times the area of the parallelogram with $\overrightarrow{AB}$ and $\overrightarrow{AD}$ as adjacent sides, then prove that $\lambda=\frac{5}{2}$

30) If $\left|\vec{A}\right|=\hat{i}+\hat{j}+\hat{k}$ and $\hat{i}=\hat{j}-\hat{k}$ are two given vector, then find a vector B satisfying the equations $\vec{A}\times\vec{B}$= $\vec{C}$ and $\vec{A}.\vec{B}$=3

31) Find the shortest distance between the following pairs of lines $\frac{x-3}{3}=\frac{y-8}{-1}=\frac{z-3}{1}$ and $\frac{x+3}{-3}=\frac{y+7}{2}=\frac{z-6}{4}$

32) Find the vector and Cartesian equation of the plane passing through the point (1,1, -1) and perpendicular to the planes x + 2y + 3z - 7 = 0 and 2x - 3y + 4z = 0

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2 MARKS VOLUME 2 BB

12th Standard

Maths

150 x 2 = 300

1) The temperature in celsius in a long rod of length 10 m, insulated at both ends, is a function of
length x given by T = x(10 − x). Prove that the rate of change of temperature at the midpoint of the
rod is zero.
2) A person learnt 100 words for an English test. The number of words the person remembers in t days after learning is given by $W(t) = 100\times(1-0.1t)^2$, $0 \le t \le 10$. What is the rate at which the person forgets the words 2 days after learning?
3) If the volume of a cube of side length x is $v = x^3$. Find the rate of change of the volume with respect to x when x = 5 units.
4) Find the angle of intersection of the curve y = sin x with the positive x -axis.
5) Find the slope of the tangent to the curves at the respective given points.
$y = x^4 + 2x^2 - x$ at x =1
6) Find the slope of the tangent to the curves at the respective given points.
$x = a\cos^3 t$, $y = b\sin^3 t$ at $t = \frac{\pi}{2}$
7) Find the point on the curve $y = x^2 - 5x + 4$ at which the tangent is parallel to the line 3x + y = 7.
8) Find the values in the interval (1, 2) of the mean value theorem satisfied by the function $f(x) = x - x^2$ for $1 \le x \le 2$
9) A truck travels on a toll road with a speed limit of 80 km/hr. The truck completes a 164 km journey in 2 hours. At the end of the toll road the trucker is issued with a speed violation ticket. Justify this using the Mean Value Theorem.
10) Suppose f(x) is a differentiable function for all x with $f'(x) \le 29$ and f(2) =17. What is the maximum value of f (7)?
11) Explain why Rolle's theorem is not applicable to the following functions in the respective intervals.
$f(x) = |\frac{1}{x}|, x \in [-1, 1]$
12) Explain why Rolle's theorem is not applicable to the following functions in the respective intervals.
$f(x) = tanx, x \in [0, \pi]$
13) Explain why Rolle's theorem is not applicable to the following functions in the respective intervals.
$f(x) = x - 2logx, x \in [2, 7]$
14) Using the Rolle's theorem, determine the values of x at which the tangent is parallel to the x -axis for the following functions:
$f(x) = x^2 - x$, $x \in [0, 1]$
15) Using the Rolle's theorem, determine the values of x at which the tangent is parallel to the x -axis for the following functions:
$f(x) = \sqrt{x} - \frac{x}{3}, x \in [0, 9]$

16) Explain why Lagrange's mean value theorem is not applicable to the following functions in the respective intervals
f(x) = |3x + 1|, x ∈ |-1, 3|
17) Find two positive numbers whose sum is 12 and their product is maximum.
18) Find two positive numbers whose product is 20 and their sum is minimum.
19) The volume of a cylinder is given by the formula $V = \pi r^2 h$. Find the greatest and least values of V if r + h = 6.
20) Compute the limit $\lim_{x\to 1}\left(\frac{x^2-3x+2}{x^2-4x+3}\right)$.
21) Compute the limit $\lim_{x\to a}\left(\frac{x^n-a^n}{x-a}\right)$
22) Evaluate the limit $\lim_{x\to 0}\left(\frac{sinmx}{x}\right)$
23) Evaluate the following limit, if necessary use l'Hôpital Rule
$\lim_{x\to 0}\frac{1-cosx}{x^2}$
24) Evaluate the following limit, if necessary use l'Hôpital Rule
$\lim_{x\to\infty}\frac{2x^2-3}{x^2-5x+3}$
25) Evaluate the following limit, if necessary use l'Hôpital Rule
$\lim_{x\to\infty}\frac{x}{logx}$

26) Evaluate the following limit, if necessary use l'Hôpital Rule
$\lim_{x\to\frac{\pi}{2}^-}\frac{secx}{tanx}$
27) Prove that the function $f(x) = x^2 + 2$ is strictly increasing in the interval (2,7) and strictly decreasing in the interval (−2, 0)
28) Prove that the function $f(x) = x^2 - 2x - 3$ is strictly increasing in $(2, \infty)$
29) Find the asymptotes of the curve $f(x) = \frac{2x^2-8}{x^2-16}$
30) Let us assume that the shape of a soap bubble is a sphere. Use linear approximation to approximate the increase in the surface area of a soap bubble as its radius increases from 5 cm to 5.2 cm. Also, calculate the percentage error.
31) A sphere is made of ice having radius 10 cm. Its radius decreases from 10 cm to 9-8 cm. Find approximations for the following:
change in the volume
32) A sphere is made of ice having radius 10 cm. Its radius decreases from 10 cm to 9-8 cm. Find approximations for the following:
change in the volume
33) The time T, taken for a complete oscillation of a single pendulum with length l, is given by the equation $T = 2\pi\sqrt{\frac{1}{g}}$, where g is a constant. Find the approximate percentage error in the calculated value of T corresponding to an error of 2 percent in the value of 1
34) Show that the percentage error in the nth root of a number is approximately $\frac{1}{n}$ times the percentage error in the number.
35) Let f , g : (a,b)→R be differentiable functions. Show that d(fg) = fdg + gdf
36) Let $g(x) = x^2 + \sin x$. Calculate the differential dg.

37) If the radius of a sphere, with radius 10 cm, has to decrease by 0 1. cm, approximately how much will its volume decrease?

38) Find differential dy for each of the following function

$y = \frac{(1-2x)^3}{3-4x}$

39) Find differential dy for each of the following function

y = (3 + sin(2x)) $^{2/3}$

40) Find differential dy for each of the following function

y = $_e x^{2-5x+7}$ cos (x^2 - 1)

41) Find df for f(x) = x^2 + x 3 and evaluate it for x = 2 and dx = 0.1

42) Find df for f(x) = x^2 + x 3 and evaluate it for x = 3 and dx = 0.02

43) Evaluate $\lim_{(x,y) \to (1,2)}$, if the limit exists, where $(x, y) = \frac{3x^2-xy}{x^2+y^2+3}$

44) Evaluate $\lim_{(x,y) \to (0,0)} cos = \left(\frac{x^3+y^3}{x+y+2}\right)$. If the unlimit exists.

45) Find the partial derivatives of the following functions at the indicated point.

f(x, y) = $3x^2$ - 2xy + y^2 + 5x + 2, (2,-5)

46) Find the partial derivatives of the following functions at the indicated point

g(x,y) =$3x^2$ + y^2 + 5x + 2, (1,-2)

47) If U(x, y, z) = $\frac{x^2+y^2}{xy} + 3z^2y$, find $\frac{\partial U}{\partial x}$; $\frac{\partial U}{\partial y}$ and $\frac{\partial U}{\partial z}$

48) If w(x, y, z) = x^2 y + y^2z + z^2x, x, y, z∈R, find the differential dw .

49) In each of the following cases, determine whether the following function is homogeneous or not. If it is so, find the degree.

f(x, y) = x^2y + $6x^3$ + 7

50) Determine whether the following function is homogeneous or not. If it is so, find the degree.$h(x, y) = \frac{6x^2y^3-\pi y^5+9x^4y}{2020x^2+2019y^2}$

51) In each of the following cases, determine whether the following function is homogeneous or not. If it is so, find the degree.

$g(x, y, z) = \frac{\sqrt{3x^2+5y^2+z^2}}{4x+7y}$

52) In each of the following cases, determine whether the following function is homogeneous or not. If it is so, find the degree.

$U(x, y, z) = xy + sin\left(\frac{y^2-2x^2}{xy}\right)$

53) Show that F(x,y) = $\frac{x^2+5xy-10y^2}{3x+7y}$ is a homogeneous function of degree 1.

54) Evaluate: $\int_0^3 (3x^2 - 4x + 5)dx$

55) Evaluate :$\int_0^1 [2x]dx$where [·] is the greatest integer function

56) Evaluate: $\int_{\frac{\pi}{2}}^{\frac{\pi}{2}}$ x cox x dx.

57) Evaluate the following definite integrals:

$\int_3^4 \frac{dx}{x^2-4}$

58) Evaluate the following definite integrals:

$\int_{-1}^{1} \frac{dx}{x^2+2x+5}$

59) Evaluate the following integrals using properties of integration:

$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (x^5 + xcos \quad x + tan^3 x + 1)dx$

60) Evaluate the following integrals using properties of integration:

$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} sin^2 x dx$

61) Evaluate $\int_b^\infty \frac{1}{a^2+x^2} dx, a > 0, b \in R$

62) Evaluate $\int_0^{\frac{\pi}{2}}$ ($\sin^2$ x + $\cos^4$ x) dx

63) Find the values of the following:

$\int_0^{\frac{\pi}{2}}$ sin ^{5}x cos^{4}xdx

64) Find the values of the following:

$\int_0^{\frac{\pi}{2}} sin^4 x x cos^6 x \quad dx$

65) Evaluate the following

$\int_0^{\pi/2} sin^{10} x \quad dx$

66) Evaluate the following

$\int_0^{\pi/2} cos^7 x \quad dx$

67) Evaluate the following

$\int_0^{\frac{\pi}{2}} sin^2 x cos^4 x dx$

68) Evaluate the following

$\int_0^1 x^2(1-x)^3 dx$

69) Evaluate the following:

$\int_0^\infty x^5 e^{-3x} dx$

70) Find, by integration, the volume of the solid generated by revolving about the x-axis, the region enclosed by y = $2x^2$, y = 0 and x = 1.

71) Find, by integration, the volume of the solid generated by revolving about the x-axis, the region enclosed by y = e^{-2x} y = 0, x = 0 and x = 1

72) Find, by integration, the volume of the solid generated by revolving about the y-axis, the region enclosed by x^2 y =1+ and y = 3.

73) Find the area bounded by y=x^2+2,x-x-axis, x=1 and x=2

74) Find the area of the region bounded by the curve y = sin x and the ordinate x=0

$x = \frac{\pi}{3}$

75) Find the area enclosed between the parabola y^2=4ax and the line x=a, x=9a.

76) Find the area bounded by y=cosx,y=x+1,y=0

77) Find the area bounded by the curve y = cosax in one arc of the curve.

78) A differential equation, determine its order, degree (if exists)

$\frac{dy}{dx} + xy = cotx$

79) A differential equation, determine its order, degree (if exists)

$\left(\frac{d^3y}{dx^3}\right)^{\frac{2}{3}} - 3\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 4 = 0$

80) A differential equation, determine its order, degree (if exists)

$$\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^2 = x sin\left(\frac{d^2y}{dx^2}\right)$$

81) A differential equation, determine its order, degree (if exists)

$$\sqrt{\frac{dy}{dx}} - 4\frac{dy}{dx} - 7x = 0$$

82) A differential equation, determine its order, degree (if exists)

$$y\left(\frac{dy}{dx}\right) = \frac{x}{\left(\frac{dy}{dx}\right)+\left(\frac{dy}{dx}\right)^3}$$

83) A differential equation, determine its order, degree (if exists)

$$x^2\frac{d^2y}{dx^2} + \left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{1}{2}} = 0$$

84) A differential equation, determine its order, degree (if exists)

$$\left(\frac{dy}{dx}\right)^3 = \sqrt{1 + \left(\frac{dy}{dx}\right)}$$

85) A differential equation, determine its order, degree (if exists)

$$\frac{d^2y}{dx^2} = xy + cos\left(\frac{dy}{dx}\right)$$

86) A differential equation, determine its order, degree (if exists)

$$\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + \int ydx = x^3$$

87) A differential equation, determine its order, degree (if exists)

$$x = e^{xy\left(\frac{dy}{dx}\right)}$$

88) Express the physical statement in the form of the differential equation.
Radium decays at a rate proportional to the amount Q present.

89) Express the physical statement in the form of the differential equation.
The population P of a city increases at a rate proportional to the product of population and to the difference between 5,00,000 and the population.

90) Express the physical statement in the form of the differential equation.
For a certain substance, the rate of change of vapor pressure P with respect to temperature T is proportional to the vapor pressure and inversely proportional to the square of the temperature.

91) Express the physical statement in the form of the differential equation.
A saving amount pays 8% interest per year, compounded continuously. In addition, the income from another investment is credited to the amount continuously at the rate of Rs 400 per year.

92) Assume that a spherical rain drop evaporates at a rate proportional to its surface area. Form a differential equation involving the rate of change of the radius of the rain drop.

93) Find the differential equation of the family of all non-vertical lines in a plane.

94) Find the differential equation of the family of all nonhorizontal lines in a plane.

95) Show that each of the following expressions is a solution of the corresponding given differential equation.
$y = 2x^2$; $xy' = 2y$

96) Show that each of the following expressions is a solution of the corresponding given differential equation.

$y = ae^{x} + be^{-x}$; $y - y = 0$

97) Find value of m so that the function $y = e^{mx}$ is a solution of the given differential equation.
$y' + 2y = 0$

98) Find value of m so that the function $y = e^{mx}$ is a solution of the given differential equation.
$y'' - 5y' + 6y = 0$

99) Show that $y = e^{-x} + mx + n$ is a solution of the differential equation $e^{x}\left(\frac{d^2y}{dx^2}\right) - 1 = 0$

100) Show that y = a cos bx is a solution of the differential equation $\frac{d^2y}{dx^2} + b^2y = 0$.

101) Determine the order and degree (if exists) of the following differential equations:
$\frac{dy}{dx} = x + y + 5$

102) Determine the order and degree (if exists) of the following differential equations:
$\left(\frac{d^4y}{dx^4}\right)^3 + 4\left(\frac{dy}{dx}\right)^7 + 6y = 5cos3x$

103) Determine the order and degree (if exists) of the following differential equations:
$\frac{d^2y}{dx^2} + 3\left(\frac{dy}{dx}\right)^2 = x^2 log\left(\frac{d^2y}{dx^2}\right)$

104) Determine the order and degree (if exists) of the following differential equations:
$3\left(\frac{d^2y}{dx^2}\right) = \left[4 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}$

105) Determine the order and degree (if exists) of the following differential equations:
$dy + (xy - \cos x)dx = 0$

106) Find the differential equation of the family of parabolas y^2 ax = 4, where a is an arbitrary constant.

107) Show that $x^2+y^2 = r^2$, where r is a constant, is a solution of the differential equation $\frac{dy}{dx} = -\frac{x}{y}$.

108) Show that $y=mx+\frac{7}{m}$, $m\neq0$ is a solution of the differential equation $xy'+7\frac{1}{y'}-y=0$.

109) Solve: $\frac{dy}{dx} = (3x+y+4)^2$.

110) Solve the differential equation:
$\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$

111) Solve the Linear differential equation:
$\frac{dy}{dx} = \frac{sin^2x}{1+x^3} - \frac{3x^2}{1+x^3}y$

112) Suppose X is the number of tails occurred when three fair coins are tossed once simultaneously. Find the values ofthe random variable X and number of points in its inverse images.

113) An urn contains 5 mangoes and 4 apples Three fruits are taken at randaom If the number of apples taken is a random variable, then find the values of the random variable and number of points in its inverse images.

114) Three fair coins are tossed simultaneously. Find the probability mass function for number of heads occurred

115) A six sided die is marked '1' on one face, '3' on two of its faces, and '5' on remaining three faces. The die is thrown twice. If X denotes the total score in two throws, find

(i) the probability mass function
(ii) the cumulative distribution function
(iii) P(4 ≤ X < 10)
(iv) P(X ≥ 6)

116) Find the probability mass function and cumulative distribution function of number of girl child in families with 4 children, assuming equal probabilities for boys and girls.

117) The cumulative distribution function of a discrete random variable is given by

$$F(x)=\begin{cases} 0 & -\infty < x < -10 \\ 0.15 & -1 \le x < 0 \\ 0.35 & 0 \le x < 1 \\ 0.60 & 1 \le x < 2 \\ 0.85 & 2 \le x < 3 \\ 1 & 3 \le x < \infty \end{cases}$$

Find
(i) the probability mass function
(ii) P(X < 1) and
(iii) P(X ~ 2

118) A random variable X has the following probability mass function.

x	1	2	3	4	5
f(x)	k^2	$2k^2$	$3k^2$	2k	3k

119) The cumulative distribution function of a discrete random variable is given by

$$F(x)=\begin{cases} 0 & \text{for } -\infty < x < 0 \\ \frac{1}{2} & \text{for } 0 \le x < 1 \\ \frac{3}{5} & \text{for } 1 \le x < 2 \\ \frac{4}{5} & \text{for } 2 \le x < 3 \\ \frac{9}{10} & \text{for } 3 \le x < 4 \\ 1 & \text{for } 4 \le x < \infty \end{cases}$$

Find
(i) the probability mass function
(ii) P(X < 3) and
(iii) P(X ~ 2).

120) The probability density function of X is given by $f(x)=\begin{cases} kxe^{-2x} & for x > 0 \\ 0 \quad for & x \le 0 \end{cases}$ Find the value opf .

121) For the random variable X with the given probability mass function as below, find the mean and variance.

$$f(x)=\left\{\frac{4-x}{6}\right. \quad x=1, 2, 3$$

122) If the probability that a fluorescent light has a useful life of at least 600 hours is 0.9, find the probabilities that among 12 such lights
(i) exactly 10 will have a useful life of at least 600 hours;

(ii) at least 11 will have a useful life of at I least 600 hours;
(iii) at least 2 will not have a useful life of at : least 600 hours.

123) A commuter train arrives punctually at a station every half hour. Each morning, a student leaves his house to the train station.Let X denote- the amount of time, in minutes that the student waits for the train from the time he reaches the train station. It is known that the pdf of X is

$$f(x) = \begin{cases} \frac{1}{3} & 0 < x < 30 \\ 0 & elsewhere \end{cases}$$

Obtain and interpret the expected value of the random variable X .

124) The time to failure in thousands of hours of an electronic equipment used in a manufactured computer has the density function $f(x) = \begin{cases} 3e^{-3x} & x > 0 \\ 0 & elsewhere \end{cases}$

Find the expected life of this electronic equipment.

125) Compute P(X = k)for the binomial distribution, B(n,p) where

n = 6,$p = \dfrac{1}{3}$,k=3

126) Compute P(X = k)for the binomial distribution, B(n,p) where

$$P(X = 10) = \binom{10}{4}\left(\frac{1}{5}\right)^4\left(1 - \frac{1}{5}\right)^{10-4}$$

127) Compute P(X = k)for the binomial distribution, B(n,p) where

n=9,$p = \dfrac{1}{2}$,k=7

128) Find the binomial distribution function for each of the following.
(i) Five fair coins are tossed once and X denotes the number of heads.
(ii) A fair die is rolled 10 times and X denotes the number of times 4 appeared.

129) Examine the binary operation (closure property) of the following operations on the respective sets (if it is not, make it binary)
a*b = a + 3ab − 5b^2;∀a,b∈Z

130) Examine the binary operation (closure property) of the following operations on the respective sets (if it is not, make it binary)

$a * b = \left(\frac{a-1}{b-1}\right), \forall a, b \in Q$

131) Verify the
(i) closure property,
(ii) commutative property,
(iii) associative property
(iv) existence of identity and
(v) existence of inverse for the arithmetic operation + on
Z_o = the set of all even integers

132) Write the statements in words corresponding to ¬p, p ∧ q , p ∨ q and q ∨ ¬p, where p is ‘It is cold’ and q is ‘It is raining.’

133) Let A = $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$be any two boolean matrices of the same type. Find AvB and A^B.

134) Determine whether $*$ is a binary operation on the sets given below.
a*b=b=a.|b| on R

135) Determine whether $*$ is a binary operation on the sets given below.
a*b=min (a,b) on A={1,2,3,4,5)

136) Determine whether $*$ is a binary operation on the sets given below.
(A*v)=a√b is binary on R

137) On Z, define ⊗by (m⊗n)=mn+nm: ∀m, n∈Z. Is ⊗binary on Z?

138) Let *be defined on R by (a*b)=a+b+ab-7. is*binary on R? If so, find $3\left(\frac{-7}{15}\right)$.

139) Let A={a+$\sqrt{5}$b:a,b∈Z}. Check whether the usual multiplication is a binary operation on A.

140) Let p: Jupiter is a planet and q: India is an island be any two simple statements. Give
verbal sentence describing each of the following statements.
(i) ¬p
(ii) p ∧ ¬q
(iii) ¬p ∨ q
(iv) p→ ¬q
(v) p↔q

141) Write each of the following sentences in symbolic form using statement variables p and q.
(i) 19 is not a prime number and all the angles of a triangle are equal.
(ii) 19 is a prime number or all the angles of a triangle are not equal
(iii) 19 is a prime number and all the angles of a triangle are equal
(iv) 19 is not a prime number

142) Determine the truth value of each of the following statements
(i) If 6 + 2 = 5 , then the milk is white.
(ii) China is in Europe or $\sqrt{3}$ is an integer
(iii) It is not true that 5 + 5 = 9 or Earth is a planet
(iv) 11 is a prime number and all the sides of a rectangle are equal

143) Which one of the following sentences is a proposition?
(i) 4 + 7 =12
(ii) What are you doing?
(iii) 3n ≤ 8,1 n ∈ N
(iv) Peacock is our national bird
(v) How tall this mountain is!

144) Fill in the following table so that the binary operation $*$ on A = {a,b,c} is commutative.

*	a	b	c
a	b		
b	c	b	a
c	a		c

145) Consider the binary operation $*$ defined on the set A = {a,b,c, d} by the following table:

*	a	b	c	d
a	a	c	b	d
b	c	d	a	a
d	d	b	a	c

Is it commutative and associative?

146) Write the converse, inverse, and contrapositive of each of the following implication.
(i) If x and y are numbers such that x = y, then $x^2 = y^2$
(ii) If a quadrilateral is a square then it is a rectangle.

147) Construct the truth table for the following statements.
$\neg p \wedge \neg q$

148) Construct the truth table for the following statements.
$\neg(p \wedge \neg q)$

149) Construct the truth table for the following statements.
$(p \vee q) \wedge \neg q$

150) Construct the truth table for the following statements.
$(\neg p \longrightarrow r) \wedge (p \leftrightarrow q)$

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2 MARKS VOLUME 2 CREATIVE

12th Standard

Maths

142 x 2 = 284

1) A particle moves in a line so that x=$\sqrt{t}$. Show that the acceleration is negative and proportional to the cube of the velocity.
2) A man 2 m high walks at a uniform speed of 5 km/ hr away from a lamp post 6 m high. Find the rate at which the length of his shadow increases?
3) At what point on the curve $y=x^2$ on [-2,2] is the tangent parallel to X-axis?
4) Find the maximum and minimum values of f(x) = |x+3| $\forall\ x \in R$.
5) Find the intervals of increasing and decreasing function for $f(x) = x^3 + 2x^2 - 1$.
6) Find x if the rate of decrease of $\frac{x^2}{2} - 2x + 5$ is twice the decrease of x.
7) Find the point on the parabola $y^2=18x$ at which the ordinate increases at twice the rate of the abscissa.
8) Find the point at which the curve $y-e^{xy}+x=0$ has a vertical tangent.
9) Find the equation of the tangent to the curve $y^2=4x+5$ and which is parallel to y=2x+7
10) Using Rolle's theorem find the value of c for f(x) = sin x in[0,2π]
11) Verify Rolle 's Theorem for $f(x) = |x-1|, O \le x \le 2$
12) If f(1)=10 and $f(x) \ge 2$ for $1 \le x \le 4$ how small can f(4) possible be?
13) Verify Lagrange's Mean Value theorem for $f(x) = \sqrt{x-2}$ in the interva [2,6]
14) Obtain Maclaurin's Series expansion for e^{2x}.
15) Expand the polynomial $f(x)=x^2-3x+2$ in power of (x-2)
16) Evaluate the following limits, if necessary using L'Hopitalrule

    (i) $\lim_{x\to 2} \frac{sin\pi x}{2-x}$

    (ii) $\frac{lim}{x \to 2} \frac{x^n - a^n}{x-2}$

    (iii) $\lim_{x\to\infty} \frac{sin\frac{2}{x}}{\frac{1}{x}}$

    (iv) $\lim_{x\to\infty} \frac{x^2}{e^x}$
17) Find the absolute extreme of the function $f(x)=x^2-2x+2$ on the closed interval [0,3]
18) Prove that the function $f(x)=2x^2+3x$ is strictly increasing on $\left[-\frac{1}{2}, \frac{1}{2}\right]$
19) Prove that the function $f(x)=e^{-x}$ is strictky increasing on [0,1]
20) Find the stationary points of each of the following functions
    (i) $f(x)=2x-3x^2$
    (ii) $f(x)=x^3-3x+1$
21) Find the critical number of $f(\theta) = \theta + sin\theta$ in [0,2π]
22) Determine the domain of concavity of the curve $y=2-x^2$

23) Determine the domain of convexity of the function $y=e^x$

24) A circular metal plate expands under heating so that its radius increases by 2%. Find the approximate increase in the area of the plate if the radius of the plate before heating is 10cm.

25) Use differentials to find $\sqrt{25.2}$

26) If f (x, y) = $2x^3 - 11x^2y + 3y^3$, prove that $x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} = 3f$

27) If of f(x, y) = $x^2 + y^3 + 2xy^2$ find f_{xx}, f_{yy}, f_{xy} and f_{yx}.

28) IF u(x, y) = $x^2 + 3xy + y^2$, x, y, ∈ R, find tha linear appraoximation for u at (2, 1)

29) If $u = log\sqrt{x^2+y^2}$ then prove that $\left(\frac{\vartheta u}{\vartheta x}\right) + \left(\frac{\vartheta u}{\vartheta y}\right) = \frac{1}{x^2+y^2}$

30) If $u=x^2y+y^2z+z^2x$, then prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = (x+y+z)^2$

31) If $u=x^2+y^2+z^2-3xyz$, then prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = 3u$

32) If $u=x^2+3xy^2+y^2$, then prove that $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$

33) If $f(x,y)=10x-x^2+xy$ find
(i) $f_x(2,6)$
(ii)$f_y(2,6)$

34) If $u = e^{\frac{x}{y}} sin\left(\frac{x}{y}\right) + e^{\frac{y}{x}} cos\left(\frac{y}{x}\right)$,then prove that

$x\frac{\vartheta u}{\vartheta x} + y\frac{\vartheta u}{\vartheta y} = 0$

35) If $w=e^{xy}$,$x=at^2$,$y=2at$, find $\frac{dw}{dt}$

36) If $w=\log(x^2+y^2)$,$x=\cos\theta$,$y=\sin\theta$, find $\frac{dw}{d\theta}$

37) If $w = e^{x^2+y^2}$,$x=\cos\theta$,$y=\sin\theta$, find $\frac{dw}{d\theta}$

38) If $w=xye^{xy}$ find $\frac{\partial^2 u}{\partial x \partial y}$

39) Using differentials, find the approximate value of $sin\left(\frac{22}{14}\right)$

40) Show that the percentage error in computing the volume of a sphere, due to an error in measuring the radius, is approximately equal to three times the percentage error in the radius.

41) The pressure P and the volume V of a gas are connected by the relation $PV^{1.4}$=constant. Find the % error in P corresponding to a devreased of $\frac{1}{2}\%$ in V.

42) If y = sin x and x changes from $\frac{\pi}{2} to \frac{22}{14}$ what is the approximate change in y.

43) Estimate $cos\left(\frac{\pi}{4}+0.01\right)-cos\left(\frac{\pi}{4}\right)$

44) Find a linear approximation to f(x)=3xe^{2x-10} at x=5

45) Without using any kind of computational aid use linear approximation to find the value of e$^{0.1}$

46) Show that (1+x)^{n}=1+nx approximately if x is close to zero.

47) Calculate df for $f=\sqrt{2x+5}$ when x = 22 and dx = 3.

48) Prove that $\int_0^{\frac{\pi}{2}} log(tan\ x)dx$

49) Evaluate $\int_0^1 \frac{e^x}{1+e^{2x}}dx$

50) Evaluate $\int_1^2 \frac{3x}{9x^2-1}dx$

51) Find the area of the region enclosed by the curve y = $\sqrt{x}$ + 1, the axis of x and the lines x=0, x=4.

52) Find the volume of the solid obtained by revolving the area of the triangle whose sides are x = 4, y = 0 and 3x - 4y = 0 about x - axis

53) Evaluate $\int_0^{\frac{\pi}{2}} log(tanx)dx=0$

54) Evaluate $\int_0^{\frac{\pi}{2}} e^{2x}cosxdx$

55) Evaluate $\int_0^{\frac{\pi}{2}} e^{3x}cosxdx$

56) Evaluate $\int_0^{\frac{\pi}{2}} e^{-x}$

57) Evaluate $\int_{-1}^1 log(\frac{3-x}{3+x})dx$

58) Evaluate $\int_0^1 \left(\frac{e^{5logx}-e^{4logx}}{e^{3logx}-e^{2logx}}\right)$

59) Evaluate $\int \sum_{r=0}^{\infty} \frac{x^r 2^r}{r!}dx$

60) Evaluate $\int e^{3x}3^{2x}5^x dx$

61) Evaluate $\int_0^{\infty} (a^{-x}-b^{-x})dx$

62) Find the slope of the tangent to the curve $y=\int_0^x \frac{dt}{1+t^3} stx=1$

63) Find the area bounded by the curve y=sin2x between the ordinates x=0.x=π and x-axis.

64) Find the volume of the solid generated when the region enclosed by $y=\sqrt{x}, y=2$ and x = 0 is revolved about y-axis.

65) Find the volume of the solid y=x^3,x=0,y=1 is revolved about the y-axis.

66) If

$$\int_0^{\infty} \frac{x^2dx}{(x^2+a^2)\left(x^2+b^2\right)(x^2+c^2)}=\frac{\pi}{2(a+b)(b+c)(c+a)}$$ then

find $\int_0^\infty \frac{dx}{(x^2+4)(x^2+9)}$

67) Evaluate $\int_0^1 \frac{e^x}{1+e^{2x}}dx$

68) Evaluate $\int_1^2 \frac{e^x}{1+e^{2x}}dx$

69) Find the area of the region enclosed by the curve $y=\sqrt{x}+1$ the axis of x and the lines x = 0 and x = 4.

70) Find the volume of the solid obtained by revolving the area of the triangle whose sides are x = 4, y = 0 and 3x – 4y = 0 about x axis.

71) Find the area of the region bounded by the curve $\sqrt{x}+\sqrt{y}=\sqrt{a}$ (x,y > 0) and the co-ordinate axes.

72) Find the area of the region bounded by the curves $y=2^x$.$y=-2x-x^2$ and the lines x=0 and x=2

73) Find the area bounded by $y=x^2+2$, x-axis, x=1 and x=2.

74) Find the area of the region bounded by the curve y=sin x and the ordinate x=0.

$x=\frac{\pi}{3}$

75) Find the area enclosed between the parabola y2=4ax and the line x=a,x=9a.

76) Find the area bounded by y=cosx,y=x+1,y=0.

77) Find the area bounded by the curve y=cosax in one arc of the curve.

78) Form the differential equation satisfied by are the straight lines in my-plane.

79) A curve passing through the origin has its slope e^x, Find the equation of the curve.

80) Solve: $\frac{dy}{dx}=1+e^{x-y}$

81) Solve: x$\frac{dy}{dx}=x+y$

82) Solve: $\frac{dy}{dx}+y=e^{-x}$

83) Determine the order and degree of $\frac{\left[1+\left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}}}{\frac{d^2y}{dx^2}}=k$

84) Find the order and degree of $\left(\frac{d^2y}{dx^2}\right)^2+cos\left(\frac{dy}{dx}\right)=0$

85) Find the order and degree of $y+\frac{dy}{dx}=\frac{1}{4}\int ydx$

86) Form the Differential Equation representing the family of curves y=Acos(x+B) where A and B are parameters.

87) Form the D.E of family of parabolas having vertex at the origin and axis along positive y-axis.

88) Form the D.E corresponding to $y=e^{mx}$ by eliminating 'm'.

89) Solve : $\frac{dy}{dx}=\frac{2x}{x^2+1}$

90) Solve : $\frac{dy}{dx} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

91) Solve: $\frac{dy}{dx} + y = 1$

92) solve:xdy+ydx=xydx

93) Define a random variable

94) How many types of random variables are there? What are they?

95) Define discrete random variable

96) Define Probability Mass Function.

97) Define Distribution function.

98) Define Probability Density function

99) Define Continuous random variable.

100) What is meant by the expected value of a random variable X?

101) Define Variance of a random variable X?

102) Prove that E(aX+b)=aE(X)+b

103) Prove that Var(x)=E(X^2)-{E(X)]^2

104) Prove that Var(ax+b)=a^2Var(X)

105) When do we say that a discrete random variable X is a binomial random variable.

106) What is the Probability Mass function of a binomial random variable?

107) The probability distribution of a random variable X is given below.

X	0	1	2	3
P(X)	K	$\frac{k}{2}$	$\frac{k}{4}$	$\frac{k}{8}$

i) Find k

ii) P(X > 2)

108) A coin is tossed twice. If X is a random variable defined as the number of heads minus the number of tails, then obtain its probability distribution.

109) A coin is tossed until a head appears or the tail appears 4 times in succession. Find the probability distribution of the number of tosses.

110) The probability distribution of a random variable X is given under :

$$P(X = x) = \begin{cases} kx^2, for\ x = 1,2,3 \\ 2kx, for\ x = 4,5,6 \\ o, otherwise \end{cases}$$

Find (i) k

(ii) E(X)

111) Is it possible that the mean of a binomial distribution is 15 and its standard deviation is 5?

112) Find the variance of the binomial distribution with parameters 8 and $\frac{1}{4}$

113) Let X be a continuous random variable with p.d.f

$f(x) = \begin{cases} 2x, 0 \le x \le 1 \\ 0, otherwise \end{cases}$ .Find E(X)

114) In an investment a man can make a profit of Rs.5000 with a probability of 0.62 or a loss of Rs.8000 with a probability of 0.38. Find the expected gain or loss %

115) The p.d.f of a continuous random variable X is

$$f(x) = \begin{cases} k, 0 \le x \le 4 \\ 0, otherwise \end{cases}$$ Find k.

116) Prove that Var(X)=E(X^2) if E(X)=0

117) Prove that Var(x+b)=Var(X)

118) Show that p v (~p) is a tautology.

119) Show that p v (q ∧ r) is a contingency.

120) In the set of integers under the operation * defined by a * b = a + b - 1. Find the identity element.

121) Let S be the set of positive rational numbers and is defined by a * b = $\frac{ab}{2}$. Then find the identity element and the inverse of 2.

122) Let G = {1, w, w^2) where w is a complex cube root of unity. Then find the universe of w^2. Under usual multiplication.

123) In a non empty set S on which a binary operation * is defined and for an element a∈S.a$\boxed{*}$a$\boxed{*}$a=e,where e is the identity element. Find the inverses of a and a$\boxed{*}$a.

124) Check whether dot product is defined on the set of vectors. Explain?

125) Is cross product commutative on the set of vectors? Justify your answer.

126) In Z, the set of integers, an operation * is defined as a$\boxed{*}$b=2(a+b). Check whether * is associative.

127) Is it possible to define a binary operation * on any non empty set S such that a * $b = \dfrac{a+b}{a-b}$,given that a and b are integers.

128) If a≡b(mod n) and b≡c(mod n),check whether a≡c(mod n).

129) If a≡b(mod n) and C≡d(mod n),check whether a+c≡(b+d) (mod n).

130) Give any four Boolean Matrices of order 2X2

131) How many Boolean matrices of order 2X2 are there?

132) A and B are Boolean matrices of order 2X2.If AVB=A, is it necessary that $B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

133) If A and B are two Boolean matrices of order 2X2 and $A \cap B = A$,is it necessary that $B = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

134) An operation * is defined as mXn=m^n-n^m.Is it binary on N?

135) Form the truth table of (~q)^p.

136) Form the truth table of (~P)→(~q).

137) Write the equivalent forms of p→q and (~p)→q.

138) Are ~pV(pVq) and pV(~pvq) tatology statements. Justify your answer.

139) Are ~pV(p^q) and p^(~p^q) contradictions.

140) p:N is divisible buy 4 and q:N is an even number. Whether p→q is true.

141) Give the truth table of ~p→~q

142) Give the truth value of (~pvq)v(~q)

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3 MARKS V2 CREATIVE

12th Standard

Maths

Exam Time : 01:40:00 Hrs | Total Marks : 100

151 x 3 = 453

1) Prove that $\frac{x}{1+x}$ < < log (1+ x) for x > 0.

2) Find the equation of normal to the cure y = $\sin^2 x$ at $\left(\frac{\pi}{3}, \frac{3}{4}\right)$.

3) Verify LMV theorem for f (x) = $x^3 - 2x^2 - x + 3$ in [0, 1].

4) The ends of a rod AB which is 5 m long moves along two grooves OX, OY which at the right angles. If A moves at a constant speed of $\frac{1}{2}$ m/sec, what is the speed of B, when it is 4m from O?

5) A ball is thrown vertically upwards, moves according to the law s = 13.8 t - 4.9 t^2 where s
is in metres and t is in seconds.
(i) Find the acceleration at t = 1
(ii) Find velocity at t = 1
(iii) Find the maximum height reached by the ball?

6) The side of a square is equal to the diameter of a circle. If the side and radius change at the same rate then find the ratio of the change of their areas.

7) The volume of a cube is increasing at the rate of $8cm^3/s$.How fast is the surface area increasing when the length of an edge is 12cm?

8) Find the equation of normal to the curve $y^4=ax^2$at(a,a)

9) Find the angle between two curves $2x^2+y^2=20$ and $x^2-4y^2+8=0$

10) Apply Rolle's theorem to find points on curve y=-1+cosx.whwre the tangent is parallel to x-axis in [0,2π]

11) Verify Rolle's theorem for f(x)=exsinx,$0 \le x \le \pi$

12) A cylindrical hole 4 mm in diameter and 12 mm deep in a metal block is reboared to increase the diameter to 4.12mm. Estimate the amount of metal removed

13) Verify Mean Value theorem for f(x)=(x-1)(x-2)(x-3) in [0,4]

14) Obtain Maclaurin's series for $\frac{1}{1+x}$

15) Write down the Taylor series expansion of the function cos x in ascending powers $x - \frac{\pi}{4}$ upto three non-zero terms.

16) Evaluate the following limits, if necessary use L'Hopitals rule
(i) $lim_{x \to 0^+} x^{sinx}$

(ii) $lim_{x \to 0} \frac{cotx}{cot2x}$

(iii) $lim_{x \to \frac{\pi}{2}^-} (tanx)^{cosx}$

17) Find the absolute extrema of the following functions on the given closed interval.

i) $f(x) = x^3 - 3x^2 + 1, \ -\frac{1}{2} \le x \le 4$

ii) $f(x) = x - 2sinx, 0 \le x \le 2\pi$

18) Prove that the function f(x)=x(x-1)(x+1) is strictly increasing in [-2,-1]

19) Find the intervals of monotonicities of the function f(x)=sinx, xƐ[0,2π]

20) Find the intervals of monotonicities and find the local extremum for the following functions

i) f(x)=20-x-x^2

ii) f(x)=x(x-1)(x+1) on [0,2]

21) Find the intervals of concavity and the point of inflection of the function f(x)=$2x^2$+$5x^2$-4x

22) Find the local extrema for the following functions using second derivate test.

(i) x^3-x

(ii) $x^3 - 3x^2 + 1, \ -\frac{1}{2} \le x \le 4$

23) Find two numbers whose sum is 100 and whose product is a maximum.

24) Find two positive numbers whose product is 100 and whose sum is minimum..

25) Find the slant asymptote for the function $f(x) = \frac{x^2 - 6x + 11}{x + 4}$

26) Use differentials to find the value of $\sqrt{0.037}$

27) Find the approximate value of f (3.02) where f(x) = $3x^2$ + 5x +3.

28) Using differentials find the approximate value of tan 46° if it is given that 1^0 = 0.01745 radians

29) If f = $\frac{x}{x^2+y^2}$ then show that = $x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y}$ = -f

30) If w = xy + z where x = cos t; y = sin t; z = t find $\frac{dw}{dt}$

31) Find the linear approximation to $g(z) = \sqrt[4]{zatz} = 2$

32) If u=log(x^2+y^2+z^2), then prove that $x\frac{\partial^2 u}{\partial z \partial x} = y\frac{\partial^2 u}{\partial z \partial x} = z\frac{\partial^2 u}{\partial x \partial y}$

33) If (x,y)=$3x^2$+$4y^2$+6xy-x^2y^2+5 then find

(i) fx(1,-1)

(ii) f_{xy}(1,1)

(iii)fxy(2,1)

34) If u=sin3xcos4y, then prove that $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$

35) If $u = \frac{x}{y^2} - \frac{y}{x^2}$ then prove that $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$

36) If u=(x-y)(y-z)(z-x), then prove that $u_x+u_y+u_z=0$

37) If v=log(tanx+tany+tanz), then prove that $\Sigma sin2x\frac{\partial v}{\partial x} = 2$

38) If $z=e^{x3}+y^3$ then prove that $x\frac{\partial z}{\partial x} + y\frac{\partial x}{\partial y} = 2$

39) If w= $\log(x^2+y^2)$ and x=rcosθ and y=rsinθ then, find $\frac{\partial w}{\partial r} and \frac{\partial w}{\partial \theta}$

40) If w=xy+z and x=cot, y=sint, z=t then find $\frac{dw}{dt}$

41) If u is a homogeneous function in x and y of degree n then prove that

$$x\frac{\partial^2 u}{\partial x^2} + y\frac{\partial^2 u}{\partial x \partial y} = (n-1)\frac{\partial u}{\partial x}$$

42) Using linear approximation find $\sqrt{0.082}$

43) Find the approximate value of $\left(\frac{17}{81}\right)^{\frac{1}{4}}$ using linear approximation.

44) Find the limit for the following if it exists $lim_{(x-y)\to(1,1)} \frac{2x^2 - xy - y^2}{x^2 - y^2}$

45) Find the limit for the following if it exists $lim_{(x,y)\to(0,0)} \frac{x^2y^2}{x^4 + 3y^4}$

46) If $w=x^2+y^2$ and $x=u^2-v^2$, y=2uv then find $\frac{\partial w}{\partial u} and \frac{\partial w}{\partial v}$

47) Evaluate : $lim_{(x,y)\to(2,0)} \frac{\sqrt{2x-y-2}}{2x-y-4}$

48) Evaluate : $lim_{(x,y,z)\to(-1,0,4)} \frac{x^2 - ze^{zy}}{6x+2y-2z}$

49)

Evaluate : $lim_{(x,y)\to(0,0)} \frac{x^2-xy}{\sqrt{x}-\sqrt{y}}$

50) z is a homogeneous function in x and y of degree n then prove that

$x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=(ax+by+n)$,where $v=ze^{ax+by}$

51)

If $w=x+2y+z^2$ and x=cos t,y=sint,z=t then find $\frac{dw}{dt}$

52) Evaluate $\int_0^{\frac{\pi}{2}} \frac{\sqrt{cot\,x}}{\sqrt{cot\,x}+\sqrt{tan\,x}}dx$

53) Evaluate $\int_{-1}^{1} log\left(\frac{2-x}{2+x}\right)dx$

54) Evaluate $\int_0^{\frac{\pi}{2}} \frac{sin\,x}{9+cos^2}dx$

55) Evaluate $\int_0^1 \sqrt{9-4x^2}dx$

56) Evaluate $\int_0^1 xe^{-2x}dx$

57)

If $f(x)=\begin{vmatrix} x+1 & 2x+1 & 3x+1 \\ 2x+1 & 3x+1 & x+1 \\ 3x+1 & x+1 & 2x+1 \end{vmatrix}$,then find $\int_0^1 f(x)dx$

[Hint:$R_2\to R_2\to R_1$;$R_2\to R_3\to R_1$]

58)

If $f(x)=\begin{vmatrix} 2cos^2x & sin2x & -sinx \\ sin2x & 2sin^2x & cosx \\ sinx & -cosx & 0 \end{vmatrix}$,then find $\int_0^{\frac{\pi}{2}} f(x)+f'(x)dx$

[Hint :f(x)=2,f'(x)=0]

59)

If $f(x)=\begin{vmatrix} cosx & 1 & 0 \\ 1 & 2cosx & 1 \\ 0 & 1 & 2cosx \end{vmatrix}$,then find $\int_0^{\frac{\pi}{2}} f(x)dx$

[Hint:f(x)=cos3x]

60)

Evaluate $\int_0^{\pi} \sqrt{1+4sin^2\frac{x}{2}-4sin\frac{x}{2}}dx$

61)

Evaluate $\int_3^6 \frac{\sqrt{x}}{\sqrt{9-x}+\sqrt{x}}dx$

62) Evaluate $\int_2^6 \frac{logx^2}{logx^2 + log\left(36 - 12x + x^2\right)}$

63) Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{1 + tan^3x}$

64) Evaluate $\int_0^1 x(1-x)^n dx$

65) Evaluate $\int_0^1 x(1-x)^{10} dx$

66) If $\int_0^\infty \frac{dx}{\left(x^2+4\right)\left(x^2+9\right)} = k\pi$ then find the value of k.

67) Find the area bounded by the parabolas $x^2 = \frac{y}{4}$ and x^2=9y and the straight line y = 2.

68) The area of the curve

69) Find the area bounded by the curve y=x|x|,x-axis and the ordinates x=-1,x=1

70) Find the area between the curve y=1-|x| and x-axis

71) Find the area of the region described by $A = \left\{(x, y), x^2 + y^2 \leq 1 \delta y^2 \leq 1 - x\right\}$

72) Find the area bounded by the curves $y=x^3$,$y=x^2$ and the ordinates x=1 and x=2.

73) Find the area bounded by the curve y=sin x,y=cosx and the y-axis

74) Find the area bounded by x=0,x=6+5y-y^2

75) Find the area bounded by y=3x and $y=x^2$

76) Evaluate $\int_0^\pi cos^3 x dx$

77) Evaluate $\int_0^{\frac{\pi}{2}} sin^0 x dx$

78) Evaluate $\int_0^{\frac{\pi}{2}} cos^{10} x dx$

79) Evaluate $\int_0^{50} [x - |x|] dx$

80) Evaluate $\int_{-2}^3 \left|1 - x^2\right| dx$

81) Evaluate $\int_0^{\frac{\pi}{2}} \frac{cos^{3/2}x}{cos^{3/2}x + sin^{3/2}x} dx$

82) Solve $\frac{dy}{dx} + \frac{y^2}{x^2} = \frac{y}{x}$

83) Form the differential equation for $y=e^{-2x}$ [A coos 3x-B sin 3x]

84) Solve: $\frac{dy}{dx}$ = $(4x+y+1)^2$

85) Solve: x$\frac{dy}{dx}$+2y=x^2

86) Solve: $\frac{dy}{dx} + y = cosx$

87) Form the D.E of the family of curves $c(y+c)^2=x^2$,where c is the parameter.

88) Form the D.E of family of curves represented by $y=c(x-c)^2$.where c is the parameter.

89) Form the D.E to $y^2=a(b-x)(b+x)$ by eliminating a and b as its parameters.

90) Find the D.E of all circles touching x-axis at the origin.

91) Find the D.E of all circles touching y-axis at the origin.

92) Obtain the D.E of all circles of radius 'r'

93) Find the D.E of all circles in the first quadrant which touch the co-ordinate axes.

94) Form the D.E corresponding to $y^2-2ay+x^2=a^2$ by eliminating 'a'.

95) Show that the function y=Acos2x-Bsin2x is a solution of the D.E $y^2+4y=0$

96) Verify that y=-x-1 is a solution of the D.E $(y-x)dy-(y^2-x^2)dx=0$

97) Solve: $\frac{dy}{dx} = \frac{1 - cosx}{1 + cosx}$

98) Solve : $\frac{dy}{dx} + 2y^2 = 0$,y=(1) and find the corresponding solution of the curve.

99) Solve :cosx(1+cosy)dx-siny(1+sinx)dy=0.

100) Solve :$(1+e^2x)dy+(1+y^2)e^xdx=0$

101) Solve :$(x^2-yx^2)dy+(y^2+x^2y^2)dx=0$

102) At any point (x, y) of a circle the slope of the tangent line segment is the slope of the line segment joining the point of contact to the point (-4,-3).Find the equation of the curve given that it passes through (-2,1).

103) Solve : $\frac{dy}{dx} - \frac{y}{x} = 2x^2, x > 0$

104) Solve : $\left(x^2 - 1\right)\frac{dy}{dx} + 2xy = \frac{1}{x^2 - 1}$

105) Solve : $\frac{dy}{dx} - y = e^x$

106) Solve : $ydx+(x-y^2)dy=0$

107) Give any three properties of distribution function.

108) Give any three properties on expectation and variance.

109) A random variable X has the following probability distribution

x	0	1	2	3	4	5	6	7
P(X)	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2+k$

Evaluate (i) k

(ii) $P(X \geq 6)$

(iii) P(0< X< 3)

110) Let X denote the number of hours you study on a Sunday. Also it is known that

$$P(X=x)=\begin{cases} 0.1 \text{ if } x=0 \\ kx \text{ if } x=1 \text{ or } 2 \\ k(5-x) \text{ if } x=3 \text{ or } 4 \\ 0, \text{otherise} \end{cases}$$

where k is a constant.

i) Find k

ii) Find the probability that you study for atleast 2 hours

iii) Find the probability that you study for atmost 2 hours.

111) Two cards are drawn successively without replacement from a well shuffled pack of 52 cards. Find the probability distribution of number of spades.

112) From a lot of 10 bulbs, which includes 3 defective bulbs, a sample of 2 bulbs is drawn at random. Find the probability distribution of defective bulbs.

113) Out of a group of 60 architects 40 are qualified and co-operative while the remaining are qualified but remain reserved. Two architects are selected from the group at random. Find the probability distribution of the number of architects who are qualified and co-operative. Which of the two values, namely co-cooperativeness or reservedness, mentioned above, do you prefer and why?

114) Four defective oranges are accidentally mixed with sixteen good ones. Three oranges are drawn from the mixed lot. Find the probability distribution of X, the number of defective oranges.

115) Two cards are drawn simultaneously from a well shuffled pack of 52 cards. Find the probability distribution of number of jacks.

116) A box contains 12 bulbs of which 3 are defective. A sample of 3 bulbs is selected from the box. Let X denote the number of defective bulbs in the sample. Find the probability distribution of X.

117) Find the variance of the probability distribution

X	0	1	2	3	4	5
P(X)	$\frac{1}{6}$	$\frac{5}{18}$	$\frac{2}{9}$	$\frac{1}{6}$	$\frac{1}{9}$	$\frac{1}{18}$

118) In a meeting, 70% of the members favour a certain proposal while remaining 30% oppose it. A member is selected at random and we let X = 0 if he opposes, and X = 1 if he is in favour. Find E(X) and Var(X).

119) Find the mean, variance and standard deviation of the number of heads in two tosses of a coin

120) In a game, a man wins Rs.5 for getting a number greater than 4 and loses Re.1 otherwise when a fair dice is thrown. The man decided to throw a die thrice but to quit as and when he gets a number greater than 4. Find the expected value of the amount he wins/loses

121) A random variable X has the following probability distribution.

x_i	-2	-1	0	1	2	3
P_i	0.1	k	0.2	2k	0.3	k

i) find k

ii) find the mean of the distribution

122) In 3 trials of a binomial distribution, the probability of 2 success is 9 times the probability of 3 success. Find the parameter of p of the distribution.

123) For 6 trials of an experiment, let X be a binomial variate which satisfies the relation 9P(X=4)=P(X=2).Find p.

124) How many times must a man toss a coin so that the probability of having atleast one head is more than 80%?

125) The probability of a shooter hitting a target is $\frac{3}{4}$.How many minimum number of times must he fire so that the probability of hitting the target atleast once is more than 0.99?

126) If the mean and variance of a binomial distribution are respectively 9 and 6, find the distribution.

127) The distribution of a continuous random variable X in range (−3,3) is given by p.d.f

$$f(x) = \begin{cases} \frac{1}{16}(3+x)^2, -3 \le x \le -1 \\ \frac{1}{16}(6-2x^2)^2, -1 \le x \le 1 \\ \frac{1}{16}(3-x)^2, 1 \le x \le 3 \end{cases}$$

Verify that the area under the curve is unity.

128) Consider a random variable X with p.d.$f(x) = \begin{cases} 3x^2, 0 < x < 1 \\ 0, otherwise \end{cases}$ Find Var(3X – 2).

129) Let X be a random variable with distribution function

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{x}{8}, & 0 \le x < 1 \\ \frac{1}{4} + \frac{x}{8}, & 1 \le x < 2 \\ \frac{3}{4} + \frac{x}{12}, & 2 \le x < 3 \end{cases}$$

Find (a)$P(1 \le X \le 2)$

(b) P(X=3)

(c) $P(1 \le X \le 3)$

130) A person tosses a coin and is to receive Rs.4 for a head and has to pay Rs.2 for a tail. Find the variance of the game.

131) Let X be a continuous random variable with $f(x) = \begin{cases} \frac{2}{x^4}, x \ge 1 \\ 0, otherwise \end{cases}$ Find the mean and the variance of X.

132) Write the truth value for each of the following statements.

(1) 3 + 5 = 8 and $\sqrt{2}$ is an irrational number.

(2) 5 is a positive integer or a square is a rectangle.

(3) Chennai is not in Tamilnadu.

133) In (z, *) where * is defined by a * b = ab, prove that * is not a binary operation on z.

134) Let G = {1, i,-1, -i} under the binary operation multiplication. Find the inverse of all the elements.

135) In (z, *) where * is defined as a * b = a + b + 2. Verify the commutative and associative axiom.

136) Construct the truth table for (-p) v (q ∧ r)

137) Give an example for (S,*) in which a^3=e where e the identity element.

138) If in a pair (S,*) where S is a nonemty set and * is a binary operation defined on S as a^2=e for all a ε s, then that * is commutative given that e is the identify element.

139) Give an example (S,*), where a^2=e for all a∈S.Given that e is identity element.

140) The pair (S,*) identify and inverse aximos, prove that * is commutative if and only if $(a*b)^2 = a^a * b^2$.

141) On the set Q of rational numbers, an operation * is defined as a*b=k(a+b) where k is a given non zero number. Is it associative

142) If on the set Q of rational numbers, a binary operation * is defined as a*b=λ(a+b) were λ is a nonzero fixed number and its given that * is associative, then the value of λ and what can we say about the operation*?

143) If a≡b (mod n) and c≡d(mod n), prove that ac≡bd(mod n).

144) If a≡b(mod n) show that a^m≡b^m (mod n).whwre m is anatural number.

145) Show that (Z_3-[0],X_3) Satisfies closure, identiy and inverse properties.

146) Show that (Z_4+$_4$) satisfies closure closure, associative and commutative properties.

147) On the set of real numbers * is defined as $a*b = k(a+b+ab)$ where k is a non zero real number.What are the conditions on R and k so that * is associative.

148) If in (S,*) satisfying closure, associative, identity and inverse axiom,$a*b^2 = a^2*b$ for some a,b∈S, then prove that a=b.

149) In in (S,*) satisfying closure, associative, identity and inverse axioms and $(a*b)^{-1} = a^{-1}*b^{-1}$ ∀a,b∈S, then prove that * is commutative.

150) If $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, B = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ and $C = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ are Boolean matrices, find

(i) AVB

(ii) AVC

(iii)AVBVC.

151) If $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ and $C = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$,find (AVB)VC

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3 MARKS V2 BB

12th Standard

Maths

Exam Time : 01:40:00 Hrs

Total Marks : 100

273 x 3 = 819

1) For the function $f(x) = x^2 \in [0, 2]$ compute the average rate of changes in the subintervals [0,0.5], [0.5,1], [1,1.5], [1.5,2] and the instantaneous rate of changes at the points x = 0.5,1,1.5, 2
2) A particle moves so that the distance moved is according to the law s(t) = $s(t) = \frac{t^3}{3} - t^2 + 3$. At what time the velocity and acceleration are zero respectively?
3) A particle is fired straight up from the ground to reach a height of s feet in t seconds, where $s(t) = 128t - 16t^2$.
   (1) Compute the maximum height of the particle reached.
   (2) What is the velocity when the particle hits the ground?
4) The price of a product is related to the number of units available (supply) by the equation Px + 3P −16x = 234, where P is the price of the product per unit in Rupees(Rs) and x is the number of units. Find the rate at which the price is changing with respect to time when 90 units are available and the supply is increasing at a rate of 15 units/week.
5) A point moves along a straight line in such a way that after t seconds its distance from the origin is $s = 2t^2 + 3t$ metres
   Find the average velocity of the points between t = 3 and t = 6 seconds.
6) A point moves along a straight line in such a way that after t seconds its distance from the origin is $s = 2t^2 + 3t$ metres
   Find the instantaneous velocities at t = 3 and t = 6 seconds.
7) If the mass m(x) (in kilograms) of a thin rod of length x (in metres) is given by, $m(x) = \sqrt{3x}$ then what is the rate of change of mass with respect to the length when it is x = 3 and x = 27 metres.
8) A stone is dropped into a pond causing ripples in the form of concentric circles. The radius r of the outer ripple is increasing at a constant rate at 2 cm per second. When the radius is 5 cm find the rate of changing of the total area of the disturbed water?
9) A beacon makes one revolution every 10 seconds. It is located on a ship which is anchored 5 km from a straight shore line. How fast is the beam moving along the shore line when it makes an angle of 45° with the shore?
10) Find the equations of tangent and normal to the curve $y = x^2 + 3x - 2$ at the point (1, 2)
11) For what value of x the tangent of the curve $y = x^3 - x^2 + x - 2$ is parallel to the line y = x.
12) Find the points on the curve $y = x^3 - x^2 + x + 3$ where the normal is parallel to the line x + y =1729.
13) Find the points on the curve $y^2 - 4xy = x^2 + 5$ for which the tangent is horizontal.

14) Find the angle between the rectangular hyperbola xy = 2 and the parabola $x^2 y + 4 = 0$

15) Show that the two curves $x^2 - y^2 = r^2$ and $xy = c^2$ where c, r are constants, cut orthogonally

16) Compute the value of 'c' satisfied by the Rolle's theorem for the function $f(x) = x^2 (1 - x)^2$, $x \in [0.1]$

17) Find the values in the interval $(\frac{1}{2}, 2)$ satisfied by the Rolle's theorem for the function $f(x) = x + \frac{1}{x}, x \in [\frac{1}{2}, 2]$

18) Compute the value of 'c' satisfied by Rolle's theorem for the function $f(x) = log(\frac{x^2+6}{5x})$ in the interval [2,3]

19) Find the absolute extrema of the following function on the given closed interval $f(x) = x^2 - 12x + 10$; [1,2]

20) Find the absolute extrema of the following function on the given closed interval $f(x) = 3x^4 - 4x^3$;[-1,2]

21) Without actually solving show that the equation $x^4+2x^3-2=0$ has only one real root in the interval (0,1).

22) Prove using the Rolle's theorem that between any two distinct real zeros of the polynomial $a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x + a_0$ there is a zero of the polynomial $n a_n x^{n-1} + (n-1) a_{n-1} x^{n-2} + \ldots + a_1$

23) Find the absolute extrema of the following functions on the given closed interval. $f(x) = 6x^{\frac{3}{4}} - 3x^{\frac{1}{3}}; [-1, 1]$

24) Find the absolute extrema of the following functions on the given closed interval.

$$f(x) = 2cosx + sin2x; \left[0, \frac{\pi}{2}\right]$$

25) Prove, using mean value theorem, that $|sin\alpha - sin\beta| \leq |\alpha - \beta|, \alpha, \beta \in R$

26) A thermometer was taken from a freezer and placed in a boiling water. It took 22 seconds for the thermometer to raise from −10°C to 100°C. Show that the rate of change of temperature at some time t is 5°C per second.

27) Using the Rolle's theorem, determine the values of x at which the tangent is parallel to the x -axis for the following functions:

$f(x) = \frac{x^2-2x}{x+2}, x \in [-1, 6]$

28) Explain why Lagrange's mean value theorem is not applicable to the following functions in the respective intervals:

$f(x) = \frac{x+1}{x}$, $x \in [-1, 2]$

29) Using the Lagrange's mean value theorem determine the values of x at which the tangent is parallel to the secant line at the end points of the given interval: $f(x) = x^3 - 3x + 2$, $x \in [-2, 2]$

30) Using the Lagrange's mean value theorem determine the values of x at which the tangent is parallel to the secant line at the end points of the given interval: $f(x) = (x - 2)(x - 7)$, $x \in [3,11]$

31) Show that the value in the conclusion of the mean value theorem for $f(x) = \frac{1}{x}$ on a closed interval of positive numbers [a,b] is $\sqrt{ab}$

32) Show that the value in the conclusion of the mean value theorem for f(x) =Ax^2 + Bx + c on any interval [a,b] is $\frac{a+b}{2}$

33) A race car driver is racing at 20th km. If his speed never exceeds 150 km/hr, what is the maximum distance he can cover in the next two hours.

34) Suppose that for a function f(x), f'(x) ≤ 1for all 1 ≤ x ≤ 4. Show that f(4) - f(1) ≤ 3.

35) Does there exist a differentiable function f(x) such that f(0) = -1, f(2) = 4 and f'(x) ≤ 2 for all x. Justify you answer.

36) Show that there lies a point on the curve f(x) = x (x + 3) $e^{\frac{\pi}{2}}$, -3 ≤ x ≤ 0 where tangent drawn is parallel to the x -axis.

37) Using mean value theorem prove that for, a > 0, b > 0, $|e^{-a} - e^{-b}| < |a - b|$.

38) Write the Taylor series expansion of $\frac{1}{x}$ about x = 2 by finding the first three non-zero terms.

39) Write the Maclaurin series expansion of the following function e^x

40) Write the Maclaurin series expansion of the following function sin x

41) Write the Maclaurin series expansion of the following function cos x

42) Write down the Taylor series expansion, of the function log x about x =1 upto three nonzero terms for x > 0.

43) Expand sin x in ascending powers x - $\frac{\pi}{4}$ upto three non-zero terms.

44) Expand the polynomial f (x) = x^2 - 3x + 2 in powers of x - 1

45) Evaluate the limit $lim_{x\to 0^+}(\frac{sin\ x}{x^2})$

46) $lim_{\theta\to 0}(\frac{1-cos\ m\theta}{1-cos\ n\theta})$ =1, then prove that, $m = \pm n$

47) Evaluate : $lim_{x\to 1^-}(\frac{log(1-x)}{cot(\pi x)})$.

48) Evaluate: $lim_{x\to 0^+}(\frac{1}{x} - \frac{1}{e^x-1})$.

49) Evaluate: $lim_{x\to\infty}(\frac{x^2+17x+29}{x^4})$.

50) Evaluate: : $lim_{x\to 0^+}$ x log x.

51) Evaluate: $lim_{x\to\infty}(\frac{e^x}{x^m}), m \in N$

52) Evaluate the following limit, if necessary use l'Hôpital Rule $lim_{x\to\infty} e^{-x}\sqrt{x}$

53) Evaluate the following limit, if necessary use l'Hôpital Rule

$$lim_{x\to 0}\left(\frac{1}{sinx} - \frac{1}{x}\right)$$

54) Evaluate the following limit, if necessary use l'Hôpital Rule

$$lim x \to 1^{+} \left(\frac{2}{x^2 - 1} - \frac{x}{x-1} \right)$$

55) Evaluate the following limit, if necessary use l'Hôpital Rule

$$lim x \to \infty \quad \left(1 + \frac{1}{x} \right)^{x}$$

56) If an initial amount A_0 of money is invested at an interest rate r compounded n times a year, the value of the investment after t years is $A = A_0 \left(1 + \frac{r}{n} \right)^{nt}$. If the interest is compounded continuously, (that is as n →∞), show that the amount after t years is $A=A_o^{ert}$.

57) Find the absolute maximum and absolute minimum values of the function f (x) = $2x^3 + 3x^2 - 12x$ on [−3, 2]

58) Find the absolute extrema of the function f (x) = 3cos x on the closed interval $[0, 2\pi]$

59) Find the intervals of monotonicity and hence find the local extrema for the function $f(x) = x^{\frac{2}{3}}$.

60) Prove that the function f (x) = x − sin x is increasing on the real line. Also discuss for the existence of local extrema.

61) Determine the intervals of concavity of the curve y = 3+ sin x .

62) Find the local extremum of the function f (x) = x^4 + x 32x

63) Find the asymptotes of the function f(x)=$\frac{1}{x}$

64) Find the slant (oblique) asymptote for the function $f(x) = \frac{x^2 - 6x + 7}{x + 5}$

65) Find the linear approximation for f(x) = $\sqrt{1 + x}, x \geq -1$ at x_0 = 3. Use the linear approximation to estimate f(3.2)

66) Use linear approximation to find an approximate value of $\sqrt{9.2}$ without using a calculator.

67) A right circular cylinder has radius r =10 cm. and height h = 20 cm. Suppose that the radius of the cylinder is increased from 10 cm to 10. 1 cm and the height does not change. Estimate the change in the volume of the cylinder. Also, calculate the relative error and percentage error.

68) Let $f(x) = \sqrt[3]{x}$. Find the linear approximation at x = 27. Use the linear approximation to approximate $\sqrt[3]{27.2}$

69) Use the linear approximation to find approximate values of $(123)^{\frac{2}{3}}$

70) Use the linear approximation to find approximate values of $\sqrt[4]{15}$

71) Use the linear approximation to find approximate values of
$\sqrt[3]{26}$

72) Find a linear approximation for the following functions at the indicated points.
$f(x) = x^3 - 5x + 12$, $x_0 = 2$

73) Find a linear approximation for the following functions at the indicated points.
$g(x) = g(x) = \sqrt{x^2 + 9}, x_0 = -4$

74) Find a linear approximation for the following functions at the indicated points.
$h(x) = \frac{x}{1+x} = \frac{1}{2}$

75) The radius of a circular plate is measured as 12.65 cm instead of the actual length 12.5 cm.find the following in calculating the area of the circular plate:
Absolute error

76) The radius of a circular plate is measured as 12.65 cm instead of the actual length 12.5 cm.find the following in calculating the area of the circular plate:
Relative error

77) The radius of a circular plate is measured as 12.65 cm instead of the actual length 12.5 cm.find the following in calculating the area of the circular plate:
Percentage error

78) Find Δf and df for the function f for the indicated values of x, Δx and compare
$f(x) = x^3 - 2x^2$; $x = 2$, $\Delta x = dx = 0.5$

79) Find Δf and df for the function f for the indicated values of x, Δx and compare
$f(x) = x^2 + 2x + 3$; $x = -0.5$, $\Delta x = dx = 0.1$

80) Assuming $\log_{10} e = 0.4343$, find an approximate value of $\log^{10} 1003$

81) The trunk of a tree has diameter 30 cm. During the following year, the circumference grew 6cm.
Approximately, how much did the tree's diameter grow?

82) The trunk of a tree has diameter 30 cm. During the following year, the circumference grew 6cm.
What is the percentage increase in area of the tree's cross-section?

83) An egg of a particular bird is very nearly spherical. If the radius to the inside of the shell is 5 mm and radius to the outside of the shell is 5.3 mm, find the volume of the shell approximately.

84) Assume that the cross section of the artery of human is circular. A drug is given to a patient to dilate his arteries. If the radius of an artery is increased from 2 mm to 2.1 mm, how much is cross-sectional area increased approximately?

85) In a newly developed city, it is estimated that the voting population (in thousands) will increase according to $V(t) = 30 + 12t^2 - t^3$, $0 \le t \le 8$ where t is the time in years. Find the approximate change in voters for the time change from 4 to $4\frac{1}{6}$ year

86) The relation between the number of words y a person learns in x hours is given by $y = 52\sqrt{x}$, $0, \le x \le 9$. What is the approximate number of words learned when x changes from
1 to 1.1 hour?

87) The relation between the number of words y a person learns in x hours is given by $y = 52\sqrt{x}$, $0, \le x \le 9$. What is the approximate number of words learned when x

changes from
4 to 4.1 hour?

88) A circular plate expands uniformly under the influence of heat. If it's radius increases from 10.5 cm to 10.75 cm, then find an approximate change in the area and the approximate percentage change in the area.

89) A coat of paint of thickness 0.2 cm is applied to the faces of a cube whose edge is 10 cm. Use the differentials to find approximately how many cubic centimeters of paint is used to paint this cube. Also calculate the exact amount of paint used to paint this cube.

90) Let f (x,y) = $\frac{3x-5y+8}{x^2+y^2+1}$ for all (x, y) $\in R^2$ Show that f is continuous on R^2

91) f(x,y) = $\frac{xy}{x^2+y^2}$, (x,y) ≠ (0,0) and f (0,0) = 0 Show that f is not continuous at f, -(0,0) and continuous at all other points of R^2

92) Consider g(x,y) = $\frac{2x^2y}{x^2+y^2}$, if (x,y) ≠ (0,0) and g(0,0) = 0 Show that g is continuous on R^2

93) Let f (x, y) = 0 if xy ≠ 0 and f (x, y) =1 if xy = 0.

(i) Calculate: $\frac{\partial f}{\partial x}(0, 0)$, $\frac{\partial f}{\partial y}(0, 0)$.

(ii) Show that f is not continuous at (0,0)

94) Let F(x, y) = x^3 y + y^2x + 7 for all (x, y)$\in R^2$. Calculate $\frac{\partial F}{\partial x}$(-1,3) and $\frac{\partial F}{\partial y}$(-2,1).

95) Let (x, y) = e^{-2y} cos(2x) for all (x, y) $\in R^2$. Prove that u is a harmonic function in R^2.

96) Let $f(x, y) = \frac{y^2-xy}{\sqrt{x}-\sqrt{y}}$ for (x, y) ≠ (0, 0). Show that $\lim_{(x, y) \to (0, 0)}$ f(x, y) = 0

97) Evaluate $\lim_{(x, y) \to (0, 0)} cos = \left(\frac{e^x siny}{y}\right)$, if the limit exists.

98) Let g(x, y) = $\frac{x^2y}{x^4+y^2}$ for (x, y) ≠ (0, 0) and f(0, 0) = 0

Show that $\lim_{(x, y) \to (1, 2)}$ g(x, y) = 0 along every line y = mx, m $\in$ R

99) Let g(x, y) = $\frac{x^2y}{x^4+y^2}$ for (x, y) ≠ (0, 0) and f(0, 0) = 0

Show that $\lim_{(x, y) \to (0, 0)}$ g(x, y) = $\frac{k}{1+k^2}$ along every parabola y = kx^2, k $\in$ R \ {0}.

100) Show that f(x, y) = $\frac{x^2-y^2}{y^2+1}$ is continuous at every (x, y) $\in R^2$

101) Let g(x, y) = $\frac{e^y sinx}{x}$, for x ≠ 0 and g(0, 0) = 1. Show that g is continuous at (0,0).

102) Find the partial derivatives of the following functions at the indicated point

h (x, y, z) = x sin (xy) + z^2x, $\left(2, \frac{\pi}{4}, 1\right)$

103) Find the partial derivatives of the following functions at the indicated point
G(x, y) = e^{x+3y} log (x^2 + y^2)

104) For each of the following functions find the f_x, f_y, and show that $f_{xy} = f_{yx}$
$f(x,y) = \cos(x^2 - 3xy)$

105) For each of the following functions find the g_{xy}, g_{xx}, g_{yy} and g_{yx}.
$g(x, y) = xe^y + 3x^2y$

106) For each of the following functions find the g_{xy}, g_{xx}, g_{yy} and g_{yx}.
$g(x, y) = \log(5x + 3y)$

107) For each of the following functions find the g_{xy}, g_{xx}, g_{yy} and g_{yx}.

108) Let $w(x, y, z) = \frac{1}{\sqrt{x^2+y^2+z^2}}$, $(x, y, z) \neq (0, 0, 0)$. Show that $\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} = 0$

109) If $v(x, y, z) = x^3 + y^3 + z^3 + xyz^3$, show that $\frac{\partial^2 v}{\partial y \partial z} = \frac{\partial^2 v}{\partial z \partial y}$

110) A firm produces two types of calculators each week, x number of type A and y number of type B. The weekly revenue and cost functions (in rupees) are $R(x, y) = 80x + 90y + 0.04xy - 0.05x^2 - 0.05y^2$ and $C(x, y) = 8x + 6y + 2000$ respectively
Find the profit function P(x, y)

111) A firm produces two types of calculators each week, x number of type A and y number of type B. The weekly revenue and cost functions (in rupees) are $R(x, y) = 80x + 90y + 0.04xy - 0.05x^2 - 0.05y^2$ and $C(x, y) = 8x + 6y + 2000$ respectively
Find $\frac{\partial P}{\partial x}$ (1200, 1800) and $\frac{\partial v}{\partial y}$ (1200, 1800)

112) Let $U(x, y, z) = x^2 - xy + 3\sin z$, $x, y, z \in R$ Find the linear approximation for U at (2,−1,0).

113) If $w(x, y) = x^3 - xy + y^2$, $x, y \in R$, find the linear approximation for w at (1,−1)

114) Let $z(x, y) = x^2y + 3xy^4$, $x, y \in R$. Find the linear approximation for z at (2, -1).

115) If $v(x,y) = x^2 - xy + \frac{1}{4}y + 7$, $x,y \in R$, find the differential dv.

116) Let W(x, y, z) = x2 - xy + 3 sin z, x, y, z ∈ R. Find the linear approximation at (2, -1, 0).

117) Let V(x,y, z) =xy+ yz + zx, x, y, z ∈ R. Find the differential dV.

118) Verify the above theorem for $F(x, y)= x^2 - 2y^2 + 2xy$ and $x(t) = \cos t$, $y(t) = \sin t$, $t \in [0, 2\pi]$

119) Let $g(x,y)= x^3 - yx + \sin(x+y)$, $x(t) = e^{3t}$, $y(t) = t^2$, $t \in R$. Find $\frac{dg}{dt}$

120) Let $g(x, y) = 2y + x^2$, $x = 2r - s$, $y = r^2 + 2s$, $r, s \in R$. Find $\frac{\partial g}{\partial r}, \frac{\partial g}{\partial s}$

121) If u (x,y) =x2y +3xy4, x = et and y = sin t, find $\frac{du}{dt}$ and evaluate It at t = 0.

122) If $u(x,y, z) = xy^2z^3$, x= sin t, y= cos t, $z = 1+ e^{2t}$, find $\frac{du}{dt}$

123) If $w (x,y, z) = x^2 + y^2 + y^2$, $x= e^t$, $y = e^t \sin t$, $z = e^t \cos t$, find $\frac{dw}{dt}$

124) Let $U(x, y, z) = xyz$, $x = e^{-t}$, $y = e^{-t} \cos t$, $z = \sin t$, $t \in R$. Find $\frac{dU}{dt}$

125) If $w(x, y) = 6x^2 - 3xy + 2y^2$, x= ety = cos s ∈ R find $\frac{dw}{ds}$, and evaluate at s = 0

126) Let $z(x, y) = x^3 - 3x^2y^3$, where x = set, $y = se^{-t}$, s, t ∈ R. Find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$

127) prove that g(x, y) = x log$\left(\frac{y}{x}\right)$ is homogeneous; what is the degree? Verify Euler's Theorem for g.

128) If u(x, y) = $\frac{x^2+y^2}{\sqrt{x+y}}$, prove that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\frac{3}{2}u$

129) Find an approximate value of $\int_1^{1.5} x dx$ by applying the left-end rule with the partition {1.1,1.2,1.3,1.4,1.5}.

130) Find an approximate value of $\int_1^{1.5} x^2 dx$ by applying the right-end rule with the partition {1.1,1.2,1.3,1.4,1.5}.

131) Find an approximate value of $\int_1^{1.5}(2-x)dx$ by applying the mid-point rule with the partition {1.1, 1.2, 1.3, 1.4, 1.5}.

132) Evaluate$\int_0^1 x dx$, as the limit of a sum.

133) Evaluate$\int_0^1 x^3 dx$, as the limit of a sum.

134) Evaluate :$\int_0^1 \frac{2x+7}{5x^2+9}dx$

135) Evaluate :$\int_0^{\frac{\pi}{2}} \frac{sec\ x\ tan\ x}{1+sec^2 x}dx$

136) Evaluate :$\int_0^9 \frac{1}{x+\sqrt{x}}dx$

137) Evaluate $\int_1^2 \frac{x}{(x+1)(x+2)}dx$

138) Evaluate: $\int_0^{\frac{\pi}{2}} \frac{cos\theta}{(1+sin\theta)(2+sin\theta)}d\theta$

139) Evaluate: $\int_0^{\frac{1}{\sqrt{2}}} \frac{sin^{-1}x}{(1-x^2)^{\frac{3}{2}}}dx$

140) Show that $\int_0^{\pi} g(sinx)dx = 2\int_0^{\frac{\pi}{2}} g(sinx)dx$, where g(sin x) is a function of sin x.

141) Show that $\int_0^{\frac{2\pi}{0}}$ g(cos x)dx = 2 $\int_0^{\pi}$ g(cosx)dx where g(cos x) is a function of cos x

142) If f (x) = f (a + x) , then $\int_0^{2a} f(x)dx = 2\int_0^a f(x)dx$

143) Evaluate: $\int_{-log2}^{log2} e^{-|x|}$ dx.

144) Evaluate: $\int_0^a \frac{fx}{fx+f(a-x)}$ dx.

145) Evaluate $\int_2^3 \frac{\sqrt{x}}{\sqrt{5-x}+\sqrt{x}}$dx.

146) Evaluate the following definite integrals:

$\int_0^1 \sqrt{\frac{1-x}{1+x}}dx$

147) Evaluate the following definite integrals:

$$\int_0^{\frac{\pi}{2}} e^x \left(\frac{1+sinx}{1+cosx} \right) dx$$

148) Evaluate the following definite integrals:

$$\int_0^{\frac{\pi}{2}} \sqrt{cos\theta} sin^3\theta d\theta$$

149) Evaluate the following definite integrals:

$$\int_0^1 \frac{1-x^2}{(1+x^2)^2} dx$$

150) Evaluate the following integrals using properties of integration:

$$\int_{-5}^{5} xcos\left(\frac{e^x-1}{e^x+1} \right) dx$$

151) Evaluate the following integrals using properties of integration:

$$\int_0^{2\pi} xlog\left(\frac{3+cos\ \ x}{3-cos\ \ x} \right) dx$$

152) Evaluate $\int_0^x x^2$cos nxdx, where n is a positive integer.

153) Evaluate $\int_0^1 e^{-2x}(1+x-2x^3)dx$

154) Evaluate: $\int_0^{2\pi} x^2 sin\ nx\ dx$ where n is a positive integer.

155) Evaluate: $\int_{-1}^{1} e^{-\lambda x}(1-x^2)dx$

156) Evaluate the following:

$$\int_0^{\frac{\pi}{2}} \frac{dx}{1+5cos^2x}$$

157) Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{4sin^2x+5cos^2x}$

158) Evaluate $\int_0^{\frac{\pi}{\sqrt{2}}} \frac{dx}{5+4sin^2x}$

159) Evaluate $\int_0^{\frac{\pi}{2}} \begin{vmatrix} cos^4x & 7 \\ sin^5x & 3 \end{vmatrix}$ dx

160) Evaluate $\int_0^{2a} x^2\sqrt{2ax-x^2}dx$

161) Evaluate $\int_0^1 x^5(1-x^2)^5 dx$

162) Evaluate $\int_0^1 x^3(1-x)^4 dx$

163) Evaluate the following

$$\int_0^{\pi/4} sin^6x \quad dx$$

164) Evaluate the following

$$\int_0^{\pi/6} sin^5 3x\ dx$$

165) Evaluate the following

$\int_0^{2\pi} sin^7 \frac{x}{4} dx$

166) Evaluate the following

$\int_{\bar{\theta}}^{\pi} sin^3\theta cos^5\theta d\theta$

167) Prove that $\int_0^\infty e^{-x} x^n dx = n!$ where n is a positive integer.

168) Evaluate $\int_0^\infty e^{-ax} x^n dx$,where a > 0 .

169) Show that Γ(n)=2$\int_0^\infty e^{-x^2} x^{2n-1} dx$

170) Evaluate $\int_0^\infty \frac{x^n}{x^x} dx$, where n is positive integer ≥ 2

171) If $\int_0^\infty e^{-ax^2} x^3 dx = 32, \alpha > 0, \quad find \alpha$

172) Find the area of the region bounded by the line 6x + 5y = 30 , x − axis and the lines x = −1 and x = 3.

173) Find the area of the region bounded by the line 7x − 5y = 35, x−axis and the lines x = −2 and x = 3.

174) Find the area of the region bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

175) Find the area of the region bounded between the parabola y^2 ax = 4 and its latus rectum

176) Find the area of the region bounded by the y -axis and the parabola x=5−4y−y^2.

177) Find the area of the region bounded by x−axis, the sine curve y = sin x, the lines x = 0 and x = 2π.

178) Find the area of the region bounded by x−axis, the curve y = cos x, the lines x = 0 and x =π.

179) Find the area of the region bounded by 3x − 2y + 6 = 0 , x = −3, x =1 and x-axis.

180) Find the area of the region bounded by 2x − y +1 = 0 , y = −1, y = 3 and y-axis

181) Find the volume of a sphere of radius a.

182) Find the volume of a right-circular cone of base radius r and height h.

183) Find the volume of the spherical cap of height h cut of from a sphere of radius r.

184) Find the volume of the solid formed by revolving the region bounded by the parabola y = x^2,x-axis, ordinates x = 0 and x = 1 about the x-axis.

185) Find the volume of the solid formed by revolving the region bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2}$1, a > b about the major axis.

186) Find, by integration, the volume of the solid generated by revolving about y-axis the region bounded between the parabola x = y^2 +1, the y-axis, and the lines y =1 and y = −1.

187) Find, by integration, the volume of the solid generated by revolving about y-axis the region bounded between the curve y=$\frac{3}{4}\sqrt{x^2 - 16}, x \geq 4$the y-axis, and the lines y =1 and y = 6.

188) Find, by integration, the volume of the solid generated by revolving about y-axis the region bounded by the curves y = log x, y = 0, x = 0 and y = 2.

189) The region enclosed between the graphs of y = x and y = x^2 is denoted by R, Find the volume generated when R is rotated through 360° about x-axis.

190) Form the differential equation of all straight lines touching the circle $x^2 + y^2 = r^2$.

191) Find the differential equation of the family of circles passing through the origin and having their centres on the x -axis.

192) Find the differential equation of the family of parabolas with vertex at (0,−1) and having
axis along the y-axis.

193) Find the differential equation corresponding to the family of curves represented by the equation $y = Ae^{8x} + Be^{-8x}$, where A and B are arbitrary constants.

194) Find the differential equation of the curve represented by $xy = ae^x + be^{-x} + x^2$.

195) The slope of the tangent to the curve at any point is the reciprocal of four times the ordinate at that point. The curve passes through (2, 5). Find the equation of the curve.

196) Show that $y = ax + \frac{b}{x}$, $x \neq 0$ is a solution of the differential equation $x^2 y'' + xy' - y = 0$.

197) Form the differential equation by eliminating the arbitrary constants A and B from y = Acos x + B sin x.

198) Find the differential equation of the family of circles passing through the points (a,0) and (−a,0).

199) Find the differential equation of the family of all ellipses having foci on the x -axis and centre at the origin.

200) Show that $y = 2(x^2-1)+Ce^{-x2}$ is a solution of the differential equation
$\frac{dy}{dx} + 2xy - 4x^3 = 0$

201) Show that y = a cos(log x) + bsin (log x), x > 0 is a solution of the differential equation $x^2 y''+ xy'+y= 0$.

202) Solve $(1 + x^2)\frac{dy}{dx} = 1 + y^2$

203) Find the particular solution of $(1+ x^3)dy - x^2 ydx =$ satisfying the condition y(1) = 2.

204) Solve $y ' = \sin^2 (x - y +)1$.

205) Solve : $\frac{dy}{dx} = \sqrt{4x + 2y - 1}$

206) If F is the constant force generated by the motor of an automobile of mass M, its velocity is given by $M \frac{dV}{dt}$=F-kV, where k is a constant. Express V in terms of t given that V = 0 when t = 0.

207) Solve the differential equation:
$ydx + (1 +x^2) \tan^{-1} xdy = 0$

208) Solve the differential equation:
$sin\frac{dy}{dx} = a, y(0) = 1$

209) Solve the differential equation:
$\frac{dy}{dx} = e^{x+y} + x^3e^y$

210) Solve the differential equation:
$(e^y+1) \cos x\, dx + e^y \sin x\, dy = 0$

211) Solve the differential equation:

$$\frac{dy}{dx} - x\sqrt{25 - x^2} = 0$$

212) Solve the differential equation:

x cos y dy = e^x(x log x + 1)dx

213) Solve the differential equation:

tan y$\frac{dy}{dx}$=cos(x+y)+cos(x-y)

214) Solve $y^2 + x^2\frac{dy}{dx} = xy\frac{dy}{dx}$

215) Solve $\frac{dy}{dx} + 2y = e^{-x}$

216) Solve: $\frac{dv}{dx} + 2y\,cot\,x = 3x^2 cosec^2 x$

217) Solve $ye^y dx=(y^3+2xe^y)dy$

218) Solve the Linear differential equation:

$$\frac{dy}{dx} + \frac{y}{xlogx} = \frac{sin2x}{logx}$$

219) Solve the Linear differential equation:

$$x\frac{dy}{dx} + y = xlogx$$

220) Solve the Linear differential equation:

$$x\frac{dy}{dx} + 2y - x^2 logx = 0$$

221) A radioactive isotope has an initial mass 200mg, which two years later is 50mg . Find the expression for the amount of the isotope remaining at any time. What is its half-life? (half-life means the time taken for the radioactivity of a specified isotope to fall to half its original value).

222) The engine of a motor boat moving at 10 m / s is shut off. Given that the retardation at any subsequent time (after shutting off the engine) equal to the velocity at that time. Find the velocity after 2 seconds of switching off the engine.

223) In a pack of 52 playing cards, two cards are drawn at random simultaneously. If the number of black cards drawn is a random variable, find the values of the random variable and number of points in its inverse images.

224) Two balls are chosen randomly from an urn containing 6 red and 8 black balls. Suppose that we win Rs. 15 for each red ball selected and we lose Rs. 10 for each black ball selected. X denotes the winning amount, then find the values of X and number of points in its inverse images.

225) A six sided die is marked '2' on one face, '3' on two ofits faces, and '4' on remaining three faces. The die is thrown twice. If X denotes the total score in two throws, find the values of the random variable and number of points in its inverse images.

226) Suppose a discrete random variable can only take the values 0, 1, and 2. The probability mass function is defined by

$$f(x) = \begin{cases} \frac{x^2+1}{k} & forx = 0, 1, 2 \\ 0 & otherwise \end{cases}$$

Find
(i) the value of k
(ii) cumulative distribution function
(iii) P(X ≤ 1).

227) The probability density function of X is

$$f(x)=\begin{cases} x & 0<x<1 \\ 2-x & 1\le x<2 \\ 0 & \text{otherwise} \end{cases}$$

Find P(0.2≤X< 0.6)

228) The probability density function of X is

$$f(x)=\begin{cases} x & 0<x<1 \\ 2-x & 1\le x<2 \\ 0 & \text{otherwise} \end{cases}$$

Find P(1.2≤X< 1.8)

229) The probability density function of X is

$$f(x)=\begin{cases} x & 0<x<1 \\ 2-x & 1\le x<2 \\ 0 & \text{otherwise} \end{cases}$$

find P(0.5≤X< 1.5)

230) For the random variable X with the given probability mass function as below, find the mean and variance.

$$f(x)=\begin{cases} \frac{1}{10} & x=2,5 \\ \frac{1}{5} & x=0,1,2,3,4 \end{cases}$$

231) For the random variable X with the given probability mass function as below, find the mean and variance \(f(x)=\begin{cases} \begin{matrix} 2(x-1) & 1

232) For the random variable X with the given probability mass function as below, find the mean and variance.

$$f(x)=\begin{cases} \frac{1}{2}e^{-\frac{x}{2}} & for \quad x>0 \\ 0 & otherwise \end{cases}$$

233) Two balls are drawn in succession without replacement from an urn containing four red balls and three black balls. Let X be the possible outcomes drawing red balls. Find the probability mass function and mean for X.

234) If μ and σ^2 are the mean and variance of the discrete random variable X, and E(X + 3)=10 and $E(X + 3)^2 = 116$, find μ and a^2•

235) Four fair coins are tossed once. Find the probability mass function, mean and variance for number of heads occurred.

236) The probability density function random variable X is given by

$$f(x)=\begin{cases} 16xe^{-4x} & for \quad x>0 \\ 0 & for \quad x\le 0 \end{cases}$$ find the mean and variance of X.

237) A lottery with 600 tickets gives one prize of Rs.200, four prizes of noo, and six prizes of Rs. 50. If the ticket costs is Rs.2, find the expected winning amount of a ticket

238) The probability that Mr.Q hits a target at any trial is $\frac{1}{4}$ .Suppose he tries at the target 10 times. Find the probability that he hits the target
(i) exactly 4 times
(ii) at least one time.

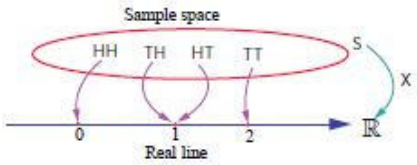
239) Using binomial distribution find the mean and variance of X for the following experiments
(i) A fair coin is tossed 100 times, and X denote the number of heads.
(ii) A fair die is tossed 240 times, and X denote the number of times that four appeared.

240) The probability that a certain kind of component will survive a electrical test is $\frac{3}{4}$. Find the probability that exactly 3 of the 5 components tested survive.

241) If X- B(n, p) such that 4P(X = 4) = P(X = 2) and n = 6 • Find the distribution, mean and standard deviation of X.

242) In a binomial distribution consisting of 5 independent trials, the probability of 1 and 2 successes are 0.4096 and 0.2048 respectively.
Find the mean and variance of the random variables.

243) Suppose two coins are tossed once. If X denotes the number of tails,
(i) write down the sample space
(ii) find the inverse image of 1
(iii) the values of the random variable and number of elements in its inverse images



A mapping $X(.)$ from S to $\mathbf{R}$

244) Suppose a pair of unbiased dice is rolled once. If X denotes the total score of two dice, write down
(i) the sample space
(ii) the values taken by the random variable X,
(iii) the inverse image of 10, and
(iv) the number of elements in inverse image of X.

245) An urn contains 2 white balls and 3 red balls. A sample of 3 balls are chosen at random from the urn. If X denotes the number of red balls chosen, find the values taken by the random variable X and its number of inverse images

246) Two balls are chosen randomly from an urn containing 6 white and 4 black balls. Suppose that we win Rs.30 for each black ball selected and we lose Rs.20 for each white ball selected. If X denotes the winning amount, then find the values of X and number of points in its inverse images.

247) Two fair coins are tossed simultaneously (equivalent to a fair coin is tossed twice). Find the probability mass function for number of heads occurred.

248) A pair of fair dice is rolled once. Find the probability mass function to get the number of fours.

249) If the probability mass function f (x) of a random variable X isx

x	1	2	3	4
f (x)	$\frac{1}{12}$	$\frac{5}{12}$	$\frac{5}{12}$	$\frac{1}{12}$

find (i) its cumulative distribution function, hence find
(ii) P(X ≤ 3) and,
(iii) P(X ≥ 2)

250) Find the probability mass function f (x) of the discrete random variable X whose cumulative distribution function F(x) is given by

$$F(x)=\begin{cases}0 & -\infty<x<-2\\ 0.25 & -2\le x<-1\\ 0.60 & -1\le x<0\\ 0.90 & 0\le x<1\\ 1 & 1\le x<\infty\end{cases}$$

Also find (i) P(X < 0) and (ii)$P(X \geq -1$

251) If X is the random variable with distribution function F(x) given by,

$$F(x)=\begin{cases}0 & x<0\\ x & 0\le x<1\\ 1 & 1\le x\end{cases}$$

then find (i) the probability density function f (x) (ii) P(0.2≤ X ≤0.7)

252) Find the mean and variance of a random variable X , whose probability density function is $f(x)=\begin{cases}\lambda e^{-2x} & for \geq 0\\ 0 & otherwise\end{cases}$

253) Verify the
(i) closure property,
(ii) commutative property,
(iii) associative property
(iv) existence of identity and
(v) existence of inverse for the arithmetic operation + on Z.

254) Verify the
(i) closure property,
(ii) commutative property,
(iii) associative property
(iv) existence of identity and
(v) existence of inverse for the arithmetic operation - on Z.

255) Verify the
(i) closure property,
(ii) commutative property,
(iii) associative property
(iv) existence of identity and

(v) existence of inverse for the arithmetic operation + on
Z_e = the set of all even integers

256) Verify
(i) closure property
(ii) commutative property, and
(iii) associative property of the following operation on the given set.
$(a*b) = a^b; \forall a,b \in N$ (exponentiation property)

257) How many rows are needed for following statement formulae?
$p \vee \neg t\ (p \vee \neg s)$

258) How many rows are needed for following statement formulae?
$((p \wedge q) \vee (\neg r \vee \neg s)) \wedge (\neg t \wedge v))$

259) Consider $p \rightarrow q$: If today is Monday, then 4 + 4 = 8.

260) Write down the
(i) conditional statement
(ii) converse statement
(iii) inverse statement, and
(iv) contrapositive statement for the two statements p and q given below.
p: The number of primes is infinite.
q: Ooty is in Kerala.

261)
Construct the truth table for $(p \overline{\vee} q) \wedge (p \overline{\vee} \neg q)$

262) Establish the equivalence property $p \rightarrow q \equiv \neg p \vee q$

263)
Let $A = \begin{pmatrix} 1 & 0\,1 & 0 \\ 0 & 1\,0 & 1 \\ 1 & 0\,0 & 1 \end{pmatrix}, B = \begin{pmatrix} 0 & 1\,0 & 1 \\ 1 & 0\,1 & 0 \\ 1 & 0\,0 & 1 \end{pmatrix}, C = \begin{pmatrix} 1 & 1\,0 & 1 \\ 0 & 1\,1 & 0 \\ 1 & 1\,1 & 1 \end{pmatrix}$ be any three boolean matrices of the same type.
Find $A \vee B$

264)
Let $A = \begin{pmatrix} 1 & 0\,1 & 0 \\ 0 & 1\,0 & 1 \\ 1 & 0\,0 & 1 \end{pmatrix}, B = \begin{pmatrix} 0 & 1\,0 & 1 \\ 1 & 0\,1 & 0 \\ 1 & 0\,0 & 1 \end{pmatrix}, C = \begin{pmatrix} 1 & 1\,0 & 1 \\ 0 & 1\,1 & 0 \\ 1 & 1\,1 & 1 \end{pmatrix}$ be any three boolean matrices of the same type.
Find $A \wedge B$

265)
Let $A = \begin{pmatrix} 1 & 0\,1 & 0 \\ 0 & 1\,0 & 1 \\ 1 & 0\,0 & 1 \end{pmatrix}, B = \begin{pmatrix} 0 & 1\,0 & 1 \\ 1 & 0\,1 & 0 \\ 1 & 0\,0 & 1 \end{pmatrix}, C = \begin{pmatrix} 1 & 1\,0 & 1 \\ 0 & 1\,1 & 0 \\ 1 & 1\,1 & 1 \end{pmatrix}$ be any three boolean matrices of the same type.
Find $(A \vee B) \wedge C$

266) Verify whether the following compound propositions are tautologies or contradictions or contingency
$(p \wedge q)\ \neg\ (p \vee q)$

267)

Let $A = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}, B = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, C = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}$ be any three boolean matrices of the same type.

Find $(A \wedge B) \vee C$

268) Verify whether the following compound propositions are tautologies or contradictions or contingency

$((p \vee q) \wedge \sim p) \rightarrow q$

269) Verify whether the following compound propositions are tautologies or contradictions or contingency

$(p \longrightarrow q) \leftrightarrow (\sim p \longrightarrow q)$

270) Show that

$\neg(p \wedge q) \equiv \neg p \vee \neg q$

271) Show that $q \rightarrow p \equiv \neg p \rightarrow \neg q$

272) Show that $p \rightarrow q$ and $q \rightarrow p$ are not equivalent

273) Show that $\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

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5 MARKS V2 BB

12th Standard

Maths

Exam Time : 01:40:00 Hrs

Total Marks : 100

226 x 5 = 1130

1) A particle moves along a horizontal line such that its position at any time $t \geq 0$ is given by $s(t) = t^3 - t^2 + t + 6\ 9\ 1$, where s is measured in metres and t in seconds?
   (1) At what time the particle is at rest?
   (2) At what time the particle changes direction?
   (3) Find the total distance travelled by the particle in the first 2 seconds.
2) If we blow air into a balloon of spherical shape at a rate of 1000^3 cm per second. At what rate the radius of the baloon changes when the radius is 7cm? Also compute the rate at which the surface area changes.
3) Salt is poured from a conveyer belt at a rate of 30 cubic metre per minute forming a conical pile with a circular base whose height and diameter of base are always equal. How fast is the height of the pile increasing when the pile is 10 metre high?
4) A road running north to south crosses a road going east to west at the point P. Car A is driving north along the first road, and car B is driving east along the second road. At a particular time car A 10 kilometres to the north of P and traveling at 80 km/hr, while car B is 15 kilometres to the easst of P and traveling at 100 km/hr. How fast is the distance between the two cars changing?
5) A camera is accidentally knocked off an edge of a cliff 400 ft high. The camera falls a distance of $s = 16t^2$ in t seconds.
   How long does the camera fall before it hits the ground?
6) A camera is accidentally knocked off an edge of a cliff 400 ft high. The camera falls a distance of $s = 16t^2$ in t seconds
   What is the average velocity with which the camera falls during the last 2 seconds?
7) A camera is accidentally knocked off an edge of a cliff 400 ft high. The camera falls a distance of $s = 16t^2$ in t seconds
   What is the instantaneous velocity of the camera when it hits the ground?
8) A particle moves along a line according to the law $s(t) = 2t^3 - 9t^2 + 12t - 4$, where $t \geq 0$. At what times the particle changes direction?
9) A particle moves along a line according to the law $s(t) = 2t^3 - 9t^2 + 12t - 4$, where $t \geq 0$. Find the total distance travelled by the particle in the first 4 seconds.
10) A particle moves along a line according to the law $s(t) = 2t^3 - 9t^2 + 12t - 4$, where $t \geq 0$.
    Find the particle's acceleration each time the velocity is zero.
11) A conical water tank with vertex down of 12 metres height has a radius of 5 metres at the top. If water flows into the tank at a rate 10 cubic m/min, how fast is the depth of the water increases when the water is 8 metres deep?
12) A ladder 17 metre long is leaning against the wall. The base of the ladder is pulled away from the wall at a rate of 5 m/s. When the base of the ladder is 8 metres from the wall.
    How fast is the top of the ladder moving down the wall?

13) A ladder 17 metre long is leaning against the wall. The base of the ladder is pulled away from the wall at a rate of 5 m/s. When the base of the ladder is 8 metres from the wall.
At what rate, the area of the triangle formed by the ladder, wall, and the floor, is changing?

14) A police jeep, approaching an orthogonal intersection from the northern direction, is chasing a speeding car that has turned and moving straight east. When the jeep is 0.6 km north of the intersection and the car is 0.8 km to the east. The police determine with a radar that the distance between them and the car is increasing at 20 km/hr. If the jeep is moving at 60 km/hr at the instant of measurement, what is the speed of the car?

15) Find the equation of the tangent and normal to the Lissajous curve given by x = 2cos3t and y = 3sin 2t, t ∈ R

16) Find the acute angle between $y = x^2$ and $y = (x - 3)^2$.

17) Find the acute angle between the curves $y = x^2$ and $x = y^2$ at their points of intersection (0,0), (1,1).

18) If the curves $ax^2+by^2=1$ and $cx^2+dy^2=1$ intersect each other orthogonally then,
$\frac{1}{a} - \frac{1}{b} = \frac{1}{c} - \frac{1}{d}$

19) Prove that the ellipse $x^2+4y^2=8$ and the hyperbola $x^2-2y^2=4$ intersect orthogonally.

20) Find the tangent and normal to the following curves at the given points on the curve $y = x^2 - x^4$ at (1, 0)

21) Find the tangent and normal to the following curves at the given points on the curve $y = x^4 + 2e^x$ at (0, 2)

22) Find the tangent and normal to the following curves at the given points on the curve
$y = x \sin x$ at $\left(\frac{\pi}{2}, \frac{\pi}{2}\right)$

23) Find the tangent and normal to the following curves at the given points on the curve
$x = \cos t$, $y = 2\sin t^2$ at $t = \frac{\pi}{3}$

24) Find the equations of the tangents to the curve $y =1+ x^3$ for which the tangent is orthogonal with the line x +12y =12

25) Find the equations of the tangents to the curve $y = \frac{x+1}{x-1}$ which are parallel to the line x + 2y = 6.

26) Find the equation of tangent and normal to the curve given by x = 7 cos t and y = 2sin t, t ∈ R at any point on the curve.

27) Prove that there is a zero of the polynomial $2x^3 - 9x^2 - 11x + 12$in the interval (2,7) given that 2 and 7 are the zeros of the polynomial $x^4 - 6x^3 - 11x^2 + 24x + 28$

28) Find the intervals of mono tonicities and hence find the local extremum for the following function:
$f(x)=2x^3+3x^2-12x$

29) Find the intervals of mono tonicities and hencefind the local extremum for the following function:

$$f(x) = \frac{x}{x-5}$$

30) Find the intervals of mono tonicities and hence find the local extremum for the following function:

$$f(x) = \frac{e^x}{1-e^x}$$

31) Find the intervals of mono tonicities and hence find the local extremum for the following function:

$$f(x) = \frac{x^3}{3} - logx$$

32) Find the intervals of mono tonicities and hence find the local extremum for the following function:
f(x) = sin x cos x + 5, x ∈ (0,2π)

33) Find intervals of concavity and points of inflexion for the following functions
$f(x) = x(x - 4)^3$

34) Find intervals of concavity and points of inflexion for the following function:
f(x) = sin x + cos x, 0< x< 2π

35) Find intervals of concavity and points of inflexion for the following function:

$$f(x) = \frac{1}{2}\left(e^x - e^{-x}\right)$$

36) Find the smallest possible value x^2+y^2 given that x +y = 10.

37) A garden is to be laid out in a rectangular area and protected by wire fence. What is the largest possible area of the fenced garden with 40 metres of wire.

38) A rectangular page is to contain 24 cm^2 of print. The margins at the top and bettemof the page are 1.5 cm and the margins at other sides of the page is 1 cm. What should be the dimensions of the page so that the area of the paper used is minimum.

39) Expand log(1+ x) as a Maclaurin's series upto 4 non-zero terms for –1 < x ≤ 1.

40) A farmer plans to fence a rectangular pasture adjacent to a river. The pasture must contain1,80,000 sq. mtrs in order to provide enough grass for herds. No fencing is needed along the river. What is the length of the I minimum needed fencing material

41) Expand tan x in ascending powers of x upto 5th power for $(-\frac{\pi}{2} < x < \frac{\pi}{2})$

42) Find the dimensions of the rectangle with maximum area that can be inscribed in a circle of radius 10 em.

43) Prove that among all the rectangles of the given perimeter, the square has the maximum area.

44) Find the dimensions of the largest rectangle that can be inscribed in a semi circle of radius rem.

45) A manufacturer wants to design an open box having a square base and a surface area of I 108 sq. em. Determine the dimensions of the box for the maximum volume.

46) A hollow cone with base radius a cm and , height b em is placed on a table. Show that the volume of the largest cylinder that can be hidden underneath is $\frac{4}{9}$ times volume of the cone.

47) Write the Maclaurin series expansion of the following function log(1 - x); -1 ≤ x < 1

48) Write the Maclaurin series expansion of the following function $\tan^{-1}(x)$; -1 ≤ x ≤ 1

49) Write the Maclaurin series expansion of the following function: $\cos^2 x$

50) Find the asymptotes of the following curve $f(x) = \frac{x^2}{x^2 - 1}$

51) Find the asymptotes of the following curve. $f(x) = \frac{x^2}{x^2 + 1}$

52) Find the asymptotes of the following curves $f(x) = \frac{3x}{\sqrt{x^2 + 2}}$

53) Find the asymptotes of the following curve $f(x) = \frac{x^2 - 6x - 1}{x + 3}$

54) Find the asymptotes of the following curves :$f(x) = \frac{x^2 + 6x - 4}{3x - 6}$

55) Sketch the graphs of the followingfunction.$y = -\frac{1}{3}\left(x^3 - 3x + 2\right)$

56) Sketch the graphs of the followingfunction.$y = x\sqrt{4 - x}$

57) Using the l'Hôpital Rule prove that, $lim x \to 0^+ (1 + x)^{\frac{1}{x}} = e$

58) Evaluate: $log(lim x \to \infty (1 + 2x)^{\frac{1}{2 log\, x}}$

59) Sketch the graphs of the followingfunction.$y = \frac{x^2 + 1}{x^2 - 4}$

60) Evaluate: $lim x \to 1\ x^{\frac{1}{1-x}}$

61) Sketch the graphs of the followingfunction $y = \frac{1}{1 + e^{-x}}$

62) Evaluate the following limit, if necessary use l'Hôpital Rule $lim x \to 0^+ x^x$

63) Sketch the graphs of the followingfunction $y = \frac{x^3}{24} - logx$

64) Evaluate the following limit, if necessary use l'Hôpital Rule $limx \to \frac{\pi}{2}(sinx)^{tanx}$

65) Evaluate the following limit, if necessary use l'Hôpital Rule $limx \to 0^+(cosx)^{\frac{1}{x^2}}$

66) Find the local extrema for the following function using second derivative test: $f(x) = -3x^5 + 5x^3$

67) Find the local extrema for the following function using second derivative test: f(x) =x logx

68) Find the local extrema for the following function using second derivative test: $f(x) = x^2 e^{-2x}$

69) Find the intervals of monotonicity and hence find the local extrema for the function f $(x) = x^2 - 4x + 4$

70) For the function $f\{x)=4x^3+3x^2-6x+1$ find the intervals of monotonicity, local extrema, intervals of concavity and points of inflection.

71) Determine the intervals of concavity of the curve $f(x) = (x-1)^3 . (x-5)$, $x \in R$ and, points of inflection if any.

72) Discuss the monotonicity and local extrema of the function $f(x) = log(1+x) - \frac{x}{1+x}, x > -1$ and hence find the domain where, $log(1+x) > \frac{x}{1+x}$

73) Find the intervals of monotonicity and local extrema of the function f (x) = x log x + 3x.

74) Find the intervals of monotonicity and local extrema of the function $f(x) = \frac{1}{1+x^2}$

75) Sketch the curve $y = f(x) = x^2 - x - 6$.

76) Find the intervals of monotonicity and local extrema of the function $f(x) = \frac{x}{1+x^2}$

77) Find the local extrema of the function $f(x)= 4x^6 - 6x^4$

78) We have a 12 square unit piece of thin material and want to make an open box by cutting small squares from the corners of our material and folding the sides up. The question is, which cut produces the box of maximum volume?

79) Find the local maximum and minimum of the function $x^2 y^2$ on the line x + y =10

80) Prove that among all the rectangles of the given area square has the least perimeter.

81) Sketch the curve $y = f(x) = x^3 - 6x - 9$

82) Sketch the curve $y = \frac{x^2-3x}{(x-1)}$

83) Find the points on the unit circle $x^2 + y^2 = 1$ nearest and farthest from (1,1).

84) A steel plant is capable of producing x tonnes per day of a low-grade steel and y tonnes per day of a high-grade steel, where $y = \frac{40-5x}{10-x}$ If the fixed market price of low-grade steel is half that of high-grade steel, then what should be optimal productions in low-grade steel and high-grade steel in order to have maximum receipts.

85) Sketch the graph of the function $y = \frac{3x}{x^2-1}$

86) Let f(x, y) = sin(xy^2) + e^{x3+5y} for all ∈ R^2. Calculate $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial^2 f}{\partial y \partial x}$ and $\frac{\partial^2 f}{\partial x \partial y}$

87) Let w(x, y) = xy+$\frac{e^y}{y^2+1}$ for all (x, y) ∈ R^2. Calculate $\frac{\partial^2 w}{\partial y \partial x}$ and $\frac{\partial^2 w}{\partial x \partial y}$

88) For each of the following functions find the f_x, f_y, and show that f_{xy} =f_{yx}

f(x,y) = $\frac{3x}{y+sinx}$

89) For each of the following functions find the f_x, f_y, and show that f_{xy} =f_{yx}

f(x,y) = $\tan^{-1}$

90) If U(x, y, z) = log (x^3 + y^3 + z^3), find $\frac{\partial U}{\partial x} + \frac{\partial U}{\partial y} + \frac{\partial U}{\partial z}$

91) If V(x,y) = e^x(x cos y - y siny), then prove that $\frac{\partial^2 V}{\partial x^2} = \frac{\partial^2 V}{\partial y^2}$ = 0

92) If w(x, y) = xy + sin (xy), then prove that $\frac{\partial^2 w}{\partial y \partial x} = \frac{\partial^2 w}{\partial x \partial y}$

93) If z(x, y) = x tan-^{1} (x, y), x = t^2, y = set, s, t ∈ R. Find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$ at s = t = 1

94) Let U(x, y) = ex sin y, where x = st^2, y = s^2 t, s, t ∈ R. Find $\frac{\partial U}{\partial s}, \frac{\partial U}{\partial t}$ and evaluate them at s = t = 1.

95) W(x, y, z) = xy + yz + zx, x =u -v, y = uv, z = u + v, u ∈ R. Find $\frac{\partial W}{\partial u}, \frac{\partial W}{\partial v}$, and evaluate them at $\left(\frac{1}{2}, 1\right)$

96) Prove that f(x, y) = x^3 - $2x^2y$ + $3xy^2$ + y^3 is taomogeneous; what is the degree? Verify fuler's Theorem for f.

97) If (x,y) = log $\left(\frac{x^2+y^2}{x+y}\right)$, prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ = 1

98) If w(x,y, z) = log $\left(\frac{5x^3y^4+7y^2xz^4-75y^3z^4}{x^2+y^2}\right)$ find $x\frac{\partial w}{\partial x} + y\frac{\partial w}{\partial y} + z\frac{\partial w}{\partial z}$

99) If u=sin^{-1} $\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, Show that x $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{1}{2}tanu$

100) Estimate the value of $\int_0^{0.5} x^2 dx$using the Riemann sums corresponding to 5 subintervals of equal width and applying (i) left-end rule (ii) right-end rule (iii) the mid-point rule.

101) Evaluate$\int_1^4 (2x^2 - 3)$ dx, as the limit of a sum

102) Evaluate the following integrals as the limits of sums.

$\int_0^1 (5x + 4)dx$

103) Evaluate the following integrals as the limits of sums.

$\int_1^2 4x^2 - 1)dx$

104) Evaluate: $\int_0^{\frac{\pi}{2}}(\sqrt{tan\quad x}+\sqrt{cot\quad x})dx$

105) Evaluate: $\int_0^{1.5}[x^2]dx$where [x] is the greatest integer function

106) Evaluate: $\int_{-4}^{4}$ [x+3]dx.

107) Show that $\int_0^{\frac{\pi}{2}}\frac{dx}{4+5sinx}$ = $\frac{1}{3}$ $\log_e 2$

108) Prove that $\int_0^{\frac{\pi}{4}}\frac{sin2xdx}{sin^4x+cos^4x}$ = $\frac{\pi}{4}$

109) Prove that $\int_0^{\frac{\pi}{4}}\frac{dx}{a^2sin^2x+b^2cos^2x}$ = $\frac{1}{ab}tan^{-1}(\frac{a}{b})$ where a,b > 0

110) Evaluate : $\int_0^{\frac{\pi}{4}}\frac{1}{sinx+cosx}$ dx

111) Evaluate$\int_0^{\pi}\frac{x}{1+sinx}$ dx

112) Prove that $\int_0^{\frac{\pi}{4}}$ log(1+tan x)dx=$\frac{\pi}{8}$ log2.

113) Show that $\int_0^1(tan^{-1}x+tan^{-1}(1-x))$ dx = $\frac{\pi}{2}$ - $\log_e 2$

114) Evaluate $\int_{-\pi}^{\pi}\frac{cos^2x}{1+a^x}$ dx

115) Evaluate the following integrals using properties of integration:
$\int_0^{2\pi}sin^4x\quad cos^3dx$

116) Evaluate the following integrals using properties of integration:
$\int_0^1|5x-3|dx$

117) Evaluate the following integrals using properties of integration:
$\int_0^{sin^2x}sin^{-1}\sqrt{t}dt+\int_0^{cos^2x}cos^{-1}\sqrt{t}dt$

118) Evaluate the following integrals using properties of integration:
$\int_0^1\frac{log(1+x)}{1+x^2}dx$

119) Evaluate the following integrals using properties of integration:
$\int_0^{\pi}\frac{xsinx}{1+sinx}dx$

120) Evaluate the following integrals using properties of integration:
$\int_{\frac{\pi}{8}}^{\frac{3\pi}{8}}\frac{1}{1+\sqrt{tanx}}dx$

121) Evaluate the following integrals using properties of integration:
$\int_0^{\pi}x\left[sin^2(sinx)+cos^2(cosx)\right]dx$

122) Evaluate the following:
$\int_0^1x^3e^{-2x}dx$

123) Evaluate the following:
$\int_0^1\frac{sin(3tan^{-1}x)tan^{-1}x}{1+x^2}dx$

124) Evaluate the following:

$$\int_0^{\frac{1}{2}} \frac{e^{a \sin^{-1}x} \sin^{-1}x}{\sqrt{1-x^2}} dx$$

125) Evaluate the following:

$$\int_0^{\frac{\pi}{2}} x^2 \cos 2x dx$$

126) Evaluate the following:

$$\int_0^{\frac{\pi}{2}} \frac{e^{-\tan x}}{\cos^6 x} dx$$

127) Find the area of the region bounded between the parabolas $y^2 x = 4$ and $x^2 y = 4$.

128) Find the area of the region bounded between the parabola $x^2=y$ and the curve $y = x$.

129) Find the area of the region bounded by $y = \cos x$, $y = \sin x$, the lines $x=\frac{\pi}{4}$ and $x=\frac{5\pi}{4}$.

130) The region enclosed by the circle $x^2 + y^2 = a^2$ is divided into two segments by the line $x = h$. Find the area of the smaller segment.

131) Find the area of the region in the first quadrant bounded by the parabola $y^2 x=4$, the line $x+y=3$ and y-axis

132) Find, by integration, the area of the region bounded by the lines $5x - 2y =15$, $x + y + 4 = 0$ and the x-axis

133) Using integration find the area of the region bounded by triangle ABC, whose vertices A, B, and C are (−1,1), (3, 2), and (0,5) respectively

134) Using integration, find the area of the region which is bounded by x-axis, the tangent and normal to the circle $x^2 y^2 + = 4$ drawn at $(1,\sqrt{3})$

135) Find the area of the region bounded by the curve $2+x-x^2+y =0$, x-axis, $x = -3$ and $x = 3$.

136) Find the area of the region bounded by the line $y = 2x + 5$ and the parabola $y = x^2 - 2x$.

137) Find the area of the region bounded between the curves $y = \sin x$ and $y = \cos x$ and the lines $x = 0$ and $x =\pi$

138) Find the area of the region bounded by $y = \tan x$, $y=\cot x$ and the lines $x = 0$,$x =\frac{\pi}{2}$,$y=0$

139) Find the area of the region bounded by the parabola $y^2 =x$ and the line $y=x-2$

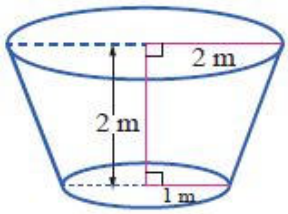
140) Father of a family wishes to divide his square field bounded by $x = 0$,$x= 4$, $y=4$ and $y=0$ along the curve $y^2 x=4$ and $x^2y=4$ into three equal parts for his wife, daughter and son. Is it possible to divide? If so, find the area to be divided among them.

141) The curve $y = (x - 2)^2+1$has a minimum point at P. A point Q on the curve is such that the slope of PQ is 2. Find the area bounded by the curve and the chord PQ.

142) Find the area of the region common to the circle $x^2 y^2 + =16$ and the parabola $y^2 x =6x$

143) Find, by integration, the volume of the container which is in the shape of a right circular conical frustum. 6. A watermelon has an ellipsoid shape which can be obtained

by



144) A watermelon has an ellipsoid shape which can be obtained by revolving an ellipse with major-axis 20 cm and minor-axis 10 cm about its major-axis. Find its volume using integration.

145) Find the differential equation of the family of all the parabolas with latus rectum 4a and whose axes are parallel to the x-axis.

146) Find the differential equations of the family of all the ellipses having foci on the y-axis and centre at the origin.

147) Show that $y = ae^{-3x} + b$, where a and b are arbitary constants, is a solution of the differential equation $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} = 0$

148) Show that the differential equation representing the family of curves $y^2 = 2a\left(x + a\frac{2}{3}\right)$ where a is a positive parameter, is $\left(y^2 - 2xy\frac{2}{3}\right)^3 = 8\left(y\frac{dy}{dx}\right)^5$.

149) Find the differential equation for the family of all straight lines passing through the origin.

150) Solve $\frac{dy}{dx} = \frac{x-y+5}{2(x-y)+7}$.

151) The velocity v , of a parachute falling vertically satisfies the equation $v\frac{dv}{dx} = g\left(1 - \frac{v^2}{k^2}\right)$, where g and k are constants. If v and x are both initially zero, find v in terms of x.

152) Find the equation of the curve whose slope is $\frac{y-1}{x^2+x}$ and which passes through the point (1,0).

153) Solve the differential equation:

$(ydx-xdy)\cot\left(\frac{x}{y}\right)=ny^2\,dx$

154) Solve the differential equation:

$\frac{dy}{dx} = tan^2(x+y)$

155) Solve $(x^2 - 3y^2)\,dx\ xydy = 0$.

156) Solve $\left(y + \sqrt{x^2+y^2}\right)dx - xdy = 0, \quad y(1) = 0$

157) Solve $(2x + 3y)dx + (y - x)dy = 0$.

158)
Solve $(1+2e^{x/y})dx+2e^{x/y}\left(1-\frac{x}{y}\right)dy=0$

159) Solve the differential equation:

$$\left[x+y \quad cos\left(\frac{y}{x}\right)\right]dx = x \quad cos\left(\frac{y}{x}\right)dy$$

160) Solve the differential equation:
$(x^3+y^3)\,dy-x^2ydx=0$

161) Solve the differential equation:

$$ye^{\frac{x}{y}}dx = \left(xe^{\frac{x}{y}}+y\right)dy$$

162) Solve the differential equation:
$2xydx + (x^2 + 2y^2)dy = 0$

163) Solve the differential equation:
$(y^2-2xy)\,dx=(x^2-2xy)\,dy$

164) Solve the differential equation:

$$x\frac{dy}{dx} = y - xcos^2\left(\frac{y}{x}\right)$$

165) Solve the differential equation:

$\left(1+3e^{\frac{y}{x}}\right)dy+3e^{\frac{y}{x}}\left(1-\frac{y}{x}\right)dx=0,$ given that y=- when x=1

166) Solve the differential equation: $(x^2+y^2)dy = xy\,dx$. It is given that $y(1) = 1$ and $y(x_0) = e$. Find the value of x_0.

167) Solve $[y(1-x \tan x)+x^2 \cos x]dx-dy=0$

168) Solve $(1+x^3)\frac{dy}{dx}+6x^2y=1+x^2$.

169) Solve the Linear differential equation:
$\cos x\frac{dy}{dx}+y \sin x=1$

170) Solve the Linear differential equation:

$$(1-x^2)\frac{dy}{dx}-xy=1$$

171) Solve the Linear differential equation:

$$\frac{dy}{dx}+\frac{y}{x}=sinx$$

172) Solve the Linear differential equation:

$$(x^2+1)\frac{d}{y}dx+2xy=\sqrt{x^2+4}$$

173) Solve the Linear differential equation:
$(2x- 10y^3)\,dy +ydx =0$

174) Solve the Linear differential equation:
$x \sin x \frac{dy}{dx} + (x \cos x + \sin x)$

175) Solve the Linear differential equation:

$$\left(y - e^{\sin^{-1}x}\right)\frac{dx}{dy} + \sqrt{1-x^2} = 0$$

176) Solve the Linear differential equation:

$$\frac{dy}{dx} + \frac{y}{(1-x)\sqrt{x}} = 1 - \sqrt{x}$$

177) Solve the Linear differential equation:

$$(1 + x + xy^2)\frac{dy}{dx} + (y + y^3) = 0$$

178) Solve the Linear differential equation:

$$(x + a)\frac{dy}{dx} - 2y = (x + a)^4$$

179) Solve the Linear differential equation:

$\frac{dy}{dx} + \frac{3y}{x} = \frac{1}{x^2}$, given that y=2 when x=1

180) The growth of a population is proportional to the number present. If the population of a colony doubles in 50 years, in how many years will the population become triple?

181) In a murder investigation, a corpse was found by a detective at exactly 8 p.m. Being alert, the detective also measured the body temperature and found it to be 70°F. Two hours later, the detective measured the body temperature again and found it to be 60°F. If the room temperature is 50°F, and assuming that the body temperature of the person before death was 98.6^{o}F, at what time did the murder occur? [log(2.43)=0.88789; log(0.5)=-0.69315]

182) A tank contains 1000 litres of water in which 100 grams of salt is dissolved. Brine (Brine is a high-concentration solution of salt (usually sodium chloride) in water) runs in a rate of 10 litres per minute, and each litre contains 5grams of dissolved salt. The mixture of the tank is kept uniform by stirring. Brine runs out at 10 litres per minute. Find the amount of salt at any time t.

183) The rate of increase in the number of bacteria in a certain bacteria culture is proportional to the number present. Given that the number triples in 5 hours, find how many bacteria will be present after 10 hours?

184) Find the population of a city at any time t, given that the rate of increase of population is proportional to the population at that instant and that in a period of 40 years the population increased from 3,00,000 to 4,00,000.

185) The equation of electromotive force for an electric circuit containing resistance and self inductance is E=Ri L$\frac{di}{dt}$, Where E is the electromotive force is given to the circuit, R the resistance and L, the coefficient of induction. Find the current i at time t when E = 0.

186) Suppose a person deposits 10,000 Indian rupees in a bank account at the rate of 5% per annum compounded continuously. How much money will be in his bank account 18 months later?

187) Assume that the rate at which radioactive nuclei decay is proportional to the number of such nuclei that are present in a given sample. In a certain sample 10% of the original number of radioactive nuclei have undergone disintegration in a period of 100 years. What percentage of the original radioactive nuclei will remain after 1000 years?

188) Water at temperature 100°C cools in 10 minutes to 80°C in a room temperature of 25°C.
Find
(i) The temperature of water after 20 minutes
(ii) The time when the temperature is 40°C

$$\left[log_e \frac{11}{15} = -0.3101; log_e 5 = 1.6094 \right]$$

189) At 10.00 A.M. a woman took a cup of hot instant coffee from her microwave oven and placed it on a nearby Kitchen counter to cool. At this instant the temperature of the coffee was 180° F, and 10 minutes later it was 160°F. Assume that constant temperature of the kitchen was 70°F.
(i) What was the temperature of the coffee at 10.15A.M.?
(ii) The woman likes to drink coffee when its temperature is between 130°F and 140°F.between what times should she have drunk the coffee?

190) A pot of boiling water at 100°C is removed from a stove at time t = 0 and left to cool in the kitchen. After 5 minutes, the water temperature has decreased to 80°C , and another 5 minutes later it has dropped to 65°C . Determine the temperature of the kitchen.

191) A tank initially contains 50 litres of pure water. Starting at time t = 0 a brine containing with 2 grams of dissolved salt per litre flows into the tank at the rate of 3 litres per minute. The mixture is kept uniform by stirring and the well-stirred mixture simultaneously flows out of the tank at the same rate. Find the amount of salt present in the tank at any time t > 0.

192) Suppose the amount of milk sold daily at a milk booth is distributed with a minimum of 200 litres and a maximum of 600 litres with probability density function

$$\begin{cases} k & 200 \le x \le 600 \\ 0 & otherwise \end{cases}$$

Find
(i) the value of k
(ii) the distribution function
(iii) the probability that daily sales will fall between 300 litres and 500 litres?

193) The probability density function of X is given

$$f(x) = \begin{cases} Ke^{\frac{-x}{3}} & for \quad x > 0 \\ 0 & for \quad x \le 0 \end{cases}$$

Find
(i) the value of k
(ii) the distribution function.
(iii) P(X < 3)
(iv) P(5 ≤X)
(v) P(X ≤ 4)

194) If X is the random variable with probability density functionj{x) given by,

$$f(x) = \begin{cases} x+1 & -1 \le x < 0 \\ -x+1 & 0 \le x < 1 \\ 0 & otherwise \end{cases}$$

then find
(i) the distribution function F(x)
(ii) P(-0.5 ≤X ≤ 0.5)

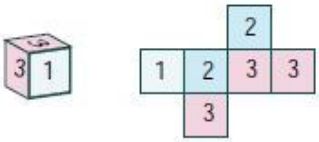
195) If X is the random variable with distribution function F(x) given by,
\(F(x)=\begin{cases} \begin{matrix} 0 & -\infty
then find (i) the probability density function f{x)
(ii) P(0.3 ≤ X ≤ 0.6)

196) A retailer purchases a certain kind of' electronic device from a manufacturer. The manufacturer ,indicates that the defective rate of the device is 5%. The inspector of the retailer randomly picks 10 items from a shipment. What is the probability that there will be
(i) at least one defective item
(ii) exactly two defective items.

197) If the probability that a fluorescent light has a useful life of at least 600 hours is 0.9, find the probabilities that among 12 such lights
(i) exactly 10 will have a useful life of at least 600 hours;
(ii) at least 11 will have a useful life of at I least 600 hours; I
(iii) at least 2 will not have a useful life of at : least 600 hours.

198) The mean and standard deviation of a binomial variate X are respectively 6 and 2. Find
(i) the probability mass function
(ii) P(X = 3)
(iii) P(X 2≤ 2).

199) A six sided die is marked ‘1’ on one face, ‘2’ on two of its faces, and ‘3’ on remaining three faces. The die is rolled twice. If X denotes the total score in two throws.
(i) Find the probability mass function.
(ii) Find the cumulative distribution function.
(iii) Find P(3 ≤ X< 6) (iv) Find P(X ≥ 4) .



200) A random variable X has the following probability mass function

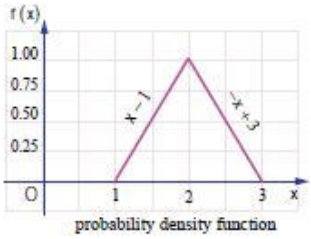
x	1	2	3	4	5	6
f(x)	k	2k	6k	5k	6k	10k

Find
(i) P(2 < X < 6)
(ii) P(2 ≤ X < 5)
(iii) P(X ≤4)
(iv) P(3 < X)

201) Find the constant C such that the function \(f(x)=\begin{cases} \begin{matrix} { Cx }^{ 2 } & 1 is a density function, and compute (i) P(1.5 < X < 3.5)

(ii) P(X ≤2)
(iii) P(3 < X) .

202) If X is the random variable with probability density functionf(x)given by,



$$f(x) = \begin{cases} x-1 & 1 \le x < 2 \\ -x+3 & 2 \le x < 3 \\ 0 & Otherwise \end{cases}$$

find (i) the distribution function F (x)
(ii) P(1.5 ≤ X ≤ 2.5)

203) The probability density function of random variable X is given by

$$f(x) = \begin{cases} k & 1 \le x \le 5 \\ 0 & otherwise \end{cases}$$ Find

(i) Distribution function
(ii) P(X < 3)
(iii) P(2 < X < 4)
(iv) P(3 ≤ X)

204) Let X be a random variable denoting the life time of an electrical equipment having probability density function

$$f(x) = \begin{cases} ke^{-2x} & forx > 0 \\ 0 & forx \le 0 \end{cases}$$

Find
(i) the value of k
(ii) Distribution function
(iii) P(X < 2)
(iv) calculate the probability that X is at least for four unit of time
(v) P(X = 3)

205) Suppose that f (x) given below represents a probability mass function

x	1	2	3	4	5	6
f(x)	c^2	$2c^2$	$3c^2$	$4c^2$	c	2c

Find
(i) the value of c
(ii) Mean and variance.

206) Two balls are chosen randomly from an urn containing 8 white and 4 black balls. Suppose that we win Rs 20 for each black ball selected and we lose Rs10 for each white ball selected. Find the expected winning amount and variance

207) A multiple choice examination has ten questions, each question has four distractors with exactly one correct answer. Suppose a student answers by guessing and if X denotes the number of correct answers, find (i) binomial distribution (ii) probability that

the student will get seven correct answers (iii) the probability of getting at least one correct answer

208) The mean and variance of a binomial variate X are respectively 2 and 1.5. Find
(i) P(X = 0)
(ii) P(X =1)
(iii) P(X ≥1)

209) On the average, 20% of the products manufactured by ABC Company are found to be defective. If we select 6 of these products at random and X denote the number of defective products find the probability that (i) two products are defective (ii) at most one product is defective (iii) at least two products are defective.

210) Identify the valid statements from the following sentences.

211) Verify
(i) closure property,
(ii) commutative property,
(iii) associative property,
(iv) existence of identity, and
(v) existence of inverse for following operation on the given set
m*n=m+n-mn; m,n ∈Z

212) Establish the equivalence property connecting the bi-conditional with conditional: $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

213) Verify
(i) closure property,
(ii) commutative property,
(iii) associative property,
(iv) existence of identity, and
(v) existence of inverse for the operation $+_5$ on Z_5 using table corresponding to addition modulo 5.

214) Verify
(i) closure property,
(ii) commutative property,
(iii) associative property,
(iv) existence of identity, and
(v) existence of inverse for the operation ×11 on a subset A = {1,3,4,5,9} of the set of remainders {0,1,2,3,4,5,6,7,8,9,10}

215) Using the equivalence property, show that $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$

216) Define an operation*on Q as follows: $a*b=\left(\frac{a+b}{2}\right)$; a,b ∈Q. Examine the closure, commutative, and associative properties satisfied by*on Q.

217) Define an operation∗ on Q as follows: $a*b=\left(\frac{a+b}{2}\right)$; a,b ∈Q. Examine the existence of identity and the existence of inverse for the operation * on Q.

218) Verify whether the following compound propositions are tautologies or contradictions or contingency
$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$

219) Let M= $\left\{\begin{pmatrix} x & x \\ x & x \end{pmatrix} : x \in R - \{0\}\right\}$ and let * be the matrix multiplication. Determine whether M is closed under $*$. If so, examine the commutative and associative properties satisfied by $*$ on M.

220) Let M= $\left\{\begin{pmatrix} x & x \\ x & x \end{pmatrix} : x \in R - \{0\}\right\}$ and let $*$ be the matrix multiplication. Determine whetherM is closed under $*$. If so, examine the existence of identity, existence of inverse properties for the operation $*$ on M.

221) Let A be Q\{1}. Define $*$ on A by x*y = x + y − xy . Is $*$ binary on A? If so, examine the commutative and associative properties satisfied by $*$ on A.

222) Check whether the statement p→(q→p) is a tautology or a contradiction without using the truth table.

223) Using truth table check whether the statements ¬(p V q) V (¬p ∧ q) and ¬p are logically equivalent.

224) Let A be Q\{1}. Define $*$ on A by x*y = x + y − xy . Is $*$ binary on A? If so, examine the existence of identity, existence of inverse properties for the operation $*$ on A.

225) Prove p⟶(q⟶r) ≡ (p ∧ q)⟶r without using truth table.

226) Prove that p→(¬q V r) ≡ ¬pV(¬qVr) using truth table.

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5 MARKS V2 CREATIVE

12th Standard

Maths

Exam Time : 01:40:00 Hrs

Total Marks : 100

119 x 5 = 595

1) If Rolle's theorem holds for f (x) = $x^3 + bx^2 + ax + 5$ on [1,3] with c= $\left(2+\frac{1}{\sqrt{3}}\right)$ find the values of a and b.

2) Find the angle of intersection of the curves $2y^2 = x^3$ and $y^2 = 32x$.

3) If f(x) = a log x + bx^2 +x has entreme values at x = - 1 and x = 2, then find a and b.

4) Prove that the semi-vertical angle of a cone of maximum volume and of given slant height is $\tan^{-1}(\sqrt{2})$.

5) Show that the curves $4x = y^2$ and 4xy = k cut at right angles if $k^2 = 512$.

6) Sand is pouring from a pipe at the rate of 12 cm^3/sec. The falling sand forms a cone on the ground in such a way that the height of the cone is always one-sixth of the radius of the base. How fast is the height of the sand cone increasing, when the height is 4 cm?

7) Gas is escaping from a spherical balloon at the rate of 900 cm^3/sec. How fast is the surface area and radius of the balloon shrinking when the radius of the balloon is 30 cm?

8) The length x of a rectangle is decreasing at the rate of 3cm/min and the width is increasing at the rate of 2cm/min. When x = 10cm, y = 6cm, find the rate of change of (i) the perimeter (ii) the area of the rectan

9) A particle moves along the curve $y = \frac{4}{3}x^3 + 5$ .Find the points on the curve at which y coordinate changes as fast as x-coordinates.

10) Find the points on the curve $y=2x^2-2x^2$ at which the tangent lines are parallel to the line y=3x-2.

11) If the curves $4x=y^2$ and 4xy=k cut at right angles show that $k^2=512$.

12) missle fired from ground level rises x metres vertically upwards in t seconds and $x = 100t - \frac{25}{2}t^2$. Find the

(i) initial velocity of the missile
(ii) the time when the height of the missile is maximum
(iii) the maximum height reached
(iv) the velocity which the missile strikes the ground.

13) Show that the equation of the normal to the curve $x=\cos^3\theta, y=a\sin^3\theta$ at 'θ' is $x\cos\theta - y\sin\theta = a\cos 2\theta$.

14) A water tank has a shape of an inverted cone with its axis vertical and vertex lower most. Its semi vertical angle is $\tan^{-1}(0.5)$. Water is poured into it at a constant rate of 5

cm3/hr. Find the rate at which the level of the water is rising at that instant when the depth of the water is 4 m.

15) Evaluate: $lim_{x \to 1} \frac{1}{x^{x-1}}$

16) If f(x)=alogx+bx^2+x has exterme at x=1, x=2 then find a and b.

17) Find the intervals of concavity and points of inflexion for f(x)=x^3-15x^2+75x-50.

18) Find the intervals for which the function f(x)=2x^2-9x^2-12x+1 is increasing or decfreasing and find the local extermems.

19) Show that the surface area of a closed cuboid with a square base and given volume is minimum, when it is a cube

20) Find the local maximum and local minimum values for f(x)=12x^2-2x^2-x^4.

21) If the curve y^2=x and xy=k are orthogonal, then prove that 8k^2-1.

22) Find the intervals of concavity and the points of inflection of f(x)=12x^2-2x^3-x^4.

23) Find the points of inflection and determine the intervals of concavity of $y = e^{-x^2}$

24) A manufacturer can sell x items at a price of rupees $\left(5 - \frac{x}{100}\right)$ each. The cost price of x items is Rs. $\left(\frac{x}{5} + 500\right)$ .Find the numbers of items he should sell to earn maximum profit.

25) Find the local maximum and local minimum values of f(x)=x^4-3x+3x^2-x.

26) Find the equation of the tangent and the normal to the curve $y = 2sin^2 3x, x = \frac{\pi}{6}$

27) If u = tan $^{-1}$ $\left(\frac{x^3+y^3}{x-y}\right)$

Prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ sin 2u.

28) Find $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y^2}$ at x = 2, y = 3 if f(x,y) = 2x^2 + 3y^2 - 2xy

29) Using differential find the approximate value of cos 61; if it is given that sin 60° = 0.86603 and 1^0 = 0.01745 radians.

30) If V = log r and r^2 = x^2 +y^2 + z^2, then prove that $\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2}{\partial z^2} = \frac{1}{r^2}$

31) If z =f(x - cy) + F (x + cy) where f and Fare any two functions and c is a constant, show that $\frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial y^2}$

32) If $u = tan^{-1}(\frac{x^2+y^2}{x-y})$,then prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{1}{2}sin2u$

33) If P=CL^{a}k^B,c > 0,$\alpha+\beta=1$, then prove that $k\frac{\partial P}{\partial k}+L\frac{\partial P}{\partial K}=P$.(Without using Euler's theorem)

34) If $u=log\sqrt{x^2+y^2+z^2}$, then prove that $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}=\frac{1}{x^2+y^2+z^2}$

35) If w=u^{2}e^v where $u=\frac{x}{y}$ and v=logx. Find $\frac{\partial w}{\partial x}$ and $\frac{\partial w}{\partial y}$

36) Verify Euler's theorem for the function $f(x,y)=\frac{1}{\sqrt{x^2+y^2}}$

37) The height of a cone is increased by k%, its semi vertical angle remaining the same. What is the approximate percentage increases in (i) T.S.A (ii) Volume assuming k is small.

38) Find $\frac{\partial w}{\partial u}, \frac{\partial w}{\partial v}$ if w=sin^{-1}(x,y) where x=u+v,y=u-v

39) If $w=x^2sin\left(\frac{x}{y}\right)+y^2cos\left(\frac{x}{y}\right)+xytan\left(\frac{x}{y}\right)$,then prove that $x\frac{\partial w}{\partial x}+y\frac{\partial w}{\partial y}=2w$

40) If

41) Find the approximate value of $\sqrt[3]{1.02}+\sqrt{1.02}$

42) Prove that $\int_0^{\frac{\pi}{2}} sin2x\ log(tan\ x)dx=0$

43) Find the area bounded by x = at^2, y = at between the ordinates corresponding to t = 1 and t = 2

44) Show that the area under the curve y = sin x and y = sin 2x between x = 0 and x = $\frac{\pi}{3}$ and x axis are as 2:3

45) Using integration, find the area of the triangle with sides y = 2x+1, y = 3x + 1 and x = 4.

46) Find the volume of the solid generated by the revolution of the loop of the curve x = t^2 y = t - $\frac{t^3}{3}$ about x-axis.

47) Find the area of the curve y^2=(x-5)^2(x-6) between
(i) x=5 and x=6
(ii) x=6 and x=7

48) Find the area of the region bounded by the curves x^2+2y^2=0 and x+3y^2=1.

49) Find the ratio of the area between the curves y=cosx and y=cos2x and x- axis from x=0 to $x=\frac{\pi}{3}$

50) Find the area bounded by the curves y=|x|-1 and y=-|x|+1

51) Find the value of 'c' for which the area bounded by the curve $y=8x^2-x^5$,the lines x=1,x=c and x-axis $\frac{16}{3}$

52) Find the area of the loop of the curve $3ay^2=x(x-a)^2$

53) Find the area of the region common to the circle $x^2+y^2=16$ and the parabola $y^2=6x$.

54) Prove that $\int_0^{\frac{\pi}{2}} sin2xlog(tanx)dx$

55) Find the area bounded by $x=at^2$,y=2at between the ordinates corresponding to t = 1 and t = 2.

56) Using integration find the area of the triangle with sides y=2x+1,y=3x+1 and x=4

57) Find the area bounded by the curve $y=xe^x$ and $y=xe^{-x}$ and the line x=1.

58) Find the area of the region bounded by $y=e^x$ and $y=e^{-x}$ and the;line x=1.

59) Find the area of the region bounded by $a^2y^2=a^2(a^2-x^2)$

60) Find the area of the region enclosed by the two circles $x^2+y^2=1$ and $(x-1)^2+y^2=1$.

61) AOB is the positive quadrant of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ where OA=a and OB=b.Find the area between the arc AB and chord AB of the elipse.

62) Show that the ratio of the area under the curve y=sinx and y=sin2x between x=0 and $x=\frac{\pi}{3}$ and x- axis are as 2 : 3.

63) Find volume of the solid generated by the revolution of the loop of the curve $x=t^2$, $y=t-\frac{t^3}{3}$ about the x-axis.

64) Find the area bounded by the curve $y2(2a-x)=x^2$ and the line x=2a.

65) Find the area bounded by the curve xy2=a2(a-x) and the y-axis

66) Find the area enclosed by the parabolas $5x^2-y=0$ and $2x^2-y+9=0$.

67) Solve: $(1 + e^{2x})\, dy + (1 + y^2)e^x\, dx = 0$ when y(0) = 1

68) A population grows at the rate of 2% per year. How long does it take for the population to double?

69) The surface area of a balloon being inflated changes at a constant rate. If initially, its radius 3 units and after 2 seconds it is 5 units, find the radius after t seconds.

70) The slope of the tangent at p(x,y) on the curve is $-\left(\frac{y+3}{x+2}\right)$. If the curve passes through the origin, find the equation of the curve.

71) Solve: $\frac{dy}{dx} = (3x+2y+1)^2$

72) Sovle $(x+2)\frac{dy}{dx} = x2+4x-9$ .Also find the domain of the function.

73) Solve : $\frac{dy}{dx} = \left(sin^2xcos^2x + xe^x\right)dx$

74) Solve : $e^{\frac{dy}{dx}} = x + 1, y(0) = 5$

75) Solve : $\left(1 + x^2\right)\frac{dy}{dx} - x = 2tan^{-1}x$

76) Solve : $\left(x^2 + x^2 + x + 1\right)\frac{dy}{dx} = 2x^2 + x$

77) Solve : $(1+y^2)(1+logx)dx+xdy=0$, given that x=1,y=1.

78) In a bank principal increases at the rate of 5% per year. In how many years Rs.1000 doubled itself.

79) Sovle : $(x+y+1)^2dy=dx, y(-1)=0$

80) Solve :$x^2dy+y(x+y)dx=0$ given that y=1 when x=1.

81) Solve :$(x^2+xy)dy=(x^2+y^2)dx$

82) Solve :$x\frac{dy}{dx}sin\left(\frac{y}{x}\right) + x - ysin\left(y\frac{y}{x}\right) = , y(1) = \frac{\pi}{2}$

83) Solve : $2x^2\left(\frac{dy}{dx}\right) - 2xy + y^2 = 0, y(e) = e$

84) Solve : $cos^2x\frac{dy}{dx} + y = tanx(0 \le x \le \frac{\pi}{2})$

85) Solve : $\frac{dy}{dx} = -\frac{x + ycos}{1 + sinx}$ .Also find the domain of the function.

86) Solve : (x-siny)dy+tanydx=0,y(0)=0

87) It is given that the rate at which some bacteria multiply is proportional to the instantaneous number present. If the original number of bacteria doubles in two hours, in how many hours will it be five times.

88) A thermometer reading 80°

89) Water at temperature 100℃ cools in 5 minutes to 80℃ in a room of temperature 30℃. Find (i) the temperature of water after 10 minutes. (ii) the time when the temperature is 40℃.

90) Four bad oranges are accidentally mixed with sixteen good ones. Find the probability distribution of bad oranges in a draw of two oranges. Also find the mean, variance and standard deviation of the distribution.

91) Two cards are drawn simultaneously from a well shuffled pack of 52 cards. Find the mean and variance of the number of red cards.

92) From a lot of 10 items containing 3 defective items, 4 items are drawn at random. Find the mean and variance of the number of defective items drawn.

93) Suppose that a pair of fair dice are tossed and let random variable X denote the sum of outcomes. Find the mean and variance of the probability distribution of X.

94) A box contains 4 red and 5 black marbles. Find the probability distribution of the red marbles in a random draw of three marbles. Also find the mean and standard deviation of the distribution

95) The mean and variance of a binomial distribution are 4 and 2 respectively. Find the probability of at least 6 success.

96) If X is a binomial random variable with mean 4 and variance 2 find $P(|X-2| \leq 2)$

97) If the sum and the product of the mean and variance of a binomial distribution are 1.8 and 0.8 respectively, find the probability distribution and the probability of at least one success.

98) The difference between the mean and variance of a binomial distribution is 1 and the difference of their squares is 11. Find the distribution.

99) If the sum of the mean and variance of a binomial distribution is 15 and the sum of their squares is 117, then find the distribution

100) In a business venture a man can make a profit of Rs.2,000 with a probability of 0.4 or have a loss of Rs.1,000 with a probability of 0.6. What is his expectation, variance and S.D of profit?

101)

The p.d.f of a continuous random variable X is $f(x) = \begin{cases} a + bx^2, 0 \leq x \leq 1 \\ 0, otherwise \end{cases}$ where a and b are some constants. Find

(a) a and b if $E(x) = \frac{3}{5}$

(b) Var (X)

102) Ten coins are tossed simultaneously. What is the probability of getting (a) exactly 6 heads (b) at least 6 heads (c) at most 6 heads?

103) The probability that an engineering college student will graduate is 0.3. Find the probability that out of 6 students (i) none (ii) one (iii) at least one will graduate.

104) The probability distribution of a random variable X is given by

X	0	1	2	3
P(X)	0.1	0.3	0.5	0.1

If $Y=X^2+3X$, find the mean and the variance of Y.

105) Show that -(p ∧ q) ≡ (~p) v (~q).

106) Construct the truth table for (p ∧ q) v r.

107) Verify (p ∧ -p) ∧ (~q ∧ p) is a tautlogy, contradiction or contingency.

108) Let S be a non-empty set and 0 be a binary operation on s defined by x 0 y = x; x, Y ∈ s. Determine whether 0 is commutative and association.

109) In (N, *) where * is defined by x * y = max (x, y) check the closure axion and identity anion.

110) M be the set of all 2X2 matrices each of whose determinant value is 1.Show that M satisfies the closure, associative, identity and inverse axioms under multiplication.

111) $M = \left\{ \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} / \alpha \neq 0 \; and \; \alpha \epsilon R \right\}$ Show that M satisfies the closure, associative, inverse, identity and commutative axioms under multiplication.

112) Show that Z7-{[0]} satisfies the closure, associative, identity, inverse and commutative axioms under multiplication modulo 5.

113) Let Q, be the set of all nonzero rational numbers and k is a nonzero fixed rational number and * be a binary operation defined as a*b=kab. Show that (Q,*) satisfies closure, associative, inverse and commutative properties.

114) Show that $(2018)^{2017}+(2020)^{2017}\equiv 0(\text{mod}2019)$.

115) Show that (Z_7-[0],X7) satisfies closure, identity, inverse and commutative properties.

116) Prove by using truth table $\sim (pV(qVr) \equiv (\sim p) \wedge (\sim q \wedge \sim r)$

117) Prove without using the truth table $\sim (pV(qVr)) \equiv (\sim pV \sim q)V(-r)$

118) Examine whether there is a nonempty subset S of the set of real numbers such that it satisfies closure, associative, identity and inverse properties under a binary operation * defined as a*b=k where k is a fixed real number.

119) Prove that $(2019)^{10}+(2020)^{10}\equiv 1025(\text{mod } 2018)$

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